Non-Formalized Research Frontiers for the Fabius Function

Six thematic syntheses awaiting Lean verification

Research quarantine for the `ProveIt` Fabius development
Eleven source notebooks in six syntheses plus a post-audit gap register

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Abstract

This volume consolidates every mathematical document formerly stored in a subdirectory of `Analysis/FabiusFunction/docs/non-formalized-research-frontiers/`. Eleven source notebooks containing natural-language proofs, symbolic derivations, numerical checks, conjectural extensions, and explanations built on top of the current Lean library are organized into six thematic syntheses followed by a post-audit gap register. They remain quarantined so that no unformalized claim can be mistaken for part of the verified primary exposition or lost while awaiting a machine-checked proof. The post-consolidation audit also records the newly formalized generic naturality, conditional affine-recursion, and exact-support API for weighted uniform series and the named geometric-q weights, series, probability law, recurrence, and reflection. Every norm-summable real weighted law with a nonzero coefficient is absolutely continuous with respect to Lebesgue measure and has null singletons. For every $|q| < 1$ the geometric law has those properties and support equal to its exact series range, and for $0 \le q < 1$ that support is $[0, 1]$. The formal API now also names its CDF, proves CDF continuity and reflection for $|q| < 1$, the exterior tails for $0 \le q < 1$, and, for $0 < q < 1$, the conditioning identities, an explicit compactly supported Radon–Nikodym density, and real-space C^∞ regularity. Its CDF and density have exact half-base bridges to the bounded Fabius function and the affine Rvachev bump. The negative- and zero-parameter density interpretation, the general-q Fourier route, and the further centered, shape, moment, transform, derivative-hierarchy, endpoint-flatness, and fixed-point layers remain in the frontier on the precise boundary recorded below.

Closely overlapping presentations have been merged rather than concatenated. Each topic uses the strongest or most complete source as its backbone, imports independent proofs, sharper constants, examples, warnings, and open obligations from the companion source, and removes repeated definitions and theorem statements. The original paths, historically recorded SHA-256 values, and repository blob identifiers remain in provenance banners and the repository ledger. Where a printed SHA-256 value cannot be reproduced from a reachable blob, it is marked unverified rather than presented as authenticated provenance. Status distinctions, citations, tables, algorithms, plots, and genuine alternative proof mechanisms are retained.

Formalization boundary. Nothing is promoted merely by appearing in this volume. A claim may enter `docs/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex` only after an exact Lean theorem has been checked for the full hypotheses and conclusion, the declaration and module have been recorded, and the corresponding frontier obligation has been dispositioned. A related definition, a weaker theorem, numerical agreement, or a paper citation is not sufficient. Some sections quote formalized inputs; that does not formalize their later deductions.

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Consolidation map and reading conventions

Each part below is a thematic synthesis. Its provenance banner names every source notebook that contributed to it and records the historically supplied SHA-256 metadata and repository blob evidence available for that source. The final register records claims removed by the primary exposition's exact-counterpart audit. Namespaced labels and bibliography keys are retained for stable cross-references, but a prefix no longer means that the surrounding prose is an untouched source block.

Notation has been standardized at the collisions that affect mathematical meaning. In particular, App_n denotes the dyadic Appell family while A_j denotes the saddle logarithmic coefficients, and cardinal weights use $\pi_{n,k}$ rather than the λ-symbol reserved for lattice centers and the lower-Lambert phase. The final saddle part keeps its historically local jet symbols p_n, d_n, E_m, but states their crosswalk explicitly so that its d_n cannot be confused with the half moments in the probability and dyadic parts. Each Fourier chapter states its exponential convention before formulas involving 2π.

Part	Principal subject	Source integration
Probability and generic engines	Probability/Laplace bridges, analytic-density obstruction, weighted paths, Toeplitz convergence, formal exp/log, and Gaussian estimates	Unique Integration material retained; its duplicate inverse and Fabius-specific saddle chapters are cross-referenced to the final part.
Repeated integration	Finite box splines, exact derivative thresholds, error kernels, extremizers, and weak-test consequences	Sharper second notebook is the backbone; historical provenance, operator, L^p, weak-error, and comparison results are imported from the first.
Exact dyadic web	Moments, Taylor reduction, Appell compression, determinant, transforms, q-binomial rows, stabilization, and Richardson limits	Later Dyadic Web is the backbone; unique Appell, simplex, determinant, prefix-block, filter, and Fourier routes are imported from q-Connections.
Dyadic endpoint asymptotics	Exact negative-Laplace decomposition, Lambert phase, vertical saddle, fixed-numerator rays, and arithmetic consequences	Later formulae-to-asymptotics notebook is the backbone; Mellin, coefficient-extraction, Bromwich, squeeze, and numerical material is imported from the bridge.
Thue–Morse convergence	Exact discrete bridge, deconvolution, sharp/all-order rates, finite certificates, endpoint identities, and acceleration	Sharper plural-rate notebook is the backbone; finite elementary bounds, zero runs, binary phase, and endpoint formulas are imported from the earlier report.
Inverse and small-argument saddle	Total inverse, exact regularity and non-elementarity, endpoint inversion, closed jets, corrections, rigorous all orders, Fourier amplitudes, and numerics	Inverse/Saddle is the backbone; phase-polynomial and right-endpoint transfers, exact coefficient means, higher explicit coefficients, Fourier amplitudes, numerical checks, traps, and the open ledger are imported from Small Argument.
Primary-exposition gap register	Claims removed by exact-counterpart audits	Post-consolidation disposition record; every candidate remains open until an exact current Lean theorem is audited.

Lean object crosswalk—inputs, not certification

Object	Representative Lean name	Principal module family
Bounded Fabius function	`fabiusReal`, `IsFabius`	`Basic.lean`, `PaperStatements.lean`
Rvachev up-function	`rvachevUp`	`Basic.lean`, `Differential.lean`
Signed global extension	`extendedFabius`	`GlobalExtension.lean`
Fractional Volterra calculus	`fractionalVolterra`, `fractionalVolterra_add`, `fractionalVolterra_affine`, `fractionalVolterra_add_one_deriv`	`FractionalVolterra.lean`, `FractionalVolterraSemigroup.lean`, `FractionalVolterraCalculus.lean`
Fractional Fabius–Rvachev shifts	`fractionalVolterra_add_one_extendedFabius_of_nonneg`, `fractionalVolterra_add_one_fabiusReal`, `fractionalVolterra_add_one_rvachevUp`	`FabiusFractionalVolterra.lean`
Pointwise analytic finite filters	`finiteAnalyticSeriesFilter`, `geometricAnalyticSeriesFilter`, `geometricLagrangeAnalyticSeriesFilter_shifted`	`AnalyticSeriesFilter.lean`
Exact dyadic evaluator	`fabiusDyadicValue`	`Arithmetic.lean`, `PaperStatements.lean`
Binary and q-binomial series	`fabiusReductionSum` and global q-series declarations	`TaylorReduction.lean`, `FabiusGlobalQBinomialSeries.lean`
Thue–Morse grid samples	`correctedPrefixGridSample`	`ThueMorseApproximation.lean`
Generic weighted uniform series and laws	`uniformProduct_map_of_ae_head_tail_recursion`, `weightedUniformSeries_map`, `weightedUniformDistribution_split_of_shift_eq_map`, `weightedUniformDistribution_absolutelyContinuous`, `weightedUniformDistribution_nullSingletonClass`, `weightedUniformDistribution_support_eq_range`, `weightedUniformDistribution_support_eq_Icc_min_max`	`WeightedUniformSeries.lean`, `WeightedUniformDistribution.lean`, `WeightedUniformSupport.lean`, `ProbabilityRepresentation.lean`

Object	Representative Lean name	Principal module family
Named geometric uniform law	`geometricUniformWeight`, `geometricUniformDistribution_selfSimilar`, `geometricUniformDistribution_absolutelyContinuous`, `geometricUniformDistribution_nullSingletonClass`, `geometricUniformDistribution_support_eq_range`, `geometricUniformDistribution_support_eq_Icc`, `weightedSumDistribution_support_eq_Icc`	`GeometricUniformLaw.lean`, `ProbabilityRepresentation.lean`
Generic continuous-CDF bridge	`continuous_cdf_of_nullSingleton`, `cdf_reflection_sub`, `measure_eq_withDensity_of_cdf_hasDerivAt`	`ContinuousCDF.lean`
Geometric uniform CDF and density	`geometricUniformCDF`, `geometricUniformDensity`, `geometricUniformCDF_eq_intervalIntegral`, `geometricUniformDistribution_eq_withDensity`, `contDiff_geometricUniformCDF`, `contDiff_geometricUniformDensity`	`GeometricUniformCDF.lean`
Half-base CDF and density bridges	`weightedSumCDF_eq_geometricUniformCDF_one_half`, `geometricUniformCDF_one_half_eq_fabiusReal`, `geometricUniformDensity_one_half_eq_rvachevUp`	`ProbabilityRepresentation.lean`
Weighted survival, lower-CDF, fractional, and inverse layer cake	`integral_smul_setIntegral_subgraph`, `integral_smul_setIntegral_supergraph`, `intervalIntegral_survival_smul_eq_integral_clamp`, `intervalIntegral_cdf_smul_eq_integral_clamp`, `fractionalVolterra_measureReal_Iic_eq_integral_posPart`, `fractionalVolterra_fabiusReal_eq_integral_posPart`, `intervalIntegral_smul_comp_fabiusInv`, `intervalIntegral_fabiusInv_eq_one_half`	`SubgraphFubini.lean`, `SurvivalLayerCake.lean`, `CDFLayerCake.lean`, `FractionalCDFLayerCake.lean`, `FabiusFractionalIntegral.lean`, `InverseLayerCake.lean`, `ProbabilityLaplaceMoments.lean`

Object	Representative Lean name	Principal module family
Fractional affine/order-raising calculus and Fabius–Rvachev shifts	fractionalVolterra_affine, fractionalVolterra_comp_affine, fractionalVolterra_add_one_deriv, fractionalVolterra_add_one_extendedFabius_of_nonneg, fractionalVolterra_add_one_fabiusReal, fractionalVolterra_add_one_rvachevUp	FractionalVolterraCalculus.lean, FabiusFractionalVolterra.lean
Absolutely-continuous inverse-pair calculus	intervalIntegral_deriv_mul_comp_add_comp_mul_deriv_of_lt_iff_lt, intervalIntegral_deriv_mul_fabiusReal_add_fabiusInv_mul_deriv, intervalIntegral_fabiusReal_add_fabiusInv_eq_one	InversePairIntegral.lean
Finite-node and geometric formal-series filters	finitePowerSeriesFilter, coeff_finitePowerSeriesFilter, finitePowerSeriesFilter_rescale, map_finitePowerSeriesFilter, coeff_finitePowerSeriesFilter_of_exact_of_le, geometricSeriesFilter, coeff_geometricSeriesFilter_of_exact, geometricSeriesFilter_eq_residual_mk, geometricLagrangeSeriesFilter_eq_residual_mk	FinitePowerSeriesFilter.lean, GeometricPowerSeriesFilter.lean
Unconditionally summable analytic-series filters	finiteAnalyticSeriesFilter, geometricAnalyticSeriesFilter, finiteAnalyticSeriesFilter_eq_tsum, finiteAnalyticSeriesFilter_eq_head_add_tail_of_exact, geometricAnalyticSeriesFilter_eq_constant_add_gaussian_tsum, geometricLagrangeAnalyticSeriesFilter_shifted	AnalyticSeriesFilter.lean

Object	Representative Lean name	Principal module family
Formal implicit roots and nonmonic Hensel lifting	`FormalImplicitRoot.exists_isRoot_sub_mem`, `FormalImplicitRoot.eq_of_isRoot_of_sub_mem`, `FormalImplicitRoot.existsUnique_isRoot_sub_mem`, `PowerSeries.Implicit.existsUnique_isRoot_constantCoeff`, `PowerSeries.Implicit.existsUnique_zeroConstant_root`, `PowerSeries.Implicit.root`, `PowerSeries.Implicit.constantCoeff_root`, `PowerSeries.Implicit.eval_root`, `PowerSeries.Implicit.eq_root`	`ImplicitPowerSeries.lean`
Totalized inverse	`fabiusInv`, `deriv_deriv_fabiusInv_eq_zero_iff`, `tendsto_deriv_fabiusInv_atTop_at_zero_right`, `fabiusInv_differentiableAt_iff`, `fabiusInv_contDiffAt_infty_iff`, `fabiusInv_analyticAt_iff`	`FabiusInverse.lean`, `InverseNotElementary.lean`
Lower-Lambert coordinate	`fabiusLambertPhase`	`FabiusLambertSaddle.lean`
All-orders saddle expansion	saddle mass/log coefficient declarations	`FabiusFullAsymptoticExpansion.lean` and related modules

Explicit nonclaims. The crosswalk identifies formal inputs and nearby APIs; it is not a claim that every formula in the corresponding part is formalized. In particular, the sharp quantitative Thue–Morse rates, repeated-integration norm identities, inverse endpoint asymptotic, general ordered-composition coefficient formulas, Fourier-amplitude estimates, and any claim marked derived, numerical, conjectural, or open remain in this frontier until an exact Lean declaration is audited. The named weights $(1-q)q^n$, their law, measure recurrence, reflection, support equal to the series range for $|q|<1$, and exact support $[0,1]$ for $0 \le q < 1$, together with absolute continuity and null singletons throughout $|q|<1$, are now exact formal inputs. So are the named CDF, its continuity and reflection for $|q|<1$, its tails for $0 \le q < 1$, and, for $0 < q < 1$, its refinement integrals, explicit compactly supported density, measure-level Radon–Nikodym identity, and real-space C^∞ regularity. What remains here is the density interpretation at $q = 0$ and at negative q, the general-q characteristic-function product, rapid decay and Fourier inversion route, named centered packaging, a closed higher-derivative hierarchy and endpoint flatness wrappers, shape theorems, transforms, moments/cumulants, and uniqueness of the affine measure fixed point.

Part I
Probability, Discrete, and Generic Asymptotic Engines

Source provenance and status. Former path: `Fabius_Integration_Research_Frontiers/Fabius_Integration_Research_Frontiers.tex`. This broad mixed-status input supplied probability bridges, discrete engines, generic saddle machinery, and an earlier inverse presentation. The inverse and Fabius-specific saddle material now live only in part VI; this part retains the reusable engines and its source-specific proofs.

Historically recorded source SHA-256: `21222ae5a8c64cf556dac562fd66943ae0b6ed881408e23e37edfa9113bdecbd`. The forensic audit did not reproduce that value from a reachable whole-file blob; the authoritative archived standalone source is Git blob `c450f8c7ab2764b02759c5295a7ae1f909c43fee`.

Topic abstract

The long article *The Fabius Function and Rvachev's Up-Function* gives a broad human-readable account of the formal development in `ProveIt`. Its breadth is also its unavoidable weakness: several new branches of the Lean library postdate the article, and several generic proof engines are named only in the module guide because they have no counterpart in the exposition. This companion supplies those missing parts.

The opening audit records a formalized redundancy in the original compact-support characterization: positivity of its dilation parameter follows from the remaining hypotheses. The analytic-locus chapter then develops the generic elementary-function obstruction and applies it to the forward Fabius family and its total inverse. The probability chapters supply the measure-theoretic bridges between the product law, its survival function, tilted Laplace moments, finite atomic laws, and half-weighted step approximants, together with the post-snapshot continuous-linear naturality, conditional operator/scalar affine-recursion, and real absolute-continuity APIs for arbitrary weighted uniform laws and the named geometric-q specialization. Two reusable discrete engines follow: weighted path sums for triangular recurrences and a finite-row Toeplitz convergence theorem. Finally, the part exposes the generic machinery used by the all-orders saddle proof: parameter-dependent Poincaré expansions, recursive exponential and logarithmic coefficients, exact finite polynomial remainders, Gaussian contraction, and uniform polynomial tails. The inverse theory, closed Fabius jets, explicit corrections, and contour assembly are cross-referenced to their single canonical treatment in part VI.

The purpose is complementarity rather than repetition. Definitions and headline theorems already proved in detail in the main article are recalled only when they are needed as input. Every substantial assertion below is either a direct human-readable rendering of a current Lean declaration, a proof-level explanation of a generic module that the main article explicitly omits, or a clearly identified elementary corollary of those declarations.

Formalization boundary. This is a research-frontier document. It preserves the source draft used during integration, including exact Lean-backed results, proof sketches, ordinary-analysis corollaries, stronger generalizations, and formulas that still lack exact declarations. Its presence under `non-formalized-research-frontiers` is intentional: no mathematical assertion in this notebook should be attributed to the formal library merely because a related definition or weaker theorem exists. The primary exposition contains the audited, declaration-backed subset. Among the obligations retained here are generic moment-matrix consequences, arbitrary-family Toeplitz wording, generic Gaussian coefficient bounds, and the geometric- q Fourier, centered-family, higher-derivative, endpoint-flatness, shape, transform, moment/cumulant, and fixed-point-uniqueness layers beyond the formalized CDF and positive-ratio smooth-density API. The explicit density's interpretation at zero and negative parameters is also not supplied by that API. The inverse endpoint and Fabius-specific coefficient obligations are tracked in part VI.

1 Scope, conventions, and the audit boundary

Let $F : \mathbb{R} \to [0,1]$ denote the bounded Fabius function, totalized in the usual way by $F(x) = 0$ for $x \le 0$ and $F(x) = 1$ for $x \ge 1$. Let

$$\mathrm{up}(x) = F(1 - |x|) \tag{1}$$

be Rvachev's compactly supported up-function, and let $\widetilde{F}$ denote the signed global extension used in the Lean development. On $[0,1]$ the basic identities are

$$F(1-x) = 1 - F(x), \qquad F'(x) = 2\,\mathrm{up}(2x-1), \qquad F'(x) = 2F(2x) \quad (0 < x < \tfrac{1}{2}). \tag{2}$$

These are inputs here, not topics to be reproved.

The current formal library also packages their exact curvature consequence:

$$F''(x) = 8\big(\mathrm{up}(4x-1) - \mathrm{up}(4x-3)\big), \tag{3}$$

with $F'' > 0$ exactly on $(0, 1/2)$, $F'' < 0$ exactly on $(1/2, 1)$, and $F'' = 0$ exactly off $(0,1)$ or at $1/2$. These are `deriv_deriv_fabiusReal`, `deriv_deriv_fabiusReal_pos_iff`, `deriv_deriv_fabiusReal_neg_iff`, and `deriv_deriv_fabiusReal_eq_zero_iff`.

The repository snapshot underlying this supplement is the following commit on the `codex/fabius-both-papers` branch, dated 25 August 2026.

`eaea1b9afe2b7ff0b7dd880bd1710e303dec6d80`.

An earlier version of the main article left four clusters to its module guide: Gaussian contraction and Gaussian tails, the formal saddle exponential and logarithm algebra, the abstract Toeplitz row lemma, and the generic weighted-path solver. Those four clusters receive complete mathematical chapters below. The non-elementarity development is also expanded here, while the inverse API is handed to part VI instead of being duplicated. The current snapshot also proves that positivity of the dilation parameter in the original compact-support characterization is redundant. Finally, several bridges are present only as one- or two-sentence transitions; they are expanded here into proof-level arguments.

1.1 Three levels of coverage

To keep the audit precise, each topic belongs to one of the following levels.

Status	Meaning in this companion
Expanded formal input	A current Lean module whose exact declarations are inputs to a longer proof-oriented or frontier discussion here: principally `ElementaryFunction.lean`, `NotElementary.lean`, `InverseNotElementary.lean`, `ProbabilityLaplaceMoments.lean`, and `FractionalVolterra.lean`, together with `FractionalVolterraSemigroup.lean`, `FractionalVolterraCalculus.lean`, `FabiusFractionalVolterra.lean`, `CDFLayerCake.lean`, `FractionalCDFLayerCake.lean`, `FabiusFractionalIntegral.lean`, `InversePairIntegral.lean`, `FinitePowerSeriesFilter.lean`, and `GeometricPowerSeriesFilter.lean`, together with `AnalyticSeriesFilter.lean`, `ThueMorseComplexProductBridge.lean`, `CauchyTransform.lean`, `CauchyCDF.lean`, and `CauchyRenormalization.lean`, together with `ContinuousCDF.lean`, `GeometricUniformCDF.lean`, and `ImplicitPowerSeries.lean`.
Missing engine	A generic theorem family that the main article deliberately leaves in the module guide: path sums, Toeplitz rows, formal coefficient recurrences, finite remainders, and Gaussian polynomial estimates.
Compressed bridge	A statement whose endpoints occur in the main article but whose intermediate measure theory, topology, or asymptotic algebra is suppressed.

1.2 Notation that must not collide

The exact half moments are denoted by d_n, following the main article. The bounded saddle exponent jets are instead denoted by

$$Z_n(t), \tag{4}$$

although their Lean declaration is `negativeLaplaceBoundedExponentJet`. This avoids using the same letter for two wholly different sequences. We write $L = \log 2$, Ψ for the smooth one-periodic correction in the negative-Laplace exponent, and $\lambda(x)$ for the exact lower-Lambert phase at small positive x.

Formal boundary convention. When a statement is a theorem in the current Lean library, the corresponding declaration is named in the margin of the discussion or in the concordance in section 5.9. When a statement is an immediate classical calculus corollary but is not presently packaged as a declaration, it is called a *corollary in ordinary analysis.* The forward and inverse second-derivative profiles, midpoint inverse jet, strict closed-half shape, and endpoint interior-derivative limits are now packaged formally; the distinction remains relevant for general higher inverse derivatives and for the transfer of the highest-jet coefficient from mass coefficients to logarithmic coefficients.

1.3 The positivity hypothesis in the original characterization is redundant

The compact-support characterization used in the first Arias de Reyna paper [3] assumes a smooth function $\phi : \mathbb{R} \to \mathbb{R}$, a real dilation parameter k, and

$$\operatorname{tsupp} \phi = [-1, 1], \qquad \phi(x) > 0 \quad (-1 < x < 1), \qquad \phi(0) = 1, \tag{5}$$

together with the derivative law

$$\phi'(x) = k\big(\phi(2x+1) - \phi(2x-1)\big). \tag{6}$$

The paper states $k > 0$ as a hypothesis. In the current formal development that field is retained for source fidelity, but a smart constructor proves that its positivity follows from the other assumptions.

Theorem 1.1 (Redundancy of scale positivity)**.** *Suppose $\phi \in C^\infty(\mathbb{R})$ satisfies (5) and (6) for some $k \in \mathbb{R}$. Then $k > 0$. Consequently these data determine an instance of the full original-characterization predicate without a separate positivity proof.*

Proof. The support identity and continuity imply $\phi(-1) = 0$: the function vanishes on $(-\infty, -1)$, and -1 is a limit point of that interval. The mean-value theorem on $[-1, 0]$ therefore gives a point $c \in (-1, 0)$ such that

$$\phi'(c) = \frac{\phi(0) - \phi(-1)}{0 - (-1)} = 1. \tag{7}$$

Now $2c - 1 < -1$, so the support condition gives $\phi(2c - 1) = 0$, whereas $-1 < 2c + 1 < 1$, so interior positivity gives $\phi(2c + 1) > 0$. Evaluating (6) at c yields

$$1 = k\phi(2c + 1). \tag{8}$$

The second factor is positive, hence $k > 0$. □

This is `IsOriginalFabius.mk_of_derivative_law`. The proof is small, but it clarifies the logical content of the original theorem: the sign of the scale is forced before the later Fourier argument identifies its exact value $k = 2$.

2 The elementary-function obstruction and inverse handoff

2.1 Canonical inverse treatment

The totalized inverse—written I in this source's audit tables and G in the canonical chapter—is treated once, in the combined inverse-and-saddle synthesis of part VI. That treatment includes the order isomorphism on $[0, 1]$, the implementation through `fabiusOrderIso`, `fabiusInv`, and `Set.IccExtend`, the four-way comparison calculus, reflection and exact dyadic inversion, transported flatness bounds, endpoint steepness, reciprocal first and second derivatives, strict closed-half curvature, and the exact differentiability, C^∞, and analytic loci. Keeping those results in a single later chapter avoids presenting the same inverse package twice; the present part retains only the generic analytic-locus obstruction that is reused for several function classes.

2.2 Elementary functions through their analytic locus

The non-elementarity proof does not attempt differential algebra or Liouville theory. It uses a topological-analytic invariant that is both stronger and easier to formalize: every one-variable elementary function is analytic on a dense open set, whereas the Fabius family is nonanalytic at every point of the relevant interval.

2.2.1 The exact formal class

Definition 2.1 (The class Elem)**.** Let Elem be the smallest class of total functions $\mathbb{R} \to \mathbb{R}$ containing all constant functions and the identity and closed under

$$+, \quad \cdot, \quad -, \quad (\cdot)^{-1}, \quad f \mapsto f^r \ (r \in \mathbb{R}), \quad \exp,\ \log,\ \sin,\ \cos,\ \arcsin,\ \arctan, \tag{9}$$

where every unary operation is applied pointwise.

This is the inductive predicate `IsElementary`. Composition is deliberately not a constructor.

Proposition 2.2 (Closure derived from the generators)**.** *The class* Elem *is closed under composition. It is consequently closed under subtraction, division, natural powers, polynomial and rational functions, square roots, absolute values, tangent, hyperbolic functions,* arccos, arsinh, *and signed odd roots.*

Proof. Induct on the construction of the outer function g. Precomposing a constant or the identity with an elementary f is elementary. Every remaining constructor commutes with precomposition: for example $(g_1 + g_2) \circ f = (g_1 \circ f) + (g_2 \circ f)$, and similarly for the unary generators. This proves closure under composition. The remaining operations are identities in the generated algebra. For example

$$|f| = \sqrt{f^2}, \qquad \tan f = \frac{\sin f}{\cos f}, \qquad \sinh f = \frac{\mathrm{e}^f - \mathrm{e}^{-f}}{2}.$$

For an odd positive integer n, the signed root may be written

$$\frac{f}{|f|}|f|^{1/n},$$

with value zero at $f = 0$ under the total reciprocal convention. □

Remark 2.3 (Totalization makes the theorem stronger). Mathlib treats reciprocal, logarithm, real power, and inverse trigonometric functions as total maps on $\mathbb{R}$; outside their classical domains they receive specified extension values. Thus theorem 2.1 is at least as large as the class represented by classical expressions on open domains. Proving that F does not belong to this larger class is a stronger statement, not a domain-convention loophole.

2.2.2 Analytic loci

For a function $f : \mathbb{R} \to \mathbb{R}$, define

$$\mathcal{A}(f) = \{x \in \mathbb{R} : f \text{ is real analytic at } x\}. \tag{10}$$

Analyticity is local, hence $\mathcal{A}(f)$ is open for every f. The substantive theorem is density for elementary functions.

The only difficult induction step is composition with a unary generator that has finitely many exceptional values. The following lemma isolates it.

Lemma 2.4 (Finite-bad-value composition principle)**.** *Let $f, g : \mathbb{R} \to \mathbb{R}$. Suppose that $\mathcal{A}(f)$ is dense and that there is a finite set $B \subset \mathbb{R}$ such that g is analytic at every point of $\mathbb{R} \setminus B$. Then $\mathcal{A}(g \circ f)$ is dense.*

Proof. Let U be a nonempty open interval. Because $\mathcal{A}(f)$ is dense and open, choose $x_0 \in U \cap \mathcal{A}(f)$. We prove that every punctured neighborhood of x_0 contains a point where $g \circ f$ is analytic.

Fix $c \in B$. The analytic function $f - c$ near x_0 has two possibilities.

1. It vanishes identically on a neighborhood of x_0. Then f is locally constant equal to c, so $g \circ f$ is locally constant and therefore analytic at x_0, regardless of whether g is analytic at c.
2. It is not locally zero. The isolated-zero theorem gives a punctured neighborhood of x_0 on which $f(x) \neq c$.

If the first alternative occurs for any $c \in B$, we are done. Otherwise intersect the finitely many punctured neighborhoods. The intersection still contains points arbitrarily close to x_0; intersect once more with the open set $U \cap \mathcal{A}(f)$. At any resulting point x, the value $f(x)$ avoids B, so g is analytic at $f(x)$ and f is analytic at x. The analytic chain rule makes $g \circ f$ analytic at x. Thus every nonempty open U meets $\mathcal{A}(g \circ f)$. □

Theorem 2.5 (Dense open analytic locus)**.** *If $f \in \mathsf{Elem}$, then $\mathcal{A}(f)$ is dense and open. Equivalently, the nonanalytic locus*

$$\mathbb{R} \setminus \mathcal{A}(f) \tag{11}$$

is closed and nowhere dense.

Proof. Openness is true for every analytic locus. Density follows by induction on the elementary construction.

Constants and the identity are analytic everywhere. For sums and products, the intersection of the two dense open analytic loci is dense and each point in the intersection is analytic for the result. Negation is immediate. For a unary generator, apply theorem 2.4 with the following exceptional sets:

Generator	Exceptional values	Analytic elsewhere
$t \mapsto t^{-1}$	$\{0\}$	yes
$t \mapsto t^r$	$\{0\}$	yes for every fixed $r \in \mathbb{R}$
$\log t$	$\{0\}$	yes under the total real-log convention
$\arcsin t$	$\{-1, 1\}$	yes
$\exp t, \sin t, \cos t, \arctan t$	$\varnothing$	everywhere

The complement of a dense open set is closed with empty interior, hence nowhere dense. □

2.2.3 The reusable obstruction

Theorem 2.6 (No agreement on an interior)**.** *Let $h : \mathbb{R} \to \mathbb{R}$ be nonanalytic at every point of* $\operatorname{int}(S)$, *where $S \subset \mathbb{R}$ has nonempty interior. No elementary function g can agree with h on all of S.*

Proof. If g were elementary, theorem 2.5 would give a point $x \in \operatorname{int}(S)$ at which g is analytic. Equality on S implies equality on a whole neighborhood of x contained in S, so analyticity transfers from g to h at x, a contradiction. □

The theorem is stronger than saying that h itself is not elementary: it rules out any piecewise repair outside S and any elementary surrogate on a substantial subset.

2.3 Strong non-elementarity of the Fabius family

The main article proves nowhere analyticity of F on $[0, 1]$, of up on $[-1, 1]$, and of the signed extension on its active interval. Combining those results with theorem 2.6 gives the new non-elementarity branch almost for free.

Theorem 2.7 (Strong local non-elementarity of F)**.** *Let $U \subset [0, 1]$ have nonempty interior. There is no $g \in \mathsf{Elem}$ such that*

$$g(x) = F(x) \qquad (x \in U). \tag{12}$$

In particular, no elementary function agrees with F on $(0, 1)$, and F is not an elementary function on $\mathbb{R}$.

Proof. The function F is nonanalytic at every point of $[0, 1]$, hence at every point of $\operatorname{int}(U)$. Apply theorem 2.6. □

Theorem 2.8 (Strong local non-elementarity of up)**.** *If $U \subset [-1, 1]$ has nonempty interior, no elementary g agrees with* up *on U. Consequently* up *is not elementary.*

Theorem 2.9 (Strong local non-elementarity of the signed extension)**.** *If $U \subset [0, 2)$ has nonempty interior, no elementary g agrees with the signed global extension $\widetilde{F}$ on U. Consequently $\widetilde{F}$ is not elementary.*

Proof of the last two theorems. The corresponding nowhere-analyticity theorems supply the hypothesis of theorem 2.6. The global conclusions follow by taking a nonempty open subinterval of the active support and observing that an elementary representation on all of $\mathbb{R}$ would in particular agree there. □

The same analytic-density obstruction applies to the totalized inverse. To avoid a second statement and proof of the inverse theorem here, section 65 records its exact analytic locus, the resulting local non-elementarity declarations, and the additional branch-domain hypotheses needed for the enlarged elementary-or-inverse class.

Remark 2.10 (Why compact support alone is not the argument). A compactly supported nonzero function cannot be globally real analytic, but elementary functions need not be analytic everywhere: $|x|$, x^{-1}, and totalized logarithms already have exceptional points. What defeats elementarity is not compact support by itself; it is nonanalyticity on an entire interval, while elementary exceptional sets are nowhere dense.

Remark 2.11 (Why this is sharper than a differential-algebra slogan). The conclusion does not depend on choosing a particular syntax for "closed form" beyond the explicit inductive class. More importantly, it is local: changing the function outside a small open interval cannot make it elementary on that interval. Thus the theorem rules out not only a global elementary formula for F, but also an elementary formula valid on any nontrivial open patch of its natural domain.

3 Probability laws and the bridges the headline formulas conceal

3.1 The product probability law in measure-theoretic form

The random-series representation in the main article is concise because it speaks in the usual probabilistic shorthand. The formal construction must establish continuity, measurability, product independence, pushforward identities, and support before it may use the same shorthand. This section records that chain explicitly.

3.1.1 Sample space and uniform convergence

Let

$$\Omega = [0,1]^{\mathbb{N}} \tag{13}$$

with its product topology and product Borel probability measure

$$\mathbf{P} = \bigotimes_{n\geq 0} \mathrm{Unif}[0,1]. \tag{14}$$

Write ω_n for the nth coordinate and define

$$X(\omega) = \sum_{n=0}^{\infty} \frac{\omega_n}{2^{n+1}}. \tag{15}$$

For every ω,

$$0 \leq \frac{\omega_n}{2^{n+1}} \leq 2^{-n-1}, \qquad \sum_{n\geq 0} 2^{-n-1} = 1. \tag{16}$$

The Weierstrass test therefore gives uniform convergence on Ω. Every partial sum is continuous, so X is continuous and hence measurable. Termwise comparison in (16) gives the pointwise support statement

$$0 \leq X(\omega) \leq 1. \tag{17}$$

Let

$$\mu = \mathbf{P} \circ X^{-1} \tag{18}$$

be the pushforward law. It is a probability measure and

$$\mu([0,1]) = 1, \qquad \mu(\mathbb{R} \setminus [0,1]) = 0, \qquad \mathrm{supp}(\mu) = [0,1]. \tag{19}$$

The Lean declarations are `weightedCoordinateSum`, `continuous_weightedCoordinateSum`, `weightedSumDistribution`, `weightedSumDistribution_Icc`, `weightedSumDistribution_compl_Icc`, `ae_weightedSumDistribution_mem_Icc`, `weightedSumDistribution_restrict_Icc`, and the exact topological-support theorem `weightedSumDistribution_support_eq_Icc`. The almost-everywhere and restriction declarations are owned upstream by `ProbabilityRepresentation.lean`.

3.1.2 Deleting the head coordinate

Define the left shift

$$(T\omega)_n = \omega_{n+1}. \tag{20}$$

The product measure is invariant under this deletion in the sense that $T_{\#}\mathbf{P} = \mathbf{P}$, and the head coordinate ω_0 is independent of the entire tail $T\omega$. Reindexing the series gives the exact pointwise split

$$X(\omega) = \frac{\omega_0 + X(T\omega)}{2}. \tag{21}$$

Consequently the pair $(\omega_0, X(T\omega))$ has law

$$\mathrm{Unif}[0,1] \otimes \mu. \tag{22}$$

Pushing (21) through this joint law gives the self-similarity identity

$$\mu = (\mathrm{Unif}[0,1] \otimes \mu) \circ \left[(u,x) \mapsto \frac{u+x}{2}\right]^{-1}. \tag{23}$$

3.1.3 Post-snapshot generic naturality and affine recursion

The formal mechanism is now independent of the dyadic weights. For weights $w : \mathbb{N} \to E$ write

$$S_w(\omega) = \sum_{n \ge 0} \omega_n w_n, \qquad \mu_w = (S_w)_*\mathbf{P}, \qquad w_n^+ = w_{n+1}.$$

First, let G, H be arbitrary measurable spaces, let $g : \Omega \to G$ be measurable, and let $\Phi : [0,1] \times G \to H$ be measurable. An almost-everywhere recursion

$$f(\omega) = \Phi\big(\omega_0, g(T\omega)\big) \quad \text{for } \mathbf{P}\text{-almost every } \omega$$

implies

$$f_*\mathbf{P} = \Phi_*\big(\mathrm{Unif}[0,1] \otimes g_*\mathbf{P}\big). \tag{24}$$

No separate measurability hypothesis on f is required; a pointwise recursion is an immediate specialization.

Now let E be a real Banach space, let F be a real normed space, let $L : E \to F$ be continuous linear, and suppose $\sum_n \|w_n\| < \infty$. The target F need not be complete. Then

$$L(S_w(\omega)) = S_{Lw}(\omega), \qquad L_*\mu_w = \mu_{Lw}, \tag{25}$$

where the law identity uses the Borel measurable structures on E and F. For every real scalar c, the stronger pointwise identity $S_{cw} = cS_w$ holds without completeness or summability. Its law-level version retains source completeness, Borel structure, and norm summability because the underlying $\mathrm{Measure.map}$ definition is totalized.

If $A : E \to E$ is continuous linear and $w_{n+1} = A(w_n)$, then

$$A_*\mu_w = \mu_{w^+}, \tag{26}$$
$$S_w(\omega) = \omega_0 w_0 + A(S_w(T\omega)), \tag{27}$$
$$\mu_w = \big((u, z) \mapsto uw_0 + Az\big)_*\big(\mathrm{Unif}[0, 1] \otimes \mu_w\big). \tag{28}$$

The scalar case $A = c\,\mathrm{id}$ requires no sign or contraction condition on c beyond the explicit norm-summability hypothesis on the displayed weights.

For real laws there is also an absolute-continuity smoothing principle. If ν is any s-finite measure on $\mathbb{R}$ and $a \neq 0$, then the image of $\mathrm{Unif}[0, 1] \otimes \nu$ under $(u, x) \mapsto ua + x$ is absolutely continuous with respect to Lebesgue measure. Consequently a norm-summable real weighted law μ_w is absolutely continuous whenever $w_n \neq 0$ for at least one index n; the nonzero coefficient need not be the head. Under the same hypotheses every singleton is μ_w-null.

The eleven public declarations in this tranche are `ProbabilityRepresentation.uniformProduct_map_of_ae_head_tail_recursion`, `ProbabilityRepresentation.uniformProduct_map_of_head_tail_recursion`, `ProbabilityRepresentation.weightedUniformSeries_map`, `ProbabilityRepresentation.weightedUniformSeries_smul`, `ProbabilityRepresentation.weightedUniformDistribution_map`, `ProbabilityRepresentation.weightedUniformDistribution_smul`, `ProbabilityRepresentation.weightedUniformDistribution_map_eq_shift`, `ProbabilityRepresentation.weightedUniformSeries_split_of_shift_eq_map`, `ProbabilityRepresentation.weightedUniformDistribution_split_of_shift_eq_map`, `ProbabilityRepresentation.weightedUniformSeries_geometric_split`, and `ProbabilityRepresentation.weightedUniformDistribution_geometric_split` in `WeightedUniformSeries.lean`. The separate dyadic bridge `ProbabilityRepresentation.weightedSumDistribution_eq_weightedUniformDistribution` in `ProbabilityRepresentation.lean` identifies the law μ above with μ_w at $w_n = 1/2/2^n$.

The two generic declarations containing the word *geometric* are conditional scalar-shift theorems. The compatibility surface additionally exports `ProbabilityRepresentation.weightedUniformSeries_smul_weights`, `ProbabilityRepresentation.uniformProduct_map_head_tail_function`, `ProbabilityRepresentation.weightedUniformDistribution_isProbabilityMeasure`, `ProbabilityRepresentation.weightedUniformDistribution_smul_weights`, `ProbabilityRepresentation.uniformProduct_map_head_tail_weightedUniformSeries`, `ProbabilityRepresentation.weightedUniformDistribution_unitInterval`, and `ProbabilityRepresentation.weightedUniformDistribution_compl_unitInterval` through `WeightedUniformSeries.lean` and `WeightedUniformDistribution.lean`. The law-level scalar wrapper is oriented as $\mu_{cw} = (x \mapsto cx)_*\mu_w$.

The four-declaration absolute-continuity tranche in `WeightedUniformDistribution.lean` consists of `ProbabilityRepresentation.uniformScaledAdd_absolutelyContinuous`, `ProbabilityRepresentation.weightedUniformDistribution_absolutelyContinuous_of_head_ne_zero`, `ProbabilityRepresentation.weightedUniformDistribution_absolutelyContinuous`, and `ProbabilityRepresentation.weightedUniformDistribution_nullSingletonClass`. The first allows an arbitrary s-finite tail law, the second assumes $w_0 \neq 0$, and the sharp final pair assumes only $\exists n,\ w_n \neq 0$; the last theorem returns Mathlib's `NullSingletonClass` explicitly.

The separate exact-support layer has ten public declarations in `WeightedUniformSupport.lean`: `ProbabilityRepresentation.isOpenPosMeasure_infinitePi`, `ProbabilityRepresentation.uniformProduct_isOpenPosMeasure`, `ProbabilityRepresentation.support_map_eq_closure_range_of_continuous`, `ProbabilityRepresentation.weightedUniformSeries_constCoordinates`, `ProbabilityRepresentation.weightedUniformDistribution_support_eq_range`, `ProbabilityRepresentation.range_weightedUnifor `mSeries_eq_Icc_min_max`, `ProbabilityRepresentation.weightedUniformDistribution_support_eq_Icc_min_max`, `ProbabilityRepresentation.range_weightedUniformSeries_eq_Icc`, `ProbabilityRepresentation.weightedUniformDistribution_support_eq_Icc`, and `ProbabilityRepresentation.weightedUniformDistribution_support_eq_unitInterval`. Thus an arbitrary norm-summable weighted law in a complete Borel real normed space has support exactly the range of its continuous series map. For arbitrary signed real weights, both range and support are exactly

$$\left[\sum_n \min\{w_n, 0\},\ \sum_n \max\{w_n, 0\}\right];$$

cancellation creates no gaps. Constant coordinate paths give the nonnegative specialization $[0, \sum_n w_n]$, including the degenerate zero-total-weight case.

There is now also a named specialization. For

$$w_n(q) = (1-q)q^n, \qquad S_q = S_{w(q)}, \qquad \mu_q = \mu_{w(q)},$$

the zero and successor coefficients are exact for every real q. Under $|q| < 1$, the weights are norm-summable with sum one, S_q is continuous and measurable, and μ_q is a probability measure satisfying

$$S_q(\omega) = (1-q)\omega_0 + qS_q(T\omega), \qquad \mu_q = \big((u, z) \mapsto (1-q)u + qz\big)_*\big(\mathrm{Unif}[0,1] \otimes \mu_q\big),$$

together with pointwise and law-level reflection. Under $0 \le q < 1$, the weights are nonnegative, the series lies in $[0, 1]$, and the law gives mass one to that interval and zero to its complement. For every $|q| < 1$, its topological support is exactly the series range, its law is absolutely continuous with respect to Lebesgue measure, and all singletons are null; neither of the latter two statements needs $q \ge 0$. Under $0 \le q < 1$, both the series range and the support are exactly $[0, 1]$.

The twenty-three declarations are `ProbabilityRepresentation.geometricUniformWeight`, `ProbabilityRepresentation.geometricUniformWeight_zero`, `ProbabilityRepresentation.geometricUniformWeight_succ`, `ProbabilityRepresentation.hasSum_geometricUniformWeight`, `ProbabilityRepresentation.summable_norm_geometricUniformWeight`, `ProbabilityRepresentation.geometricUniformWeight_nonneg`, `ProbabilityRepresentation.geometricUniformSeries`, `ProbabilityRepresentation.continuous_geometricUniformSeries`, `ProbabilityRepresentation.measurable_geometricUniformSeries`, `ProbabilityRepresentation.geometricUniformSeries_split`, `ProbabilityRepresentation.geometricUniformSeries_reflect`, `ProbabilityRepresentation.geometricUniformSeries_mem_Icc`, `ProbabilityRepresentation.geometricUniformDistribution`, `ProbabilityRepresentation.geometricUniformDistribution_isProbabilityMeasure`, `ProbabilityRepresentation.geometricUniformDistribution_absolutelyContinuous`, `ProbabilityRepresentation.geometricUniformDistribution_nullSingletonClass`, `ProbabilityRepresentation.uniformProduct_map_head_tail_geometricUniformSeries`, `ProbabilityRepresentation.geometricUniformDistribution_selfSimilar`, `ProbabilityRepresentation.geometricUniformDistribution_reflection`, `ProbabilityRepresentation.geometricUniformDistribution_Icc`, `ProbabilityRepresentation.geometricUniformDistribution_compl_Icc`, `ProbabilityRepresentation.geometricUniformDistribution_support_eq_range`, and `ProbabilityRepresentation.geometricUniformDistribution_support_eq_Icc` in `GeometricUniformLaw.lean`. At $q = 1/2$, `ProbabilityRepresentation.weightedCoordinateSum_eq_geometricUniformSeries_one_half` and `ProbabilityRepresentation.weightedSumDistribution_eq_geometricUniformDistribution_one_half` identify the dyadic series and law with this specialization, while `ProbabilityRepresentation.weightedSumDistribution_support_eq_Icc` records the exact support of the dyadic law.

At compiler-validated checkpoint `e6f535e5d`, two further modules promote the real-space CDF and positive-ratio density layer. Write

$$F_q(x) = \mathrm{cdf}(\mu_q)(x), \qquad f_q(x) = \frac{F_q(x/q) - F_q((x-(1-q))/q)}{1-q}. \tag{29}$$

The definitions are total for real q. The CDF is monotone, measurable, nonnegative, and at most one for every parameter. When $|q| < 1$, it is continuous and satisfies

$$F_q(1-x) = 1 - F_q(x), \qquad F_q(1/2) = 1/2. \tag{30}$$

For $0 \le q < 1$, including $q = 0$, it has the exact tails

$$F_q(x) = 0 \quad (x \le 0), \qquad F_q(x) = 1 \quad (x \ge 1). \tag{31}$$

For $0 < q < 1$, the conditioning and ordinary interval-integral identities are

$$F_q(x) = \int_0^1 F_q\left(\frac{x-(1-q)u}{q}\right) \mathrm{d}u, \tag{32}$$

$$= \frac{q}{1-q} \int_{(x-(1-q))/q}^{x/q} F_q(t)\, \mathrm{d}t. \tag{33}$$

On the same strict range, $F_q'(x) = f_q(x)$ at every real point. The density is continuous, nonnegative and symmetric about $1/2$, vanishes on both exterior half-lines, has ordinary support contained in $(0, 1)$, topological support contained in $[0, 1]$, and compact support. The exact measure identity and smoothness conclusions are

$$\mu_q = (\mathrm{Leb} : \mathrm{Measure}(\mathbb{R}))\ \mathrm{withDensity}\ \big(x \mapsto \mathrm{ofReal}(f_q(x))\big), \qquad F_q, f_q \in C^\infty(\mathbb{R}). \tag{34}$$

The proof of C^∞ bootstraps the real-space affine derivative refinement; it does not formalize the report's Fourier-product argument.

Exact CDF/density API crosswalk. The generic module `ContinuousCDF.lean` exports `continuous_cdf_of_nullSingleton`, `cdf_reflection_sub`, and `measure_eq_withDensity_of_cdf_hasDerivAt`. The first converts null singletons into CDF continuity, the second converts affine reflection invariance into CDF reflection, and the third identifies an everywhere pointwise CDF derivative with the measure's Lebesgue density.

All of the following names lie in `GeometricUniformCDF.lean` and the namespace `Fabius.ProbabilityRepresentation`. The total definition and universal CDF surface are `geometricUniformCDF`, `monotone_geometricUniformCDF`, `geometricUniformCDF_nonneg`, `geometricUniformCDF_le_one`, and `measurable_geometricUniformCDF`. The $|q| < 1$ continuity, reflection, and midpoint statements are `continuous_geometricUniformCDF`, `geometricUniformCDF_reflection`, and `geometricUniformCDF_one_half`. The $0 \le q < 1$ tails are `geometricUniformCDF_zero_of_nonpos` and `geometricUniformCDF_one_of_one_le`.

For $0 < q < 1$, the refinement, density, and derivative declarations are `geometricUniformCDF_eq_integral`, `geometricUniformCDF_eq_intervalIntegral`, `geometricUniformDensity`, `geometricUniformCDF_hasDerivAt`, `deriv_geometricUniformCDF`, `continuous_geometricUniformDensity`, and `geometricUniformDensity_nonneg`. Exterior vanishing and support are `geometricUniformDensity_zero_of_nonpos`, `geometricUniformDensity_zero_of_one_le`, `support_geometricUniformDensity_subset_Ioo`, `support_geometricUniformDensity_subset_Icc`, `tsupport_geometricUniformDensity_subset_Icc`, and `geometricUniformDensity_hasCompactSupport`. Finally, `geometricUniformDensity_reflection`, `geometricUniformDistribution_eq_withDensity`, `contDiff_geometricUniformCDF`, and `contDiff_geometricUniformDensity` give density symmetry, the Radon–Nikodym identification, and smoothness of every finite order.

The half-base crosswalk in `ProbabilityRepresentation.lean` is exact at every real argument: `weightedSumCDF_eq_geometricUniformCDF_one_half` identifies the two CDF definitions, `geometricUniformCDF_one_half_eq_fabiusReal` identifies that CDF with every bounded Fabius representative, and `geometricUniformDensity_one_half_eq_rvachevUp` proves

$$f_{1/2}(x) = 2\,\mathrm{up}(2x - 1). \tag{35}$$

Remaining geometric-q frontier. The promotion is deliberately not read past its hypotheses. The law μ_q is absolutely continuous for every $|q| < 1$, but the selected total function f_q is proved to be its Radon–Nikodym density only when $0 < q < 1$; its density interpretation at $q = 0$ and at negative q remains open in this interface. The following report layers also remain in the research quarantine:

- the general-q characteristic-function product, quantitative rapid decay, and the Fourier-inversion construction of smooth densities;
- separately named centered variables and densities and their affine transport package;
- a closed all-order derivative-refinement hierarchy and explicit endpoint-flatness wrappers, beyond the formal existence of all finite derivatives;
- log-concavity, strictness, central plateaux and other parameter-dependent shape claims;
- transform formulas, Bernoulli cumulants, Bell moments, and related asymptotic or Gaussian-limit statements; and
- uniqueness of the affine measure fixed point.

Proposition 3.1 (The smoothing equation for the CDF)**.** *Let $H(y) = \mu((-\infty, y])$. Then for every $y \in \mathbb{R}$,*

$$H(y) = \int_0^1 H(2y - u)\,\mathrm{d}u. \tag{36}$$

For $0 \le y \le 1/2$ this reduces to

$$H(y) = \int_0^{2y} H(t)\,\mathrm{d}t. \tag{37}$$

Proof. Condition on the head coordinate in (21). For a fixed u,

$$\mathbb{P}\left(\frac{u + X'}{2} \le y\right) = \mathbb{P}(X' \le 2y - u) = H(2y - u),$$

where X' has law μ and is independent of u. Integrate over $u \in [0, 1]$. If $0 \le y \le 1/2$, support implies $H(2y - u) = 0$ when $u > 2y$; substitute $t = 2y - u$ and use the reflection identity established below, or reverse the interval directly. □

3.1.4 Coordinate reflection

Let $R : \Omega \to \Omega$ be

$$(R\omega)_n = 1 - \omega_n. \tag{38}$$

Lebesgue measure on $[0, 1]$ is invariant under $u \mapsto 1 - u$, so the infinite product measure is invariant under R. Moreover

$$X(R\omega) = \sum_{n \ge 0} \frac{1 - \omega_n}{2^{n+1}} = 1 - X(\omega). \tag{39}$$

Thus the law is reflection invariant:

$$(x \mapsto 1 - x)_{\#}\mu = \mu. \tag{40}$$

Conditioning on the tail X', the head coordinate U_0 is uniform, so the law of $(U_0+X')/2$ is absolutely continuous with respect to Lebesgue measure. This smoothing step is now formalized by `ProbabilityRepresentation.geometricUniformDistribution_absolutelyContinuous` at $q = 1/2$; `ProbabilityRepresentation.geometricUniformDistribution_nullSingletonClass` packages the resulting null singletons. Thus H is continuous. Reflection invariance then gives

$$H(1 - y) = 1 - H(y). \tag{41}$$

The fixed-point uniqueness theorem for the bounded Fabius equation now identifies

$$H = F \quad \text{on all of } \mathbb{R}. \tag{42}$$

The phrase "on all of $\mathbb{R}$" includes the two constant tails; it is not a reference to the signed extension $\widetilde{F}$.

Theorem 3.2 (Probability representation, global form)**.** *For every $y \in \mathbb{R}$,*

$$F(y) = \mathbb{P}\left(\sum_{n\geq 0} \frac{U_n}{2^{n+1}} \leq y\right), \tag{43}$$

where the U_n are independent uniform variables on $[0, 1]$. Equivalently,

$$\mathrm{up}(x) = \mu((-\infty, 1 - |x|]) \tag{44}$$

for every $x \in \mathbb{R}$.

3.2 The survival-function Fubini identity

The main article's probability chapter records the scalar differentiable bridge between integration against the Fabius law and an ordinary integral against up. The formal development now also contains the stronger Banach-valued partial and full layer-cake forms recorded below.

For $t \geq 0$, support and (42) give

$$\mu((t, \infty)) = 1 - F(t) = \mathrm{up}(t). \tag{45}$$

The equality remains correct for $t > 1$, when both sides are zero.

Theorem 3.3 (Expectation from the survival function)**.** *Let $g, g' : \mathbb{R} \to \mathbb{R}$ be continuous and suppose*

$$g(x) - g(0) = \int_0^x g'(t)\,\mathrm{d}t \qquad (0 \leq x \leq 1). \tag{46}$$

Then

$$\int_{[0,1]} g(x)\,\mathrm{d}\mu(x) = g(0) + \int_0^1 g'(t)\,\mathrm{up}(t)\,\mathrm{d}t. \tag{47}$$

Proof. Put

$$A = \{(t, x) \in [0, 1] \times \mathbb{R} : t < x\}, \qquad K(t, x) = \mathbf{1}_A(t, x)g'(t). \tag{48}$$

Continuity of g' on the compact interval makes K integrable with respect to $\mathrm{Leb}_{[0,1]} \otimes \mu$. Integrating first in t and using (46) gives, for $x \in [0, 1]$,

$$\int_0^1 K(t, x)\,\mathrm{d}t = \int_0^x g'(t)\,\mathrm{d}t = g(x) - g(0). \tag{49}$$

Integrating first in x gives

$$\int K(t,x)\,\mathrm{d}\mu(x) = g'(t)\mu((t,\infty)) = g'(t)\,\mathrm{up}(t) \tag{50}$$

by (45). Fubini equates the two iterated integrals. Since μ is a probability measure supported on $[0,1]$, the integral of the constant $g(0)$ is $g(0)$, and rearrangement yields (47). $\square$

Remark 3.4 (Endpoint conventions disappear under integration). The survival function uses the open ray (t,∞). The CDF uses $(-\infty,t]$. Their complements are exact, and no endpoint atom enters because μ has a continuous CDF. Likewise, replacing $(0,1)$ by any of the standard half-open versions changes no Lebesgue integral. The formal proof nevertheless records these null-set conversions explicitly.

Theorem 3.5 (Banach-valued survival layer cake). *Let E be a real Banach space and $0 \le c \le 1$. If $k : \mathbb{R} \to E$ is interval-integrable on $[0,c]$, then*

$$\int_0^c \mathrm{up}(t)k(t)\,\mathrm{d}t = \int_{\mathbb{R}} \left(\int_0^{\min\{z,c\}} k(t)\,\mathrm{d}t\right) \mathrm{d}\mu(z). \tag{51}$$

If k is interval-integrable on $[0,1]$, then

$$\int_0^1 \mathrm{up}(t)k(t)\,\mathrm{d}t = \int_{\mathbb{R}} \left(\int_0^{z} k(t)\,\mathrm{d}t\right) \mathrm{d}\mu(z). \tag{52}$$

No positivity hypothesis is imposed on k; in particular $E = \mathbb{C}$ gives the complex-kernel form.

The exact specializations are `ProbabilityRepresentation.intervalIntegral_rvachevUp_smul_eq_integral_min` and `ProbabilityRepresentation.intervalIntegral_rvachevUp_smul_eq_integral` in `ProbabilityLaplaceMoments.lean`. Their reusable finite-measure versions—including the support-free clamped identity—are the three `intervalIntegral_survival_smul_eq_integral_clamp`, `intervalIntegral_survival_smul_eq_integral_min_of_ae_mem_Icc`, and `intervalIntegral_survival_smul_eq_integral_of_ae_mem_Icc` declarations in `SurvivalLayerCake.lean`, all derived from `integral_smul_setIntegral_subgraph` in `SubgraphFubini.lean`.

Theorem 3.6 (Atom-preserving lower-CDF and fractional Fabius layer cake). *For a finite measure ν on $\mathbb{R}$, $a \le b$, and a Banach-valued map k interval-integrable on $[a,b]$,*

$$\int_a^b \nu((-\infty,t])k(t)\,\mathrm{d}t = \int_{\mathbb{R}} \left(\int_{\min\{\max\{z,a\},b\}}^{b} k(t)\,\mathrm{d}t\right) \mathrm{d}\nu(z). \tag{53}$$

For the partial identity on $[c,b]$, only $c \le b$ and the one-sided almost-everywhere bound $z \le b$ are needed. Compact support on $[a,b]$ gives the convenient partial corollary and the full terminal-primitive identity. No atomlessness hypothesis is needed. For the Fabius law μ, every $\alpha > 0$, and every $x \in \mathbb{R}$,

$$I^{\alpha}_{0+}F(x) = \frac{1}{\Gamma(\alpha+1)} \int_{\mathbb{R}} (x-z)_+^{\alpha}\,\mathrm{d}\mu(z) = \frac{1}{\Gamma(\alpha+1)} \int_{\Omega} (x - X(\omega))_+^{\alpha}\,\mathrm{d}\mathbf{P}(\omega). \tag{54}$$

At $x = 1$, reflection of μ turns the stopped power into the ordinary moment:

$$I^{\alpha}_{0+}F(1) = \frac{1}{\Gamma(\alpha+1)} \int_{\mathbb{R}} z^{\alpha}\,\mathrm{d}\mu(z) = \frac{1}{\Gamma(\alpha+1)} \int_{\Omega} X(\omega)^{\alpha}\,\mathrm{d}\mathbf{P}(\omega). \tag{55}$$

The closed-supergraph engine is `integral_smul_setIntegral_supergraph`. Its four finite-measure forms in `CDFLayerCake.lean` are `intervalIntegral_cdf_smul_eq_integral_clamp`, `intervalIntegral_cdf_smul_eq_integral_max_of_ae_le`, `intervalIntegral_cdf_smul_eq_integral_max_of_ae_mem_Icc`, and `intervalIntegral_cdf_smul_eq_integral_of_ae_mem_Icc`. The support-free clamped, lower-supported positive-part, probability-CDF positive-part, and compactly supported endpoint-power consequences in `FractionalCDFLayerCake.lean` are `fractionalVolterra_measureReal_Iic_eq_integral_clamp`, `fractionalVolterra_measureReal_Iic_eq_integral_posPart`, `fractionalVolterra_cdf_eq_integral_posPart`, and `fractionalVolterra_measureReal_Iic_eq_integral_rpow_of_ae_mem_Icc`. Finally, `ProbabilityRepresentation.fractionalVolterra_fabiusReal_eq_integral_posPart`, `ProbabilityRepresentation.fractionalVolterra_fabiusReal_eq_uniformProduct_integral_posPart`, `ProbabilityRepresentation.fractionalVolterra_fabiusReal_one_eq_integral_rpow`, and `ProbabilityRepresentation.fractionalVolterra_fabiusReal_one_eq_uniformProduct_integral_rpow` in `FabiusFractionalIntegral.lean` are exactly the two Fabius displays above. These are positive-real integral identities, not complex-order continuations, fractional derivatives, Caputo formulas, or folded-Rvachev, quantile, or inverse-function specializations.

Remark 3.7 (Formalized inverse layer-cake boundary). The same subgraph engine proves a real- or complex-weighted, Banach-valued inverse-clock identity on compact rectangles. For the Fabius clocks its explicit-primitive and pointwise-C^1 forms are `intervalIntegral_smul_intervalIntegral_fabiusInv` and `intervalIntegral_smul_comp_fabiusInv` in `InverseLayerCake.lean`. The real weighted-area specialization is `intervalIntegral_mul_fabiusInv_eq`. The module also exports `intervalIntegral_fabiusInv_eq_intervalIntegral_rvachevUp` and `intervalIntegral_fabiusInv_eq_one_half`, which identify the inverse area with the unit-interval up area and with $1/2$. These are full-interval theorems. Separately, `intervalIntegral_deriv_mul_comp_add_comp_mul_deriv_of_lt_iff_lt` in `InversePairIntegral.lean` packages arbitrary real absolutely-continuous outer functions on an arbitrary ordered compact rectangle whose two clocks preserve the displayed intervals and satisfy the strict inverse-order relation. Clock measurability is derived, not assumed, and degenerate ordered intervals are included. Its Fabius wrapper and area-sum specialization are `intervalIntegral_deriv_mul_fabiusReal_add_fabiusInv_mul_deriv` and `intervalIntegral_fabiusReal_add_fabiusInv_eq_one`. These declarations still do not package a variable-upper-endpoint area identity, improper singular-endpoint extensions, higher or fractional inverse primitives, or complex Mellin continuation.

Remark 3.8 (Formalized positive-real fractional Volterra boundary). At compiled full-facade checkpoint `e6f535e5d`, the definition `fractionalVolterra` in `FractionalVolterra.lean` is a total Gamma-normalized oriented interval integral for a real order, arbitrary real base and endpoint, and an input valued in any real normed space. The unconditional declarations `fractionalVolterra_self`, `fractionalVolterra_congr`, `fractionalVolterra_smul`, `fractionalVolterra_one`, and `fractionalVolterra_nat_succ` give the base-point value, locality on the closed unoriented endpoint interval, real-scalar compatibility, the ordinary order-one primitive, and exact agreement with `normalizedVolterra` at every positive natural order. In the classical regime $0 < \alpha$ and $a \le x$, a continuous input has an interval-integrable fractional kernel by `intervalIntegrable_fractionalVolterra_kernel`, and `fractionalVolterra_add_input` gives additivity for two such continuous inputs. For $0 < \alpha$, $0 < \beta$, and $s \le x$, `intervalIntegrable_fractionalVolterra_betaKernel` in `FractionalVolterraSemigroup.lean` proves interval integrability of the raw shifted beta kernel, including the degenerate case $s = x$. When $s < x$, `intervalIntegral_fractionalVolterra_betaKernel` proves

$$\int_s^x (x-u)^{\alpha-1}(u-s)^{\beta-1}\,\mathrm{d}u = (x-s)^{\alpha+\beta-1}\frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}. \tag{56}$$

and `intervalIntegral_fractionalVolterra_normalizedBetaKernel` proves

$$\int_s^x \frac{(x-u)^{\alpha-1}}{\Gamma(\alpha)} \frac{(u-s)^{\beta-1}}{\Gamma(\beta)} \, \mathrm{d}u = \frac{(x-s)^{\alpha+\beta-1}}{\Gamma(\alpha+\beta)}. \tag{57}$$

If the target is complete, $a \le x$, and f is continuous on $[a, x]$, then `fractionalVolterra_add` gives

$$I_{a+}^{\alpha+\beta} f(x) = I_{a+}^{\alpha} \left[u \mapsto I_{a+}^{\beta} f(u) \right] (x). \tag{58}$$

For a complete target, $v \in E$, and $a < x$, the same module proves

$$I_{a+}^{\alpha} \left[t \mapsto \frac{(t-a)^{\beta-1}}{\Gamma(\beta)} v \right] (x) = \frac{(x-a)^{\alpha+\beta-1}}{\Gamma(\alpha+\beta)} v, \qquad \alpha, \beta > 0, \tag{59}$$

$$I_{a+}^{\alpha} [t \mapsto (t-a)^{\rho} v] (x) = \frac{\Gamma(\rho+1)(x-a)^{\alpha+\rho}}{\Gamma(\alpha+\rho+1)} v, \qquad \alpha > 0,\ \rho > -1. \tag{60}$$

The constant specialization includes $a = x$:

$$I_{a+}^{\alpha} [t \mapsto v](x) = \frac{(x-a)^{\alpha}}{\Gamma(\alpha+1)} v \qquad (\alpha > 0,\ a \le x). \tag{61}$$

The exact declarations are `fractionalVolterra_normalized_rpow_smul`, `fractionalVolterra_rpow_smul`, and `fractionalVolterra_const`. The companion `FractionalVolterraCalculus.lean` proves increasing-affine covariance for arbitrary real order, $0 < c$, and $a \le x$, without regularity, integrability, or completeness:

$$I_{(ca+d)+}^{\alpha} f(cx+d) = c^{\alpha} I_{a+}^{\alpha} [t \mapsto f(ct+d)](x). \tag{62}$$

Its inverse-smul form is `fractionalVolterra_comp_affine`. For a Banach-valued continuous g with right-sided interior derivative g', assuming g' is interval-integrable, `fractionalVolterra_add_one_deriv` gives

$$I_{a+}^{\alpha+1} g'(x) = I_{a+}^{\alpha} g(x) - \frac{(x-a)^{\alpha}}{\Gamma(\alpha+1)} g(a) \qquad (\alpha > 0,\ a \le x), \tag{63}$$

including $a = x$; `fractionalVolterra_add_one_deriv_of_eq_zero` removes the boundary term when $g(a) = 0$. The three declarations in `FabiusFractionalVolterra.lean` then prove

$$I_{0+}^{\alpha+1} \mathcal{F}(x) = 2^{\alpha} I_{0+}^{\alpha} \mathcal{F}(x/2), \qquad \alpha > 0,\ x \ge 0, \tag{64}$$

$$I_{0+}^{\alpha+1} F(x) = 2^{\alpha} I_{0+}^{\alpha} F(x/2), \qquad \alpha > 0,\ 0 \le x \le 1, \tag{65}$$

$$I_{-1+}^{\alpha+1} \operatorname{up}(x) = 2^{\alpha} I_{0+}^{\alpha} F((x+1)/2), \qquad \alpha > 0,\ x \ge -1. \tag{66}$$

They are `fractionalVolterra_add_one_extendedFabius_of_nonneg`, `fractionalVolterra_add_one_fabiusReal`, and `fractionalVolterra_add_one_rvachevUp`. No order-zero operator law, reversed-endpoint fractional interpretation or covariance, nonpositive-scale covariance, semigroup theorem for merely interval-integrable inputs or noncomplete targets, Caputo or other fractional derivative, complex-order theorem, hierarchy-specific semigroup, iterated fractional shift, shifted dyadic-lattice or endpoint-up wrapper, or inverse-function fractional specialization is included. The positive-real Fabius-CDF specialization is recorded separately in theorem 3.6.

Remark 3.9 (Formalized unconditional analytic-filter boundary). At the same compiled checkpoint, `AnalyticSeriesFilter.lean` turns the finite moment algebra of a row (ξ_i, w_i) into an exact pointwise series theorem. If each weighted sampled series $\sum_{m \ge 0} w_i \big((\xi_i z)^m a_m \big)$ is unconditionally summable, then `finiteAnalyticSeriesFilter_eq_tsum` proves

$$\sum_i \sum_{m \ge 0} w_i \big((\xi_i z)^m a_m \big) = \sum_{m \ge 0} \left(\sum_i w_i \xi_i^m \right) z^m a_m. \tag{67}$$

The generic statement works over a commutative semiring acting on a normed additive group, and a zero weight needs no convergence at its node. Exactness through degree p splits off the reproduced finite head; at target zero the head is exactly a_0. For a geometric row over a commutative ring with continuous constant scalar multiplication, the finite row-moment factor in degree $p+1+r$ is

$$(-1)^p q^{\binom{p+1}{2}} \begin{bmatrix} p+r \\ p \end{bmatrix}_q . \tag{68}$$

This is `geometricAnalyticSeriesFilter_eq_constant_add_gaussian_tsum`. Assuming $j \mapsto q^j$ is injective on $\{0,\dots,p\}$, the geometric-Lagrange and shifted-scale wrappers over a scalar field impose sample summability only where the corresponding weight is nonzero. These are Mathlib `Summable`/`tsum` identities and hence unconditional and pointwise. They do not cover conditionally-only convergent boundary series, prove radius or uniform convergence or quantitative norm/error bounds, determine residual signs or asymptotic acceleration rates, instantiate an actual sinc product, or supply a formal-`PowerSeries`-evaluation bridge.

3.3 Raw, endpoint, and tilted Laplace moments

The main article develops half moments and the negative generating function from the analytic side. The new probability–Laplace module proves that those quantities are exactly the moments of the product law.

3.3.1 Three moment families

For a probability measure ν supported on $[0,1]$, define

$$U_n(\nu) = \int_{[0,1]} (1-x)^n \, \mathrm{d}\nu(x), \tag{69}$$

$$L_k(\nu;s) = \int_{[0,1]} \mathrm{e}^{-sx} x^k \, \mathrm{d}\nu(x), \tag{70}$$

$$J_k(s) = \int_0^1 t^k \, \mathrm{up}(t) \mathrm{e}^{-st} \, \mathrm{d}t. \tag{71}$$

For real s, the integrand in (70) is nonnegative, so

$$L_k(\nu;s) \ge 0. \tag{72}$$

Apply theorem 3.3 to $g(x) = \mathrm{e}^{-sx} x^k$. For $k=0$ one obtains

$$L_0(\mu;s) = 1 - sJ_0(s). \tag{73}$$

For $k \ge 0$, applying it to $g(x) = \mathrm{e}^{-sx} x^{k+1}$ gives

$$L_{k+1}(\mu;s) = (k+1)J_k(s) - sJ_{k+1}(s). \tag{74}$$

These equations are exactly the defining clauses of `fabiusLaplaceMoment`.

Theorem 3.10 (Law moments equal analytic Fabius moments). *For every $k \in \mathbb{N}$ and $s \in \mathbb{R}$,*

$$L_k(\mu;s) = M_k(s), \tag{75}$$

where

$$M_0(s) = G(-s), \tag{76}$$

$$M_{k+1}(s) = (k+1)J_k(s) - sJ_{k+1}(s) \tag{77}$$

are the tilted Fabius moments and G is the generating function of the main article. In particular $M_k(s) \ge 0$ for real s.

Proof. The case $k=0$ is the integration-by-parts identity (73), which is also the negative generating-function formula. The successor case is (74). □

3.3.2 Differential recurrence

Differentiation under the compact integral in (71) is justified by a uniform exponential majorant in a neighborhood of each real s. It gives

$$J_k'(s) = -J_{k+1}(s). \tag{78}$$

Substituting into (76)–(77) yields the cleaner recurrence

$$M_k'(s) = -M_{k+1}(s). \tag{79}$$

Iteration gives

$$M_k^{(n)}(s) = (-1)^n M_{k+n}(s). \tag{80}$$

Thus all M_k are C^∞ on $\mathbb{R}$, and the negative generating function is completely monotone:

$$(-1)^n \frac{\mathrm{d}^n}{\mathrm{d}s^n} G(-s) = M_n(s) \geq 0. \tag{81}$$

3.3.3 Reflection and the half moments

Reflection invariance (40) gives

$$\int (1-x)^n \, \mathrm{d}\mu(x) = \int x^n \, \mathrm{d}\mu(x). \tag{82}$$

At zero tilt the analytic moment normalization is the exact rational half-moment sequence d_n:

$$M_n(0) = d_n. \tag{83}$$

Combining the identifications yields the full probability interpretation.

Theorem 3.11 (Probability meaning of the half moments)**.** *For every* $n \in \mathbb{N}$*,*

$$d_n = \mathbb{E}[X^n] = \mathbb{E}[(1-X)^n] = \int_{[0,1]} x^n \, \mathrm{d}\mu(x) = \int_{[0,1]} (1-x)^n \, \mathrm{d}\mu(x). \tag{84}$$

Moreover

$$\frac{\mathrm{d}^n}{\mathrm{d}s^n} G(-s)\bigg|_{s=0} = (-1)^n d_n. \tag{85}$$

Proof. Use (75) at $s = 0$, then (83) and (82). The derivative identity is (80) with $k = 0$ and $s = 0$. □

The equality $d_n = \mathbb{E}[X^n]$ is not merely an interpretation of a previously known rational sequence. It is the bridge that licenses positivity, log-convexity, moment-matrix positivity, and Laplace comparison arguments without rebuilding them from the recurrence for d_n. The formal library keeps the bridge in a separate module precisely so the probabilistic and saddle branches can import it independently.

The probability chapter hands its zero-th tilted moment directly to the endpoint asymptotic synthesis. At $k = 0$, (75) identifies the zero-th tilted law moment with $M_0(s) = G(-s)$. For $s > 0$, the notation of part IV calls this same negative Laplace transform $B(s)$; its independent-coordinate product is already recorded in (467). Thus the exact negative-Laplace product decomposed there is the Laplace transform of the same product-space law, rather than a second independent model.

3.4 From atomic laws to half-endpoint histograms

The main article introduces the polynomial step approximants $\mathcal{W}_n$ and then moves quickly to their convergence. The formal proof has a nontrivial middle layer: the coefficients first define an atomic probability measure; each atom is smeared across a centered cell; interval integrals of the resulting histogram are sandwiched between masses of slightly shrunken and enlarged intervals.

3.4.1 Half-endpoint indicators are ordinary indicators almost everywhere

For $a < b$, define

$$\mathbf{1}^*_{[a,b]}(x) = \begin{cases} 1, & a < x < b, \\ \frac{1}{2}, & x = a \text{ or } x = b, \\ 0, & \text{otherwise.} \end{cases} \tag{86}$$

The two endpoint values are essential for pointwise gluing of adjacent cells, but irrelevant to Lebesgue integration.

Lemma 3.12 (Almost-everywhere replacement)**.** *For every $a, b \in \mathbb{R}$,*

$$\mathbf{1}^*_{[a,b]} = \mathbf{1}_{(a,b]} \quad \textit{Lebesgue-almost everywhere.} \tag{87}$$

Consequently

$$\int_{\mathbb{R}} \mathbf{1}^*_{[a,b]}(x)\,\mathrm{d}x = \max(b-a, 0), \tag{88}$$

and, for $c \le d$,

$$\int_c^d \mathbf{1}^*_{[a,b]}(x)\,\mathrm{d}x = \max\bigl(\min(d,b) - \max(c,a), 0\bigr). \tag{89}$$

Proof. The functions differ only at a and b, a null set. The first integral is the length of $(a,b]$. The second is the length of $(c,d] \cap (a,b] = (\max(c,a), \min(d,b)]$, with truncation at zero when the interval is empty. ☐

3.4.2 Atomic measure, cell geometry, and exact normalization

At level n, let $x_{n,m}$ be the location of the mth polynomial atom and let $p_{n,m} \ge 0$ be its normalized mass, with

$$\sum_m p_{n,m} = 1. \tag{90}$$

The corresponding atomic law is

$$\nu_n = \sum_m p_{n,m}\delta_{x_{n,m}}. \tag{91}$$

Let

$$h_n = 2^{-n-1} \tag{92}$$

be the half-width of a cell and

$$C_{n,m} = [x_{n,m} - h_n, x_{n,m} + h_n]. \tag{93}$$

Its full width is $2h_n = 2^{-n}$. Smearing the atom uniformly over its cell produces the histogram

$$\mathcal{W}_n(x) = 2^n \sum_m p_{n,m} \mathbf{1}^*_{C_{n,m}}(x). \tag{94}$$

By (88), each term has integral $p_{n,m}$, hence

$$\int_{\mathbb{R}} \mathcal{W}_n(x)\,\mathrm{d}x = 1. \tag{95}$$

The pointwise half weights ensure that when two cells share an endpoint, the two half contributions add to the correct plateau value.

3.4.3 The overlap sandwich

For a fixed interval $[a,b]$ with $a \le b$, let

$$\ell_{n,m}(a,b) = |[a,b] \cap C_{n,m}| = \max\bigl(\min(b, x_{n,m} + h_n) - \max(a, x_{n,m} - h_n), 0\bigr). \tag{96}$$

Then

$$\int_a^b \mathcal{W}_n(x)\,\mathrm{d}x = \sum_m p_{n,m}\, 2^n \ell_{n,m}(a,b). \tag{97}$$

The normalized overlap $2^n \ell_{n,m}$ lies in $[0,1]$.

If $x_{n,m} \in (a + h_n, b - h_n]$, then the entire cell lies inside $[a,b]$, so the normalized overlap is 1. If the cell meets $[a,b]$, then its center lies in $(a - h_n, b + h_n]$. Therefore, term by term,

$$\mathbf{1}_{(a+h_n, b-h_n]}(x_{n,m}) \le 2^n \ell_{n,m}(a,b) \le \mathbf{1}_{(a-h_n, b+h_n]}(x_{n,m}). \tag{98}$$

Multiplication by $p_{n,m}$ and summation gives the central bridge.

Theorem 3.13 (Atomic–histogram interval sandwich). *For $a \le b$,*

$$\nu_n((a + h_n, b - h_n]) \le \int_a^b \mathcal{W}_n(x)\,\mathrm{d}x \le \nu_n((a - h_n, b + h_n]). \tag{99}$$

This is the exact content of the inner- and outer-overlap lemmas in `StepMeasureBridge.lean`. Its importance is conceptual: it converts weak convergence of measures into convergence of integrals of discontinuous histograms without claiming pointwise convergence of densities from weak convergence alone.

3.4.4 Portmanteau plus a moving-boundary squeeze

The atomic laws ν_n coincide with the finite convolution laws and converge weakly to the absolutely continuous law with density up. Because that limit gives zero mass to every singleton, the frontier of $(a,b]$ is null. Portmanteau therefore gives

$$\nu_n((a,b]) \longrightarrow \int_a^b \mathrm{up}(x)\,\mathrm{d}x. \tag{100}$$

The intervals in (99) move with n, so one more squeeze is needed. Fix $\varepsilon > 0$ and choose $\delta > 0$ with $4\delta < \varepsilon$. Eventually $h_n \le \delta$, and hence

$$(a + \delta, b - \delta] \subset (a + h_n, b - h_n], \tag{101}$$

$$(a - h_n, b + h_n] \subset (a - \delta, b + \delta]. \tag{102}$$

The bound $0 \le \mathrm{up} \le 1$ gives

$$0 \le \int_a^b \mathrm{up} - \int_{a+\delta}^{b-\delta} \mathrm{up} \le 2\delta, \tag{103}$$

$$0 \le \int_{a-\delta}^{b+\delta} \mathrm{up} - \int_a^b \mathrm{up} \le 2\delta. \tag{104}$$

Apply (100) to the two fixed comparison intervals and combine with (99).

Theorem 3.14 (Interval-integral convergence of the step approximants). *For every $a, b \in \mathbb{R}$,*

$$\int_a^b \mathcal{W}_n(x)\,\mathrm{d}x \longrightarrow \int_a^b \mathrm{up}(x)\,\mathrm{d}x. \tag{105}$$

Proof. The preceding fixed-δ sandwich proves the result for $a \le b$. If $b < a$, reverse the orientation of both interval integrals. □

Warning 3.15 (The logical order cannot be shortened). Weak convergence of ν_n does not by itself imply pointwise convergence of the histograms $\mathcal{W}_n$, nor even convergence of their values at a single point. The formal route is

$$\begin{aligned}\text{weak convergence} &\Longrightarrow \text{null-boundary interval masses}\\ &\Longrightarrow \text{moving-cell sandwich} \Longrightarrow \text{interval-integral convergence.}\end{aligned}$$

Other parts of the development then use shape, refinement, Fourier decay, or difference quotients to obtain stronger function-level convergence.

4 The reusable discrete engines

4.1 Weighted paths solve triangular recurrences

The exact inverse-power recurrence for $F(2^{-n})$ is triangular. The main article states its composition formula, but the Lean proof first establishes a generic solver for every semiring-valued triangular recurrence. That abstraction explains both the form of the answer and the uniqueness argument.

4.1.1 Compositions as increasing paths

A composition of n is an ordered list of positive integers

$$c = (r_1, \dots, r_m), \qquad r_1 + \cdots + r_m = n. \tag{106}$$

Its partial sums

$$s_0 = 0, \qquad s_j = r_1 + \cdots + r_j \quad (1 \le j \le m) \tag{107}$$

form a strictly increasing path

$$0 = s_0 < s_1 < \cdots < s_m = n. \tag{108}$$

Conversely, every such path determines the positive increments $r_j = s_j - s_{j-1}$.

Let R be a semiring and let

$$w : \mathbb{N} \times \mathbb{N} \to R \tag{109}$$

be an edge-weight function. Define

$$W_w(c) = \prod_{j=1}^{m} w(s_{j-1}, s_j), \tag{110}$$

$$P_w(n) = \sum_{c \models n} W_w(c). \tag{111}$$

For $n = 0$, there is one empty composition and its empty product is 1, so

$$P_w(0) = 1. \tag{112}$$

4.1.2 Last-edge decomposition

Every nonempty path to n has a unique last predecessor $k < n$. Deleting its final edge gives a path to k, and appending the edge $k \to n$ reverses the operation. Therefore

Theorem 4.1 (Last-edge decomposition)**.** *For $n > 0$,*

$$P_w(n) = \sum_{k=0}^{n-1} P_w(k)w(k,n). \tag{113}$$

Proof. Partition the finite set of compositions of n by the sum k of all blocks except the last. For fixed k, deleting the final block is a bijection with compositions of k. Path-weight multiplicativity gives

$$W_w(c \parallel (n-k)) = W_w(c)w(k,n).$$

Summing first over compositions of k and then over $k < n$ gives (113). □

One can instead partition by the number of edges. Let $P_{w,r}(n)$ be the contribution of compositions with exactly r blocks. Uniformly in n,

$$P_w(n) = \sum_{r=0}^{n} P_{w,r}(n), \tag{114}$$

and for $n > 0$ the zero-edge term vanishes, so the range may be written $1 \le r \le n$.

4.1.3 The generic triangular solver

Theorem 4.2 (Path-sum solution of a triangular recurrence)**.** *Let $x : \mathbb{N} \to R$ satisfy*

$$x_0 = b, \tag{115}$$

$$x_n = \sum_{k=0}^{n-1} x_k w(k,n) \qquad (n > 0). \tag{116}$$

Then

$$x_n = bP_w(n) \qquad (n \ge 0). \tag{117}$$

Consequently the recurrence and initial value determine x uniquely.

Proof. Use strong induction on n. At $n = 0$, (112) gives $bP_w(0) = b = x_0$. For $n > 0$, insert the induction hypotheses into (116):

$$x_n = \sum_{k<n} bP_w(k)w(k,n) = b\sum_{k<n} P_w(k)w(k,n) = bP_w(n),$$

where the last equality is theorem 4.1. Any two solutions therefore equal the same path sum at each index. □

Remark 4.3. No subtraction, division, topology, or order is used. The theorem works in an arbitrary semiring. This is why the formal development separates it from the rational Fabius specialization.

4.2 Specialization to the inverse-dyadic recurrence

The Fabius specialization is stated only once, as (319) in the exact dyadic part. There it receives both the infinite-product/geometric-simplex derivation and the independent recurrence-path derivation obtained by specializing theorem 4.2. Keeping the generic solver here preserves the reusable semiring argument without repeating the same inverse-dyadic theorem and its small numerical cases.

4.3 A finite-row Toeplitz convergence theorem

The complex-shift and discrete-limit formulas involve a row of coefficients depending on n, not a fixed summability kernel. The main article verifies the Fabius specialization but omits the abstract theorem. The needed result is a finite-row version of the Toeplitz summability principle in a normed ring.

4.3.1 Abstract statement

Let K be a normed ring. For each n, let J_n be a finite index set, let $w_{n,j} \in K$, and let $q(n,j) \in \mathbb{N}$ be the sampled sequence index. Assume:

$$\sum_{j\in J_n} w_{n,j} = 1, \tag{118}$$

$$\sum_{j\in J_n} \|w_{n,j}\| \leq C \tag{119}$$

for one constant $C \geq 0$, and assume that the sampled indices move uniformly into every tail:

$$\forall N\ \exists n_0\ \forall n \geq n_0\ \forall j \in J_n, \qquad q(n,j) \geq N. \tag{120}$$

Theorem 4.4 (Finite-row Toeplitz convergence)**.** *If $H_m \to L$ in K, then*

$$\sum_{j\in J_n} w_{n,j} H_{q(n,j)} \longrightarrow L. \tag{121}$$

Proof. Fix $\varepsilon > 0$. Since $H_m \to L$, choose N such that

$$\|H_m - L\| < \frac{\varepsilon}{C+1} \qquad (m \geq N). \tag{122}$$

Choose n_0 from (120). For $n \geq n_0$, use the row-mass identity to subtract L:

$$\sum_{j\in J_n} w_{n,j} H_{q(n,j)} - L = \sum_{j\in J_n} w_{n,j} \big(H_{q(n,j)} - L\big).$$

Therefore

$$\begin{aligned}\left\| \sum_{j\in J_n} w_{n,j} H_{q(n,j)} - L \right\| &\leq \sum_{j\in J_n} \|w_{n,j}\|\, \|H_{q(n,j)} - L\| \\ &< \frac{\varepsilon}{C+1} \sum_{j\in J_n} \|w_{n,j}\| \\ &\leq \varepsilon \frac{C}{C+1} < \varepsilon.\end{aligned}$$

This proves convergence. □

Remark 4.5. The weights need not be positive, the rows need not be nested, and no entrywise limit of $w_{n,j}$ is required. The theorem uses exactly three facts: row mass one, uniformly bounded total variation, and uniform migration of all sampled indices into the tail.

4.4 The Fabius specialization and its canonical exact treatment

The abstract theorem above is the reusable normed-ring engine. Its Fabius specialization is developed only once in part III: the half-base q-binomial rows have mass one, uniformly bounded total variation, and sampled indices $2n - j$ escaping every finite prefix. The shifted-spline definition there also records the indispensable half-cell cutoff

$$\left\lfloor 2^p x - \frac{1}{2} \right\rfloor + 1 = \left\lfloor 2^p x + \frac{1}{2} \right\rfloor_+ ,$$

which converts an inclusive upper index into a natural range length without creating a spurious summand below the first cell. The exact-dyadic chapter then strengthens the old coarse convergence statement to an explicit $O_Q(4^{-n})$ estimate, exact stabilization at dyadic inputs, and the synchronized q-Richardson limits. Those sharper statements, rather than a second copy of the row calculation, are the canonical specialization of theorem 4.4 in this synthesis.

5 The formal and analytic machinery of the all-orders saddle expansion

5.1 Parameter-dependent Poincaré expansions

The endpoint asymptotic is not an ordinary expansion with constant coefficients. Its coefficients retain a bounded one-periodic dependence on the exact Lambert phase. The formal library therefore uses an asymptotic API in which coefficient families may depend on the asymptotic parameter, provided that dependence remains bounded.

Let ℓ be a filter on a parameter space A, let $\sigma : A \to \mathbb{K}$ be a scale tending to zero, and let $f : A \to E$ take values in a normed vector space over the normed field $\mathbb{K}$. For coefficient families $c_k : A \to E$, put

$$S_N[f](a) = \sum_{k=0}^{N-1} \sigma(a)^k c_k(a). \tag{123}$$

Definition 5.1 (Full parameter-dependent expansion)**.** We write

$$f(a) \sim_\ell \sum_{k\geq 0} \sigma(a)^k c_k(a) \tag{124}$$

when

1. every c_k is bounded along ℓ, equivalently $c_k = O_\ell(1)$;
2. for every $N \in \mathbb{N}$,
$$f - S_N[f] = O_\ell(\sigma^N). \tag{125}$$

This is `HasAsymptoticExpansion`. Boundedness of the coefficients is not redundant: if arbitrary unbounded parameter dependence were allowed, powers of σ could be moved between the "coefficient" and the scale, destroying the meaning of order.

5.1.1 Elementary algebra of full expansions

Suppose f and g have expansions with the same scale. Direct manipulation of finite partial sums proves:

$$f + g \sim \sum_{k\geq 0} \sigma^k (c_k + d_k), \tag{126}$$

$$f - g \sim \sum_{k\geq 0} \sigma^k (c_k - d_k), \tag{127}$$

$$\alpha f \sim \sum_{k\geq 0} \sigma^k (\alpha c_k). \tag{128}$$

The API also supports transport across eventual equality and pullback along a map tending to the source filter. These apparently routine lemmas are what let the final proof switch between complex and real saddle ratios, replace exact identities after a threshold, and move from the logarithmic coordinate to $x \downarrow 0$ without reopening every remainder estimate.

5.1.2 Flat functions

Definition 5.2 (Flat relative to a scale)**.** A function r is flat relative to σ along ℓ if

$$r = O_\ell(\sigma^N) \qquad \text{for every } N \in \mathbb{N}. \tag{129}$$

Theorem 5.3 (Flat remainders are invisible to all coefficients)**.** *A flat r has the full expansion with every coefficient zero. Consequently, if*

$$f \sim \sum_{k\geq 0} \sigma^k c_k, \tag{130}$$

then

$$f + r \sim \sum_{k\geq 0} \sigma^k c_k, \qquad f - r \sim \sum_{k\geq 0} \sigma^k c_k. \tag{131}$$

The implications hold in both directions.

Proof. For the zero-coefficient expansion, the Nth partial sum is identically zero, so the remainder condition is exactly (129). Add or subtract this expansion coefficientwise from the expansion of f. Reversibility follows by moving r to the other side. □

This theorem is the formal reason that the forward negative-Laplace tail can be restored after the central saddle expansion without changing any algebraic-order coefficient.

5.1.3 Exact logarithmic coordinates

Define

$$x(t) = 2^{-t}, \qquad t(x) = -\log_2 x \quad (x > 0). \tag{132}$$

These maps are exact inverses on $x > 0$, and

$$t \to +\infty \iff x(t) \downarrow 0, \qquad x \downarrow 0 \iff t(x) \to +\infty. \tag{133}$$

Theorem 5.4 (Full-expansion coordinate equivalence)**.** *For functions f, σ, c_k on the positive half-line,*

$$f(2^{-t}) \sim_{t\to\infty} \sum_{k\geq 0} \sigma(2^{-t})^k c_k(2^{-t}) \tag{134}$$

if and only if

$$f(x) \sim_{x\downarrow 0} \sum_{k\geq 0} \sigma(x)^k c_k(x). \tag{135}$$

Proof. Pull back the full expansion along either coordinate map. The two composites are eventually identical on their respective filters by (132); boundedness and every remainder estimate therefore transfer in both directions. □

5.2 Recursive coefficients of a formal exponential

The saddle integrand contains the exponential of a formal perturbation. Rather than invoke Bell polynomials or repeatedly expand a universal power series, the Lean development uses a finite recurrence whose nth output depends only on the first n input coefficients.

Let R be a commutative $\mathbb{Q}$-algebra, and let

$$E(u) = \sum_{j\geq 1} E_j u^j \tag{136}$$

be a formal exponent with zero constant term. Define $a_n \in R$ recursively by

$$a_0 = 1, \tag{137}$$

$$a_{n+1} = \frac{1}{n+1}\sum_{j=0}^{n}(j+1)E_{j+1}a_{n-j}. \tag{138}$$

Theorem 5.5 (Correctness and uniqueness of the exponential recurrence)**.** *The power series*

$$A(u) = \sum_{n\geq 0} a_n u^n \tag{139}$$

satisfies

$$A'(u) = E'(u)A(u), \qquad A(0) = 1. \tag{140}$$

It is the unique formal power series with these two properties, and therefore

$$A(u) = \exp(E(u)). \tag{141}$$

Proof. The coefficient of u^n in A' is $(n+1)a_{n+1}$. The coefficient of u^n in $E'A$ is

$$\sum_{j=0}^{n}(j+1)E_{j+1}a_{n-j},$$

so (140) is coefficientwise equivalent to (138). Conversely the differential equation and initial value determine a_0, then a_1, then a_2, and so on, proving uniqueness. The universal formal exponential satisfies the same equation and initial value. □

The first coefficients are

$$a_1 = E_1, \tag{142}$$

$$a_2 = E_2 + \frac{1}{2}E_1^2, \tag{143}$$

$$a_3 = E_3 + E_1E_2 + \frac{1}{6}E_1^3, \tag{144}$$

$$a_4 = E_4 + E_1E_3 + \frac{1}{2}E_2^2 + \frac{1}{2}E_1^2E_2 + \frac{1}{24}E_1^4. \tag{145}$$

Equation (145) is the exact finite identity used for the second saddle correction.

5.2.1 Structural laws

The recurrence is preferable to a black-box exponential because it exposes several laws needed downstream.

Proposition 5.6 (Finite dependence)**.** *The coefficient a_n depends only on $E_1, \dots, E_n$. If two exponent families agree through order n, their nth exponential coefficients agree.*

Proposition 5.7 (Naturality)**.** *For every morphism of commutative $\mathbb{Q}$-algebras $\varphi : R \to S$,*

$$\varphi(a_n(E)) = a_n(\varphi \circ E). \tag{146}$$

In particular, evaluation of polynomial- or function-valued coefficients commutes with the recurrence.

Proposition 5.8 (Rescaling, sums, and parity)**.** *Let $c \in R$ be central.*

1. *If $\widetilde{E}_j = c^j E_j$, then*
$$a_n(\widetilde{E}) = c^n a_n(E). \tag{147}$$

2. *If E and F are two exponent families, then*
$$a_n(E + F) = \sum_{r+s=n} a_r(E) a_s(F). \tag{148}$$

3. *If $E_j(-v) = (-1)^j E_j(v)$, then*
$$a_n(-v) = (-1)^n a_n(v). \tag{149}$$

Proof architecture. Naturality is induction through finite sums and rational scalar multiplication. Rescaling follows from the recurrence because every summand has total degree $(j+1)+(n-j) = n+1$. The addition law is the coefficient form of $\exp(E + F) = \exp(E) \exp(F)$. Parity is the specialization $c = -1$ of rescaling after pointwise reflection in v. □

5.3 Recursive coefficients of a formal logarithm

Let
$$A(u) = 1 + \sum_{n\geq 1} a_n u^n. \tag{150}$$

We seek
$$B(u) = \log A(u) = \sum_{n\geq 1} b_n u^n. \tag{151}$$

Instead of expanding the universal logarithm, use

$$A(u)B'(u) = A'(u). \tag{152}$$

Solving the coefficient of u^n for b_{n+1} gives

$$b_0 = 0, \tag{153}$$
$$b_{n+1} = a_{n+1} - \frac{1}{n+1} \sum_{j=0}^{n-1} (n-j) b_{n-j} a_{j+1}. \tag{154}$$

The empty sum at $n = 0$ gives $b_1 = a_1$.

Theorem 5.9 (Correctness and uniqueness of the logarithm recurrence). *If $a_0 = 1$, the series generated by (154) is the unique series with zero constant term satisfying (152); hence it is the formal logarithm of A.*

Proof. The coefficient of u^n in AB' contains the term $(n+1)b_{n+1}$ from $a_0 b'$, plus terms involving only $b_1, \dots, b_n$. Equating it with $(n+1)a_{n+1}$ and dividing by $n+1$ gives (154). This triangular relation proves uniqueness. The formal logarithm satisfies $B(0) = 0$ and $B'A = A'$. □

The first coefficients are

$$b_1 = a_1, \tag{155}$$

$$b_2 = a_2 - \frac{1}{2}a_1^2, \tag{156}$$

$$b_3 = a_3 - a_1 a_2 + \frac{1}{3}a_1^3, \tag{157}$$

$$b_4 = a_4 - a_1 a_3 - \frac{1}{2}a_2^2 + a_1^2 a_2 - \frac{1}{4}a_1^4. \tag{158}$$

As with the exponential recurrence, b_n depends only on $a_1, \dots, a_n$, commutes with $\mathbb{Q}$-algebra morphisms and evaluation, obeys the rescaling law

$$a_j \mapsto c^j a_j \quad \Longrightarrow \quad b_n \mapsto c^n b_n, \tag{159}$$

and preserves parity.

Remark 5.10 (Why a_0 is a hypothesis, not an input). The recursive definition (154) never reads a_0; it is meaningful for every sequence. Correctness as a logarithm, however, requires $a_0 = 1$. Separating the recurrence from its normalization makes finite coefficient calculations easier while keeping the theorem honest.

5.3.1 From mass expansions to logarithmic expansions

Suppose a positive real function has a full expansion

$$R(a) \sim 1 + \sum_{n\geq 1} \sigma(a)^n a_n(a), \qquad \sigma(a) \to 0, \tag{160}$$

with bounded coefficient families. Finite polynomial remainders, discussed in the next section, show that for each N,

$$\log R(a) = \sum_{n=1}^{N-1} \sigma(a)^n b_n(a) + O(\sigma(a)^N), \tag{161}$$

where b_n is generated by (154). The positivity hypothesis fixes the real branch and is eventually automatic when $R \to 1$.

5.4 Exact finite remainders from formal coefficient identities

Formal power-series equality says that two expressions have the same coefficients below a chosen order. An asymptotic proof needs an evaluated identity with an explicit factor of the first omitted power. The module `SaddleExpansionFiniteRemainder.lean` provides exactly that bridge without invoking an infinite series.

Fix $L \geq 1$ and truncate the exponent:

$$E_{<L}(X) = \sum_{k=1}^{L-1} E_k X^k. \tag{162}$$

Truncate the scalar exponential polynomial at the same order and substitute:

$$T_L(X) = \sum_{q=0}^{L-1} \frac{E_{<L}(X)^q}{q!}. \tag{163}$$

Let

$$A_{<L}(X) = \sum_{k=0}^{L-1} a_k X^k \tag{164}$$

be the polynomial formed from the recursive exponential coefficients.

Theorem 5.11 (Exact divisibility of the finite defect)**.** *If $E_0 = 0$, then*

$$X^L \mid \big(T_L(X) - A_{<L}(X)\big). \tag{165}$$

Equivalently, there is a canonical polynomial $Q_L(X)$ such that

$$T_L(X) = A_{<L}(X) + X^L Q_L(X). \tag{166}$$

Proof. The coefficient of X^k in T_L agrees with that of the full formal substitution $\exp(E(X))$ for every $k < L$: terms of scalar exponential degree $q \geq L$ cannot contribute below degree L because E has zero constant term, and truncating E itself does not alter coefficients below L. By theorem 5.5, that common coefficient is a_k. Hence the first L coefficients of the defect vanish, which is equivalent to divisibility by X^L. The formal construction takes Q_L to be the result of applying polynomial division by X exactly L times. □

Evaluating at $x \in R$ gives the identity actually used in estimates:

$$\sum_{q=0}^{L-1} \frac{1}{q!} \left(\sum_{k=1}^{L-1} E_k x^k \right)^q = \sum_{k=0}^{L-1} a_k x^k + x^L Q_L(x). \tag{167}$$

The quotient Q_L is a polynomial in x whose coefficients are algebraic expressions in the finitely many parameters $E_1, \dots, E_{L-1}$. If those parameter families are uniformly bounded, then Q_L is uniformly bounded on a fixed compact x-window. Thus the exact factorization (167) turns formal coefficient agreement into a uniform $O(x^L)$ estimate with no appeal to convergence of an infinite formal series.

5.5 Gaussian contraction: integration becomes polynomial algebra

After the saddle variable is scaled by $v = \sqrt{\lambda}\,\theta$, the leading kernel is the standard Gaussian

$$\gamma(v) = \mathrm{e}^{-v^2/2}. \tag{168}$$

Every finite reference term is $\gamma(v)$ times a complex polynomial. The generic contraction module turns its whole-line integral into an algebraic linear functional.

5.5.1 Gaussian moments

Define the unnormalized moments

$$M_n^{\mathrm{G}} = \int_{\mathbb{R}} \mathrm{e}^{-v^2/2} v^n \, \mathrm{d}v. \tag{169}$$

Every moment is absolutely integrable. Symmetry gives

$$M_{2j+1}^{\mathrm{G}} = 0. \tag{170}$$

Integration by parts, with $u = v^{n+1}$ and $\mathrm{d}w = -v\mathrm{e}^{-v^2/2} \, \mathrm{d}v$, gives

$$M_{n+2}^{\mathrm{G}} = (n+1) M_n^{\mathrm{G}}. \tag{171}$$

Since

$$M_0^{\mathrm{G}} = \sqrt{2\pi}, \tag{172}$$

the normalized moments

$$\mu_n = \frac{M_n^{\mathrm{G}}}{\sqrt{2\pi}} \tag{173}$$

satisfy

$$\mu_0 = 1, \quad \mu_1 = 0, \quad \mu_{n+2} = (n+1)\mu_n. \tag{174}$$

Thus

$$\mu_{2j} = (2j-1)!!, \qquad \mu_{2j+1} = 0. \tag{175}$$

The first even values needed below are

$$\mu_4 = 3, \quad \mu_6 = 15, \quad \mu_8 = 105, \quad \mu_{10} = 945, \quad \mu_{12} = 10395. \tag{176}$$

5.5.2 The contraction functional

For $p(X) = \sum_k p_k X^k \in \mathbb{C}[X]$, define

$$\mathcal{G}[p] = \sum_k p_k \mu_k. \tag{177}$$

This is a $\mathbb{C}$-linear map $\mathbb{C}[X] \to \mathbb{C}$.

Theorem 5.12 (Gaussian polynomial contraction)**.** *For every complex polynomial p,*

$$\int_{\mathbb{R}} \mathrm{e}^{-v^2/2} p(v) \, \mathrm{d}v = \sqrt{2\pi} \, \mathcal{G}[p]. \tag{178}$$

In particular

$$\mathcal{G}[X^{2j}] = (2j-1)!!, \qquad \mathcal{G}[X^{2j+1}] = 0. \tag{179}$$

Proof. Expand p as a finite sum of monomials, interchange that finite sum with the integral, and use (173). Integrability follows coefficientwise from the absolute Gaussian moments. ☐

The same coefficientwise expansion yields the norm estimate

$$|\mathcal{G}[p]| \le \sum_k |p_k| \mu_{2\lceil k/2 \rceil}, \tag{180}$$

with odd coefficients contributing zero in the exact functional. In the formal proof a slightly more uniform support sum is used so it composes smoothly with polynomial families.

Remark 5.13 (Parity is an algebraic annihilator)*.* The vanishing of odd Gaussian moments is not merely a simplification after integration. It is the mechanism that removes every odd power of the small parameter $\varepsilon = \lambda^{-1/2}$ from the mass expansion. The exponent coefficient of order m has parity m in v, the formal exponential recurrence preserves total parity, and (179) kills every odd result. The final expansion is therefore in integer powers of λ^{-1}.

5.6 Uniform Gaussian tails for polynomial references

Whole-line contraction is useful only after the central-window integral has been replaced by a whole-line integral with a negligible error. The omitted Gaussian-tail modules provide a single coefficient weight that controls every polynomial reference.

5.6.1 A factorial coefficient weight

For $p(X) = \sum_k p_k X^k$, define

$$\mathfrak{W}(p) = \sum_{k \in \operatorname{supp} p} |p_k| \, 8k!. \tag{181}$$

It is finite and nonnegative.

Lemma 5.14 (Monomial Gaussian tail)**.** *For $A \geq 4$ and $k \in \mathbb{N}$,*

$$\int_{|v|>A} \mathrm{e}^{-v^2/2} |v|^k \, \mathrm{d}v \leq 8k! \, \frac{\mathrm{e}^{-A^2/4}}{A}. \tag{182}$$

Proof architecture. Split the kernel as $\mathrm{e}^{-v^2/2} = \mathrm{e}^{-v^2/4}\mathrm{e}^{-v^2/4}$. The factor $|v|^k \mathrm{e}^{-v^2/4}$ is controlled uniformly by a factorial bound, while the remaining Gaussian tail is estimated by integrating the derivative of $\mathrm{e}^{-v^2/4}$ and using $|v| \geq A$. The constants are chosen so the result holds simultaneously for every $k \geq 0$ once $A \geq 4$; no optimization is needed in the saddle application. □

Theorem 5.15 (Polynomial Gaussian tail)**.** *For every $p \in \mathbb{C}[X]$ and $A \geq 4$,*

$$\int_{|v|>A} \left| \mathrm{e}^{-v^2/2} p(v) \right| \mathrm{d}v \leq \mathfrak{W}(p) \frac{\mathrm{e}^{-A^2/4}}{A}. \tag{183}$$

Proof. Use $|p(v)| \leq \sum_{k \in \operatorname{supp} p} |p_k| |v|^k$, integrate the finite sum, and apply theorem 5.14 coefficient-wise. □

5.6.2 The all-orders central radius

For expansion order N and scale parameter $b \geq 1$, set

$$A_N(b) = \sqrt{32(N+1)\log b}. \tag{184}$$

Then

$$\mathrm{e}^{-A_N(b)^2/4} = b^{-8(N+1)}. \tag{185}$$

The exponent $8(N+1)$ is much stronger than the requested algebraic order $N+1$, leaving ample room for the factor $1/A_N(b)$ and for bounded coefficient weights.

Theorem 5.16 (All-orders polynomial tail)**.** *Let $b(a) \to +\infty$ along a filter and let $p_a \in \mathbb{C}[X]$ satisfy*

$$\mathfrak{W}(p_a) = O(1). \tag{186}$$

Then for every fixed N,

$$\int_{|v|>A_N(b(a))} \left| \mathrm{e}^{-v^2/2} p_a(v) \right| \mathrm{d}v = O\big(b(a)^{-(N+1)}\big). \tag{187}$$

Proof. Eventually $A_N(b(a)) \geq 4$ and $b(a) \geq 1$. Apply theorem 5.15, substitute (185), and use

$$b^{-8(N+1)} A_N(b)^{-1} \leq b^{-(N+1)}.$$

Multiplication by the bounded family (186) preserves the big-O rate. □

Remark 5.17. The theorem is formulated for a moving polynomial family. This is essential: the saddle reference polynomial depends periodically on the exact phase. Periodicity and smoothness make its finitely many coefficient functions bounded, hence its factorial weight is bounded.

5.7 Handoff to the Fabius-specific saddle synthesis

The preceding Poincaré, formal exponential/logarithm, finite-remainder, and Gaussian modules are deliberately generic. Their Fabius-specific instantiation is assembled once in part VI, after the exact lower-Lambert phase and periodic negative-Laplace decomposition of part IV. There the closed shifted-Stirling jets feed the two-monomial exponent coefficients, the formal exponential produces the saddle-mass coefficients, Gaussian contraction removes the local variable, and the recursive logarithm produces the periodic corrections. The same chapter gives the explicit first and second corrections and the rigorous all-orders central/minor-arc/flat-tail assembly.

This separation also fixes the claim boundary. The mass top-jet theorem is formalized; the transfer to the logarithmic highest-Ψ-derivative coefficient is retained as a derived statement. Likewise, the expansion is stated in the exact phase: replacing that phase by a truncated elementary approximation inside periodic coefficients requires a separate composition theorem. The full formulas, status ledger, and remaining obligations are therefore kept with the canonical application rather than repeated here.

Supplementary material from this synthesis

5.8 Executable recurrences in mathematical pseudocode

The following algorithms are not substitutes for the proofs above. They are compact implementations of the finite algebra that appears repeatedly in the formal development. All arrays are zero-indexed.

Algorithm 5.18 (Weighted path sum)**.** The dynamic program below computes $P_w(n)$ without enumerating compositions. It is the triangular recurrence proved equivalent to the composition sum in theorem 4.2.

```
function PathSums(N, weight):
    P[0] := 1
    for n := 1 to N:
        P[n] := 0
        for k := 0 to n - 1:
            P[n] := P[n] + P[k] * weight(k, n)
    return P

function FabiusInverseDyadic(N):
    weight(k, n) := 1 / ((2^n - 1) * factorial(n - k + 1))
    a := PathSums(N, weight)
    for n := 0 to N:
        value[n] := 2^(-binomial(n, 2)) * a[n]
    return value
```

Algorithm 5.19 (Formal exponential coefficients)**.** Given $E[1], \ldots, E[N]$ in a commutative $\mathbb{Q}$-algebra, this returns the first N coefficients of $\exp(\sum_{j\geq 1} E[j]u^j)$.

```
function ExpCoeff(E, N):
    a[0] := 1
    for n := 0 to N - 1:
        s := 0
        for j := 0 to n:
            s := s + (j + 1) * E[j + 1] * a[n - j]
        a[n + 1] := s / (n + 1)
    return a
```

Algorithm 5.20 (Formal logarithm coefficients)**.** The input is a unit series $1 + \sum_{j\geq 1} a[j]u^j$. The output satisfies $\log(1 + \sum a[j]u^j) = \sum b[j]u^j$.

```
function LogCoeff(a, N):
    b[0] := 0
    for n := 0 to N - 1:
        s := 0
        for j := 0 to n - 1:
            s := s + (n - j) * b[n - j] * a[j + 1]
        b[n + 1] := a[n + 1] - s / (n + 1)
    return b
```

Algorithm 5.21 (Normalized Gaussian contraction)**.** The moment table can be built once to the degree of the input polynomial.

```
function GaussianMoments(D):
    mu[0] := 1
    if D >= 1: mu[1] := 0
    for n := 0 to D - 2:
        mu[n + 2] := (n + 1) * mu[n]
    return mu

function GaussianContract(coeff, D):
    mu := GaussianMoments(D)
    result := 0
    for k := 0 to D:
        result := result + coeff[k] * mu[k]
    return result
```

Algorithm 5.22 (Saddle coefficient pipeline)**.** This is the finite algebra at fixed logarithmic order N. The analytic proof is what justifies applying it to the central integral with a uniform remainder.

```
function SaddleCoefficients(N, phase):
    # 1. Build bounded exponent jets Z[0 .. 2N - 1].
    Z := ClosedJets(2*N - 1, phase)

    # 2. Build exponent polynomials E[1 .. 2N].
    for m := 1 to 2*N:
        E[m] := i^m * (Z[m - 1] * X^m / factorial(m)
                       + (-1)^(m + 1) * X^(m + 2) / (m + 2))

    # 3. Exponentiate formally in epsilon.
    a := ExpCoeff(E, 2*N)

    # 4. Contract even epsilon orders against the normalized Gaussian.
    B[0] := 1
    for j := 1 to N:
        B[j] := GaussianContract(coefficients(a[2*j]), degree(a[2*j]))

    # 5. Convert mass coefficients to logarithmic coefficients.
    A := LogCoeff(B, N)
    return (B, A)
```

5.9 Lean declaration and module concordance

File names below are relative to

`Analysis/FabiusFunction/Lean/FabiusFunction/`.

The table lists the principal declarations, not every supporting lemma.

Module	Content explained in this companion
`OriginalCharacterization.lean`	`IsOriginalFabius.mk_of_derivative_law`: the support/mean-value argument showing that scale positivity follows from the other original-characterization assumptions; section 1.3.
`FabiusInverse.lean`	The complete order, calculus, curvature, exact-locus, and endpoint-steepness interface is consolidated in part VI; section 2.1 is the handoff.

Module	Content explained in this companion
`ElementaryFunction.lean`	The inductive predicate `IsElementary`, composition closure, derived elementary operations, analytic loci, the finite-bad-value composition theorem, and the dense-open analytic-locus theorem; section 2.2.
`NotElementary.lean`	No elementary function agrees with a bounded Fabius function, up, or the signed extension on a set with nonempty interior; section 2.3.
`InverseNotElementary.lean`	Exact analytic locus of the total inverse and exclusion from both the elementary class and the class enlarged by admissible inverse branches; section 2.3.
`ProbabilityRepresentation.lean`	The product sample space, uniformly convergent coordinate sum, pushforward law, the exact dyadic-to-generic bridge `ProbabilityRepresentation.weightedSumDistribution_eq_weightedUniformDistribution`, the two half-base geometric bridges, head–tail split, coordinate reflection, the exact support theorem `ProbabilityRepresentation.weightedSumDistribution_support_eq_Icc`, and identification of the CDF with F; section 3.1.
`WeightedUniformSeries.lean`	The eleven-declaration generic tranche: almost-everywhere and pointwise head–tail map recursion, continuous-linear and scalar naturality, shifted-law transport, and operator/scalar affine recursions; section 3.1.3.
`WeightedUniformDistribution.lean`	The compatibility head–tail, probability, scalar-law, and unit-interval surface, followed by the four-declaration s-finite uniform-smoothing, sharp weighted-law absolute-continuity, and null-singleton tranche; section 3.1.3.
`WeightedUniformSupport.lean`	The ten-declaration exact-support tranche: open positivity for arbitrary products, continuous-pushforward support, support equal to the series range, exact signed-real min/max intervals, and the nonnegative and unit-mass specializations; section 3.1.3.
`GeometricUniformLaw.lean`	The twenty-three-declaration named geometric weight, series, and law API: summability and mass, continuity and measurability, affine splitting, reflection, absolute continuity, null singletons, and support equal to the series range for $\lvert q\rvert < 1$, with exact support $[0, 1]$ for $0 \le q < 1$; section 3.1.3.
`ContinuousCDF.lean`	The three reusable real-CDF bridges: continuity from null singletons, CDF reflection from affine reflection invariance, and identification of an everywhere pointwise CDF derivative as a Lebesgue Radon–Nikodym density; section 3.1.3.
`GeometricUniformCDF.lean`	The named geometric CDF and density API: continuity and reflection for $\lvert q\rvert < 1$, tails for $0 \le q < 1$, and, for $0 < q < 1$, conditioning and interval-integral refinement, everywhere differentiation, nonnegativity, symmetry, compact support, the exact `withDensity` identity, and real-space C^∞ regularity; section 3.1.3.
`SubgraphFubini.lean`	`integral_smul_setIntegral_subgraph` and `integral_smul_setIntegral_supergraph`, the real- or complex-weighted Banach-valued open-subgraph and atom-preserving closed-supergraph exchanges over arbitrary s-finite measures; section 3.2.
`SurvivalLayerCake.lean`	The support-free clamped and interval-supported partial/full Banach-valued survival identities; theorem 3.5.

Module	Content explained in this companion
`CDFLayerCake.lean`, `FractionalCDFLayerCake.lean`	The support-free clamped, one-sided-upper-supported partial, compactly supported partial, and full Banach-valued lower-CDF identities, followed by the positive-real clamped, lower-supported positive-part, probability-CDF, and compact endpoint-power fractional forms; theorem 3.6.
`FabiusFractionalIntegral.lean`	The distribution-law and product-sample-space positive-real stopped-power formulas for every real endpoint, together with both unit-endpoint ordinary-moment forms; theorem 3.6.
`InverseLayerCake.lean`	Generic inverse-clock and Fabius explicit-primitive/pointwise-C^1 layer cake, the full weighted inverse-area identity, equality with the up area, and the inverse mean $1/2$; section 3.2.
`InversePairIntegral.lean`	The real two-function absolutely-continuous inverse-pair identity on arbitrary ordered compact rectangles, with clock monotonicity and measurable clamped representatives derived from interval preservation and the strict order equivalence; its Fabius wrapper and area sum; section 3.2.
`CauchyTransform.lean`, `CauchyCDF.lean`, `CauchyRenormalization.lean`	The ordinary up-law and Fabius Cauchy–Stieltjes transforms on their slit domains; three atom-exact compact-support Cauchy–CDF integration-by-parts theorems; and five renormalization theorems giving the two affine domain maps, the exact Rvachev-transform DDE, and its all-order finite Thue–Morse derivative orbit. No logarithmic fixed point, higher-kernel integral identity, named Stieltjes DDE, boundary/Plemelj theorem, moment/Laurent expansion, generalized order, or Jacobi/Padé result is asserted.
`FractionalVolterra.lean`, `FractionalVolterraSemigroup.lean`	The total real-order Gamma-normalized operator, its unconditional endpoint/locality/scalar, order-one, and positive-natural-order rules, and its positive-order ordered-endpoint continuous-kernel integrability and input additivity; raw shifted-beta-kernel integrability for positive orders and ordered endpoints $s \leq x$; the raw and normalized values under a strict endpoint inequality; and additive composition of positive orders for continuous Banach-valued inputs on ordered compact intervals. Normalized shifted powers, arbitrary shifted real powers in the sharp range $\rho > -1$, and constants have their exact Gamma-quotient values. No corresponding semigroup for merely interval-integrable inputs or noncomplete targets, fractional derivative, complex order, order-zero law, reversed-endpoint fractional interpretation, or hierarchy-specific semigroup; theorem 3.8.
`FractionalVolterraCalculus.lean`, `FabiusFractionalVolterra.lean`	Increasing-affine covariance for arbitrary real order, positive scale, and ordered endpoints; positive-order Banach-valued integration by parts across an interval-integrable right interior derivative with its exact left-boundary term; and the three positive-real single-step order shifts for the signed and bounded Fabius functions and up. These modules export four and three theorems, respectively, but no nonpositive-scale/reversed covariance, fractional derivative or Caputo operator, complex-order continuation, iterated/lattice or endpoint-up wrapper, or inverse specialization; theorem 3.8.

Module	Content explained in this companion
`ProbabilityLaplaceMoments.lean`	The survival function, partial/full Rvachev survival layer cake, differentiable compact-support Fubini identity, raw and endpoint moment identifications, zero-tilt normalization, and positivity; the dyadic almost-sure support and restriction declarations are owned upstream by `ProbabilityRepresentation.lean`; sections 3.2 and 3.3.
`LaplaceMoments.lean`	`tiltedSurvivalMoment`, `fabiusLaplaceMoment`, the differential recurrence, iterated derivatives, smoothness, and evaluation at zero tilt; section 3.3.
`StepMeasureBridge.lean`	Almost-everywhere replacement of half-endpoint indicators, exact overlap integrals, atomic measure comparisons, the shrunken/enlarged interval sandwich, portmanteau, and convergence of every interval integral; section 3.4.
`FabiusInverseDyadicClosedForm.lean`	Compositions, path weights, last-edge decomposition, generic triangular solver, uniqueness, and the exact composition formula for $F(2^{-n})$; sections 4.1 and 4.2.
`FabiusDiscreteLimitToeplitz.lean`	Corrected prefix length, exact q-binomial row mass, total-variation bound 16, the generic finite-row Toeplitz theorem, and the specialization $q(n, j) = 2n - j$; sections 4.3 and 4.4.
`FinitePowerSeriesFilter.lean`, `GeometricPowerSeriesFilter.lean`	The coefficientwise diagonal action of arbitrary finite-node weighted rescale filters, rescaling and coefficient-ring naturality, exactness through the prescribed order, and the denominator-free Gaussian residual multiplier for geometric filters, including the distinct-node Lagrange specialization. These are formal power-series identities only and assert no analytic convergence, norm bound, or remainder estimate.
`AnalyticSeriesFilter.lean`	The definitions `finiteAnalyticSeriesFilter` and `geometricAnalyticSeriesFilter`; diagonal summability and evaluation `summable_finiteAnalyticSeriesFilter_diagonal` and `finiteAnalyticSeriesFilter_eq_tsum`; the exact head/tail decompositions `finiteAnalyticSeriesFilter_eq_head_add_tail_of_exact` and `finiteAnalyticSeriesFilter_eq_constant_add_tail_of_exact_zero`; and the exact Gaussian tails `geometricAnalyticSeriesFilter_eq_constant_add_gaussian_tsum`, `geometricLagrangeAnalyticSeriesFilter_eq_constant_add_gaussian_tsum`, and `geometricLagrangeAnalyticSeriesFilter_shifted`. Every sampled-series assumption is unconditional and explicit; no conditional/boundary convergence, radius or uniform convergence, error/norm or asymptotic estimate, positivity, or Fabius-specific acceleration claim is made.
`ImplicitPowerSeries.lean`	One distinguished root definition and eight nonmonic formal implicit-root theorems. The three `FormalImplicitRoot` results give existence, uniqueness, and unique existence in a complete adic commutative ring from an approximate root and a derivative unit modulo the ideal (standalone uniqueness uses the stated Jacobson-radical bound). The six `PowerSeries.Implicit` declarations give arbitrary-base and zero-constant lifting, the distinguished root, its constant coefficient and evaluation equation, and uniqueness over an arbitrary commutative coefficient ring. No concrete inverse-Fabius germ, analytic convergence, plateau localization, flat-remainder, or quantile theorem is asserted.

Module	Content explained in this companion
`ThueMorseComplexProductBridge.lean`	Total complex finite Thue–Morse sinc and negative-Laplace block identities at every level, including the removable origin, their quotient forms away from zero, origin simplification laws, and the exact finite Fourier–Laplace rotation. No infinite-product convergence, analytic logarithm, tail estimate, affine lacunary cosine companion, or measure-specific transform theorem is asserted.
`SaddleExpansionAlgebra.lean`	`expCoeff`, its differential equation, uniqueness, naturality, rescaling, addition, parity, and `HasAsymptoticExpansion`; sections 5.1 and 5.2.
`SaddleLogExpansionAlgebra.lean`	`logCoeff`, the identity $AB' = A'$, low-order formulas, uniqueness, naturality, rescaling, and parity; section 5.3.
`SaddleExpansionFiniteRemainder.lean`	Finite exponent substitution, equality of coefficients below the truncation order, divisibility of the defect by X^L, and the evaluated exact remainder; section 5.4.
`SaddleExpansionSmallArgument.lean`	Exact transfer of full expansions between $t \to \infty$ and $x = 2^{-t} \downarrow 0$; section 5.1.
`SaddleExpansionFlat.lean`	Zero-coefficient expansion of all-orders-flat errors and invariance of every coefficient under addition or subtraction of such an error; sections 5.1 and 5.7.
`GaussianPolynomialContraction.lean`	Gaussian integrability, odd moments, integration-by-parts recurrence, normalized moments, linear polynomial contraction, and the whole-line integral identity; section 5.5.
`GaussianPolynomialTail.lean`	The factorial coefficient weight and explicit complementary-interval tail bound; section 5.6.
`GaussianPolynomialTailAllOrders.lean`	The radius $\sqrt{32(N+1)\log b}$ and arbitrary algebraic tail order for bounded polynomial families; section 5.6.
`FabiusSaddle*.lean`	The closed jets, shifted-Stirling conversion, explicit low-order corrections, top mass jet, and the central/tail/all-orders Fabius application are mapped in the canonical concordance of section 77; this part supplies only the generic coefficient, remainder, and Gaussian engines used there.

5.10 Audit matrix against the main article

Topic	Status before this companion	Added here
Original scale positivity	Assumed in the source statement	Mean-value proof that the dilation parameter is automatically positive before its exact value $k = 2$ is derived.
Total inverse I	Absent	Canonical treatment consolidated in part VI; this part retains only the analytic-density obstruction shared with the forward functions.
Elementary function class	Absent	Exact inductive closure, totalization convention, finite-bad-value composition, and dense-open analytic locus.
Non-elementarity	Absent	No local agreement on any set with nonempty interior for F, up, the signed extension, or the total inverse, including the enlarged admissible inverse-branch class.
Product probability law	Headline proof only	Uniform convergence and measurability, head–tail pushforward identity, coordinate reflection, and exact dyadic support $[0, 1]$.

Topic	Status before this companion	Added here
Generic weighted-law naturality, absolute continuity, and support	Not recorded in the pinned companion snapshot	Almost-everywhere and pointwise head–tail recursion, continuous-linear and scalar naturality, shifted-law transport, conditional operator/scalar affine fixed-point laws, and the seven-name compatibility surface; a four-name tranche gives s-finite uniform smoothing, sharp nontrivial weighted-law absolute continuity, and null singletons; a separate ten-name tranche gives support equal to the series range, exact signed-real min/max intervals, and nonnegative specializations.
Named geometric-q law	Not recorded in the pinned companion snapshot	Exact weights, summability and mass, series and probability law, affine self-similarity, reflection, absolute continuity, null singletons, and support equal to the series range for $\lvert q\rvert < 1$, exact support $[0, 1]$ for $0 \leq q < 1$, a named CDF continuous and symmetric for $\lvert q\rvert < 1$, its exact nonnegative-range tails, and, for $0 < q < 1$, refinement integrals, an explicit compactly supported Radon–Nikodym density, and real-space C^{∞} regularity. Series, law, CDF, and density half-base bridges are formal. The zero/negative parameter density interpretation, general-q Fourier route, centered package, closed higher-derivative and endpoint-flatness wrappers, shape, transforms, moments/cumulants, and fixed-point uniqueness remain outside the formal boundary.
Probability–Laplace bridge	Absent	Generic Banach-valued partial and full survival layer cake, inverse layer cake and exact inverse area, identification of raw, endpoint, and tilted moments, complete monotonicity, and half-moment interpretation.
Atomic-to-step bridge	Compressed	Half-endpoint null-set conversion, exact cell overlap, shrunken/enlarged interval sandwich, portmanteau, and moving-boundary squeeze.
Weighted path solver	Explicitly omitted	Generic semiring theorem, uniqueness, edge-count decomposition, Fabius specialization, and worked values.
Finite-row Toeplitz theorem	Explicitly omitted	Abstract normed-ring proof, exact Fabius row mass, uniform variation bound, tail-index condition, and corrected prefix rounding.
Parameter-dependent full expansions	Only summarized	Exact API, coefficient boundedness, closure, flat errors, and coordinate transfer.
Formal exponential/logarithm	Explicitly omitted	Recurrences, formal differential equations, uniqueness, low orders, naturality, rescaling, addition, and parity.
Finite formal-to-analytic remainder	Absent	Exact divisibility by the first omitted power and evaluated polynomial remainder.
Pointwise analytic finite filters	Absent	Unconditional diagonalization under weighted sample-series summability, exact finite heads and tails, and Gaussian residual tails for geometric and geometric-Lagrange rows. Conditional boundary series, uniform estimates, signs/rates, and sinc-product instantiation remain outside the formal result.

Topic	Status before this companion	Added here
Gaussian contraction/tails	Explicitly omitted	Moment recurrence, contraction functional, factorial coefficient weight, explicit tail, and all-orders radius.
Fabius-specific saddle application	Split across overlapping notebooks	Consolidated once in part VI, including the closed jets, two explicit corrections, rigorous all-orders assembly, exact-phase warning, higher symbolic coefficients, and numerical claim boundary.

The following substantial topics are intentionally not repeated because the main article already treats them at proof level: construction and uniqueness of F and up; the original Rvachev characterization apart from the redundancy isolated in section 1.3; the infinite sinc product and Fourier decay; support, positivity, shape, and nowhere analyticity; exact moment recurrences and denominator arithmetic; dyadic rational evaluation, q-Pochhammer and q-binomial formulas; Thue–Morse interpolation and iterated difference identities; Legendre expansions and least-squares approximation; and the computational API. Their use as input above is always signposted.

5.11 Notation crosswalk

This companion	Main-article notation	Principal Lean declaration
$I(y)$	inverse of bounded F	`fabiusInv`
μ	law of the weighted uniform sum	`weightedSumDistribution`
$J_k(s)$	tilted survival moment	`tiltedSurvivalMoment`
$M_k(s)$	tilted probability/Fabius moment	`fabiusLaplaceMoment`
d_n	exact half moment	`halfMoment`
$P_w(n)$	weighted composition/path sum	generic path-sum definition in `FabiusInverseDyadicClosedForm.lean`
σ	abstract small scale	first argument of `HasAsymptoticExpansion`
a_n	formal exponential coefficient	`SaddleExpansion.expCoeff`
b_n	formal logarithm coefficient	`SaddleExpansion.logCoeff`
$\mathcal{G}[p]$	normalized Gaussian contraction	`SaddleExpansion.gaussianPolynomialContraction`
$Z_n(t)$	bounded exponent jet $\widetilde{\mathcal{Z}}_n$	`negativeLaplaceBoundedExponentJet`
$B_j(t)$	normalized saddle mass coefficient	`fabiusSaddleMassCoefficient`
$A_j(t)$	logarithmic saddle coefficient	`fabiusSaddleLogCoefficient`
$\lambda(x)$	exact lower-Lambert phase	`fabiusLambertPhase`
$\mathcal{M}(x)$	corrected sharp Lambert main	`fabiusSharpLambertMain`
$\mathcal{S}(x)$	normalized real saddle mass/ratio	`fabiusSaddleRatio` and the kernel-mass bridge

5.12 A compact dependency diagram

The proof dependencies of the material unique to this companion can be read from the following diagram. An arrow means that the target uses the source as a mathematical input, not merely as a

Lean import.

$$
\begin{aligned}
&\text{support + interior positivity + mean-value theorem + derivative law} \longrightarrow k > 0,\\
&\text{strict monotonicity + surjectivity + flatness} \longrightarrow \text{total inverse} \longrightarrow \text{endpoint steepness},\\
&\text{nowhere analyticity + dense elementary analytic locus} \longrightarrow \text{strong non-elementarity},\\
&\text{product law} \longrightarrow \text{survival identity} \longrightarrow \text{Laplace/raw/endpoint moment bridge},\\
&\text{weak convergence + null boundaries + cell geometry} \longrightarrow \text{step-integral convergence},\\
&\text{triangular recurrence} \longrightarrow \text{weighted paths} \longrightarrow \text{composition formula},\\
&q\text{-binomial row mass + variation + tail indices} \longrightarrow \text{discrete Toeplitz limit},\\
&\text{closed jets} \longrightarrow \text{exponent coefficients} \longrightarrow \text{formal exponential}\\
&\qquad\qquad \longrightarrow \text{Gaussian contraction} \longrightarrow \text{mass coefficients},\\
&\text{mass coefficients} \longrightarrow \text{formal logarithm} \longrightarrow \text{log coefficients},\\
&\text{finite remainders + central Taylor + Gaussian tails + minor arcs}\\
&\qquad\qquad \longrightarrow \text{all-orders saddle mass},\\
&\text{all-orders mass + exact reduction + flat tail + coordinate transfer}\\
&\qquad\qquad \longrightarrow \text{full exact-phase expansion.}
\end{aligned}
$$

Topic references

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[2] J. Arias de Reyna, On the Fabius function, *The American Mathematical Monthly* **89** (1982), 291–298.

[3] J. Arias de Reyna, An infinitely differentiable function with compact support: Definition and properties, arXiv:1702.05442.

[4] J. Arias de Reyna, Arithmetic of the Fabius function, arXiv:1702.06487, version 3 (2017).

[5] V. Reshetnikov, *The Fabius Function and Rvachev's Up-Function: Exact Arithmetic, Probability, Fourier Analysis, and Corrected Sharp Asymptotics*, `ProveIt`, `Analysis/FabiusFunction/docs/Fabius_Function_and_Rvachev_Up/`.

[6] V. Reshetnikov, `ProveIt`: formal development of the Fabius function and Rvachev's up-function, GitHub repository, snapshot `eaea1b9afe2b7ff0b7dd880bd1710e303dec6d80`, 25 August 2026, https://github.com/VladimirReshetnikov/ProveIt/tree/eaea1b9afe2b7ff0b7dd880bd1710e303dec6d80/Analysis/FabiusFunction.

Part II
Repeated Integration and Rvachev's Up-Function

Synthesized provenance and status. This part merges the former `Repeated_Integration_and_Rvachev_Up/Repeated_Integration_and_Rvachev_Up.tex` and `Rvachev_Up_from_Repeated_Integration-2/Rvachev_Up_from_Repeated_Integration.tex`. The second source supplies the canonical sharp threshold $n \geq r+2$, the first admissible-stage argument, the complete comparison-kernel data, and the rigorous all-orders proof. The first source contributes the 2012 Stack Exchange provenance, the weak-test and L^p corollaries, the operator interpretation, and the comparison with ordinary truncation. The resulting sharp identities and expansions remain natural-language proofs and formal obligations.

Former source SHA-256 values: `eadcb4b414ac93723a91acbd5062d44340f78134a2cecbc18f7d7ba67eb2c9be` and `fad16072df30d9e6eb5df03da57dd217769180730581f1dfd19fa5be0d16b262`.

Topic abstract

A Math Stack Exchange question asks for a closed form for the sequence

$$f_0(x) = \frac{1}{2}\mathbf{1}_{(-1,1)}(x), \qquad f_n(x) = 2\int_{-1}^{x} \big(f_{n-1}(2t+1) - f_{n-1}(2t-1)\big)\,\mathrm{d}t.$$

The first integration produces the piecewise linear tent $f_1(x) = (1-|x|)_+$, and the subsequent iterates visually converge very rapidly to Rvachev's compactly supported C^∞ up-function. This report gives a complete finite-stage description and a sharp convergence theorem.

The integral operator is exactly a probabilistic averaging operator: if g is the density of Y and U is uniform on $[-1,1]$, then the next density is that of $(Y+U)/2$. Consequently, for $n \geq 1$,

$$f_n = u_{2^{-1}} * u_{2^{-2}} * \cdots * u_{2^{-n}} * u_{2^{-n}}, \qquad u_a(x) = \frac{1}{2a}\mathbf{1}_{[-a,a]}(x).$$

This identifies f_n as a nonuniform box spline. Collapsing its general inclusion–exclusion formula with the Thue–Morse generating polynomial yields, for $n \geq 1$,

$$f_n(x) = \frac{2^{n(n+1)/2-1}}{n!}\sum_{j=0}^{2^n}\big(\tau_n(j) - \tau_n(j-1)\big)\left(x+1-\frac{j}{2^{n-1}}\right)_+^n,$$

where $\tau_n(j) = (-1)^{\mathrm{wt}_2(j)}$ on $0 \leq j < 2^n$ and is zero outside that range. The nth derivative is therefore a rescaled Thue–Morse step function.

The principal quantitative result derived here is exact, rather than merely asymptotic. For every fixed $r \geq 0$ and every $n \geq r+2$,

$$\boxed{\left\| f_n^{(r)} - \mathrm{up}^{(r)} \right\|_\infty = \frac{2^{\binom{r+3}{2}}}{9\,4^n}.}$$

Thus, once the rth derivative has passed its first admissible spline stage, every further integration divides its uniform error by exactly four. In the indexing that starts with the piecewise linear tent $g_0 = f_1$, this becomes

$$\|g_m - \mathrm{up}\|_\infty = \frac{2}{9\,4^m} \qquad (m \geq 1),$$

while the exceptional initial error is $\|g_0 - \mathrm{up}\|_\infty = 13/72$. The proof is based on an exact Peano-kernel identity. The comparison kernel between a uniform random variable and the up-distributed random variable has both integral and maximum equal to $1/9$; the finite partial density has disjoint extremal plateaux whose height is known exactly. Their combination gives both the upper bound and matching extremizers.

6 The problem, the indexing, and the answer

This construction was posed independently on Mathematics Stack Exchange in 2012 [1]. Nathan Grigg found the exact finite-stage Heaviside/Thue–Morse expression [2]; the probability and box-spline viewpoints below explain its coefficients and, more importantly, expose the sharp convergence law. The iteration is not ordinary truncation of the dyadic random series. It replaces the omitted infinite tail by one uniform box of exactly the same support radius, preserving support, mass, symmetry, and mean while leaving a controlled variance defect.

The recurrence considered in the Math Stack Exchange question [1, 2] is

$$f_0(x) = \frac{1}{2}\mathbf{1}_{(-1,1)}(x), \qquad f_n(x) = 2\int_{-1}^{x} \big(f_{n-1}(2t+1) - f_{n-1}(2t-1)\big)\,\mathrm{d}t. \tag{188}$$

Changing the values of f_0 at $x = \pm 1$ changes neither any later iterate nor any of the norm statements below. We therefore use whichever endpoint convention is convenient.

There is a small indexing issue in the informal description of the construction. The literal seed f_0 in (188) is piecewise constant; the first iterate is the piecewise linear function

$$f_1(x) = (1 - |x|)_+ \,. \tag{189}$$

When one says that the up-function arises by "repeatedly integrating a piecewise linear function," the natural indexing is therefore

$$g_m = f_{m+1}, \qquad g_0(x) = (1 - |x|)_+ \,. \tag{190}$$

Both indexings will be retained: f_n matches the question, while g_m matches the piecewise-linear wording.

6.1 Main finite-stage formula

Write $\mathrm{wt}_2(j)$ for the number of 1's in the binary expansion of j. For $n \ge 0$ define the zero-extended finite Thue–Morse word

$$\tau_n(j) = \begin{cases} (-1)^{\mathrm{wt}_2(j)}, & 0 \le j < 2^n, \\ 0, & \text{otherwise}, \end{cases} \qquad \Delta\tau_n(j) = \tau_n(j) - \tau_n(j-1). \tag{191}$$

The accepted Stack Exchange answer gives a formula equivalent to the following one; the probabilistic and spline derivation in section 9 explains why its coefficients are first differences of the Thue–Morse word.

Theorem 6.1 (Exact nth iterate). *For every $n \ge 1$ and $x \in \mathbb{R}$,*

$$\boxed{f_n(x) = \frac{2^{n(n+1)/2-1}}{n!} \sum_{j=0}^{2^n} \Delta\tau_n(j) \left(x + 1 - \frac{j}{2^{n-1}}\right)_+^n .} \tag{192}$$

Equivalently,

$$f_n(x) = \frac{1}{2n!\,2^{n(n-1)/2}} \sum_{j=0}^{2^n} \Delta\tau_n(j)\,(2^n x + 2^n - 2j)_+^n \,. \tag{193}$$

The knots are

$$-1, -1 + 2^{1-n}, \ldots, 1 - 2^{1-n}, 1. \tag{194}$$

On every open knot interval f_n is a polynomial of degree n, and globally $f_n \in C^{n-1}(\mathbb{R})$.

In the tent indexing (190), theorem 6.1 reads

$$\boxed{g_m(x) = \frac{2^{m(m+3)/2}}{(m+1)!} \sum_{j=0}^{2^{m+1}} \Delta\tau_{m+1}(j) \left(x + 1 - \frac{j}{2^m}\right)_+^{m+1}.} \tag{195}$$

Thus the mth repeated integration of the tent is a degree-$(m+1)$ spline on the dyadic mesh of width 2^{-m}.

6.2 Main sharp convergence theorem

For a bounded function h let $\|h\|_\infty = \sup_{x\in\mathbb{R}} |h(x)|$. When $h \in C^r$, we use the max version of the C^r norm,

$$\|h\|_{C^r} = \max_{0\le j\le r} \left\|h^{(j)}\right\|_\infty. \tag{196}$$

Theorem 6.2 (Exact derivative error). *Let $r \ge 0$. The first iterate belonging to C^r is f_{r+1}, and its rth-derivative error is*

$$\left\|f_{r+1}^{(r)} - \mathrm{up}^{(r)}\right\|_\infty = \frac{13}{72}\, 2^{\binom{r+1}{2}}. \tag{197}$$

For every $n \ge r+2$,

$$\boxed{\left\|f_n^{(r)} - \mathrm{up}^{(r)}\right\|_\infty = \frac{2^{\binom{r+3}{2}}}{9\,4^n}.} \tag{198}$$

All of these bounds are attained at the same 2^{r+2} points

$$c_{r+2,j} = -1 + \frac{2j+1}{2^{r+2}}, \qquad 0 \le j < 2^{r+2}. \tag{199}$$

At the first admissible stage,

$$f_{r+1}^{(r)}(c_{r+2,j}) - \mathrm{up}^{(r)}(c_{r+2,j}) = (-1)^{\mathrm{wt}_2(j)} \frac{13}{72} 2^{\binom{r+1}{2}}, \tag{200}$$

and for $n \ge r+2$,

$$f_n^{(r)}(c_{r+2,j}) - \mathrm{up}^{(r)}(c_{r+2,j}) = (-1)^{\mathrm{wt}_2(j)} \frac{2^{\binom{r+3}{2}}}{9\,4^n}. \tag{201}$$

The exact quartering begins immediately after the exceptional first C^r spline:

$$\frac{\left\|f_{r+2}^{(r)} - \mathrm{up}^{(r)}\right\|_\infty}{\left\|f_{r+1}^{(r)} - \mathrm{up}^{(r)}\right\|_\infty} = \frac{4}{13}, \qquad \frac{\left\|f_{n+1}^{(r)} - \mathrm{up}^{(r)}\right\|_\infty}{\left\|f_n^{(r)} - \mathrm{up}^{(r)}\right\|_\infty} = \frac{1}{4} \quad (n \ge r+2). \tag{202}$$

For the tent indexing this yields the particularly simple uniform result

$$\|g_0 - \mathrm{up}\|_\infty = \frac{13}{72}, \qquad \boxed{\|g_m - \mathrm{up}\|_\infty = \frac{2}{9\,4^m} \quad (m \ge 1).} \tag{203}$$

Thus every integration after the first one gains exactly two binary digits in the sharp uniform error.

The approximation in this report is distinct from the centered finite-CDF spline already discussed in the companion article. Here f_n is itself a probability *density*, and the last unresolved tail is replaced by a scaled uniform variable. The true tail and the uniform replacement have equal mass and equal mean. The first surviving discrepancy is therefore quadratic in the tail scale 2^{-n}, which explains the factor 4^{-n}; the comparison-kernel argument below determines the coefficient exactly.

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7 The integration operator is a probability operator

For $a > 0$, let

$$u_a(x) = \frac{1}{2a}\mathbf{1}_{[-a,a]}(x) \tag{204}$$

be the density of a random variable uniform on $[-a, a]$. Let T denote the operator in (188):

$$(Th)(x) = 2\int_{-1}^{x} \big(h(2t+1) - h(2t-1)\big)\,\mathrm{d}t. \tag{205}$$

Lemma 7.1 (Moving-window and convolution forms)**.** *Suppose h is integrable and supported in $[-1, 1]$. Then*

$$(Th)(x) = \int_{2x-1}^{2x+1} h(s)\,\mathrm{d}s = 2(h * u_1)(2x). \tag{206}$$

Proof. In the first term of (205), put $s = 2t + 1$; in the second, put $s = 2t - 1$. This gives

$$(Th)(x) = \int_{-1}^{2x+1} h(s)\,\mathrm{d}s - \int_{-3}^{2x-1} h(s)\,\mathrm{d}s.$$

The support assumption allows the lower endpoint -3 in the second integral to be replaced by -1, after which the difference is the integral over $[2x - 1, 2x + 1]$. The convolution identity follows directly from the definition of u_1. ☐

Proposition 7.2 (Probabilistic meaning)**.** *If h is the density of a random variable Y, and U is independent and uniform on $[-1, 1]$, then Th is the density of*

$$\frac{Y + U}{2}. \tag{207}$$

In particular, T preserves nonnegativity, total mass one, evenness, and support in $[-1, 1]$.

Proof. The density of $Y + U$ is $h * u_1$. Scaling a random variable by $1/2$ multiplies its density by 2 and evaluates it at $2x$, giving exactly (206). The listed properties are immediate. ☐

Let $U_1, U_2, \ldots$ be independent random variables, each uniform on $[-1, 1]$. Start with $X_0 = U_1$ and define recursively

$$X_n = \frac{X_{n-1} + U_{n+1}}{2}. \tag{208}$$

Then f_n is the density of X_n.

Theorem 7.3 (Finite random-sum representation)**.** *For every $n \geq 1$,*

$$X_n \overset{d}{=} \sum_{k=1}^{n} 2^{-k} U_k + 2^{-n} U_{n+1}. \tag{209}$$

Consequently,

$$f_n = u_{2^{-1}} * u_{2^{-2}} * \cdots * u_{2^{-n}} * u_{2^{-n}}. \tag{210}$$

Proof. Expanding (208) gives weights $2^{-n}, 2^{-n}, 2^{-(n-1)}, \ldots, 2^{-1}$ on $n + 1$ independent copies of U. Because the variables are identically distributed, the two copies of 2^{-n} may be placed last, giving (209). Convolution of the corresponding scaled uniform densities gives (210). ☐

Remark 7.4 (The natural Markov chain need not converge pathwise). The almost-sure convergence obtained by reordering the summands into the nested series is a coupling of the *laws*. The forward recursion (208) continually inserts fresh noise of size $1/2$, so it should instead be viewed as a Markov chain converging to its stationary law. Reordering the i.i.d. summands produces the nested series coupling; it does not assert pathwise convergence of the original forward chain.

Several facts that are less transparent in the integral recursion are now immediate:

$$f_n \geq 0, \qquad \int_{\mathbb{R}} f_n(x)\,\mathrm{d}x = 1, \qquad f_n(-x) = f_n(x), \qquad \operatorname{supp} f_n = [-1, 1]. \tag{211}$$

Moreover, for $n \geq 1$,

$$f_n(0) = \int_{-1}^{1} f_{n-1}(s)\,\mathrm{d}s = 1. \tag{212}$$

8 The limit is Rvachev's up-function

Consider the almost surely convergent random series

$$X = \sum_{k=1}^{\infty} 2^{-k} U_k. \tag{213}$$

Its support is $[-1, 1]$. Denote its density by up. Existence and smoothness also follow from the Fourier product below; they are classical properties of the Rvachev–Arias de Reyna construction [3].

8.1 Self-similarity and the differential equation

Splitting off the first summand in (213) gives the distributional fixed-point identity

$$X \overset{d}{=} \frac{U + X'}{2}, \tag{214}$$

where X' is an independent copy of X. By theorem 7.2, its density is a fixed point of T:

$$\operatorname{up}(x) = \int_{2x-1}^{2x+1} \operatorname{up}(s)\,\mathrm{d}s. \tag{215}$$

Differentiating gives

$$\operatorname{up}'(x) = 2\big(\operatorname{up}(2x+1) - \operatorname{up}(2x-1)\big). \tag{216}$$

Also $\operatorname{up}(0) = \int_{-1}^{1} \operatorname{up} = 1$. The even compactly supported smooth solution of (216) normalized by $\operatorname{up}(0) = 1$ is Rvachev's up-function. Thus the repeated-integration limit is not merely analogous to up; it is exactly up with the normalization used in the companion article.

8.2 Fourier product

Use the non-unitary characteristic Fourier transform

$$\mathscr{F}h(t) = \int_{\mathbb{R}} h(x)\mathrm{e}^{-\mathrm{i}tx}\,\mathrm{d}x, \qquad \operatorname{sinc} z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0. \end{cases} \tag{217}$$

Then $\mathscr{F}u_a(t) = \operatorname{sinc}(at)$. From (210),

$$\mathscr{F}f_n(t) = \operatorname{sinc}(2^{-n}t) \prod_{k=1}^{n} \operatorname{sinc}(2^{-k}t) = \left(\prod_{k=1}^{n-1} \operatorname{sinc}(2^{-k}t)\right) \operatorname{sinc}^2(2^{-n}t). \tag{218}$$

Under the symmetric convention used in the Stack Exchange question, the right side is multiplied by $(2\pi)^{-1/2}$, exactly reproducing the conjectured formula there. Taking the limit gives

$$\mathscr{F}\operatorname{up}(t) = \prod_{k=1}^{\infty} \operatorname{sinc}(2^{-k}t). \tag{219}$$

The product is rapidly decreasing together with all its derivatives; Fourier inversion therefore gives a C^∞ density. The product also makes the finite-depth correction transparent: f_n has the first n factors of the limiting product, with the last factor repeated once.

8.3 Relation with the bounded Fabius function

Let $\xi_k = (U_k + 1)/2$, so that the ξ_k are independent uniform variables on $[0, 1]$. Then

$$Y = \sum_{k=1}^{\infty} 2^{-k} \xi_k = \frac{X + 1}{2}. \tag{220}$$

The cumulative distribution function of Y is the bounded Fabius function F. Hence

$$\mathbb{P}(X \le x) = F\left(\frac{x+1}{2}\right), \tag{221}$$

and, using $F(1 - y) = 1 - F(y)$,

$$\mathbb{P}(X > t) = F\left(\frac{1-t}{2}\right), \qquad 0 \le t \le 1. \tag{222}$$

On the two halves of the support,

$$\mathrm{up}(x) = \begin{cases} F(x+1), & -1 \le x \le 0, \\ F(1-x), & 0 \le x \le 1. \end{cases} \tag{223}$$

This is exactly the normalization used in the existing exposition.

9 Exact finite splines and the Thue–Morse collapse

The convolution formula (210) already gives a closed form in the language of box splines. To obtain the explicit truncated-power expression, we first record the general one-dimensional inclusion–exclusion formula.

9.1 A general nonuniform box-spline formula

Lemma 9.1 (Density of a sum of centered uniforms)**.** *Let V_i be independent and uniform on $[-a_i, a_i]$, where $a_i > 0$ and $1 \le i \le m$. Put $A = a_1 + \cdots + a_m$. The density $b_{\boldsymbol{a}}$ of $V_1 + \cdots + V_m$ is*

$$b_{\boldsymbol{a}}(x) = \frac{1}{2^m (m-1)! \prod_{i=1}^m a_i} \sum_{\varepsilon \in \{0,1\}^m} (-1)^{|\varepsilon|} \left(x + A - 2 \sum_{i=1}^m \varepsilon_i a_i \right)_+^{m-1}. \tag{224}$$

Proof. Translate V_i to $W_i = V_i + a_i$, which is uniform on $[0, 2a_i]$. The distributional derivative of the density of W_i is $(\delta_0 - \delta_{2a_i})/(2a_i)$. After convolving the m factors and integrating $m - 1$ times from the left endpoint, one obtains the usual inclusion–exclusion formula for the density of $W_1 + \cdots + W_m$. Translating the sum back by A gives (224). Equivalently, one may verify the formula by taking Laplace transforms. □

For f_n, the $m = n + 1$ half-widths are

$$2^{-1}, 2^{-2}, \ldots, 2^{-(n-1)}, 2^{-n}, 2^{-n}. \tag{225}$$

Their sum is 1, and their product is $2^{-n(n+3)/2}$. Thus the scalar prefactor in (224) becomes

$$\frac{2^{n(n+1)/2-1}}{n!}. \tag{226}$$

It remains to combine subsets that produce the same knot.

9.2 The subset-sum polynomial

After multiplication by 2^{n-1}, the possible increments $2a_i$ in (224) are

$$2^{n-1}, 2^{n-2}, \ldots, 2, 1, 1. \tag{227}$$

The signed subset-sum generating polynomial is therefore

$$\begin{aligned} B_n(z) &= (1-z)^2 \prod_{r=1}^{n-1} (1 - z^{2^r}) \\ &= (1-z) \prod_{r=0}^{n-1} (1 - z^{2^r}). \end{aligned} \tag{228}$$

The classical finite Thue–Morse identity is

$$\prod_{r=0}^{n-1} (1 - z^{2^r}) = \sum_{j=0}^{2^n-1} (-1)^{\mathrm{wt}_2(j)} z^j = \sum_{j\in\mathbb{Z}} \tau_n(j) z^j. \tag{229}$$

Multiplication by $1 - z$ takes first differences of the coefficient sequence:

$$B_n(z) = \sum_{j=0}^{2^n} \Delta\tau_n(j) z^j. \tag{230}$$

This explains the otherwise rather mysterious coefficient $c_n(j) - c_n(j-1)$ in the Stack Exchange answer: the duplicated smallest uniform interval contributes the extra factor $1 - z$, turning the Thue–Morse word into its first difference.

Proof of theorem 6.1. Insert the widths (225) into theorem 9.1. The subset sum indexed by ε equals $j/2^{n-1}$ after grouping terms with the same integer j; by (230), the total sign of that group is $\Delta\tau_n(j)$. The scalar is (226). This gives (192). Factoring 2^{-n} from each truncated power gives (193). □

9.3 Derivatives, knots, and exact smoothness

For $0 \le r \le n-1$, differentiating (192) gives

$$f_n^{(r)}(x) = \frac{2^{n(n+1)/2-1}}{(n-r)!} \sum_{j=0}^{2^n} \Delta\tau_n(j) \left(x + 1 - \frac{j}{2^{n-1}} \right)_+^{n-r}. \tag{231}$$

The formula is also valid away from the knots for $r = n$. If

$$-1 + \frac{j}{2^{n-1}} < x < -1 + \frac{j+1}{2^{n-1}}, \qquad 0 \le j < 2^n, \tag{232}$$

then the first-difference sum telescopes and gives

$$\boxed{f_n^{(n)}(x) = 2^{n(n+1)/2-1} \tau_n(j).} \tag{233}$$

Thus the highest ordinary derivative is literally a Thue–Morse step word. Because this step function has jumps at the active knots, f_n is C^{n-1} but not C^n globally. Each application of T adds exactly one degree and exactly one order of classical smoothness.

9.4 The first iterates

The first three functions are

$$f_0(x) = \frac{1}{2}\mathbf{1}_{(-1,1)}(x), \tag{234}$$

$$f_1(x) = (1 - |x|)_+ \,, \tag{235}$$

$$f_2(x) = \begin{cases} 1 - 2x^2, & |x| \le \frac{1}{2}, \\ 2(1 - |x|)^2, & \frac{1}{2} \le |x| \le 1, \\ 0, & |x| \ge 1. \end{cases} \tag{236}$$

The next one already displays the full Thue–Morse organization compactly:

$$f_3(x) = \frac{16}{3}\sum_{j=0}^{8} \Delta\tau_3(j)\left(x + 1 - \frac{j}{4}\right)_+^3. \tag{237}$$

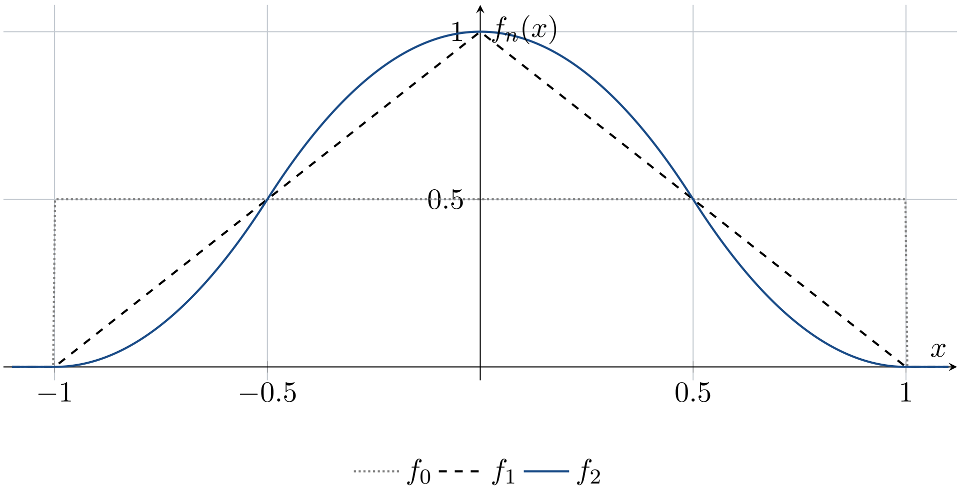

Figure 1: The rectangle, the piecewise linear tent, and the quadratic spline. Later iterates add one polynomial degree and one smoothness order at each step.

10 Replacing the last tail: the exact error decomposition

Put

$$S_n = \sum_{k=1}^{n} 2^{-k} U_k, \qquad a_n = 2^{-n}, \tag{238}$$

and let p_n be the density of S_n. Let X' be an independent copy of the limiting random variable X. The finite and infinite laws split as

$$X_n \overset{d}{=} S_n + a_n U, \qquad X \overset{d}{=} S_n + a_n X'. \tag{239}$$

If

$$\rho_a(x) = a^{-1}\,\mathrm{up}(x/a) \tag{240}$$

then (239) is exactly

$$f_n = p_n * u_{a_n}, \qquad \mathrm{up} = p_n * \rho_{a_n}. \tag{241}$$

The repeated-integration approximation therefore does one very specific thing: it keeps the first n dyadic uniform coordinates exact and replaces the remaining self-similar tail $a_n X'$ by a single uniform coordinate $a_n U$.

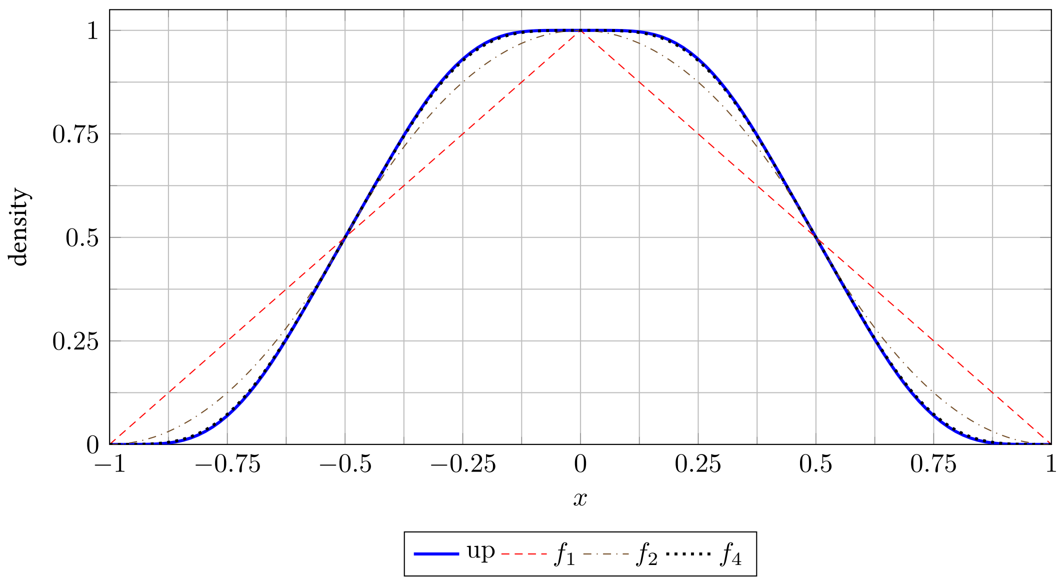

Figure 2: The first smooth finite box splines and their limit. The plotted up-function is a high-stage numerical evaluation used only for illustration.

10.1 The comparison kernel

Definition 10.1 (Uniform-versus-up Peano kernel)**.** For $t \in \mathbb{R}$, define

$$K(t) = \mathbb{E}\,(U - t)_+ - \mathbb{E}\,(X - t)_+\,. \tag{242}$$

The kernel contains every constant needed for the sharp error theorem. We first calculate its basic invariants.

Lemma 10.2 (Second and first absolute moments)**.** *The limiting random variable X satisfies*

$$\mathbb{E}X^2 = \frac{1}{9}, \qquad \mathbb{E}|X| = \frac{5}{18}. \tag{243}$$

Consequently,

$$F\left(\frac{1}{4}\right) = \frac{5}{72}. \tag{244}$$

Proof. Independence and centering in (213) give

$$\mathbb{E}X^2 = \frac{1}{3}\sum_{k=1}^{\infty} 4^{-k} = \frac{1}{9}.$$

For fixed $x \in [-1, 1]$ and U uniform on $[-1, 1]$,

$$\mathbb{E}\big(|x + U| \mid x\big) = \frac{1}{2}\int_{-1}^{1} |x + u|\,\mathrm{d}u = \frac{1 + x^2}{2}.$$

Using $X \overset{d}{=} (X' + U)/2$ gives

$$\mathbb{E}|X| = \frac{1}{2}\mathbb{E}|X' + U| = \frac{1}{4}\big(1 + \mathbb{E}X^2\big) = \frac{5}{18}.$$

By symmetry, $\mathbb{E}X_+ = \frac{1}{2}\mathbb{E}|X| = 5/36$. On the other hand, (222) and $F'(y) = 2F(2y)$ imply

$$\mathbb{E}X_+ = \int_0^1 F\left(\frac{1-t}{2}\right)\,\mathrm{d}t = 2\int_0^{1/2} F(y)\,\mathrm{d}y = 2F\left(\frac{1}{4}\right),$$

which proves (244). ☐

Theorem 10.3 (Exact kernel data)**.** *The kernel K is even, continuous, supported in $[-1,1]$, and nonnegative. For $0 \le t \le 1$,*

$$K(t) = \frac{(1-t)^2}{4} - 2F\left(\frac{1-t}{4}\right). \tag{245}$$

Moreover,

$$\boxed{\int_{-1}^{1} K(t)\,\mathrm{d}t = \frac{1}{9}, \qquad \|K\|_\infty = K(0) = \frac{1}{9},} \tag{246}$$

while

$$\|K'\|_\infty = \frac{13}{72}. \tag{247}$$

Proof. Evenness follows from symmetry and the fact that U and X have the same mean. Both variables are supported in $[-1,1]$, so K vanishes outside that interval.

For nonnegativity, let ϕ be convex. From the fixed-point identity $X \overset{d}{=} (X'+U)/2$ and convexity,

$$\mathbb{E}\phi(X) \le \frac{1}{2}\mathbb{E}\phi(X') + \frac{1}{2}\mathbb{E}\phi(U).$$

Since X' has the same law as X, this rearranges to $\mathbb{E}\phi(X) \le \mathbb{E}\phi(U)$. Thus X is below U in convex order. Applying this to $\phi_t(y) = (y-t)_+$ gives $K(t) \ge 0$.

For $0 \le t \le 1$,

$$\mathbb{E}\,(U-t)_+ = \frac{(1-t)^2}{4}.$$

Also, by (222),

$$\begin{aligned}\mathbb{E}\,(X-t)_+ &= \int_t^1 \mathbb{P}(X>s)\,\mathrm{d}s = \int_t^1 F\left(\frac{1-s}{2}\right)\mathrm{d}s \\ &= 2\int_0^{(1-t)/2} F(y)\,\mathrm{d}y = 2F\left(\frac{1-t}{4}\right),\end{aligned}$$

which proves (245).

The Peano identity proved in theorem 10.4 below, applied to $h(y) = y^2/2$, gives

$$\int K(t)\,\mathrm{d}t = \frac{1}{2}\left(\mathbb{E}U^2 - \mathbb{E}X^2\right) = \frac{1}{2}\left(\frac{1}{3} - \frac{1}{9}\right) = \frac{1}{9}.$$

At the origin, theorem 10.2 gives

$$K(0) = \mathbb{E}U_+ - \mathbb{E}X_+ = \frac{1}{4} - \frac{5}{36} = \frac{1}{9}.$$

It remains to see that this is the maximum. Differentiating (245) gives

$$K'(t) = F\left(\frac{1-t}{2}\right) - \frac{1-t}{2}. \tag{248}$$

On $[0,1/2]$, the Fabius function is convex: $F'(y) = 2F(2y)$ is increasing. Since $F(0) = 0$ and $F(1/2) = 1/2$, convexity places its graph below the chord, so $F(y) \le y$. Therefore $K'(t) \le 0$ for $t \ge 0$, and $K(0)$ is the maximum.

Finally, put $y = (1-t)/2$ in (248). The magnitude of K' is the maximum of $y - F(y)$ on $[0,1/2]$. Its derivative is $1 - 2F(2y)$, which vanishes exactly at $y = 1/4$ because $F(1/2) = 1/2$. Thus

$$\|K'\|_\infty = \frac{1}{4} - F\left(\frac{1}{4}\right) = \frac{1}{4} - \frac{5}{72} = \frac{13}{72}.$$

□

The equality between the mass and the maximum in (246) is the key numerical coincidence behind the clean final rate. The mass controls all sufficiently smooth stages; the maximum controls the borderline stage at which a distributional box-spline derivative becomes an ordinary continuous error function.

10.2 Peano identity and the pointwise error formula

Lemma 10.4 (Peano identity)**.** *Let h' be absolutely continuous on $[-1,1]$. Then*

$$\mathbb{E}h(U) - \mathbb{E}h(X) = \int_{-1}^{1} K(t)h''(t)\,\mathrm{d}t. \tag{249}$$

Proof. For $y \in [-1,1]$,

$$h(y) = h(-1) + h'(-1)(y+1) + \int_{-1}^{1} (y-t)_+\, h''(t)\,\mathrm{d}t.$$

Take expectations first with $y = U$ and then with $y = X$. The constant terms cancel, and the linear terms cancel because both random variables have mean zero. Fubini's theorem then gives (249). □

Apply the lemma with $h(y) = p_n^{(r)}(x - a_n y)$. Whenever $n \geq r+3$, the needed weak second derivative is bounded and the result is an ordinary integral.

Proposition 10.5 (Exact pointwise error identity)**.** *Let $r \geq 0$ and $n \geq r+3$. Then for every $x \in \mathbb{R}$,*

$$f_n^{(r)}(x) - \mathrm{up}^{(r)}(x) = a_n^2 \int_{-1}^{1} K(t) p_n^{(r+2)}(x - a_n t)\,\mathrm{d}t. \tag{250}$$

Proof. From (241),

$$f_n^{(r)}(x) = \mathbb{E}p_n^{(r)}(x - a_n U), \qquad \mathrm{up}^{(r)}(x) = \mathbb{E}p_n^{(r)}(x - a_n X).$$

Use theorem 10.4; differentiating twice with respect to its argument contributes the factor a_n^2. □

There is also a distributional form that remains valid at the borderline stage. Define

$$G_a(x) = aK(x/a). \tag{251}$$

Since K'' is the difference between the densities of U and X in the distributional sense,

$$u_a - \rho_a = \mathrm{D}^2 G_a. \tag{252}$$

Consequently,

$$f_n - \mathrm{up} = p_n * \mathrm{D}^2 G_{a_n}. \tag{253}$$

This identity will handle the first two smoothness thresholds without any appeal to a classical second derivative of p_n.

It also gives a sharp weak-test estimate that is useful even when a uniform density norm is not the natural metric.

Corollary 10.6 (Sharp weak error)**.** *For every $\phi \in C^2(\mathbb{R})$ with bounded second derivative,*

$$|\mathbb{E}\phi(X_n) - \mathbb{E}\phi(X)| \leq \frac{4^{-n}}{9} \left\|\phi''\right\|_\infty. \tag{254}$$

Before taking absolute values the difference is nonnegative for convex ϕ. The constant is sharp: equality holds for $\phi(x) = x^2$.

Proof. Pair (253) with ϕ and move the two distributional derivatives onto ϕ. The prefix density has mass one, while G_{a_n} is nonnegative and has mass $4^{-n}/9$ by (246). Convexity gives the sign. For $\phi(x) = x^2$, the identity is exactly the variance difference $\mathbb{E}X_n^2 - \mathbb{E}X^2 = 2\,4^{-n}/9$. ☐

Lemma 10.7 (Summatory Thue–Morse cancellation). *Let $\varepsilon_j = (-1)^{\mathrm{wt}_2(j)}$. Then for every $J \geq 0$,*

$$\sum_{j=0}^{J} \varepsilon_j = \begin{cases} \varepsilon_{J/2}, & J \text{ even}, \\ 0, & J \text{ odd}. \end{cases} \tag{255}$$

In particular, every prefix sum has absolute value at most one.

Proof. The identities $\varepsilon_{2q} = \varepsilon_q$ and $\varepsilon_{2q+1} = -\varepsilon_q$ make every complete adjacent pair cancel. An even endpoint leaves the last even term unpaired. ☐

11 Sharp derivative geometry of the partial densities

The pointwise identity (250) becomes sharp because the derivatives of p_n are disjoint signed translates of a single tail density, each having a flat top of known width and height.

For $m \geq 0$, define the dyadic centers

$$c_{m,j} = -1 + \frac{2j+1}{2^m}, \qquad 0 \leq j < 2^m. \tag{256}$$

For $m = 0$, this is the singleton $\{0\}$. At the top piecewise-polynomial derivative order, derivatives below are understood through their bounded almost-everywhere representatives; values at the finitely many knots may be chosen arbitrarily between the adjacent limits and do not affect the L^∞ norm used in this section.

Theorem 11.1 (Derivative decomposition and exact plateaux). *Let $0 \leq m \leq n-1$. Let $q_{m,n}$ be the density of the tail sum*

$$R_{m,n} = \sum_{k=m+1}^{n} 2^{-k} U_k. \tag{257}$$

Then, almost everywhere,

$$p_n^{(m)}(x) = 2^{\binom{m}{2}} \sum_{j=0}^{2^m-1} (-1)^{\mathrm{wt}_2(j)} q_{m,n}(x - c_{m,j}). \tag{258}$$

The translated supports in this sum have disjoint interiors. Moreover,

$$\left\| p_n^{(m)} \right\|_\infty = 2^{\binom{m+1}{2}}, \tag{259}$$

and on the central interval of each translated tail,

$$p_n^{(m)}(x) = (-1)^{\mathrm{wt}_2(j)} 2^{\binom{m+1}{2}} \quad \textit{whenever} \quad |x - c_{m,j}| < 2^{-n}. \tag{260}$$

Proof. In the distributional sense,

$$\mathrm{D}u_{2^{-k}} = 2^{k-1}\big(\delta_{-2^{-k}} - \delta_{2^{-k}}\big). \tag{261}$$

Differentiate the first m factors in $p_n = u_{2^{-1}} * \cdots * u_{2^{-n}}$. The product of the scalar factors is

$$\prod_{k=1}^{m} 2^{k-1} = 2^{\binom{m}{2}}.$$

The 2^m signed sums of the locations $\pm 2^{-k}$ are precisely the centers $c_{m,j}$, in increasing order, and the sign is $(-1)^{\mathrm{wt}_2(j)}$. Convolution with the undifferentiated tail gives (258).

Adjacent centers are separated by 2^{1-m}. The support radius of $q_{m,n}$ is

$$\sum_{k=m+1}^{n} 2^{-k} = 2^{-m} - 2^{-n},$$

so its support diameter is $2^{1-m} - 2^{1-n}$, strictly smaller than the center spacing. Hence the translated interiors are disjoint.

Scaling shows

$$q_{m,n}(x) = 2^m p_{n-m}(2^m x). \tag{262}$$

The density p_s has supremum one: its first factor $u_{1/2}$ has height one, and convolution with a probability density cannot increase the supremum. It actually equals one on $|x| < 2^{-s}$, because the remaining factors have total support radius $1/2 - 2^{-s}$. Thus $q_{m,n}$ has height 2^m and is constant at that height on $|x| < 2^{-n}$. Multiplying by $2^{\binom{m}{2}}$ proves (259) and (260). □

At the missing endpoint $m = n$, the same calculation gives a signed Dirac comb:

$$\mathrm{D}^n p_n = 2^{\binom{n}{2}} \sum_{j=0}^{2^n-1} (-1)^{\mathrm{wt}_2(j)} \delta_{c_{n,j}}. \tag{263}$$

The centers are spaced by $2^{1-n} = 2a_n$.

11.1 A sharp derivative norm for the up-function

The finite plateau geometry also recovers a useful sharp fact about the limit itself.

Corollary 11.2 (Exact derivative peaks of up)**.** *For every $m \geq 0$,*

$$\boxed{\left\| \mathrm{up}^{(m)} \right\|_\infty = 2^{\binom{m+1}{2}}.} \tag{264}$$

More precisely,

$$\mathrm{up}^{(m)}(c_{m,j}) = (-1)^{\mathrm{wt}_2(j)} 2^{\binom{m+1}{2}} \qquad (0 \leq j < 2^m). \tag{265}$$

Proof. Take $n \geq m+1$. Since $\mathrm{up} = p_n * \rho_{a_n}$,

$$\mathrm{up}^{(m)}(x) = \int p_n^{(m)}(x-y)\rho_{a_n}(y)\,\mathrm{d}y.$$

Convolution with a probability density cannot exceed $\left\| p_n^{(m)} \right\|_\infty$, giving the upper bound. At $x = c_{m,j}$, the whole support $[-a_n, a_n]$ of ρ_{a_n} lies in the constant plateau (260), so the convolution equals the signed plateau height. This proves both statements. □

Corollary 11.3 (Higher-order flatness at the peak grid)**.** *For every $q \geq 0$, every $0 \leq j < 2^q$, and every $k > q$,*

$$\mathrm{up}^{(k)}(c_{q,j}) = 0. \tag{266}$$

Proof. At $q = 0$, the only point is $c_{0,0} = 0$. On the right of zero, $\mathrm{up}(x) = F(1-x)$, and the smooth bounded Fabius function is constant on $[1, \infty)$; therefore all positive-order derivatives of up vanish at zero.

Let $q \geq 1$, put $a = 2^{-q}$, and write $\mathrm{up} = p_q * \rho_a$. For $\ell = k - q \geq 1$,

$$\mathrm{up}^{(k)} = (\mathrm{D}^q p_q) * \rho_a^{(\ell)}.$$

Insert the signed Dirac comb (263). At $x = c_{q,j}$ every off-diagonal translate is evaluated at distance at least $2a$, outside the support $[-a, a]$ of ρ_a. The diagonal translate is $\rho_a^{(\ell)}(0) = a^{-\ell-1}\,\mathrm{up}^{(\ell)}(0) = 0$ by the first paragraph. Thus every term vanishes. □

12 The exact convergence rate

We now combine the nonnegative kernel of section 10 with the exact plateaux of section 11.

12.1 All stages after the first admissible one

Theorem 12.1 (Sharp L^∞ error for every derivative)**.** *Let $r \geq 0$ and $n \geq r+2$. Then*

$$\left\| f_n^{(r)} - \mathrm{up}^{(r)} \right\|_\infty = \frac{4^{-n}}{9} \left\| \mathrm{up}^{(r+2)} \right\|_\infty = \frac{2^{\binom{r+3}{2}}}{9\,4^n}. \tag{267}$$

At the grid (199), the signed equality (201) holds.

Proof. Put $m = r+2$.

First suppose $n \geq m+1$. From (250), the nonnegativity and mass of K, and (259),

$$\begin{aligned} |f_n^{(r)}(x) - \mathrm{up}^{(r)}(x)| &\leq a_n^2 \int K(t)\,\mathrm{d}t \left\| p_n^{(m)} \right\|_\infty \\ &= 4^{-n} \cdot \frac{1}{9} \cdot 2^{\binom{m+1}{2}}. \end{aligned}$$

At $x = c_{m,j}$, every point $c_{m,j} - a_n t$ with $-1 < t < 1$ lies in the constant plateau (260). Hence no inequality remains:

$$f_n^{(r)}(c_{m,j}) - \mathrm{up}^{(r)}(c_{m,j}) = (-1)^{\mathrm{wt}_2(j)} 4^{-n} 2^{\binom{m+1}{2}} \int K(t)\,\mathrm{d}t.$$

This proves the result for $n \geq m+1$.

It remains to treat the borderline case $n = m$. Use (253) and move all $m = r+2$ derivatives to p_m:

$$f_m^{(r)} - \mathrm{up}^{(r)} = (\mathrm{D}^m p_m) * G_{a_m}.$$

By (263),

$$f_m^{(r)}(x) - \mathrm{up}^{(r)}(x) = 2^{\binom{m}{2}} a_m \sum_{j=0}^{2^m-1} (-1)^{\mathrm{wt}_2(j)} K\left(\frac{x - c_{m,j}}{a_m}\right). \tag{268}$$

The copies of K have support radius a_m and their centers are $2a_m$ apart, so their interiors are disjoint. Therefore

$$\left\| f_m^{(r)} - \mathrm{up}^{(r)} \right\|_\infty = 2^{\binom{m}{2}} a_m \|K\|_\infty = 2^{\binom{m}{2} - m} \frac{1}{9}.$$

Since $\binom{m}{2} - m = \binom{m+1}{2} - 2m$, this is exactly the right side of (267) with $n = m$. At $x = c_{m,j}$, use $K(0) = 1/9$ to obtain the signed formula. □

The two cases of the proof use different geometric data:

- for $n \geq r+3$, the error averages a constant extremal plateau and uses $\int K = 1/9$;
- for $n = r+2$, the error is a disjoint sum of rescaled kernel copies and uses $\|K\|_\infty = 1/9$.

The equality of those two kernel constants is exactly what makes the same formula valid at the threshold and beyond it.

12.2 The first C^r spline

The first iterate with r continuous derivatives is f_{r+1}. Its error has a different constant, controlled by K' rather than by K.

Theorem 12.2 (Exceptional first stage). *For every $r \geq 0$,*

$$\left\| f_{r+1}^{(r)} - \mathrm{up}^{(r)} \right\|_\infty = 2^{\binom{r+1}{2}} \left\| K' \right\|_\infty = \frac{13}{72} 2^{\binom{r+1}{2}}. \tag{269}$$

The supremum is attained at every point $c_{r+2,k}$, and the signed value is (200).

Proof. Set $n = r+1$. In (253), the total derivative order is $r+2 = n+1$. Move n derivatives to p_n and one to G_{a_n}. Since $G'_{a_n}(x) = K'(x/a_n)$, (263) gives

$$f_n^{(r)}(x) - \mathrm{up}^{(r)}(x) = 2^{\binom{n}{2}} \sum_{j=0}^{2^n-1} (-1)^{\mathrm{wt}_2(j)} K' \left(\frac{x - c_{n,j}}{a_n} \right). \tag{270}$$

Again the supports have disjoint interiors, so the supremum is $2^{\binom{n}{2}} \|K'\|_\infty$. Substitute $n = r+1$ and (247).

For the signed values, (248) gives $K'(-1/2) = 13/72$ and $K'(1/2) = -13/72$. The two points $c_{n,j} \mp a_n/2$ are respectively $c_{n+1,2j}$ and $c_{n+1,2j+1}$; using $\tau_{n+1}(2j) = \tau_n(j)$ and $\tau_{n+1}(2j+1) = -\tau_n(j)$ in (270) proves (200). □

12.3 Exact C^r convergence

Because the constants in (267) increase strictly with the derivative order, the highest derivative controls the max norm (196).

Corollary 12.3 (Exact C^r norm). *For every $r \geq 0$,*

$$\|f_{r+1} - \mathrm{up}\|_{C^r} = \frac{13}{72} 2^{\binom{r+1}{2}}, \tag{271}$$

and for every $n \geq r+2$,

$$\boxed{\|f_n - \mathrm{up}\|_{C^r} = \frac{2^{\binom{r+3}{2}}}{9\,4^n}.} \tag{272}$$

Thus, for every fixed r, the tail of the sequence lies in $C^r(\mathbb{R})$ and converges to up *in the C^r norm; the error is exactly geometric from the second admissible stage onward. No finite f_n is itself C^∞, so this is deliberately stated as eventual convergence in every finite smoothness norm rather than literally as convergence of a sequence inside the Fréchet space $C^\infty(\mathbb{R})$.*

Proof. For $n \geq r+2$, the constants in (267) grow strictly with the derivative order, since the ratio of the order-$(j+1)$ constant to the order-j constant is 2^{j+3}. Hence the rth derivative controls the maximum in (196). At the exceptional stage $n = r+1$, the assertion is immediate for $r = 0$. For $r \geq 1$, the largest lower-order error is the order-$(r-1)$ error, and its ratio to the order-r error in (269) is

$$\frac{2^{\binom{r+2}{2}}/(9\,4^{r+1})}{(13/72)2^{\binom{r+1}{2}}} = \frac{2^{2-r}}{13} < 1.$$

Thus the highest available derivative controls the max norm at the first stage as well. □

For the tent indexing $g_m = f_{m+1}$, this becomes the following direct answer to the piecewise-linear formulation.

Corollary 12.4 (Repeated integration starting from the tent). *Let* $g_0(x) = (1-|x|)_+$ *and* $g_{m+1} = Tg_m$. *For every* $r \geq 0$,

$$\|g_r - \mathrm{up}\|_{C^r} = \frac{13}{72} 2^{\binom{r+1}{2}}, \tag{273}$$

and, for every $m \geq r+1$,

$$\boxed{\|g_m - \mathrm{up}\|_{C^r} = \frac{2^{\binom{r+3}{2}-2}}{9\,4^m}.} \tag{274}$$

In particular, (203) *holds.*

12.4 The uniform errors and exact extremal values

For $r = 0$, the complete startup behavior is

$$\|f_0 - \mathrm{up}\|_\infty = \frac{1}{2}, \qquad \|f_1 - \mathrm{up}\|_\infty = \frac{13}{72}, \qquad \|f_n - \mathrm{up}\|_\infty = \frac{8}{9\,4^n} \quad (n \geq 2). \tag{275}$$

The value for f_0 is immediate from $\mathrm{up}(0) = 1$ and $0 \leq \mathrm{up} \leq 1$. For f_1, put $y = 1 - |x|$. By (223),

$$f_1(x) - \mathrm{up}(x) = y - F(y), \tag{276}$$

whose absolute maximum is attained at $y = 1/4$ and equals $1/4 - F(1/4) = 13/72$.

The four extremizers from (199) are $\pm 1/4, \pm 3/4$. Since

$$\mathrm{up}(\pm 3/4) = F(1/4) = \frac{5}{72}, \qquad \mathrm{up}(\pm 1/4) = 1 - F(1/4) = \frac{67}{72}, \tag{277}$$

we obtain exact finite-stage values for every $n \geq 2$:

$$f_n(\pm 3/4) = \frac{5}{72} + \frac{8}{9\,4^n}, \tag{278}$$

$$f_n(\pm 1/4) = \frac{67}{72} - \frac{8}{9\,4^n}. \tag{279}$$

The first few are shown in table 7.

Table 7: Sharp uniform errors and two extremal values.

n	$\|f_n - \mathrm{up}\|_\infty$	$f_n(3/4)$	$f_n(1/4)$
0	$1/2$	–	–
1	$13/72$	$1/4$	$3/4$
2	$1/18$	$1/8$	$7/8$
3	$1/72$	$1/12$	$11/12$
4	$1/288$	$7/96$	$89/96$
5	$1/1152$	$9/128$	$119/128$
6	$1/4608$	$107/1536$	$1429/1536$

13 The asymptotic error profile

The exact norm formula gives the rate, but the Peano identity also identifies the whole limiting error shape.

Theorem 13.1 (Uniform leading profile). *For every fixed* $r \geq 0$,

$$4^n \big(f_n^{(r)} - \mathrm{up}^{(r)}\big) \longrightarrow \frac{1}{9}\,\mathrm{up}^{(r+2)} \qquad \textit{uniformly on } \mathbb{R}. \tag{280}$$

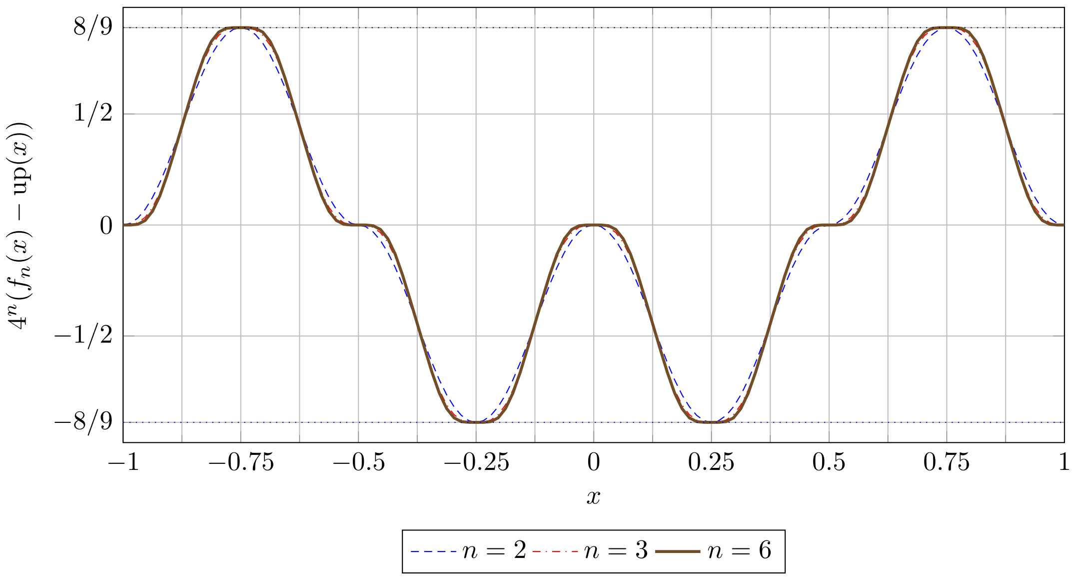

Figure 3: The errors after multiplication by 4^n. Their shape approaches $\mathrm{up}''/9$, while the exact extrema remain fixed at $\pm 8/9$ from the second stage onward. Numerical curves are shown only as an illustration of the proved identities.

Proof. Put $m = r + 2$. From (250),

$$4^n\big(f_n^{(r)}(x) - \mathrm{up}^{(r)}(x)\big) = \int K(t) p_n^{(m)}(x - a_n t)\,\mathrm{d}t.$$

For fixed m, $p_n^{(m)} \to \mathrm{up}^{(m)}$ uniformly. One direct estimate follows from $\mathrm{up} = p_n * \rho_{a_n}$ and the mean-value theorem:

$$\left\| p_n^{(m)} - \mathrm{up}^{(m)} \right\|_\infty \le a_n \mathbb{E}|X| \left\| p_n^{(m+1)} \right\|_\infty,$$

for all sufficiently large n. The same derivative bound shows that replacing $p_n^{(m)}(x - a_n t)$ by $p_n^{(m)}(x)$ costs $O(a_n)$ uniformly. Finally use $\int K = 1/9$. □

Combining (280) with (264) predicts the sharp constant in (267). The exact theorem says more: the norm of the leading profile is already the exact norm of the full error for every $n \ge r + 2$.

There is an operator-theoretic reason for the factor $1/4$. Differentiating the fixed-point identity $T\,\mathrm{up} = \mathrm{up}$ gives

$$T\big(\mathrm{up}^{(r)}\big) = 2^{-r}\,\mathrm{up}^{(r)}. \tag{281}$$

Thus up'' is an eigenfunction with eigenvalue $1/4$. Matching the tail mass and mean annihilates the modes of orders zero and one, so the second-derivative mode is the first one that survives.

Corollary 13.2 (L^p leading-profile asymptotics)**.** *For every fixed $r \ge 0$ and $1 \le p < \infty$,*

$$\lim_{n\to\infty} 4^n \left\| f_n^{(r)} - \mathrm{up}^{(r)} \right\|_{L^p} = \frac{1}{9} \left\| \mathrm{up}^{(r+2)} \right\|_{L^p}. \tag{282}$$

For $p = \infty$, the corresponding norm equality is already exact for every $n \ge r + 2$.

Proof. Use the uniform convergence in (280) and the common compact support for finite p; use (267) for $p = \infty$. □

13.1 A full even-derivative expansion

The Fourier product provides an all-orders refinement. Put

$$\Phi(z) = \prod_{k=1}^{\infty} \operatorname{sinc}(z/2^k), \qquad A(z) = \frac{\operatorname{sinc} z}{\Phi(z)}. \tag{283}$$

The quotient is analytic near the origin (indeed on $|z| < 2\pi$) and meromorphic globally. Wherever $\Phi(2^{-n}t) \neq 0$ one has

$$\mathscr{F} f_n(t) = \mathscr{F}\,\mathrm{up}(t) A(2^{-n}t), \tag{284}$$

and at the exceptional frequencies the product on the right is interpreted by its continuous finite-product extension. Only the Taylor germ at zero is used below. It is convenient to encode that germ by

$$B(w) = A(\mathrm{i}w) = \frac{\sinh w/w}{\displaystyle\prod_{k=1}^{\infty} \frac{\sinh(w/2^k)}{w/2^k}} = \sum_{m=0}^{\infty} \beta_m w^{2m}. \tag{285}$$

The first coefficients are

$$\beta_0 = 1, \qquad \beta_1 = \frac{1}{9}, \qquad \beta_2 = \frac{2}{2025}, \qquad \beta_3 = -\frac{1}{2679075}, \qquad \beta_4 = \frac{58}{10247461875}. \tag{286}$$

Proposition 13.3 (All-orders C^r expansion)**.** *For fixed integers $r \geq 0$ and $M \geq 1$,*

$$f_n^{(r)} = \sum_{m=0}^{M-1} \beta_m 4^{-mn}\,\mathrm{up}^{(r+2m)} + O_{r,M}(4^{-Mn}) \tag{287}$$

uniformly on $\mathbb{R}$ as $n \to \infty$.

Proof. Choose R with $1 < R < 2\pi$ and put $a = 2^{-n}$. On $|z| \leq R$, Taylor's theorem gives

$$A(z) = \sum_{m=0}^{M-1} \beta_m (-1)^m z^{2m} + O_{M,R}(|z|^{2M}). \tag{288}$$

On the low-frequency region $|t| \leq R/a$, multiply this by $(\mathrm{i}t)^r \mathscr{F}\,\mathrm{up}(t)$. Since $\mathscr{F}\,\mathrm{up}$ is a Schwartz function, the L^1 norm of the resulting remainder is at most

$$C_{r,M,R}\, a^{2M} \int_{\mathbb{R}} |t|^{r+2M} |\mathscr{F}\,\mathrm{up}(t)|\,\mathrm{d}t = O_{r,M}(a^{2M}).$$

The sign in (288) is exactly the sign in $(\mathrm{i}t)^{2m} = (-1)^m t^{2m}$, so Fourier inversion produces the displayed derivatives of up.

It remains to control $|t| > R/a$. From (218) and $|\operatorname{sinc} y| \leq \min(1, |y|^{-1})$, for $|t| \geq R2^n$,

$$|\mathscr{F} f_n(t)| \leq 2^{n(n+1)/2+n} |t|^{-(n+1)}. \tag{289}$$

Hence, for fixed r and large n,

$$\int_{|t|>R2^n} |t|^r |\mathscr{F} f_n(t)|\,\mathrm{d}t \leq C_r R^{r-n} 2^{-n^2/2+(r+3/2)n},$$

which is smaller than a^{2M} for all sufficiently large n. The high-frequency tails of the finitely many $\mathrm{up}^{(r+2m)}$ terms are also $O_{r,M}(a^{2M})$, again because $\mathscr{F}\,\mathrm{up}$ is Schwartz. Fourier inversion now proves the uniform remainder in (287). □

For explicit coefficient generation, write

$$\log B(w) = \sum_{m=1}^{\infty} \lambda_m w^{2m}. \tag{290}$$

With the standard Bernoulli numbers B_{2m},

$$\lambda_m = \frac{2^{2m-1} B_{2m}}{m(2m)!} \frac{4^m - 2}{4^m - 1}. \tag{291}$$

Indeed,

$$\log \frac{\sinh w}{w} = \sum_{m \geq 1} \frac{2^{2m-1} B_{2m}}{m(2m)!} w^{2m}, \qquad \sum_{k \geq 1} 4^{-mk} = \frac{1}{4^m - 1};$$

subtracting the logarithms of the denominator factors in (285) gives (291). The coefficients β_m are then obtained recursively from

$$\beta_0 = 1, \qquad \beta_m = \frac{1}{m} \sum_{j=1}^{m} j \lambda_j \beta_{m-j}. \tag{292}$$

Thus the first terms of (287) are

$$f_n^{(r)} = \mathrm{up}^{(r)} + \frac{\mathrm{up}^{(r+2)}}{9\,4^n} + \frac{2\,\mathrm{up}^{(r+4)}}{2025\,16^n} - \frac{\mathrm{up}^{(r+6)}}{2679075\,64^n} + O_r(256^{-n}). \tag{293}$$

At the extremal grid (199), all higher derivative terms vanish by theorem 11.3; there the leading term is not merely asymptotic but exactly equal to the full error, as (201) states.

14 Comparison with ordinary truncation

The ordinary prefix density p_n and the repeated-integration density f_n retain the same first n dyadic boxes, but they handle the remaining tail differently.

	Ordinary truncation p_n	Repeated integration f_n
Random variable	S_n	$S_n + 2^{-n}U$
Number of boxes	n	$n+1$
Support radius	$1 - 2^{-n}$	1 exactly
Tail model	omitted	uniform on the exact tail support
Mean error	zero by symmetry	zero by symmetry
Leading smooth error	uncorrected tail scale	variance defect, scale 4^{-n}
Sharp uniform law	not the law studied here	$8/(9\,4^n)$ for $n \geq 2$

The additional uniform box is therefore a support-and-moment correction rather than a negligible finite-product artifact. If one could also replace the tail by a positive, compactly supported law with matching variance, the up$''$ term would disappear and the next formal mode suggests a 16^{-n} scheme. This variance-matched construction is a conjectural direction; no such positive scheme is asserted here.

15 Exact computation and numerical use

A piecewise-polynomial representation also admits a simple primitive recursion. If A_{n-1} is a continuous primitive of f_{n-1}, then

$$f_n(x) = A_{n-1}(2x+1) - A_{n-1}(2x-1). \tag{294}$$

It preserves exact rational coefficients and raises the polynomial degree by one, making it convenient when every cell, rather than one point, is required.

15.1 A direct exact-arithmetic evaluator

Formula (192) immediately gives an exact rational evaluator at rational arguments. The following compact Python implementation uses arbitrary-precision fractions. It is intended as executable notation rather than as the fastest large-n algorithm.

Listing 1: Exact evaluation of the finite iterates.

```python
from fractions import Fraction
from math import factorial


def tau(n: int, j: int) -> int:
   if j < 0 or j >= (1 << n):
      return 0
   return -1 if (j.bit_count() & 1) else 1


def positive_power(x: Fraction, n: int) -> Fraction:
   return x**n if x > 0 else Fraction(0)


def f(n: int, x: Fraction) -> Fraction:
   x = Fraction(x)
   if n == 0:
      return Fraction(1, 2) if abs(x) < 1 else Fraction(0)

   coefficient = Fraction(
      1 << (n * (n + 1) // 2 - 1), factorial(n)
   )
   mesh = Fraction(1, 1 << (n - 1))

   return coefficient * sum(
      (tau(n, j) - tau(n, j - 1))
      * positive_power(x + 1 - j * mesh, n)
      for j in range((1 << n) + 1)
   )
```

Only indices satisfying $j < 2^{n-1}(x+1)$ contribute, so the loop may be truncated at the active cell. Derivatives are obtained by replacing $n!$ with $(n-r)!$ and the power n with $n-r$, as in (231).

15.2 Conditioning and stable evaluation

The truncated-power formula is exact but can exhibit severe cancellation at large n: individual terms are much larger than their sum. Three alternatives are preferable in floating-point work.

(i) Use the convolution description (210) and propagate a piecewise-polynomial representation from one level to the next.

(ii) Use a B-spline/de Boor representation on the dyadic knot sequence rather than the truncated-power basis.

(iii) In Fourier space, evaluate the finite sinc product (218) and invert it with a quadrature adapted to a compactly supported smooth target.

For certified approximation of up, the sharp error gives an immediate stopping rule. To obtain uniform error at most ε from the tent-indexed sequence, it is necessary and sufficient that

$$m \geq \left\lceil \frac{1}{2} \log_2 \left(\frac{2}{9\varepsilon} \right) \right\rceil \qquad (m \geq 1). \tag{295}$$

For a fixed derivative order r, replace the numerator $2/9$ by $2^{\binom{r+3}{2}-2}/9$ and also require $m \geq r+1$.

16 How this fits the existing ProveIt development

The present approximation is closely related to, but not identical with, two families already developed in the repository.

- The finite convolution and polynomial machinery in `FabiusFunction/EarlyApproximants.lean` provides reusable measure-theoretic and coefficient infrastructure.
- The Thue–Morse spline machinery in `FabiusFunction/FabiusUniformSpline.lean` already contains many of the finite-power sum and block-translation lemmas needed to identify the top derivative pattern.
- The centered finite-CDF spline in the main article approximates the bounded Fabius function. The sequence here approximates the up-density itself and has a different tail replacement and a different sharp rate.

A natural formalization could be organized as follows.

(1) Define the probability operator T and prove $Th(x) = \int_{2x-1}^{2x+1} h$ for supported integrable h.

(2) Define X_n recursively and prove the finite random-sum identity (209); derive (210) at the level of measures.

(3) Introduce the nonuniform box spline and prove the general inclusion–exclusion lemma (224).

(4) Reuse the finite Thue–Morse product to prove (230), then obtain the exact iterate and derivative formulas.

(5) Define K by (242). Prove convex domination with the one-line Jensen argument, then formalize its mass, maximum, and derivative maximum.

(6) Prove the derivative decomposition (258), support separation, and the exact plateaux.

(7) Combine the Peano identity with the plateau lemma, splitting the proof into the ordinary case $n \geq r+3$ and the distributional threshold $n = r+2$.

The analytic core of the sharp theorem is therefore modest. The technically delicate parts for Lean are likely to be bookkeeping for weak derivatives of interval densities and the transfer between measure convolution and classical derivatives. One can avoid most of that friction by proving the borderline formulas directly from finite signed Dirac combs and using ordinary absolutely continuous derivatives only after the threshold.

17 Conclusions

The repeated-integration construction has three mutually reinforcing exact descriptions:

(i) an averaging recursion $X \mapsto (X+U)/2$;

(ii) a convolution of dyadically scaled uniform densities;

(iii) a dyadic polynomial spline whose top derivative is a Thue–Morse word.

These descriptions identify the limit as Rvachev's up-function and reduce the finite-stage closed form to a single first-difference Thue–Morse sum. More unexpectedly, the same probabilistic representation yields a completely sharp convergence theorem. The finite stage and the limit differ only in the last scaled tail: a uniform variable replaces an up-distributed variable. Their mass and mean agree, so the error starts at second order in 2^{-n}. The comparison kernel quantifies this replacement exactly, while the partial box spline supplies disjoint extremal plateaux. The outcome is the exact identity

$$\left\| f_n^{(r)} - \mathrm{up}^{(r)} \right\|_\infty = \frac{2^{\binom{r+3}{2}}}{9\,4^n} \qquad (n \geq r+2),$$

not just an upper bound or an asymptotic equivalent.

In the most literal piecewise-linear formulation, starting from the tent and integrating again and again, the answer is especially crisp: after the initial tent, the uniform error is exactly $2/(9\,4^m)$. Every additional integration divides the worst-case error by four, and the points at which that error is attained are fixed dyadic points rather than moving with the iteration.

Topic references

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Part III
The Dyadic Web of the Fabius Function

Synthesized provenance and status. This part merges the former `Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Connections.tex` and `Fabius_Dyadic_Formulae_and_Alternative_Representations/Fabius_Dyadic_Formulae_and_Alternative_Representations.tex`. The Dyadic Web supplies the canonical master-coefficient, exact-stabilization, and q-Richardson framework. The earlier source contributes the Appell compression, ordered-composition and Hessenberg formulas, binary prefix rearrangement, limiting filter, and integral-transform views. The former coarse $O(2^{-n})$ discrete estimate is omitted in favor of the explicit $O_Q(4^{-n})$ theorem below. Claims without an exact audited Lean declaration remain frontier obligations.

Former source SHA-256 values: `81a0a911ac7ab28e12c6b87ccaf76ffccaf32e48f873abff18fb7a2d01bcf3e5` and `462276b10fcd32b0446deb7cfedc4ec07c2ae55dbd333d5ff9b1d98f07df89e1`.

Topic abstract

Exact values of the Fabius function at dyadic rationals appear in several forms that look unrelated: a Thue–Morse sum corrected by the even moments of Rvachev's up-function, a finite half-base q-binomial expression with a seemingly arbitrary complex translation, a bit-recursive Taylor reduction, an absolutely convergent binary series at arbitrary argument, and a finite q-weighted discrete approximant whose limit is the signed global Fabius function. This companion article identifies the common algebra behind these formulae and makes their connections explicit.

The central object is the coefficient

$$[X^n]\big(B_m(X)\mathscr{A}(X)\big), \qquad B_m(X) = \sum_{h<m} (-1)^{\mathrm{wt}_2(h)} \mathrm{e}^{(m-1-h)X},$$

where $\mathscr{A}$ is the exponential moment series of the Fabius probability law. Direct expansion gives the classical moment formula; probabilistic interpretation gives an expectation formula; Cauchy's formula gives a contour representation; and geometric divided differences through $-1, -2, \dots, -2^n$ give the q-binomial formula. The same coefficient also produces new finite identities: an exact moment-corrected expansion in centered splines, its reciprocal deconvolution, and a residue formula for the Gaussian interpolation row.

At arbitrary arguments the two infinite constructions have different architectures. The binary series is a genuine telescoping sum whose summands are individually independent of the complex shift. The discrete formula is instead a Toeplitz transform of shifted splines; its finite rows generally depend on the shift, although their limit does not. Writing the row generating polynomial as

$$W_n(z) = \frac{(z/2; 1/2)_n}{(1/2; 1/2)_n}$$

reveals it as a q-Richardson filter: it preserves constants and annihilates the geometric error modes $2^{-p}, 2^{-2p}, \dots, 2^{-np}$. This yields an explicit uniform $\mathcal{O}(4^{-n})$ error bound. Most importantly, at a dyadic input $m/2^s$ the discrete approximant stabilizes exactly for every row $n \geq s$; thus the "limit formula" literally becomes the exact dyadic q-formula after the denominator has been resolved. A synchronized nearest-dyadic version provides a second, termwise translation-invariant discrete limit.

18 Scope, conventions, and the formula map

This article is a companion rather than a replacement for the comprehensive exposition [12]. It assumes the construction and basic analytic theory developed there and concentrates on one cluster of results: exact dyadic evaluation, binary reduction, finite half-base q-calculus, and discrete limits. The proofs formalized in Lean under `Analysis/FabiusFunction`[11] are the source of the exact identities used below. Several consequences in this article are then derived directly from those identities; a formalization map is given in section 27.

We use three related functions, with the same convention as the companion article:

- F is the bounded Fabius distribution function, equal to 0 on $(-\infty, 0]$ and to 1 on $[1, \infty)$;
- up is the even compactly supported Rvachev density on $[-1, 1]$;
- $\mathcal{F}$ is the signed global extension satisfying $\mathcal{F}'(x) = 2\mathcal{F}(2x)$ on $\mathbb{R}$.

They agree on $[0, 1]$. Put

$$\varepsilon_r = (-1)^{\mathrm{wt}_2(r)}, \qquad c_k = \int_{-1}^{1} u^{2k}\,\mathrm{up}(u)\,\mathrm{d}u,$$

and let d_n denote the half moments normalized by

$$F(2^{-n}) = 2^{-\binom{n}{2}} \frac{d_n}{n!}. \tag{296}$$

The normalized moment series is

$$\mathscr{A}(X) = \sum_{n\geq 0} d_n \frac{X^n}{n!}. \tag{297}$$

We reserve q for the fixed base $q = 1/2$ and use $Q \in \mathbb{C}$ for a freely chosen translation. This distinction is essential: in the expressions $(a; q)_n$ and $(r - 2^k + Q)^{n+k}$ the two symbols play completely different roles.

18.1 Five presentations of the same value

The identities studied below fall into five families. Their superficial differences are summarized in table 8; the rest of the article explains the arrows between rows.

The main structural conclusions can be stated before any calculation.

The common core. The exact dyadic formulae are all coefficient extractors applied to the same entire function $B_m\mathscr{A}$. The binary series repeatedly reuses its Taylor coefficients at successive binary scales. The discrete formula applies a normalized q-Pochhammer polynomial to the sequence of finite-spline degrees. At a dyadic input these two arbitrary-argument mechanisms both become finite: the binary series terminates, while the discrete rows stabilize exactly.

18.2 The exact dyadic formula taken as input

For $0 \leq m \leq 2^n$ the moment formula proved in the formal development is

$$\boxed{F\left(\frac{m}{2^n}\right) = \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{m-1} \varepsilon_h \sum_{k=0}^{\lfloor n/2\rfloor} \binom{n}{2k} c_k (2m - 2h - 1)^{n-2k}.} \tag{298}$$

Presentation	Range and status	Principal coefficients	Mechanism
Moment/Thue–Morse formula	finite and exact at $m/2^n$	c_k and ε_h	coefficient expansion in the centered probability law
Half-base q-formula	finite and exact at $m/2^n$	$\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}$ and $(1/2;1/2)_n$	geometric divided difference after block cancellation
Binary Taylor series	infinite and absolutely convergent for $x \geq 0$; finite at dyadics	binary prefixes and d_j	exact residual telescope along the binary expansion
Global q-binary series	same outer series as the preceding row	a finite q-identity inside each scale	termwise replacement of a Taylor polynomial
Discrete q-limit	one finite row for each n, convergent for all $x \in \mathbb{R}$	normalized Gaussian row $\omega_{n,j}$	Toeplitz transform of shifted finite splines

Table 8: The formula families. The two infinite constructions are not partial-sum versions of one another.

The same finite expression, interpreted with $\mathcal{F}$ on the left, is valid for every $m, n \in \mathbb{N}$. The q-binomial version is

$$\mathcal{F}(m/2^n) = \frac{1}{2^{n^2}(1/2;1/2)_n} \sum_{k=0}^{n} \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(n+k)!} \times \sum_{r=0}^{m2^k-1} \varepsilon_r (r - m2^k + Q)^{n+k}, \tag{299}$$

valid for every $Q \in \mathbb{C}$ and independent of Q as a complete finite expression. A central goal below is to explain why (298) and (299) are two readings of one coefficient.

19 The master coefficient and its probabilistic meaning

19.1 The probability law behind the moment series

Let $U_1, U_2, \ldots$ be independent random variables, each uniformly distributed on $[0,1]$, and set

$$Y = \sum_{j=1}^{\infty} \frac{U_j}{2^j}, \qquad Z = 2Y - 1. \tag{300}$$

Then $Y \in [0,1]$, its distribution function is F, and Z has density up on $[-1,1]$. Consequently

$$d_n = \mathbb{E}(Y^n), \qquad c_k = \mathbb{E}(Z^{2k}), \qquad \mathbb{E}(Z^{2k+1}) = 0. \tag{301}$$

The moment generating function of one uniform variable is

$$E_0(X) = \frac{\mathrm{e}^X - 1}{X} = \int_0^1 \mathrm{e}^{Xu}\,\mathrm{d}u,$$

with the removable value $E_0(0) = 1$. Independence gives the locally uniformly convergent entire product

$$\mathscr{A}(X) = \mathbb{E}(\mathrm{e}^{XY}) = \prod_{j=1}^{\infty} E_0(X/2^j). \tag{302}$$

Separating the first factor, or merely replacing X by $2X$ in the product, gives the refinement equation

$$\mathscr{A}(2X) = E_0(X)\mathscr{A}(X). \tag{303}$$

The centered moment series

$$\mathscr{C}(T) = \mathbb{E}(\mathrm{e}^{TZ}) = \sum_{k\geq 0} c_k \frac{T^{2k}}{(2k)!} \tag{304}$$

satisfies the exact change of coordinates

$$\boxed{\mathscr{A}(X) = \mathrm{e}^{X/2}\mathscr{C}(X/2).} \tag{305}$$

The familiar hyperbolic-sine form follows by writing $E_0(t) = \mathrm{e}^{t/2}\sinh(t/2)/(t/2)$:

$$\mathscr{A}(X) = \mathrm{e}^{X/2}\prod_{j=1}^{\infty}\frac{\sinh(X/2^{j+1})}{X/2^{j+1}}, \tag{306}$$

$$\mathscr{C}(T) = \prod_{j=1}^{\infty}\frac{\sinh(T/2^{j})}{T/2^{j}}. \tag{307}$$

Thus the moment series used in exact arithmetic is the real-exponential counterpart of the sinc product in Fourier analysis.

19.2 The Thue–Morse prefix series

For $m \in \mathbb{N}$ define

$$B_m(X) = \sum_{h=0}^{m-1} \varepsilon_h \mathrm{e}^{(m-1-h)X}, \qquad B_0(X) = 0. \tag{308}$$

This finite exponential polynomial records the numerator m and nothing else. Its two factors have distinct roles:

- B_m carries the finite Thue–Morse prefix and therefore the binary numerator;
- $\mathscr{A}$ carries the universal Fabius moments and therefore the analytic law.

Their product is the promised master object.

Theorem 19.1 (Master coefficient identity)**.** *For every $m, n \in \mathbb{N}$,*

$$\boxed{\mathcal{F}\Big(\frac{m}{2^n}\Big) = 2^{-\binom{n}{2}}[X^n]\big(B_m(X)\mathscr{A}(X)\big).} \tag{309}$$

If $0 \leq m \leq 2^n$, the left side is the bounded value $F(m/2^n)$.

Proof. Put $t_h = 2m - 2h - 1$. By (304),

$$\sum_{k=0}^{\lfloor n/2\rfloor}\binom{n}{2k}c_k t_h^{n-2k} = n![W^n]\big(\mathrm{e}^{t_h W}\mathscr{C}(W)\big).$$

Using (305) at $X = 2W$ gives $\mathscr{C}(W) = \mathrm{e}^{-W}\mathscr{A}(2W)$. Since $t_h - 1 = 2(m-1-h)$,

$$\begin{aligned} n![W^n]\big(\mathrm{e}^{t_h W}\mathscr{C}(W)\big) &= n![W^n]\big(\mathrm{e}^{2(m-1-h)W}\mathscr{A}(2W)\big) \\ &= n!2^n[X^n]\big(\mathrm{e}^{(m-1-h)X}\mathscr{A}(X)\big). \end{aligned}$$

Multiply by ε_h, sum over $h < m$, and insert the result in (298). The power of two becomes $2^{n-\binom{n+1}{2}} = 2^{-\binom{n}{2}}$. □

The theorem compresses the double sum in (298) into one coefficient, but it also expands naturally in several different directions.

Corollary 19.2 (Expectation form). *For every $m, n \in \mathbb{N}$,*

$$\mathcal{F}(m/2^n) = \frac{2^{-\binom{n}{2}}}{n!} \sum_{h=0}^{m-1} \varepsilon_h \, \mathbb{E}\big(m - 1 - h + Y\big)^n \tag{310}$$

$$= \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{m-1} \varepsilon_h \, \mathbb{E}\big(2m - 2h - 1 + Z\big)^n. \tag{311}$$

Proof. Because $\mathscr{A}(X) = \mathbb{E}(\mathrm{e}^{XY})$,

$$[X^n]\big(\mathrm{e}^{aX}\mathscr{A}(X)\big) = \frac{1}{n!}\mathbb{E}(a+Y)^n.$$

Apply this termwise to B_m. The second form follows from $m-1-h+Y = (2m-2h-1+Z)/2$. □

Expanding the centered expectation in (311), using symmetry to discard the odd moments, recovers (298) immediately. In this sense the Thue–Morse/moment formula is simply the binomial theorem applied inside one probability expectation.

Corollary 19.3 (Cauchy coefficient representation). *For any $\rho > 0$,*

$$\mathcal{F}(m/2^n) = \frac{2^{-\binom{n}{2}}}{2\pi \mathrm{i}} \int_{|X|=\rho} \frac{B_m(X)\mathscr{A}(X)}{X^{n+1}} \, \mathrm{d}X. \tag{312}$$

This contour formula is elementary, but useful conceptually: the q-binomial formula will turn out to be a second residue computation, now in an auxiliary geometric node variable.

19.3 The numerator and the law separate completely

Expanding (309) without centering gives another exact finite form:

$$\mathcal{F}(m/2^n) = \frac{2^{-\binom{n}{2}}}{n!} \sum_{h=0}^{m-1} \varepsilon_h \sum_{j=0}^{n} \binom{n}{j} d_j (m-1-h)^{n-j}. \tag{313}$$

The d_j are universal and may be precomputed once. The numerator enters only through the finite power sums $\sum_{h<m} \varepsilon_h (m-1-h)^r$. This is often a more transparent bridge to the Taylor reduction series than the centered c_k formula, because the reduction polynomials themselves are written in the d_j.

Remark 19.4 (Two independent cancellations). The Thue–Morse prefix in (313) may annihilate low ordinary powers when m is a complete binary block. The q-binomial formula introduces a second cancellation across the geometric scale index. These cancellations live in different variables and should not be conflated: one acts on h or r inside a binary block, the other on the scale node -2^k.

20 Moment coordinates and alternative exact representations

20.1 The triangular transforms between c_k and d_n

The affine relation $Z = 2Y - 1$ gives both directions of the moment change of coordinates.

Proposition 20.1 (Moment transforms)**.** *For every $n, k \in \mathbb{N}$,*

$$d_n = 2^{-n} \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k, \tag{314}$$

$$c_k = \sum_{j=0}^{2k} (-1)^j 2^j \binom{2k}{j} d_j. \tag{315}$$

Proof. Since $Y = (1+Z)/2$, expand Y^n and use the vanishing odd moments of Z. Conversely expand $(2Y-1)^{2k}$; because $2k$ is even, the sign of the term of degree j is $(-1)^j$. □

The first formula explains the odd-looking shift $2m - 2h - 1$ in (298): it is precisely the centered coordinate obtained after converting the half moments to the even moments of up.

20.2 An infinite-cube composition formula

Each factor in (302) has the expansion

$$E_0(X/2^j) = \sum_{r \geq 0} \frac{X^r}{2^{jr}(r+1)!}.$$

Taking the coefficient of X^n gives an exact positive series over finite-support weak compositions.

Proposition 20.2 (Composition representation of the half moments)**.** *For every $n \in \mathbb{N}$,*

$$d_n = n! \sum_{\substack{r_1, r_2, \ldots \geq 0 \\ r_1 + r_2 + \cdots = n}} \prod_{j \geq 1} \frac{2^{-jr_j}}{(r_j+1)!}. \tag{316}$$

Only finitely many r_j are nonzero in each summand, and the series converges absolutely. Equivalently,

$$d_n = \lim_{J \to \infty} \int_{[0,1]^J} \left(\sum_{j=1}^{J} \frac{u_j}{2^j} \right)^n \mathrm{d}u_1 \cdots \mathrm{d}u_J. \tag{317}$$

This gives a q-free, manifestly positive representation of every inverse-dyadic value:

$$F(2^{-n}) = \frac{2^{-\binom{n}{2}}}{n!} d_n = 2^{-\binom{n}{2}} \sum_{r_1 + r_2 + \cdots = n} \prod_{j \geq 1} \frac{2^{-jr_j}}{(r_j+1)!}. \tag{318}$$

Grouping the nonzero entries of the weak composition gives a finite ordered-composition form. If $\rho = (r_1, \ldots, r_\ell) \vDash n$ and $s_j = r_1 + \cdots + r_j$, then

$$\boxed{F(2^{-n}) = 2^{-\binom{n}{2}} \sum_{\rho \vDash n} \prod_{j=1}^{\ell} \frac{1}{(2^{s_j}-1)(r_j+1)!}.} \tag{319}$$

There is also an independent recurrence-path proof. Apply theorem 4.2 to the normalized recurrence with

$$w(k,n) = \frac{1}{(2^n - 1)(n-k+1)!}, \qquad 0 \leq k < n.$$

A path $0 = s_0 < s_1 < \cdots < s_\ell = n$ corresponds to the composition $r_j = s_j - s_{j-1}$; its chronological edge product is exactly the product in (319). Thus the geometric-simplex calculation below and the generic triangular solver give two genuinely different derivations of the one canonical formula. Indeed, choosing distinct active scales $1 \le j_1 < \cdots < j_\ell$ produces the geometric simplex sum

$$\sum_{1\le j_1<\cdots<j_\ell} 2^{-\sum_i j_i r_i} = \prod_{h=1}^{\ell} \frac{1}{2^{t_h}-1}, \qquad t_h = r_h + \cdots + r_\ell. \tag{320}$$

To prove it, write $j_i = i + g_1 + \cdots + g_i$ with $g_h \ge 0$. The independent geometric series contribute $(1-2^{-t_h})^{-1}$, and the initial powers cancel because $\sum_h t_h = \sum_i i r_i$; reversing the composition changes the tail sums t_h into the prefix sums s_j.

The same triangular recurrence has a compact determinant form. Put $a_n = d_n/n!$, $a_0 = 1$, and

$$\beta_{n,k} = \frac{1}{(2^n-1)(n-k+1)!}, \qquad 0 \le k < n. \tag{321}$$

Let H_n be the lower-Hessenberg matrix whose last-row pattern is $(\beta_{n,0},\dots,\beta_{n,n-1})$, whose entries below the superdiagonal are $\beta_{i,j}$, and whose superdiagonal entries are -1. Then

$$\boxed{F(2^{-n}) = 2^{-\binom{n}{2}} \det H_n.} \tag{322}$$

Expansion along the last row gives $\det H_n = \sum_{k<n} \beta_{n,k} \det H_k$, exactly the recurrence for a_n. Expanding the determinant by Hessenberg paths recovers (319).

20.3 Cumulants, Bernoulli numbers, and Bell polynomials

The infinite product also gives a compact partition formula. With Bernoulli numbers in the convention $B_2 = 1/6$, $B_4 = -1/30$, one has

$$\log \frac{e^X - 1}{X} = \frac{X}{2} + \sum_{r\ge1} \frac{B_{2r}}{2r(2r)!} X^{2r}. \tag{323}$$

Summing this identity over the dyadic scales in (302) yields

$$\log \mathscr{A}(X) = \frac{X}{2} + \sum_{r\ge1} \frac{B_{2r}}{2r(2r)!(2^{2r}-1)} X^{2r}. \tag{324}$$

Thus the cumulants κ_j of Y are

$$\kappa_1 = \frac{1}{2}, \qquad \kappa_{2r} = \frac{B_{2r}}{2r(2^{2r}-1)}, \qquad \kappa_{2r+1} = 0 \quad (r \ge 1). \tag{325}$$

Theorem 20.3 (Bell-polynomial representation)**.** *Let $\mathcal{B}_n$ be the complete exponential Bell polynomial. Then*

$$d_n = \mathcal{B}_n(\kappa_1,\dots,\kappa_n), \tag{326}$$

and explicitly

$$d_n = n! \sum_{\substack{m_1,\dots,m_n\ge0 \\ 1m_1+2m_2+\cdots+nm_n=n}} \prod_{r=1}^{n} \frac{1}{m_r!}\left(\frac{\kappa_r}{r!}\right)^{m_r}. \tag{327}$$

Consequently

$$\boxed{F(2^{-n}) = \frac{2^{-\binom{n}{2}}}{n!} \mathcal{B}_n(\kappa_1,\dots,\kappa_n).} \tag{328}$$

Proof. Equation (324) is the standard cumulant expansion $\log \mathscr{A}(X) = \sum_{j\geq 1} \kappa_j X^j/j!$. Exponentiating and comparing coefficients is exactly the defining generating identity for the complete Bell polynomials. □

The representation is useful when Bernoulli arithmetic is already available. It also gives the moment recurrence

$$d_{n+1} = \sum_{j=0}^{n} \binom{n}{j} \kappa_{j+1} d_{n-j}, \tag{329}$$

while (303) gives the rational recurrence

$$(2^n - 1)d_n = \sum_{r=1}^{n} \binom{n}{r} \frac{d_{n-r}}{r+1} \qquad (n \geq 1). \tag{330}$$

n	d_n	c_n	$F(2^{-n})$
0	1	1	1
1	$1/2$	$1/9$	$1/2$
2	$5/18$	$19/675$	$5/72$
3	$1/6$	$583/59535$	$1/288$
4	$143/1350$	$132809/32531625$	$143/2073600$
5	$19/270$	$46840699/24405225075$	$19/33177600$

Table 9: The columns use different index conventions: c_n is the $2n$th centered moment, whereas d_n is the nth half moment.

20.4 Exact moment correction of the matched spline

The centered finite spline of degree $p \geq 1$ is

$$S_p(x) = \frac{(-1)^p}{2^{\binom{p}{2}} p!} \sum_{r=0}^{N_p(x)-1} \varepsilon_r \left(r - 2^p x + \frac{1}{2}\right)^p, \qquad N_p(x) = \max\left(0, \left\lfloor 2^p x + \frac{1}{2} \right\rfloor\right). \tag{331}$$

At degree zero set

$$S_0(x) = \sum_{r=0}^{N_0(x)-1} \varepsilon_r. \tag{332}$$

The companion article proves the global estimate

$$\sup_{x\in\mathbb{R}} |S_p(x) - \mathcal{F}(x)| \leq 2^{-p}. \tag{333}$$

At the matched grid point $m/2^p$ the cutoff is m and

$$S_p(m/2^p) = \frac{1}{2^{\binom{p+1}{2}} p!} \sum_{h=0}^{m-1} \varepsilon_h (2m - 2h - 1)^p. \tag{334}$$

The $k = 0$ part of (298) is exactly this matched spline. The remaining moments supply a finite correction hierarchy.

Theorem 20.4 (Moment-corrected matched-spline identity)**.** *For all* $m, n \in \mathbb{N}$,

$$\boxed{\mathcal{F}(m/2^n) = \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{c_k}{(2k)!\, 2^{k(2n-2k+1)}} S_{n-2k}\Big(\frac{m}{2^{n-2k}}\Big).} \tag{335}$$

For $0 \le m \le 2^n$ *the left side is* $F(m/2^n)$. *The spline arguments on the right are understood globally and may lie outside* $[0, 1]$.

Proof. In (298) put $p = n - 2k$. By (334),

$$\sum_{h=0}^{m-1} \varepsilon_h (2m - 2h - 1)^p = 2^{\binom{p+1}{2}} p!\, S_p(m/2^p).$$

Moreover

$$\frac{1}{n!}\binom{n}{2k} p! = \frac{1}{(2k)!},$$

and

$$\binom{n+1}{2} - \binom{p+1}{2} = k(2n - 2k + 1).$$

Substitution gives (335). □

The formula has three noteworthy features.

1. Its first term is the obvious finite-spline approximation $S_n(m/2^n)$.
2. The correction drops the spline degree by two at each step, reflecting the evenness of up and the disappearance of odd centered moments.
3. The numerator m remains fixed, while the denominator is coarsened from 2^n to 2^{n-2k}. Thus the formula couples degree reduction to dyadic rescaling.

Corollary 20.5 (Exact defect of the matched spline)**.** *For every* $m, n \in \mathbb{N}$,

$$\mathcal{F}(m/2^n) - S_n(m/2^n) = \sum_{k=1}^{\lfloor n/2 \rfloor} \frac{c_k}{(2k)!\, 2^{k(2n-2k+1)}} S_{n-2k}\Big(\frac{m}{2^{n-2k}}\Big). \tag{336}$$

This is stronger than an error estimate: it identifies the error exactly as a finite sequence of lower-degree global splines. The small factor in the first correction is $c_1/(2 \cdot 2^{2n-1}) = 1/(9 \cdot 2^{2n})$.

20.5 The reciprocal correction transform

The previous relation is triangular and can be inverted. Define reciprocal centered moments γ_k by

$$\frac{1}{\mathscr{C}(T)} = \sum_{k \ge 0} \gamma_k \frac{T^{2k}}{(2k)!}. \tag{337}$$

Thus

$$\gamma_0 = 1, \qquad \gamma_n = -\sum_{k=0}^{n-1} \binom{2n}{2k} \gamma_k c_{n-k} \quad (n \ge 1), \tag{338}$$

and the first values are

$$1, \quad -\frac{1}{9}, \quad \frac{31}{675}, \quad -\frac{2347}{59535}, \quad \frac{1904369}{32531625}, \ldots. \tag{339}$$

Theorem 20.6 (Reciprocal moment deconvolution)**.** *For every $m, n \in \mathbb{N}$,*

$$\boxed{S_n(m/2^n) = \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{\gamma_k}{(2k)!\,2^{k(2n-2k+1)}} \mathcal{F}\Big(\frac{m}{2^{n-2k}}\Big)\,.} \tag{340}$$

Proof. For a fixed numerator m, introduce

$$P_p(m) = \sum_{h=0}^{m-1} \varepsilon_h (2m-2h-1)^p, \qquad G_p(m) = 2^{\binom{p+1}{2}} p!\,\mathcal{F}(m/2^p).$$

The exact moment formula says, in exponential generating functions,

$$\sum_{p\geq 0} G_p(m)\frac{T^p}{p!} = \mathscr{C}(T) \sum_{p\geq 0} P_p(m) \frac{T^p}{p!}.$$

Multiply by $1/\mathscr{C}(T)$, compare coefficients, and then use $P_p(m) = 2^{\binom{p+1}{2}} p! S_p(m/2^p)$. The same triangular exponent simplification as in theorem 20.4 gives (340). □

Equations (335) and (340) are a finite convolution/deconvolution pair. They show that the exact dyadic values and the matched centered splines carry precisely the same information once the universal moment series $\mathscr{C}$ is known.

First bridge to the discrete theory. The centered spline is not merely an approximation from which exact dyadic arithmetic is separate. At a matched grid point, the exact value is obtained by a finite universal moment correction of that spline, and the correction is invertible. Later, the half-base q-Pochhammer row will give a second exact correction, using consecutive spline degrees at the *same* argument rather than parity-separated degrees at rescaled arguments.

21 The half-base q-formula as geometric coefficient extraction

21.1 The Gaussian row is a divided difference

For a general base q, write

$$(a;q)_n = \prod_{j=0}^{n-1} (1 - aq^j), \tag{341}$$

and let $\begin{bmatrix} n \\ k \end{bmatrix}_q$ denote the Gaussian binomial coefficient. At $q = 1/2$ the finite q-binomial theorem is

$$(z;1/2)_n = \sum_{k=0}^{n} (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} z^k. \tag{342}$$

Put

$$x_k = -2^k, \qquad w_{n,k} = (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2}, \qquad \mathfrak{m}_n = \prod_{\ell=1}^{n} (2^\ell - 1). \tag{343}$$

Then

$$\sum_{k=0}^{n} w_{n,k} x_k^d = \begin{cases} 0, & 0 \leq d < n, \\ \mathfrak{m}_n, & d = n, \end{cases} \tag{344}$$

and

$$\frac{w_{n,k}}{\mathfrak{m}_n} = \frac{1}{\prod_{\ell\neq k}(x_k - x_\ell)}, \qquad (1/2;1/2)_n = 2^{-\binom{n+1}{2}} \mathfrak{m}_n. \tag{345}$$

Equivalently, the half-base coefficients contain only integral Gaussian arithmetic and an explicit power of two:

$$\begin{bmatrix} n \\ k \end{bmatrix}_{1/2} = 2^{-k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_2. \tag{346}$$

Thus the functional

$$\mathfrak{D}_n[P] = \frac{1}{\mathfrak{m}_n} \sum_{k=0}^{n} w_{n,k} P(-2^k) \tag{347}$$

is the divided difference $[-1, -2, \ldots, -2^n]P$. For every polynomial of degree at most n, it returns the leading coefficient.

Proposition 21.1 (Residue form of the geometric extractor). *Let*

$$\Pi_n(z) = \prod_{k=0}^{n} (z + 2^k),$$

and let P be a polynomial of degree at most n. If Γ encloses all nodes $-1, -2, \ldots, -2^n$, then

$$[z^n]P(z) = \mathfrak{D}_n[P] = \frac{1}{2\pi \mathrm{i}} \int_\Gamma \frac{P(z)}{\Pi_n(z)} \, \mathrm{d}z. \tag{348}$$

Proof. The residues at the simple poles $x_k = -2^k$ are $P(x_k)/\Pi_n'(x_k) = P(x_k)/\prod_{\ell \neq k}(x_k - x_\ell)$, giving the barycentric sum in (347). Lagrange interpolation identifies that sum with the leading coefficient. □

Remark 21.2 (The same extractor at a general geometric base). For $0 < q < 1$, let the nodes be $1, q^{-1}, \ldots, q^{-n}$. Lagrange interpolation gives, for every polynomial P of degree at most n,

$$[t^n]P(t) = \frac{(-1)^n q^{\binom{n+1}{2}}}{(q;q)_n} \sum_{k=0}^{n} (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q P(q^{-k}). \tag{349}$$

The corresponding barycentric denominator is

$$\frac{1}{\prod\limits_{j \neq k} (q^{-k} - q^{-j})} = \frac{(-1)^{n-k} q^{\binom{n+1}{2} + \binom{k}{2}}}{(q;q)_n} \begin{bmatrix} n \\ k \end{bmatrix}_q. \tag{350}$$

Equivalently, the finite q-binomial theorem kills every lower monomial after the specialization $z = q^{-m}$, while the surviving top term is normalized by

$$(q^{-n}; q)_n = (-1)^n q^{-\binom{n+1}{2}} (q;q)_n. \tag{351}$$

Thus the half-base stencil above is the $q = 1/2$ instance of a generic geometric divided difference, not an isolated Gaussian-binomial coincidence.

The q-symbols in the dyadic formula therefore have a concrete and limited task: they are the closed form of a divided-difference stencil on a geometric grid. No infinite basic-hypergeometric summation is involved.

21.2 The parametric coefficient polynomial

Introduce an auxiliary node variable z and define

$$\mathscr{R}_m(z;X) = \frac{\mathrm{e}^{-X} B_m(zX)\mathscr{A}(zX)}{\mathscr{A}(-X)}, \qquad r_{m,n}(z) = [X^n]\mathscr{R}_m(z;X). \tag{352}$$

Because $\mathscr{A}(-X)$ is a unit with constant term 1, this is a well-defined formal series and an entire analytic germ. More importantly, $r_{m,n}$ is a polynomial in z.

Lemma 21.3 (Node polynomial and its leading coefficient)**.** *For fixed m, n, the polynomial $r_{m,n}(z)$ has degree at most n and*

$$[z^n]r_{m,n}(z) = [X^n]\big(B_m(X)\mathscr{A}(X)\big) = 2^{\binom{n}{2}}\mathcal{F}(m/2^n). \tag{353}$$

Proof. Write $\mathrm{e}^{-X}/\mathscr{A}(-X) = \sum_{j\geq 0} u_j X^j$ with $u_0 = 1$. Since $B_m(zX)\mathscr{A}(zX) = (B_m\mathscr{A})(zX)$, the coefficient of X^n is a polynomial in z of degree at most n. Its z^n term can only use the constant u_0, so its leading coefficient is $[X^n](B_m\mathscr{A})$. Apply theorem 19.1. □

The exact dyadic value is now the leading coefficient of a degree-n polynomial sampled at $-1, -2, \ldots, -2^n$. Applying theorem 21.1 gives a compact alternative to the displayed q-sum.

Corollary 21.4 (Geometric-node contour representation)**.** *For every $m, n \in \mathbb{N}$,*

$$\boxed{\mathcal{F}(m/2^n) = \frac{2^{-\binom{n}{2}}}{2\pi\mathrm{i}} \int_\Gamma \frac{r_{m,n}(z)}{\prod_{k=0}^{n}(z+2^k)}\,\mathrm{d}z,} \tag{354}$$

where Γ encloses the geometric nodes.

Together, (312) and (354) exhibit two independent Cauchy extractions:

coefficient in the analytic variable X	leading coefficient in the geometric node z
$[X^n](B_m\mathscr{A})$	$[z^n]r_{m,n}(z)$
ordinary Cauchy kernel X^{-n-1}	barycentric kernel $\Pi_n(z)^{-1}$.

The finite q-binomial formula is what results when the second contour is evaluated by residues and each node value is converted to a Thue–Morse block sum.

21.3 Why the node values are Thue–Morse power sums

For a complete binary block set

$$C_k(X) = \sum_{r=0}^{2^k-1} \varepsilon_r \mathrm{e}^{(r-2^k)X}. \tag{355}$$

The digit factorization $\sum_{r<2^k} \varepsilon_r z^r = \prod_{j=0}^{k-1}(1-z^{2^j})$ gives

$$C_k(X) = (-1)^k 2^{\binom{k}{2}} X^k \mathrm{e}^{-X} \prod_{j=0}^{k-1} E_0(-2^j X). \tag{356}$$

Hence C_k has a zero of exact order k. The refinement (303), iterated k times, gives

$$\frac{\mathrm{e}^{-X}\mathscr{A}(-2^k X)}{\mathscr{A}(-X)} = \mathrm{e}^{-X}\prod_{j=0}^{k-1} E_0(-2^j X). \tag{357}$$

Combining (356) and (357) identifies the node specialization of (352) with a normalized complete block.

There is also a useful differential-operator reading of this cancellation. For a polynomial or entire function P, set

$$(\mathfrak{T}_k P)(c) = \sum_{r=0}^{2^k-1} \varepsilon_r P(c+r). \tag{358}$$

Writing $D = \mathrm{d}/\mathrm{d}c$, the binary digits of r give

$$\mathfrak{T}_k = \prod_{j=0}^{k-1} \left(1 - e^{2^j D}\right). \tag{359}$$

Every factor starts with $-2^j D$, so degrees below k vanish and the first surviving moment is

$$\sum_{r=0}^{2^k-1} \varepsilon_r r^k = (-1)^k 2^{\binom{k}{2}} k!. \tag{360}$$

More generally,

$$c \longmapsto \frac{1}{(n+k)!} \sum_{r<2^k} \varepsilon_r (r+c)^{n+k} \tag{361}$$

has degree at most n, with leading coefficient

$$[c^n] \left(\frac{1}{(n+k)!} \sum_{r<2^k} \varepsilon_r (r+c)^{n+k} \right) = \frac{(-1)^k 2^{\binom{k}{2}}}{n!}. \tag{362}$$

With $\operatorname{sinhc}(z) = \sinh(z)/z$, its midpoint-symmetric factorization is

$$\mathfrak{T}_k = (-1)^k 2^{\binom{k}{2}} D^k e^{(2^k-1)D/2} \prod_{j=0}^{k-1} \operatorname{sinhc}(2^{j-1} D). \tag{363}$$

This exposes a kth derivative followed by an even differential correction about the block midpoint, explaining why centered and translated forms arise naturally.

For a general numerator, the long block factors as

$$\sum_{r=0}^{m2^k-1} \varepsilon_r \mathrm{e}^{(r-m2^k)X} = B_m(-2^k X) C_k(X). \tag{364}$$

After division by the forced factor $(-1)^k 2^{\binom{k}{2}} X^k$, the coefficient of X^n is

$$\frac{(-1)^k 2^{-\binom{k}{2}}}{(n+k)!} \sum_{r=0}^{m2^k-1} \varepsilon_r (r - m2^k)^{n+k}. \tag{365}$$

Multiplication by the interpolation weight $w_{n,k}$ cancels the sign and contributes a second factor $2^{-\binom{k}{2}}$. Their product is the characteristic $4^{-\binom{k}{2}}$ of (299). Finally (345) converts $\mathfrak{m}_n$ into $2^{n^2}(1/2;1/2)_n$ after the factor $2^{\binom{n}{2}}$ from theorem 21.3 is included. This recovers the exact q-formula with $Q = 0$.

21.4 Translation invariance is lower-degree annihilation

The shift Q enters a block generating function by multiplication with e^{QX}. If $P_{m,k}(X)$ denotes the normalized long-block series after its forced X^k has been removed, then the q-numerator is a weighted sum of $[X^n](\mathrm{e}^{QX}P_{m,k}(X))$. Expanding the exponential,

$$[X^n](\mathrm{e}^{QX}P_{m,k}(X)) = \sum_{s=0}^{n} \frac{Q^{n-s}}{(n-s)!}[X^s]P_{m,k}(X). \tag{366}$$

For each $s < n$, the Gaussian row sees a polynomial of geometric-node degree at most s and annihilates it. Only the $s = n$ term survives; it has no Q factor.

Proposition 21.5 (Finite translation invariance; Lean formalized). *For fixed m, n, the polynomial*

$$Q \longmapsto \sum_{k=0}^{n} \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{m2^k-1} \varepsilon_r (r - m2^k + Q)^{n+k} \tag{367}$$

is constant.

The constant-polynomial identity and its coefficientwise form are the exact declarations `qBinomialThueMorseDyadicTranslatedFormulaPolynomial_eq_const` and `qBinomialThueMorseDyadicTranslatedFormulaPolynomial_coeff` in `FabiusDyadicQBinomialScalar.lean`.

This is stronger than saying that a limit is independent of Q: every coefficient of $Q, Q^2, \ldots$ vanishes in the finite expression itself. The distinction becomes important in section 23, where the finite discrete rows generally do depend on Q.

21.5 A useful abstract summary

The exact q-formula is a composition of two annihilators:

1. the complete Thue–Morse block has a zero of order k, removing all inner polynomial degrees below k;
2. the Gaussian row on -2^k removes all remaining node degrees below n and returns the leading degree n.

The exponent $n+k$ in the inner power is therefore forced: k degrees are consumed by the binary block, leaving exactly n degrees for the geometric extractor. The formula is a hybrid of automatic-sequence cancellation, ordinary exponential generating functions, and finite q-calculus; it is not a terminating basic-hypergeometric summation in the usual ${}_r\phi_s$ sense [7, 9].

22 Binary Taylor reduction and the global series

22.1 The finite reduction polynomial

The inverse-dyadic moments assemble into the polynomial

$$\mathcal{T}_m(y) = \sum_{j=0}^{m} 2^{\binom{j+1}{2}-\binom{m-j}{2}} \frac{d_{m-j}}{(m-j)!}\frac{y^j}{j!}. \tag{368}$$

The signed global extension satisfies the exact Taylor-block identity

$$\mathcal{F}(2^{-m}+y) = -\mathcal{F}(y) + \mathcal{T}_m(y), \qquad 0 \le y < 2^{-m}. \tag{369}$$

This is the elementary step behind both the terminating dyadic recursion and the infinite binary telescope.

22.2 Appell compression of every Taylor block

To avoid collision with the later saddle coefficients, write

$$\mathrm{App}_m(\theta) = \mathbb{E}(Y+\theta)^m = \sum_{j=0}^{m} \binom{m}{j} d_{m-j}\theta^j. \tag{370}$$

Its exponential generating function is $\sum_m \mathrm{App}_m(\theta)X^m/m! = e^{\theta X}\mathscr{A}(X)$; hence it is an Appell family,

$$\mathrm{App}'_m(\theta) = m\,\mathrm{App}_{m-1}(\theta), \qquad \mathrm{App}_m(\theta) = (-1)^m\,\mathrm{App}_m(-1-\theta). \tag{371}$$

The full Appell addition law is

$$\mathrm{App}_m(\theta+c) = \sum_{j=0}^{m} \binom{m}{j} \mathrm{App}_{m-j}(\theta)c^j. \tag{372}$$

The irregular powers of two in (368) collapse under the natural tail scale:

$$\boxed{\mathcal{T}_m(2^{-m}\theta) = 2^{-\binom{m}{2}}\frac{\mathrm{App}_m(\theta)}{m!}.} \tag{373}$$

Equivalently, at an arbitrary scale,

$$\mathcal{T}_m(y) = 2^{-\binom{m}{2}}\frac{\mathrm{App}_m(2^m y)}{m!}. \tag{374}$$

Indeed, after substituting $y = 2^{-m}\theta$, the exponent in every jth term is $\binom{j+1}{2} - \binom{m-j}{2} - mj = -\binom{m}{2}$. For $0 \le \theta < 1$, support and positivity give $0 \le \mathrm{App}_m(\theta) \le 2^m$.

For orientation, the first rows are

$$\begin{aligned}
\mathrm{App}_0(\theta) &= 1,\\
\mathrm{App}_1(\theta) &= \theta + \frac{1}{2},\\
\mathrm{App}_2(\theta) &= \theta^2 + \theta + \frac{5}{18},\\
\mathrm{App}_3(\theta) &= \theta^3 + \frac{3}{2}\theta^2 + \frac{5}{6}\theta + \frac{1}{6},\\
\mathrm{App}_4(\theta) &= \theta^4 + 2\theta^3 + \frac{5}{3}\theta^2 + \frac{2}{3}\theta + \frac{143}{1350}.
\end{aligned}$$

For $x \ge 0$ set

$$p_m(x) = \lfloor 2^m x\rfloor, \qquad y_m(x) = x - \frac{p_m(x)}{2^m}, \qquad 0 \le y_m(x) < 2^{-m}, \tag{375}$$

and put

$$p^*_{m-1}(x) = \begin{cases} \lfloor x/2\rfloor, & m = 0,\\ p_{m-1}(x), & m \ge 1. \end{cases} \tag{376}$$

Define

$$\alpha_m(x) = (-1)^{\mathrm{wt}_2(p_m(x))}\big(2p^*_{m-1}(x) - p_m(x)\big). \tag{377}$$

This coefficient is 0 unless the newly exposed binary digit is 1, in which case it is a sign.

Theorem 22.1 (Binary reduction series). *For every $x \geq 0$,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{\infty} \alpha_m(x) \mathcal{T}_m(y_m(x)).} \tag{378}$$

The series converges absolutely, and its partial sum through $m = N$ has the uniform remainder bound

$$\left| \mathcal{F}(x) - \sum_{m=0}^{N} \alpha_m(x) \mathcal{T}_m(y_m(x)) \right| \leq 2^{1-N} \qquad (N \geq 1). \tag{379}$$

The proof is an exact residual telescope. If $R_m(x) = (-1)^{\mathrm{wt}_2(p_m(x))} \mathcal{F}(y_m(x))$, then

$$\alpha_m(x) \mathcal{T}_m(y_m(x)) = R_{m-1}(x) - R_m(x) \quad (m \geq 1),$$

with a separately correct scale-zero term. Since $0 \leq y_m < 2^{-m}$ and $|\mathcal{F}(y)| \leq 2y$ near the origin, $|R_m| \leq 2^{1-m}$.

22.3 The sparse digit form on the unit interval

For $0 \leq x < 1$, write the floor-selected binary expansion

$$x = 0.\delta_1 \delta_2 \delta_3 \cdots = \sum_{m \geq 1} \delta_m 2^{-m}, \qquad \delta_m \in \{0, 1\}, \tag{380}$$

where p_m is the integer represented by the first m digits. Let $s_m = \delta_1 + \cdots + \delta_m$ and let

$$t_m = \sum_{\ell > m} \delta_\ell 2^{-\ell} = y_m(x) \tag{381}$$

be the unscaled tail. If $\delta_m = 1$, then $p_m = 2p_{m-1} + 1$ and

$$\alpha_m = (-1)^{s_m}(-1) = (-1)^{s_{m-1}};$$

if $\delta_m = 0$, then $\alpha_m = 0$.

Corollary 22.2 (Digit-sparse series). *For $0 \leq x < 1$,*

$$\boxed{F(x) = \sum_{\substack{m \geq 1 \\ \delta_m = 1}} (-1)^{s_{m-1}} \mathcal{T}_m(t_m).} \tag{382}$$

At a dyadic rational the floor convention chooses the terminating binary expansion, so the sum is finite and contains exactly one term for each set bit of the numerator.

If $\theta_m = 2^m t_m = \{2^m x\}$, the Appell compression gives the equivalent digit-sparse form

$$F(x) = \sum_{\substack{m \geq 1 \\ \delta_m = 1}} (-1)^{s_{m-1}} \frac{2^{-\binom{m}{2}}}{m!} \operatorname{App}_m(\theta_m). \tag{383}$$

It converges uniformly on $0 \leq x < 1$. If S_N denotes its truncation through $m = N$, then

$$\boxed{|F(x) - S_N(x)| \leq 2 \frac{2^{N+1-\binom{N+1}{2}}}{(N+1)!}.} \tag{384}$$

The ratio of consecutive majorants is $2^{1-m}/(m+1) \leq 1/2$; the digit series therefore has a super-exponential certificate, substantially sharper than the generic 2^{1-N} residual bound needed for the all-real telescope.

This is the most direct connection between the fast bit-recursive evaluator and the arbitrary argument series: the finite recursion is not a separate algorithm pasted onto the infinite formula; it is literally the same telescope when the binary tail eventually becomes zero.

22.4 Direct one-row Appell and nested digit–q forms

The triangular reconstruction below is not the only way to insert the Gaussian row. Retaining the Appell variable throughout the finite identity gives the direct centered formula.

Theorem 22.3 (Centered Gaussian-binomial representation of the Appell row). *For every $n \geq 0$ and $\theta \in \mathbb{C}$,*

$$\boxed{\frac{\mathrm{App}_n(\theta)}{n!} = \frac{2^{-\binom{n+1}{2}}}{(1/2;1/2)_n} \sum_{k=0}^{n} \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{2^k-1} \varepsilon_r \big(r - 2^k(1+\theta)\big)^{n+k}.} \tag{385}$$

At $\theta = 0$, multiplication by $2^{-\binom{n}{2}}$ recovers the centered inverse-dyadic formula. More generally, a common translation may be inserted in every inner block.

Theorem 22.4 (Raw translated Appell representation). *For every $n \geq 0$ and $\theta, \eta \in \mathbb{C}$,*

$$\boxed{\frac{\mathrm{App}_n(\theta)}{n!} = \frac{(-1)^n 2^{-\binom{n+1}{2}}}{(1/2;1/2)_n} \sum_{k=0}^{n} \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{2^k-1} \varepsilon_r \big(r + \eta + 2^k\theta\big)^{n+k}.} \tag{386}$$

The right-hand side is independent of η.

For a fixed outer scale n, the translation must be common to every k-term in that row. Different outer scales may use different translations η_n without changing any term; this is a genuine translation gauge and can reduce intermediate powers in floating-point work, but varying the shift within one k-row invalidates the cancellation.

Termwise substitution into (383) gives, for $0 \leq x < 1$,

$$\begin{aligned} F(x) = \sum_{\substack{m\geq 1 \\ \delta_m = 1}} (-1)^{s_{m-1}} \frac{1}{2^{m^2}(1/2;1/2)_m} \sum_{k=0}^{m} \frac{\begin{bmatrix} m \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(m+k)!} \\ \times \sum_{r=0}^{2^k-1} \varepsilon_r \big(r - 2^k(1+\theta_m)\big)^{m+k}, \end{aligned} \tag{387}$$

and, for any fixed $\eta \in \mathbb{C}$,

$$\begin{aligned} F(x) = \sum_{\substack{m\geq 1 \\ \delta_m = 1}} (-1)^{s_{m-1}} \frac{1}{(-2)^{m^2}(1/2;1/2)_m} \sum_{k=0}^{m} \frac{\begin{bmatrix} m \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(m+k)!} \\ \times \sum_{r=0}^{2^k-1} \varepsilon_r \big(r + \eta + 2^k\theta_m\big)^{m+k}. \end{aligned} \tag{388}$$

Each mth summand equals the corresponding Appell summand before any infinite summation is interchanged. At a dyadic input the series terminates, so these are finite exact identities.

22.5 Replacing each Taylor coefficient by a finite q-identity

For $Q \in \mathbb{C}$ define the finite inverse-block numerator

$$\mathcal{N}_n^{(Q)} = \sum_{k=0}^{n} \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{2^k-1} \varepsilon_r (r - 2^k + Q)^{n+k}. \tag{389}$$

The exact one-block q-formula is equivalent to

$$\frac{\mathcal{N}_n^{(Q)}}{2^{\binom{n}{2}}(1/2;1/2)_n} = \frac{2^n d_n}{n!}, \tag{390}$$

which is independent of Q. Insert these finite identities into the polynomial (368):

$$\mathcal{Q}_m^{(Q)}(y) = 2^{-\binom{m+1}{2}} \sum_{n=0}^{m} \frac{(2^{m+1}y)^{m-n}}{(m-n)!} \frac{\mathcal{N}_n^{(Q)}}{2^{\binom{n}{2}}(1/2;1/2)_n}. \tag{391}$$

Theorem 22.5 (q-Taylor identity and global q-series)**.** *For every m, y, and $Q \in \mathbb{C}$,*

$$\mathcal{Q}_m^{(Q)}(y) = \mathcal{T}_m(y). \tag{392}$$

Consequently, for fixed Q and every $x \geq 0$,

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{\infty} \alpha_m(x) \mathcal{Q}_m^{(Q)}(y_m(x)),} \tag{393}$$

with absolute convergence.

The key word is *termwise*: every finite polynomial $\mathcal{Q}_m^{(Q)}$ already equals $\mathcal{T}_m$ and is already independent of Q before the outer infinite sum is considered. The q-symbols do not create the convergence of the outer series. They merely provide an alternative finite presentation of each Taylor block.

22.6 The dyadic triangle of finite representations

Suppose $x = m/2^s \in [0,1]$ is written with a terminating binary expansion. Then three different finite mechanisms evaluate the same rational number:

1. (298) sums the Thue–Morse prefix $0 \leq h < m$ and applies the even-moment correction in one shot;
2. (299) uses one geometric row $0 \leq k \leq s$ and two nested cancellations;
3. (382) walks only through the set bits of m, using inverse-dyadic Taylor data.

The finite sums have no natural term-by-term identification. Their equality is mediated by the master coefficient (309) and by the exact Taylor reduction (369). A fourth finite representation, obtained from the discrete limit, appears in the next section.

The combinatorial bridge between the one-shot and set-bit organizations is an exact prefix-block decomposition. Write $M/2^N = 0.\delta_1 \ldots \delta_N{}_2$ and $p_m = \lfloor 2^m M/2^N \rfloor$. For any function P on the nonnegative integers and $k \geq 0$,

$$\sum_{r=0}^{M2^k-1} \varepsilon_r P(r) = \sum_{\substack{1\leq m\leq N\\ \delta_m=1}} \varepsilon_{p_{m-1}} \sum_{s=0}^{2^{N-m+k}-1} \varepsilon_s P\Big(p_{m-1}2^{N-m+k+1} + s\Big). \tag{394}$$

The interval $[0, M)$ is the disjoint union of the blocks selected by the set bits. After scaling by 2^k, each block begins at a multiple of twice its length, so adding the local coordinate creates no binary carry and its Thue–Morse sign factors. This explains why the one-shot q formula and the sparse Appell formula group the same exact quantity so differently.

23 Discrete q-limits, exact stabilization, and extrapolation

23.1 Shifted splines and the finite row

For $p \geq 1$, $Q \in \mathbb{C}$, and $x \in \mathbb{R}$, define

$$S_p^{(Q)}(x) = \frac{(-1)^p}{2^{\binom{p}{2}} p!} \sum_{r=0}^{N_p(x)-1} \varepsilon_r (r - 2^p x + Q)^p. \tag{395}$$

At $Q = 1/2$ this is the centered spline S_p. For $x \geq 0$, the formal development proves the exact fixed-cutoff Taylor relation and the audited bound

$$\left|S_p^{(Q)}(x) - S_p(x)\right| \leq 2^{-(p-1)} \mathrm{e}^{|Q-1/2|}. \tag{396}$$

This is `norm_fabiusComplexShiftSpline_sub_center_le_half_pow_mul_exp` in `Fabius ComplexShiftSpline.lean`, with hypotheses $1 \leq p$ and $0 \leq x$. When $x \leq 0$, both shifted splines vanish by `fabiusComplexShiftSpline_eq_zero_of_nonpos`; an all-real bound follows by a direct case split, but is not presently packaged as one public theorem. The sharper estimate used in the frontier calculation below is

$$\left|S_p^{(Q)}(x) - S_p(x)\right| \leq 2^{-(p-1)} \left(\mathrm{e}^{|Q-1/2|} - 1\right). \tag{397}$$

It is a natural-language candidate, not presently an exact Lean theorem; its formalization obligation and the $p = 2$ edge-case warning are recorded in the Primary Exposition Gap Register at the end of this volume. Consequently (398), theorem 23.5, and theorem 23.6 retain frontier status with their printed sharp constants. Together with (333),

$$\left|S_p^{(Q)}(x) - \mathcal{F}(x)\right| \leq (2\mathrm{e}^{|Q-1/2|} - 1)2^{-p}. \tag{398}$$

The discrete q-approximant is

$$\mathcal{D}_n^{(Q)}(x) = \frac{1}{2^{n^2}(1/2;1/2)_n} \sum_{k=0}^{n} \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(n+k)!} \times \sum_{r=0}^{N_{n+k}(x)-1} \varepsilon_r (r - 2^{n+k}x + Q)^{n+k}. \tag{399}$$

For every fixed Q and every real x, the Lean development proves

$$\mathcal{D}_n^{(Q)}(x) \longrightarrow \mathcal{F}(x). \tag{400}$$

The notation strongly resembles the exact dyadic formula (299). The following result makes the resemblance literal.

23.2 At dyadic inputs the limit stabilizes exactly

Theorem 23.1 (Finite stabilization on the dyadic grid)**.** *Let $x = m/2^s$ with $m, s \in \mathbb{N}$. Then, for every $n \geq s$ and every $Q \in \mathbb{C}$,*

$$\boxed{\mathcal{D}_n^{(Q)}(x) = \mathcal{F}(x).} \tag{401}$$

If $0 \leq x \leq 1$, the right side is $F(x)$.

Proof. For $n \geq s$ put $M = m2^{n-s}$, so that $x = M/2^n$. For every $0 \leq k \leq n$,

$$2^{n+k}x = M2^k \in \mathbb{N}, \qquad N_{n+k}(x) = \left\lfloor M2^k + \frac{1}{2} \right\rfloor = M2^k.$$

Therefore (399) becomes exactly the dyadic formula (299) with numerator M and denominator exponent n. That formula is valid for unreduced dyadic representations and for every complex translation Q. □

This is the sharpest connection between the exact and limiting formulae: the discrete formula extends the exact dyadic row to arbitrary x, and on a dyadic grid it needs no limiting process once the row depth reaches the denominator depth.

Corollary 23.2 (Exact finite shifted-spline extrapolation)**.** *For $x = m/2^s$, $n \geq s$, and every $Q \in \mathbb{C}$,*

$$\mathcal{F}(x) = \sum_{j=0}^{n} \omega_{n,j} S^{(Q)}_{2n-j}(x), \tag{402}$$

where the weights $\omega_{n,j}$ are defined in (404) below.

Thus every dyadic value admits an exact representation as a finite signed combination of consecutive spline degrees, all evaluated at the same point. Each shifted spline may depend strongly on Q, but the complete combination does not.

23.3 The row polynomial

Reindexing (399) by $j = n - k$ gives the exact identity

$$\mathcal{D}^{(Q)}_n(x) = \sum_{j=0}^{n} \omega_{n,j} S^{(Q)}_{2n-j}(x), \tag{403}$$

with

$$\omega_{n,j} = \frac{(-1)^j \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} 2^{-\binom{j+1}{2}}}{(1/2; 1/2)_n}. \tag{404}$$

The full row is encoded by one normalized q-Pochhammer polynomial.

Proposition 23.3 (Generating polynomial of the Toeplitz row)**.** *Define*

$$W_n(z) = \sum_{j=0}^{n} \omega_{n,j} z^j. \tag{405}$$

Then

$$\boxed{W_n(z) = \frac{(z/2; 1/2)_n}{(1/2; 1/2)_n} = \prod_{\ell=1}^{n} \frac{1 - z/2^\ell}{1 - 1/2^\ell}.} \tag{406}$$

Consequently

$$W_n(1) = 1, \tag{407}$$

$$W_n(2^r) = 0 \qquad (1 \leq r \leq n), \tag{408}$$

$$\sum_{j=0}^{n} |\omega_{n,j}| = W_n(-1) = \frac{(-1/2; 1/2)_n}{(1/2; 1/2)_n}, \tag{409}$$

$$\sum_{j=0}^{n} |\omega_{n,j}| 2^j = W_n(-2) = \frac{(-1; 1/2)_n}{(1/2; 1/2)_n}. \tag{410}$$

Proof. Apply the finite q-binomial theorem to $(z/2;1/2)_n$. Its coefficient of z^j is

$$(-1)^j 2^{-\binom{j}{2}} 2^{-j} \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} = (-1)^j 2^{-\binom{j+1}{2}} \begin{bmatrix} n \\ j \end{bmatrix}_{1/2},$$

which gives (406) after division by $(1/2;1/2)_n$. The evaluations at $1, 2^r, -1, -2$ follow from the product and from the alternating signs of the coefficients. □

Current formal status. For rational z, the full identity (406) is `sum_range_discreteLimitWeight_mul_pow`, and the exact dyadic root locus (408) is `sum_range_discreteLimitWeight_mul_pow_eq_zero_iff`. The mass and unweighted variation evaluations are respectively `sum_range_discreteLimitWeight` and `sum_abs_discreteLimitWeight`, all in `FabiusDiscreteLimitToeplitz.lean`. The weighted variation at -2, the logarithmic-derivative moment formulas below, and the limiting filter are not claimed as named Lean counterparts here.

Logarithmic differentiation at $z = 1$ records the row's first signed moment:

$$\sum_{j=0}^{n} j\omega_{n,j} = W_n'(1) = -\sum_{\ell=1}^{n} \frac{1}{2^\ell - 1}. \tag{411}$$

Further logarithmic differentiation gives every higher factorial row moment. The coefficients also have the finite symmetric form

$$\omega_{n,j} = \frac{(-1)^j 2^{-\binom{j+1}{2}}}{(1/2;1/2)_j (1/2;1/2)_{n-j}}. \tag{412}$$

For each fixed j this gives the limit

$$\omega_{\infty,j} = \frac{(-1)^j 2^{-\binom{j+1}{2}}}{(1/2;1/2)_j (1/2;1/2)_\infty}, \tag{413}$$

with stationary generating filter

$$W_\infty(z) = \frac{(z/2;1/2)_\infty}{(1/2;1/2)_\infty}. \tag{414}$$

It has unit mass and finite total variation. This limiting filter is a useful diagnostic, although the finite roots in (408) are what produce exact q-Richardson cancellation.

The familiar row-sum identity and bounded variation are therefore only two evaluations of a single polynomial. The roots $2, 4, \ldots, 2^n$ contain additional information that the basic Toeplitz proof does not need.

23.4 A q-Pochhammer filter in the degree variable

Fix x and Q and regard $p \mapsto S_p^{(Q)}(x)$ as a sequence. Let the backward shift L act by $\mathsf{L}S_p^{(Q)} = S_{p-1}^{(Q)}$. Then (403) and (406) can be written as

$$\boxed{\mathcal{D}_n^{(Q)}(x) = W_n(\mathsf{L}) S_{2n}^{(Q)}(x) = \prod_{\ell=1}^{n} \frac{1 - 2^{-\ell}\mathsf{L}}{1 - 2^{-\ell}} S_{2n}^{(Q)}(x).} \tag{415}$$

Each factor preserves a constant sequence because its value at $\mathsf{L} = 1$ is 1. Meanwhile the zero $W_n(2^r) = 0$ means that it removes a geometric error mode with ratio 2^{-r} in the degree.

Proposition 23.4 (q-Richardson exactness). *Suppose a rational-valued model sequence has the form*

$$u_p = L + \sum_{r=1}^{n} a_r 2^{-rp}. \tag{416}$$

Then

$$\sum_{j=0}^{n} \omega_{n,j} u_{2n-j} = L. \tag{417}$$

Proof. The constant contributes $LW_n(1) = L$. The rth error contributes

$$a_r 2^{-2nr} \sum_{j=0}^{n} \omega_{n,j} 2^{rj} = a_r 2^{-2nr} W_n(2^r) = 0.$$

□

Current formal status. The module-valued theorem `discreteLimitWeight_qRichardson_exact_module` in `FabiusDiscreteLimitToeplitz.lean` holds for sequences in any additive commutative monoid carrying a $\mathbb{Q}$-module structure. It needs no target multiplication, topology, order, or norm: the rational row weights and dyadic mode coefficients act only by scalar multiplication. Its single-mode core is `sum_range_discreteLimitWeight_mul_geometricMode_smul_eq_zero`. The theorem `discreteLimitWeight_qRichardson_exact` is the $M = \mathbb{Q}$ specialization displayed above; its zero-based coefficient a_r multiplies the mode $2^{-(r+1)p}$. These theorems transport cancellation at the rational dyadic nodes; they do not classify zeros at arbitrary scalar arguments.

No claim is needed that the actual spline error has a finite expansion of the form (416). The proposition explains the algebraic design of the row: it is a geometric-grid analogue of Richardson extrapolation, with the normalized q-Pochhammer polynomial as its error-annihilating filter.

23.5 A quantitative 4^{-n} convergence bound

The bounded-variation proof of (400) uses only that all sampled degrees $2n-j$ tend to infinity. Retaining their actual sizes gives a stronger quantitative consequence.

Theorem 23.5 (Explicit discrete-row error; frontier sharpening). *For $n \geq 1$, $Q \in \mathbb{C}$, and every $x \in \mathbb{R}$, let*

$$H_n = \frac{(-1; 1/2)_n}{(1/2; 1/2)_n}. \tag{418}$$

Then

$$\boxed{|\mathcal{D}_n^{(Q)}(x) - \mathcal{F}(x)| \leq H_n \big(2\mathrm{e}^{|Q-1/2|} - 1\big) 4^{-n}.} \tag{419}$$

Moreover $H_n \leq 64$, so the convergence is uniformly $\mathcal{O}_Q(4^{-n})$ on the whole real line. At the centered shift,

$$|\mathcal{D}_n^{(1/2)}(x) - \mathcal{F}(x)| \leq H_n 4^{-n}. \tag{420}$$

Proof. Apply (398) with $p = 2n - j$ in (403):

$$\begin{aligned} |\mathcal{D}_n^{(Q)}(x) - \mathcal{F}(x)| &\leq \big(2\mathrm{e}^{|Q-1/2|} - 1\big) \sum_{j=0}^{n} |\omega_{n,j}| 2^{-(2n-j)} \\ &= \big(2\mathrm{e}^{|Q-1/2|} - 1\big) 4^{-n} \sum_{j=0}^{n} |\omega_{n,j}| 2^{j}. \end{aligned}$$

Use (410). Finally

$$H_n = \frac{2}{1-2^{-n}} \frac{(-1/2;1/2)_{n-1}}{(1/2;1/2)_{n-1}} \le 4 \cdot 16 = 64,$$

where the last factor is the uniform variation bound proved in the formal development. □

The same argument isolates the finite translation dependence.

Corollary 23.6 (Decay of the shift dependence)**.** *For every $x \in \mathbb{R}$ and $n \ge 1$,*

$$|\mathcal{D}_n^{(Q)}(x) - \mathcal{D}_n^{(1/2)}(x)| \le 2H_n\big(\mathrm{e}^{|Q-1/2|} - 1\big)4^{-n}. \tag{421}$$

Thus the finite rows can depend dramatically on Q at small n, but the dependence is uniformly suppressed at a geometric rate. This is analytically different from theorem 21.5, where the finite dyadic expression is exactly constant in Q.

23.6 A synchronized nearest-dyadic q-limit

There is a simpler way to turn the exact dyadic formula into a limit at arbitrary argument. For $x \ge 0$ define the nearest nonnegative dyadic numerator

$$M_n(x) = \max\left(0, \left\lfloor 2^n x + \frac{1}{2} \right\rfloor\right) = N_n(x), \qquad \widetilde{x}_n = \frac{M_n(x)}{2^n}. \tag{422}$$

Use this one numerator at every inner scale:

$$\widetilde{\mathcal{D}}_n^{(Q)}(x) = \frac{1}{2^{n^2}(1/2;1/2)_n} \sum_{k=0}^{n} \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(n+k)!} \times \sum_{r=0}^{M_n(x)2^k-1} \varepsilon_r (r - M_n(x)2^k + Q)^{n+k}. \tag{423}$$

Unlike (399), every finite row is now an exact dyadic formula.

Theorem 23.7 (Synchronized exact-grid representation)**.** *For every n, $Q \in \mathbb{C}$, and $x \ge 0$,*

$$\boxed{\widetilde{\mathcal{D}}_n^{(Q)}(x) = \mathcal{F}\left(\frac{M_n(x)}{2^n}\right).} \tag{424}$$

In particular, the left side is exactly independent of Q. On $0 \le x \le 1$,

$$\left|\widetilde{\mathcal{D}}_n^{(Q)}(x) - F(x)\right| \le 2^{-n}, \tag{425}$$

and hence

$$F(x) = \lim_{n\to\infty} \widetilde{\mathcal{D}}_n^{(Q)}(x). \tag{426}$$

Proof. Equation (424) is (299) with numerator $M_n(x)$. For $x \in [0,1]$, $|\widetilde{x}_n - x| \le 2^{-n-1}$ and the bounded Fabius function is 2-Lipschitz, giving (425). □

The same construction applied to (298) gives a purely moment/Thue–Morse limit:

$$F(x) = \lim_{n\to\infty} \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{M_n(x)-1} \varepsilon_h \times \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k (2M_n(x) - 2h - 1)^{n-2k}, \tag{427}$$

again with error at most 2^{-n} on $[0,1]$. This representation contains no q-symbols at all: it is exact dyadic arithmetic followed by nearest-grid convergence.

The synchronized limit is conceptually simple but converges at the basic grid rate $\mathcal{O}(2^{-n})$. The asynchronous row (399) is adapted to the natural spline degree $n+k$ at each scale and, by theorem 23.5, achieves the stronger uniform bound $\mathcal{O}_Q(4^{-n})$. The normalized q-Pochhammer row acts as an acceleration filter.

23.7 What fails away from a dyadic point

Let

$$\theta_n(x) = 2^n x - M_n(x), \qquad -\frac{1}{2} \le \theta_n(x) < \frac{1}{2}. \tag{428}$$

Then for every $k \ge 0$,

$$N_{n+k}(x) = M_n(x)2^k + \left\lfloor 2^k\theta_n(x) + \frac{1}{2} \right\rfloor, \tag{429}$$

and the shifted coordinate in the inner power is

$$r - 2^{n+k}x + Q = r - M_n(x)2^k + \big(Q - 2^k\theta_n(x)\big). \tag{430}$$

Thus the source discrete row differs from the synchronized exact dyadic row in exactly two ways:

1. the complete block of length $M_n 2^k$ acquires a boundary defect $\lfloor 2^k\theta_n + 1/2 \rfloor$;
2. the common translation Q is replaced by the scale-dependent translation $Q - 2^k\theta_n$.

At a resolved dyadic point, $\theta_n = 0$, so both defects disappear simultaneously; this is another proof of theorem 23.1. Away from the dyadic grid the shifts vary with k, so the finite common-translation invariance of theorem 21.5 cannot be applied across the row.

23.8 Two infinite constructions that share a limit but not terms

The global binary q-series (393) and the discrete limit (400) both evaluate $\mathcal{F}$, but their relation is not that of an infinite series and its partial sums.

Feature	Binary q-series	Discrete q-limit
Outer index	binary scale m	approximation row n
Finite inner object	$\mathcal{Q}_m^{(Q)} = \mathcal{T}_m$	shifted splines $S_{2n-j}^{(Q)}$
Shift dependence	absent in every summand	generally present in every finite row
Convergence source	exact residual telescope	Toeplitz summability of spline convergence
At dyadic x	outer series terminates at the last set bit	rows stabilize exactly once n reaches the denominator depth
Natural error bound	2^{1-N} after scale N	$H_n(2e^{\lvert Q-1/2\rvert} - 1)4^{-n}$

Table 10: The two arbitrary-argument q constructions.

Warning 23.8. It is therefore incorrect to call $\mathcal{D}_n^{(Q)}$ the nth partial sum of the global q-binomial series. The two families use the same finite half-base identities, but they insert them into different analytic frameworks.

23.9 The first rows

For reference, the first normalized Toeplitz rows are

$$\begin{aligned}
(\omega_{1,j})_{j=0}^{1} &= (2, -1),\\
(\omega_{2,j})_{j=0}^{2} &= (8/3, -2, 1/3),\\
(\omega_{3,j})_{j=0}^{3} &= (64/21, -8/3, 2/3, -1/21),\\
(\omega_{4,j})_{j=0}^{4} &= (1024/315, -64/21, 8/9, -2/21, 1/315).
\end{aligned}$$

Their ordinary and 2^j-weighted variations are shown in table 11.

n	$\sum_j \lvert\omega_{n,j}\rvert$	$H_n = \sum_j \lvert\omega_{n,j}\rvert 2^j$
1	3	4
2	5	8
3	45/7	80/7
4	51/7	96/7
5	1683/217	3264/217

Table 11: Two evaluations of the row polynomial: $W_n(-1)$ and $W_n(-2)$.

24 A unified dyadic identity and a worked example

24.1 The complete equality chain

Let $x = m/2^s \in [0, 1]$, use the terminating binary digits $x = 0.\delta_1 \cdots \delta_s$, and let $N \geq s$. Combining the preceding results gives the following chain of exact finite expressions:

$$\begin{aligned}
F(x) &= 2^{-\binom{s}{2}}[X^s]\big(B_m(X)\mathscr{A}(X)\big) && (431)\\
&= \frac{2^{-\binom{s+1}{2}}}{s!}\sum_{h=0}^{m-1}\varepsilon_h\sum_{k=0}^{\lfloor s/2\rfloor}\binom{s}{2k}c_k(2m-2h-1)^{s-2k} && (432)\\
&= \frac{1}{2^{s^2}(1/2;1/2)_s}\sum_{k=0}^{s}\frac{\left[{s \atop k}\right]_{1/2}}{4^{\binom{k}{2}}(s+k)!}\sum_{r=0}^{m2^k-1}\varepsilon_r(r-m2^k+Q)^{s+k} && (433)\\
&= \sum_{k=0}^{\lfloor s/2\rfloor}\frac{c_k}{(2k)!2^{k(2s-2k+1)}}S_{s-2k}\Big(\frac{m}{2^{s-2k}}\Big) && (434)\\
&= \sum_{\substack{1\leq j\leq s\\ \delta_j=1}}(-1)^{\delta_1+\cdots+\delta_{j-1}}\mathcal{T}_j\left(\sum_{\ell>j}\delta_\ell 2^{-\ell}\right) && (435)\\
&= \sum_{j=0}^{N}\omega_{N,j}S^{(Q)}_{2N-j}(x). && (436)
\end{aligned}$$

The letter k has unrelated local meanings in different lines; this is unavoidable because one line indexes even moments and another indexes geometric scales. The equalities are not termwise transformations. They are different factorizations of the same exact value:

- (431) is the common scalar invariant;
- (432) expands the probability law in centered moments;

- (433) extracts the same coefficient by geometric interpolation;
- (434) corrects one matched spline through even moments;
- (435) telescopes through the set bits;
- (436) extrapolates consecutive spline degrees at the same point.

24.2 The value at $5/16$

Take

$$x = \frac{5}{16} = 0.0101_2, \qquad F(x) = \frac{305857}{2073600} \approx 0.1475004822530864.$$

This one value illustrates every bridge above.

24.2.1 Master coefficient and moment formula

The prefix polynomial is

$$B_5(X) = \mathrm{e}^{4X} - \mathrm{e}^{3X} - \mathrm{e}^{2X} + \mathrm{e}^{X} - 1. \tag{437}$$

Using $d_0 = 1, d_1 = 1/2, d_2 = 5/18, d_3 = 1/6, d_4 = 143/1350$ in $\mathscr{A}$ gives

$$[X^4](B_5\mathscr{A}) = \frac{305857}{32400}. \tag{438}$$

Since $2^{-\binom{4}{2}} = 1/64$, (309) gives (24.2). Equivalently, with $c_1 = 1/9$ and $c_2 = 19/675$,

$$F(5/16) = \frac{2^{-10}}{4!} \sum_{h=0}^{4} \varepsilon_h \left[t_h^4 + 6c_1 t_h^2 + c_2\right], \qquad t_h = 9 - 2h. \tag{439}$$

24.2.2 Moment correction of the matched spline

The three spline values required by (335) are

$$S_4(5/16) = \frac{1205}{8192}, \qquad S_2(5/4) = \frac{15}{16}, \qquad S_0(5) = -1. \tag{440}$$

Hence

$$\begin{aligned} F(5/16) &= S_4(5/16) + \frac{1}{2304} S_2(5/4) + \frac{19}{16588800} S_0(5) \\ &= \frac{1205}{8192} + \frac{5}{12288} - \frac{19}{16588800} = \frac{305857}{2073600}. \end{aligned} \tag{441}$$

The matched degree-four spline is already close, but the exact discrepancy is resolved by two universal lower-degree corrections.

24.2.3 The two set bits

The binary digits equal 1 only at positions 2 and 4. The reduction polynomial at the first position is

$$\mathcal{T}_2(y) = \frac{5}{72} + y + 4y^2, \tag{442}$$

and $\mathcal{T}_4(0) = F(1/16) = 143/2073600$. The signs are $+$ at position 2 and $-$ at position 4, so

$$\begin{aligned} F(5/16) &= \mathcal{T}_2(1/16) - \mathcal{T}_4(0) \\ &= \frac{85}{576} - \frac{143}{2073600} = \frac{305857}{2073600}. \end{aligned} \tag{443}$$

The prefix formula sums five Thue–Morse terms; the binary formula needs only the two set bits.

24.2.4 Exact q-extrapolation of centered splines

At row $N = 4$, (436) reads

$$F(5/16) = \frac{1024}{315} S_8(5/16) - \frac{64}{21} S_7(5/16) + \frac{8}{9} S_6(5/16) \\ - \frac{2}{21} S_5(5/16) + \frac{1}{315} S_4(5/16). \tag{444}$$

The individual spline values are

$$S_4 = \frac{1205}{8192}, \qquad S_5 = \frac{289799}{1966080}, \qquad S_6 = \frac{13917509}{94371840}, \\ S_7 = \frac{74236247}{503316480}, \qquad S_8 = \frac{395939357}{2684354560}, \tag{445}$$

all evaluated at $5/16$. None equals $F(5/16)$, but the q-Pochhammer filter combines them exactly.

24.2.5 Finite shift dependence before stabilization

The first discrete rows illustrate the difference between exact finite translation invariance and asymptotic translation independence.

n	$\mathcal{D}_n^{(0)}(5/16)$	$\mathcal{D}_n^{(1/2)}(5/16)$	$\mathcal{D}_n^{(3)}(5/16)$
1	0.156250000	0.156250000	3.906250000
2	0.151041667	0.147460938	−0.531250000
3	0.147505374	0.147501983	0.125788484
4	0.147500482	0.147500482	0.147500482

Table 12: The row becomes exactly equal to $F(5/16)$ at $n = 4$, independently of Q.

The exact dyadic q-formula itself is independent of Q already at row four, although its outer summands vary substantially. Their decimal contributions are:

scale k	$Q = 0$	$Q = 1/2$	$Q = 3$
0	0.000626240	0.000466967	0.000000000
1	−0.016345021	−0.014038037	−0.005184307
2	0.141657926	0.131089095	0.084488329
3	−0.467452954	−0.449506045	−0.365062120
4	0.489014292	0.479488502	0.433258580
sum	0.147500482	0.147500482	0.147500482

Table 13: Different translated decompositions of the same exact dyadic value.

25 Fourier representation

The exact-arithmetic network has a complementary global integral form. For $Z = 2Y - 1$, independence gives

$$\widehat{\mathrm{up}}(\xi) = \prod_{j=0}^{\infty} \operatorname{sinc}\left(\frac{\pi\xi}{2^j}\right), \qquad \widehat{f}(\xi) = \int_{\mathbb{R}} f(t)\mathrm{e}^{-2\pi\mathrm{i}\xi t}\,\mathrm{d}t. \tag{446}$$

Integrating the Fourier inversion formula from the symmetry center gives, for $0 < x < 1$,

$$\boxed{F(x) = \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \frac{\sin(2\pi\xi(2x-1))}{\xi}\prod_{j=0}^\infty \operatorname{sinc}\left(\frac{\pi\xi}{2^j}\right)\,\mathrm{d}\xi.} \tag{447}$$

The rapid decay of the sinc product makes this a convergent integral. Its rationality at dyadic x is a nonlocal cancellation, in contrast with the manifestly finite Appell and q formulae.

An arbitrary-argument contour representation comes from the same moment product. If

$$L_Y(s) = \mathbb{E}e^{-sY} = \mathscr{A}(-s) = \prod_{j=1}^\infty \frac{1-e^{-s/2^j}}{s/2^j}, \tag{448}$$

then, for $\sigma > 0$ and $0 < x < 1$, the frontier Bromwich formula is

$$\boxed{F(x) = \frac{1}{2\pi\mathrm{i}}\lim_{T\to\infty}\int_{\sigma-\mathrm{i}T}^{\sigma+\mathrm{i}T} \frac{e^{sx}}{s}\prod_{j=1}^\infty \frac{1-e^{-s/2^j}}{s/2^j}\,\mathrm{d}s.} \tag{449}$$

Thus one product has three complementary readings: coefficient extraction gives moments and exact dyadic values, logarithmic expansion gives cumulants and Bell-polynomial formulae, and contour inversion gives $F(x)$ directly. This arbitrary-x representation is distinct from the inverse-dyadic Bromwich-to-coefficient collapse in the next part.

26 Computational and conceptual consequences

26.1 Which representation is best for which task?

The formulae are mathematically equivalent at dyadic inputs, but not computationally interchangeable.

Task	Recommended representation	Reason
Prove rationality or denominator facts	master coefficient or moment formula	all arithmetic is finite and the universal moments are explicit rationals
Evaluate one dyadic with a sparse numerator	binary Taylor reduction	the number of reductions is the number of set bits
Evaluate many numerators at one depth	precompute d_j or c_k, then use prefix recurrences	universal moment data are reused
Explain the q-coefficients	geometric divided difference and residue form	Gaussian coefficients are recognized as interpolation weights, not mysterious special-function decorations
Approximate arbitrary x with a simple certified error	centered spline S_p	direct global error 2^{-p} and one finite Thue–Morse sum
Accelerate arbitrary-argument spline convergence	discrete row $\mathcal{D}_n^{(Q)}$	exact q-Pochhammer filter and uniform $\mathcal{O}_Q(4^{-n})$ bound
Use only exact dyadic arithmetic in a limiting scheme	synchronized row $\widetilde{\mathcal{D}}_n^{(Q)}$ or synchronized moment limit	every finite stage is an exact dyadic value and is translation-independent
Trace dependence on binary digits	digit-sparse series	one nonzero summand per exposed 1-bit

Table 14: Representation selection. The q-formula is structurally illuminating but is usually not the cheapest naive exact evaluator.

The exact q-formula contains inner blocks of length $m2^k$, so a direct implementation can be expensive. Its value is primarily structural: it exposes geometric interpolation, translation invariance, and the extrapolation filter. For exact numerical evaluation, the bit recursion or the master coefficient with precomputed moments is generally preferable.

For an exact dyadic $x = M/2^N$, a sparse-bit evaluator first computes the universal moments $d_0, \dots, d_N$ from (330), evaluates the required Appell rows by Horner's rule or (370), and sums only over the set bits in (383). A straightforward implementation uses $O(N^2)$ rational arithmetic operations; this is an arithmetic-operation count, not a bit-complexity bound. It avoids the exponentially long inner Thue–Morse blocks in the closed q formula.

For a non-dyadic argument, the same digit–Appell series skips zero bits and has the superexponential certificate (384). The centered nested q-series can be poorly conditioned because large powers cancel. The common-translation gauge in (386) may reduce those intermediate magnitudes, although exact rational or ball arithmetic remains preferable; again, the chosen translation must be common within each finite row.

26.2 A compact exact-arithmetic scheme

Equation (330) computes all half moments with rational arithmetic, after which (313) evaluates a dyadic value.

Algorithm 26.1 (Moment-prefix evaluator)**.** Given $m, n \in \mathbb{N}$:

1. Set $d_0 = 1$ and, for $1 \le j \le n$, compute
$$d_j = \frac{1}{2^j - 1} \sum_{r=1}^{j} \binom{j}{r} \frac{d_{j-r}}{r+1}.$$
2. Form
$$V = \sum_{h=0}^{m-1} \varepsilon_h \sum_{j=0}^{n} \binom{n}{j} d_j (m-1-h)^{n-j}.$$
3. Return $2^{-\binom{n}{2}} V/n!$.

The result is $\mathcal{F}(m/2^n)$ and is $F(m/2^n)$ when $0 \le m \le 2^n$.

A literal pseudocode version is included for clarity.

Listing 2: Exact dyadic evaluation from half moments.

```
d[0] := 1
for j := 1,...,n do
    s := 0
    for r := 1,...,j do
        s := s + binomial(j,r) * d[j-r] / (r+1)
    end
    d[j] := s / (2^j - 1)
end

V := 0
for h := 0,...,m-1 do
    inner := 0
    for j := 0,...,n do
        inner := inner + binomial(n,j) * d[j] * (m-1-h)^(n-j)
    end
    V := V + (-1)^(binary_weight(h)) * inner
end
return V / (2^(n*(n-1)/2) * n!)
```

The centered-moment formula can be evaluated similarly. Its advantage is parity: only $\lfloor n/2 \rfloor + 1$ moments occur. The half-moment formula aligns more directly with the binary Taylor polynomials.

26.3 Exact arithmetic checks implied by the connections

The network of representations supplies a strong verification suite. For chosen m, n one can independently check:

1. the master coefficient against the centered-moment formula;
2. the half-base q expression at two or more translations Q;
3. the terminating binary reduction when $m/2^n \in [0, 1]$;
4. the moment-corrected matched-spline identity;
5. the exact q-extrapolation row at any order $N \geq n$;
6. representation independence under $(m, n) \mapsto (2m, n + 1)$.

These checks stress different cancellation patterns. Agreement is therefore more informative than recomputing the same formula in a rearranged order.

The last item is now more than a check: `qBinomialThueMorseDyadicFormula_refine` proves it exactly. The stronger `qBinomialThueMorseDyadicTranslatedFormula_eq_of_rat_eq` allows any two numerator–exponent pairs representing the same rational dyadic value and permits different rational translations on the two sides. The translated refinement and half-shifted variants are proved in `FabiusQBinomialFormula.lean`. These theorem-level invariances do not formalize the separate finite-row stabilization question in section 23.

26.4 Endpoint and convention warnings

Several small conventions are mathematically active.

Warning 26.2 (The scale-zero term). The global binary series begins at $m = 0$. It happens to vanish for $0 \leq x < 1$, but it equals 1 at $x = 1$ and controls the signed behavior on integer blocks. A series beginning at $m = 1$ is valid only on the half-open unit interval.

Warning 26.3 (Integer exponents). The exponent $\binom{j+1}{2} - \binom{m-j}{2}$ in (368) is an integer and may be negative. Replacing integer subtraction by truncated natural subtraction changes the polynomial.

Warning 26.4 (Centered powers are signed). Expressions such as $(r - m2^k + Q)^{n+k}$ live in a characteristic-zero scalar field. The quantity $r - m2^k$ is deliberately negative on much of the block. Truncated subtraction in $\mathbb{N}$ destroys the formula.

Warning 26.5 (Binary expansions at dyadics). The floor prefixes $p_m = \lfloor 2^m x \rfloor$ select the terminating expansion at a dyadic rational. Replacing it by the alternative expansion ending in infinitely many 1-digits changes the termwise digit series even though the represented real number is unchanged.

27 Relation to the Lean development

The principal source theorems are formalized in the following modules. Filenames are relative to `Analysis/FabiusFunction`.

Module	Role in the present article
`DyadicClosedForm.lean`	Exact Thue–Morse/even-moment formula for arbitrary dyadic numerators.

Module	Role in the present article
`HalfQBinomial.lean`	Finite q-Pochhammer and Gaussian-binomial calculus at $q = 1/2$, including the interpolation annihilation row.
`ThueMorseExponential.lean`	Complete-block exponential factorization, Prouhet cancellation, and translation of inner powers.
`FabiusQBinomialFormula.lean`	Centered and translated finite q-binomial formulas, arbitrary numerators, common-translation invariance, and invariance under equal rational dyadic representations.
`FabiusQBinomialTaylor.lean`	Identification of the finite q expression with the Taylor reduction polynomial.
`FabiusQBinomialScalarFormula.lean` and `FabiusDyadicQBinomialScalar.lean`	Scalar normalizations, real/complex translated forms, and the constant-polynomial coefficient theorem.
`FabiusRawQBinomialFormula.lean` and `FabiusRawQBinomialScalar.lean`	Raw inverse-dyadic forms and their scalar versions.
`FabiusBinaryReductionSeries.lean`	Exact binary residual telescope, absolute convergence, and uniform tail bound.
`FabiusParityPowerSeries.lean`	Corrected scale-zero parity-power presentation.
`FabiusGlobalQBinomialSeries.lean`	Termwise substitution of finite q-identities into the binary telescope.
`FabiusComplexShiftSpline.lean`	Shifted finite splines, fixed-cutoff Taylor expansion, and uniform shift bound.
`FabiusDiscreteLimitToeplitz.lean`	Row mass, exact total variation, uniform variation bound, and finite-row Toeplitz theorem.
`FabiusDiscreteLimitIntegration.lean`	Reindexing of the discrete formula and convergence to the signed global function.
`TaylorReduction.lean`	Binary Taylor-reduction polynomials, their exact analytic identity, and finite termination on dyadic inputs.
`FabiusUniformSpline.lean`	Centered finite-power spline approximants and the uniform analytic error estimate used by the discrete rows.

Except where explicitly marked below as frontier-dependent, the following statements are direct paper-level corollaries or reorganizations of those formalized results:

- the expectation and two contour representations (310), (312), and (354);
- the composition and Bell-polynomial formulae for d_n;
- the moment-corrected spline transform and its reciprocal (335)–(340);
- exact finite stabilization of the discrete row at dyadic arguments;
- the row polynomial W_n, the shift-operator form, and the q-Richardson interpretation;
- the explicit $\mathcal{O}_Q(4^{-n})$ bound (419), which remains a frontier sharpening because it depends on the unformalized shifted-spline estimate (397), whose proof obligation and edge-case warning are recorded in the Primary Exposition Gap Register;
- the synchronized nearest-dyadic q- and moment-limits.

Apart from that frontier sharpening, the listed statements require no new unproved property of the Fabius function; each direct corollary follows by finite algebra, coefficient extraction, or the already formalized uniform spline estimates. The direct corollaries are marked here to distinguish conceptual reorganization from theorem names that already occur verbatim in the repository.

28 Conclusion

The exact dyadic, global-series, and discrete-limit formulae are best understood not as a collection of special tricks but as a small number of reusable operators acting in different variables.

The product $B_m\mathscr{A}$ separates numerator arithmetic from the universal probability law. Expanding its coefficient in centered moments gives the classical Thue–Morse formula; reading it as an expectation gives a probabilistic representation; taking a Cauchy integral gives an analytic one. Introducing the geometric node z turns the same coefficient into the leading term of a degree-n polynomial. The Gaussian row at base $1/2$ is then simply the barycentric divided difference on $-1, -2, \dots, -2^n$, and the q-Pochhammer symbol is the node polynomial of that grid.

The centered finite spline sits directly inside the exact arithmetic. Its matched-grid value is the zeroth centered-moment term, and the remaining moments form an exact, invertible defect correction. A second correction uses the normalized q-Pochhammer polynomial in the spline degree: it preserves constants, annihilates geometric error modes, and yields exact recovery at dyadic points. The resulting discrete row is therefore naturally viewed as a q-Richardson extrapolator, not merely as an opaque Toeplitz average.

Finally, the two arbitrary-argument q constructions have distinct logical roles. The global q-series is the binary residual telescope with an alternative finite formula inserted into each summand. The discrete limit is a finite extrapolation row whose shift dependence disappears only asymptotically. Their meeting point is the dyadic grid: there the binary series terminates, the discrete row stabilizes, the synchronized row is exact, and all roads return to the same master coefficient.

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Part IV
From Exact Dyadic Formulae to the Small-Argument Asymptotic

Synthesized provenance and status. This part merges the former `Fabius_Dyadic_Asymptotic_Bridge/Fabius_Dyadic_Asymptotic_Bridge.tex` and `Fabius_Dyadic_Formulae_to_Asymptotics/Fabius_Dyadic_Formulae_to_Asymptotics.tex`. The latter supplies the canonical normalization, moment–Laplace route, exact-phase theorem, status ledger, and Lean correspondence. The former contributes the Mellin calculation, vertical coefficient extraction, fixed and moderately growing numerator rays, exact Bromwich-to-coefficient collapse, and continuum-boundary analysis. Repeated recurrence, composition, and q-binomial preliminaries now refer to part III. The vertical and growing-prefix routes remain natural-language formalization obligations.

Former source SHA-256 values: `c5ad7c5298d958ab63f459a3e246c2293d3020525223bc05ef2d0e1edab58f10` and `6e98efd3afc402159a7f080e8604c951d5bc51be4a73383da67c1241d46d54ca`.

Topic abstract

The exact arithmetic of the Fabius function and its endpoint asymptotics are usually presented as two nearly disjoint subjects. At reciprocal powers of two one has rational recurrences, composition sums, Thue–Morse power sums, and a finite half-base q-binomial formula; near the origin one has a negative-Laplace product, a lower-Lambert saddle, and a logarithmically periodic correction. This report supplies the missing bridge.

The central exact observation is that the apparently formidable q-binomial numerator at $F(2^{-n})$ is not a new asymptotic object. After choosing a convenient scalar translation, the inner Thue–Morse block becomes a coefficient of the ratio $A(-2^kX)/A(-X)$, where

$$A(z) = \sum_{n\geq 0} a_n z^n = \prod_{j\geq 1} \frac{\mathrm{e}^{z/2^j} - 1}{z/2^j}, \qquad F(2^{-n}) = 2^{-\binom{n}{2}} a_n.$$

The Gaussian row at $q = 1/2$ is exactly the divided-difference functional on the geometric nodes $-1, -2, \ldots, -2^n$; it annihilates all lower powers and extracts the leading coefficient. Consequently the complete finite numerator collapses identically to $\mathfrak{m}_n a_n$, where $\mathfrak{m}_n = \prod_{j=1}^{n}(2^j - 1)$. This "decompression" proves that the q-Pochhammer, q-binomial, Thue–Morse, recurrence, composition, and generating-product formulae all encode the same coefficient sequence.

Two asymptotic derivations are then developed. The fully rigorous real-variable route writes the half moment as $d_n = \mathbb{E}(1-X)^n$, compares it through second order with the negative Laplace transform $A(-n) = \mathbb{E}\mathrm{e}^{-nX}$, and uses the exact quadratic-plus-periodic decomposition of $\log A(-s)$. A complementary coefficient-saddle route applies Cauchy's formula directly to $a_n = [z^n]A(z)$; it reaches the same lower-Lambert coordinate and provides an independent explanation of every cancellation. If $L = \log 2$ and λ_n is the large solution of

$$\lambda_n - \frac{\log \lambda_n}{L} = n, \qquad \lambda_n = -\frac{1}{L} W_{-1}(-L2^{-n}),$$

then the resulting sharp dyadic formula is

$$\log F(2^{-n}) = -\frac{L}{2}\lambda_n^2 + \left(1 + \frac{L}{2}\right)\lambda_n - \frac{1}{2}\log \lambda_n + C_F + \Psi(\lambda_n) + \mathcal{O}(n^{-1}),$$

where Ψ is a smooth, nonconstant, mean-zero, one-periodic function with explicit Gamma–zeta Fourier coefficients. Expanding the Lambert phase yields

$$\log F(2^{-n}) = -\frac{L}{2}n^2 - n\log n + \left(1+\frac{L}{2}\right)n - \frac{\log^2 n}{2L} + C_F - \frac{\log^2 n}{2L^2 n} + \Psi(\lambda_n) + \mathcal{O}(n^{-1}).$$

The exact phase is essential: replacing it by $\log_2 n$ leaves a generally larger $(\log n)/n$ term. Corollaries give equally sharp asymptotics for the recurrence sequence, the half moments, the composition sum, and the literal q-binomial numerator. Numerical checks against exact rational values confirm the constant, phase, and first saddle correction.

29 The missing bridge

The Fabius function admits two descriptions that look, at first sight, almost antagonistic. Its values at dyadic rationals are algebraic and finite: every value is rational; there are terminating recurrences; the Thue–Morse sequence provides finite signed power sums; and a half-base Gaussian-binomial row packages those sums into a compact formula. Its endpoint asymptotic is analytic and infinite: one studies a Laplace product, separates a periodic renormalization of a dyadic harmonic sum, and evaluates a saddle determined by the lower real branch of Lambert's W-function.

The two subjects are not merely compatible. They are two coordinate systems on the same object. The exact formulae contain the asymptotic information in a highly compressed form, and the purpose of this report is to make the decompression explicit.

29.1 What is proved

There are four levels of connection.

1. The reciprocal-power value $F(2^{-n})$ is exactly a half moment divided by a factorial and a triangular power of two. This already gives the quadratic logarithmic decay.

2. After factorial normalization, all reciprocal-power values are coefficients of one entire function A. The recurrence, composition sum, and infinite product are equivalent presentations of A.

3. The finite q-binomial/Thue–Morse formula can be reduced identically to the same coefficient $a_n = [z^n]A(z)$. No limiting argument is involved. The q-binomial row is a geometric divided difference; its role is to undo the blockwise Thue–Morse cancellation.

4. The product for $A(-s)$ has an exact quadratic-plus-periodic logarithm. Transferring from $A(-n)$ to the endpoint moment, or applying a coefficient saddle directly to A, yields the corrected lower-Lambert asymptotic.

The first, second, third, and the full real-variable fourth level are represented in the Lean formalization under `Analysis/FabiusFunction`. The coefficient-saddle calculation in Section 35 is included as an independent conventional-mathematics derivation; it is useful because it starts from the coefficient formula itself and exposes the same cancellations with very little probability language.

29.2 Headline notation and result

Throughout,

$$L = \log 2. \tag{450}$$

Let P be the periodic correction in the exact negative-Laplace decomposition, let

$$\overline{P} = \int_0^1 P(u)\,\mathrm{d}u, \qquad \Psi(u) = P(u) - \overline{P}, \tag{451}$$

and put

$$A_0 = \frac{6\gamma^2 + 12\gamma_1 - \pi^2}{12}, \qquad C_F = \frac{A_0}{L} - \frac{7L}{12} - \frac{1}{2}\log \pi. \tag{452}$$

Here γ is Euler's constant and γ_1 is the first Stieltjes constant. For every sufficiently large integer n, let λ_n be the large positive solution of

$$\lambda_n 2^{-\lambda_n} = 2^{-n}, \qquad \text{equivalently} \qquad \lambda_n - \frac{\log \lambda_n}{L} = n. \tag{453}$$

Theorem 29.1 (Sharp reciprocal-power asymptotic). *As $n \to \infty$,*

$$\boxed{\log F(2^{-n}) = -\frac{L}{2}\lambda_n^2 + \left(1 + \frac{L}{2}\right)\lambda_n - \frac{1}{2}\log \lambda_n + C_F + \Psi(\lambda_n) + \mathcal{O}(n^{-1}).} \tag{454}$$

Equivalently,

$$\begin{aligned} \log F(2^{-n}) = &-\frac{L}{2}n^2 - n\log n + \left(1 + \frac{L}{2}\right) n - \frac{\log^2 n}{2L} + C_F \\ &- \frac{\log^2 n}{2L^2 n} + \Psi(\lambda_n) + \mathcal{O}(n^{-1}). \end{aligned} \tag{455}$$

The function Ψ is smooth, mean-zero, one-periodic, and nonconstant. Its Fourier coefficients are explicit.

The rest of the report derives Theorem 29.1 from the exact arithmetic formulae rather than taking it as an externally supplied endpoint theorem.

The most important conceptual point is that the periodic term is already present in the exact dyadic values. It does not arise because one interpolates between dyadic points. The fractional parts of the exact saddle coordinates λ_n drift through the logarithmic period, so the single rational sequence $F(2^{-n})$ samples the entire periodic correction.

---

30 Exact dyadic inputs from the preceding part

The exact algebra needed here is developed once in part III. In the present notation,

$$d_n = \mathbb{E}X^n = \mathbb{E}(1-X)^n, \qquad a_n = \frac{d_n}{n!}, \qquad A(z) = \sum_{n\geq 0} a_n z^n = \mathscr{A}(z), \tag{456}$$

with

$$F(2^{-n}) = 2^{-\binom{n}{2}} a_n, \tag{457}$$

$$A(z) = \prod_{j\geq 1} \frac{e^{z/2^j} - 1}{z/2^j}, \tag{458}$$

$$A(2z) = \frac{e^z - 1}{z} A(z), \qquad A(z) = e^z A(-z). \tag{459}$$

The first equality is (296); the product and refinement are (302)–(303). The recurrence, ordered-composition formula, Mersenne normalization, and general-numerator master coefficient are respectively (330), (319), (345), and (309).

The normalized endpoint and generating-product identifications are the exact declarations `fabius_inverse_two_pow_eq_recurrenceSequence`, `halfMomentGeneratingSeries_eq_fabiusRecurrenceSequence_series`, `complexGeneratingFunction_eq_fabiusRec` `urrenceSequence_series`, and `fabiusRecurrenceSequence_series_neg_eq_tprod` in `FabiusRecurrenceSequence.lean`.

For later asymptotic bookkeeping, let $\mathfrak{m}_n = \prod_{j=1}^{n}(2^j - 1)$. The literal inverse-power half-base q-numerator is the finite geometric extractor applied to the degree-n coefficient polynomial, and the exact collapse reads

$$\mathcal{N}_n = \mathfrak{m}_n a_n. \tag{460}$$

This is the $m = 1$ specialization of the general formula in section 21; it is not a separate asymptotic input. Thus every recurrence, composition, moment, and literal q consequence below is a corollary of one coefficient sequence rather than a parallel derivation.

31 The first asymptotic already visible in the exact values

Before resolving the periodic fluctuation, the endpoint identity alone gives the principal quadratic law. Taking logarithms in (457),

$$\log F(2^{-n}) = \log d_n - \log(n!) - \binom{n}{2} L. \tag{461}$$

Because $0 \leq \mathrm{up} \leq 1$,

$$d_n \leq n \int_0^1 t^{n-1}\,\mathrm{d}t = 1. \tag{462}$$

A fixed positive portion of the mass on $[0, 1/2]$ gives, for $n \geq 1$,

$$2^{-(n+1)} \leq d_n. \tag{463}$$

Combining these bounds with elementary factorial estimates yields the explicit inequality

$$\left|\log F(2^{-n}) + \frac{L}{2}n^2\right| \leq 3n\log(n+1), \qquad n \geq 1. \tag{464}$$

Therefore

$$\lim_{n\to\infty} \frac{\log F(2^{-n})}{n^2} = -\frac{L}{2}. \tag{465}$$

This is already a derivation of the leading endpoint asymptotic from exact arithmetic. It also explains why neither the factorial nor the half moment can change the coefficient of n^2: $\log(n!) = \mathcal{O}(n \log n)$ and $\log d_n = \mathcal{O}(n)$ under the crude bounds. To recover the $-n\log n$ term, the $\log^2 n$ term, the constant, and the periodic fluctuation, however, one must understand d_n much more sharply.

32 The exact negative-Laplace decomposition

The sharp asymptotic enters through the same function A. For $s > 0$ put

$$B(s) = A(-s) = \mathbb{E}\mathrm{e}^{-sX}, \qquad \Lambda(s) = \log B(s). \tag{466}$$

From (458),

$$B(s) = \prod_{j=1}^{\infty} \frac{1 - \mathrm{e}^{-s/2^j}}{s/2^j}, \tag{467}$$

and hence

$$\Lambda(s) = \sum_{j=1}^{\infty} \kappa(s/2^j), \qquad \kappa(u) = \log \frac{1 - \mathrm{e}^{-u}}{u}. \tag{468}$$

This is a dyadic harmonic sum. Its dilation equation is exact:

$$\Lambda(2s) = \Lambda(s) + \log(1 - \mathrm{e}^{-s}) - \log s. \tag{469}$$

32.1 Periodization of the dilation equation

Define the forward logarithmic tail

$$T(s) = \sum_{m=0}^{\infty} \log(1 - \mathrm{e}^{-s2^m}) < 0, \qquad \mathcal{E}(s) = -T(s) > 0. \tag{470}$$

Since $2^m \geq m + 1$,

$$0 \leq \mathcal{E}(s) \leq \frac{\mathrm{e}^{-s}}{(1 - \mathrm{e}^{-s})^2}, \tag{471}$$

and in particular $\mathcal{E}(s) \leq 4\mathrm{e}^{-s}$ for $s \geq L$. For $u \in \mathbb{R}$, set

$$P(u) = \Lambda(2^u) + \frac{L}{2}(u^2 - u) + T(2^u). \tag{472}$$

Theorem 32.1 (Exact quadratic-plus-periodic decomposition). *The function P is smooth and one-periodic. For every $s > 0$,*

$$\boxed{\Lambda(s) = -\frac{(\log s)^2}{2L} + \frac{1}{2}\log s + P(\log_2 s) + \mathcal{E}(s).} \tag{473}$$

All derivatives of P are bounded.

Proof. The forward tail satisfies

$$T(2s) = T(s) - \log(1 - \mathrm{e}^{-s}).$$

Under $u \mapsto u + 1$, the change in the quadratic term of (472) is

$$\frac{L}{2}\big((u+1)^2 - (u+1) - (u^2 - u)\big) = Lu = \log s.$$

This cancels the changes in (469) and in T, proving $P(u+1) = P(u)$. Rearrangement gives (473). Local uniform convergence of the defining series permits differentiation; periodicity then makes every derivative bounded. □

The drift $-(\log s)^2/(2L)$ is the analytic counterpart of the triangular power $2^{-\binom{n}{2}}$ in the exact endpoint formula. The periodic remainder is the residual freedom in solving the dilation equation on logarithmic scale.

32.2 Mean, Fourier modes, and the asymptotic constant

Let $\overline{P}$ and Ψ be as in (451). The Mellin transform of the kernel makes the normalization explicit. For $-1 < \Re z < 0$,

$$\widetilde{\kappa}(z) = \int_0^\infty \kappa(u) u^{z-1}\,\mathrm{d}u = -\Gamma(z)\zeta(1+z), \tag{474}$$

$$\widetilde{\Lambda}(z) = \frac{2^z}{1 - 2^z}\big(-\Gamma(z)\zeta(1+z)\big). \tag{475}$$

The first identity follows by integration by parts and the subtracted Bose integral. The geometric factor is the Mellin image of the same dyadic scale that produced the Mersenne factors in the exact formulas. Its poles at $2\pi\mathrm{i}k/L$ give the Fourier modes; the finite part at zero gives the mean

$$\overline{P} = \frac{A_0}{L} - \frac{L}{12}, \tag{476}$$

where A_0 is defined in (452). With

$$\chi_k = \frac{2\pi \mathrm{i} k}{L}, \tag{477}$$

the Fourier coefficients of the centered function are

$$\widehat{\Psi}(0) = 0, \qquad \boxed{\widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{L}} \quad (k \neq 0). \tag{478}$$

The Gamma factor has no zeros, and ζ has no zeros on $\Re z = 1$ away from its pole at 1, so every nonzero Fourier coefficient is nonzero. Thus Ψ is genuinely nonconstant. Stirling's vertical estimate for Γ makes the coefficients exponentially small; in fact the oscillation has amplitude only a few parts in 10^6, but it cannot be deleted at an error scale tending to zero.

The Gaussian normalization later contributes $-\frac{1}{2}\log(2\pi)$. Therefore

$$\overline{P} - \frac{1}{2}\log(2\pi) = \frac{A_0}{L} - \frac{7L}{12} - \frac{1}{2}\log\pi = C_F. \tag{479}$$

Numerically,

$$\overline{P} = -1.10904536543418668\ldots, \qquad C_F = -2.02798389863885942\ldots. \tag{480}$$

33 Sharp route I: endpoint moments versus Laplace moments

This route begins with the exact rational object $d_n = \mathbb{E}(1-X)^n$ and converts it into the negative-Laplace product at the natural scale $s = n$. It is the route closest to the current Lean bridge modules.

33.1 Second-order kernel comparison

For $0 \le x \le 1$ and $n \ge 1$, the logarithmic expansion

$$n\log(1-x) = -nx - \frac{n}{2}x^2 + \mathcal{O}(nx^3)$$

is useful only after the large-x region is controlled uniformly. A convenient global form is

$$\left|(1-x)^n - \mathrm{e}^{-nx}\left(1 - \frac{n}{2}x^2\right)\right| \le 16\mathrm{e}^{-nx}(nx^3 + n^2x^4). \tag{481}$$

Integrating against the law of X gives

$$d_n = B(n) - \frac{n}{2}\mathbb{E}\big(X^2\mathrm{e}^{-nX}\big) + R_n, \tag{482}$$

$$|R_n| \le 16\left(n\mathbb{E}(X^3\mathrm{e}^{-nX}) + n^2\mathbb{E}(X^4\mathrm{e}^{-nX})\right). \tag{483}$$

Let $q(s) = \Lambda(s) = \log B(s)$. The normalized tilted second moment is

$$\frac{\mathbb{E}(X^2\mathrm{e}^{-sX})}{B(s)} = q''(s) + q'(s)^2. \tag{484}$$

The corresponding third- and fourth-moment identities are polynomials in $q', q'', q^{(3)}, q^{(4)}$. Differentiation of (473), together with the exponentially small tail, shows that these normalized moments have precisely the inverse powers needed to make (483) a relative $\mathcal{O}(n^{-1})$ error after the second term is retained.

Proposition 33.1 (Endpoint–Laplace logarithmic transfer). *As $n \to \infty$,*

$$\boxed{\log d_n = q(n) - \frac{n}{2}\big(q''(n) + q'(n)^2\big) + \mathcal{O}(n^{-1}).} \tag{485}$$

Proof architecture. Divide (482) by $B(n)$. The second-order displacement is

$$\alpha_n = \frac{n}{2}\big(q''(n) + q'(n)^2\big) = \mathcal{O}(\log^2 n/n),$$

and the normalized remainder is $\mathcal{O}(n^{-1})$. Thus $d_n/B(n) = 1 - \alpha_n + \mathcal{O}(n^{-1})$. The local estimate $\log(1-y) = -y + \mathcal{O}(y^2)$ applies; since $\alpha_n^2 = \mathcal{O}(\log^4 n/n^2) = \mathcal{O}(n^{-1})$, it gives (485). □

33.2 Differentiating the exact periodic decomposition

Write

$$\ell = \log s, \qquad u = \log_2 s = \frac{\ell}{L}. \tag{486}$$

Ignoring an exponentially small differentiated tail, (473) gives

$$q'(s) = \frac{-\ell/L + 1/2 + \Psi'(u)/L}{s} + \mathcal{O}(\mathrm{e}^{-s}), \tag{487}$$

and

$$q''(s) = \frac{\ell/L - 1/2 - 1/L - \Psi'(u)/L + \Psi''(u)/L^2}{s^2} + \mathcal{O}(\mathrm{e}^{-s}). \tag{488}$$

Squaring (487) and adding (488), the terms linear in ℓ that do not involve Ψ' cancel. Since Ψ' and Ψ'' are bounded,

$$\frac{s}{2}\big(q''(s) + q'(s)^2\big) = \frac{\ell^2}{2L^2 s} - \frac{\ell\,\Psi'(u)}{L^2 s} + \mathcal{O}(s^{-1}). \tag{489}$$

This cancellation is the analytic reason that the first nonperiodic endpoint correction is $-\log^2 n/(2L^2 n)$ rather than a mixture of $\log^2 n/n$ and $\log n/n$ terms.

Substituting (473) and (489) into (485) yields the first sharp form.

Theorem 33.2 (Half moments at the logarithmic phase)**.** *Let $u_n = \log_2 n$. Then*

$$\begin{aligned}\log d_n = {} & -\frac{\log^2 n}{2L} + \frac{1}{2}\log n + \overline{P} + \Psi(u_n) \\ & -\frac{\log^2 n}{2L^2 n} + \frac{\log n}{L^2 n}\Psi'(u_n) + \mathcal{O}(n^{-1}).\end{aligned} \tag{490}$$

Combining (490) with Stirling's formula and (461) gives

$$\begin{aligned}\log F(2^{-n}) = {} & -\frac{L}{2}n^2 - n\log n + \left(1 + \frac{L}{2}\right)n - \frac{\log^2 n}{2L} + C_F \\ & -\frac{\log^2 n}{2L^2 n} + \Psi(u_n) + \frac{\log n}{L^2 n}\Psi'(u_n) + \mathcal{O}(n^{-1}).\end{aligned} \tag{491}$$

Formula (491) has already recovered the constant and the periodic function from the exact endpoint values. The derivative term is not an extra oscillation; it is the first-order displacement from the naive phase u_n to the exact Lambert phase.

34 The lower-Lambert phase and the derivative cancellation

34.1 The exact phase

The natural saddle coordinate at $x = 2^{-n}$ is

$$\lambda_n = -\frac{1}{L}W_{-1}(-L2^{-n}), \tag{492}$$

where W_{-1} is the lower real branch of Lambert's function [5]. It is the large solution of (453). Its fixed-point form is

$$\lambda_n = n + \log_2 \lambda_n. \tag{493}$$

The standard lower-Lambert expansion gives

$$\begin{aligned}\lambda_n &= n + \frac{\log n}{L} + \frac{\log n}{L^2 n} + \frac{\log n - \frac{1}{2}\log^2 n}{L^3 n^2} \\ &\quad + \frac{\log n - \frac{3}{2}\log^2 n + \frac{1}{3}\log^3 n}{L^4 n^3} + \mathcal{O}\left(\frac{\log^4 n}{n^4}\right).\end{aligned} \tag{494}$$

For the periodic phase transfer only the first displacement is needed. Define

$$\delta_n = \log_2 \lambda_n - \log_2 n. \tag{495}$$

Then

$$\delta_n = \frac{\log n}{L^2 n} + \mathcal{O}\left(\frac{\log^2 n}{n^2}\right). \tag{496}$$

34.2 Why Ψ is evaluated at λ_n

Since n is an integer, periodicity and (493) imply the exact identity

$$\Psi(\lambda_n) = \Psi(\lambda_n - n) = \Psi(\log_2 \lambda_n). \tag{497}$$

The fixed-point step is the exact declaration `dyadicLambertPhase_fixedPoint` in `FabiusLambertPhase.lean`; the periodic rewrite uses `negativeLaplacePsi_periodic` in `PeriodicMean.lean`. Taylor expansion at $u_n = \log_2 n$, with uniformly bounded second derivative, gives

$$\begin{aligned}\Psi(\lambda_n) &= \Psi(u_n + \delta_n) \\ &= \Psi(u_n) + \frac{\log n}{L^2 n}\Psi'(u_n) + \mathcal{O}(n^{-1}).\end{aligned} \tag{498}$$

Indeed, the error in (496) contributes $\mathcal{O}(\log^2 n/n^2) = \mathcal{O}(n^{-1})$, and the quadratic Taylor remainder contributes the same or less. Substitution of (498) into (491) proves (455).

This calculation explains exactly why the phase may not be replaced by $\log_2 n$ at the claimed precision. One would omit the term

$$\frac{\log n}{L^2 n}\Psi'(\log_2 n), \tag{499}$$

which is generally of order $(\log n)/n$, not $1/n$. The tiny amplitude of Ψ affects numerical visibility, but not the asymptotic order.

Warning 34.1 (Phase precision is amplified). The leading exponent contains $-L\lambda^2/2$. An error $\Delta\lambda$ in the phase changes the exponent by approximately $-L\lambda\Delta\lambda$. Therefore deriving an elementary expansion with remainder $\mathcal{O}(n^{-1})$ requires the Lambert phase to one order beyond a naive term count. Keeping λ_n exact inside Ψ avoids this bookkeeping hazard.

34.3 Compact versus expanded form

Define

$$\mathcal{M}_n = -\frac{L}{2}\lambda_n^2 + \left(1 + \frac{L}{2}\right)\lambda_n - \frac{1}{2}\log \lambda_n + C_F + \Psi(\lambda_n). \tag{500}$$

Substitution of (494) and collection of terms gives

$$\begin{aligned}\mathcal{M}_n = & -\frac{L}{2}n^2 - n\log n + \left(1+\frac{L}{2}\right)n - \frac{\log^2 n}{2L} + C_F \\ & -\frac{\log^2 n}{2L^2 n} + \Psi(\lambda_n) + \mathcal{O}(n^{-1}).\end{aligned} \tag{501}$$

Thus (454) and (455) are equivalent at the stated precision. The compact formula is structurally cleaner and is the one that survives unchanged for arbitrary small real arguments.

35 Sharp route II: a coefficient saddle from the exact generating product

The endpoint–Laplace route starts from $d_n = \mathbb{E}(1-X)^n$. There is a second route beginning directly with the recurrence coefficient $a_n = [z^n]A(z)$. It is particularly well suited to the question posed here because a_n is exactly what remains after the q-binomial formula is decompressed.

35.1 Positive-axis growth of A

The reflection identity (459) and Theorem 32.1 give, for $r > 0$,

$$\begin{aligned}K(r) := \log A(r) &= r + \Lambda(r) \\ &= r - \frac{\log^2 r}{2L} + \frac{1}{2}\log r + P(\log_2 r) + \mathcal{E}(r).\end{aligned} \tag{502}$$

The reference saddle is chosen by ignoring the bounded derivative of the periodic term:

$$r - \frac{\log r}{L} = n. \tag{503}$$

Its large solution is exactly $r = \lambda_n$. The exact coefficient saddle for $rK'(r) = n$ differs from λ_n by only $\mathcal{O}(1)$, because

$$rK'(r) = r - \frac{\log r}{L} + \frac{1}{2} + \frac{1}{L}P'(\log_2 r) + \mathcal{O}(re^{-r}). \tag{504}$$

An $\mathcal{O}(1)$ displacement of a saddle of curvature $\asymp 1/r$ changes the logarithmic coefficient estimate by only $\mathcal{O}(1/r)$.

35.2 Vertical extraction and the normalized saddle mass

The probability representation gives a stronger contour than the circular sketch. For $n \geq 1$ and every $c > 0$,

$$a_n = \frac{1}{2\pi\mathrm{i}}\int_{c-\mathrm{i}\infty}^{c+\mathrm{i}\infty} \frac{A(z)}{z^{n+1}}\,\mathrm{d}z. \tag{505}$$

The integral is absolutely convergent because $|A(c+\mathrm{i}t)| \leq A(c)$ and $|c+\mathrm{i}t|^{-n-1}$ is integrable. Fubini and the inverse-Laplace kernel recover $\mathbb{E}X^n/n! = a_n$, so this is an exact consequence of the moment law rather than an inversion assumption for F.

Set $c = \lambda_n$ and write $z = \lambda_n(1+\mathrm{i}\theta)$. Define

$$\mathcal{C}_n = \sqrt{\frac{\lambda_n}{2\pi}}\int_{-\infty}^{\infty} \frac{A(\lambda_n(1+\mathrm{i}\theta))}{A(\lambda_n)}(1+\mathrm{i}\theta)^{-n-1}\,\mathrm{d}\theta. \tag{506}$$

Then the exact coefficient identity is

$$F(2^{-n}) = \frac{2^{-\binom{n}{2}} A(\lambda_n)\lambda_n^{-n}}{\sqrt{2\pi\lambda_n}}\,\mathcal{C}_n. \tag{507}$$

Substituting the periodic decomposition and using $\log_2 \lambda_n = \lambda_n - n$ gives the exact separation

$$\boxed{\log F(2^{-n}) - \mathcal{M}_n = \mathcal{E}(\lambda_n) + \log \mathcal{C}_n.} \tag{508}$$

Thus all algebraic normalization, the constant, and the periodic phase have been removed before any approximation is made.

35.3 Local phase, domination, and all orders

Put $u = \mathrm{Log}(1 + \mathrm{i}\theta)$. On a fixed central neighborhood, the exponent in (506) satisfies

$$\begin{aligned}\Phi_\lambda^{\mathrm{coef}}(\theta) &:= \log \frac{A(\lambda(1+\mathrm{i}\theta))}{A(\lambda)} - (n+1)u \\ &= \lambda(\mathrm{i}\theta - u) - \frac{u}{2} - \frac{u^2}{2L} + \Psi\Big(\lambda + \frac{u}{L}\Big) - \Psi(\lambda) + \mathcal{E}_\lambda^{\mathrm{coef}}(\theta),\end{aligned} \tag{509}$$

where every fixed derivative of the last term is $O(\mathrm{e}^{-\lambda})$. This is the same scaled local phase as in the continuous Bromwich analysis. Globally,

$$\left|\frac{A(\lambda(1+\mathrm{i}\theta))}{A(\lambda)}(1+\mathrm{i}\theta)^{-n-1}\right| \le (1+\theta^2)^{-(n+1)/2}, \tag{510}$$

so the vertical contour supplies its own Gaussian central bound and integrable tails.

With $\theta = v/\sqrt{\lambda}$, the phase begins with $-v^2/2$; the $\lambda^{-1/2}$ contribution is odd and integrates to zero. Gaussian contraction gives

$$\log \mathcal{C}_n = \frac{A_1(\lambda_n)}{\lambda_n} + O(\lambda_n^{-2}), \tag{511}$$

where A_1 is displayed in (534). More generally, expanding (509), exponentiating, and integrating its even Gaussian moments yields, for every fixed N,

$$\log \mathcal{C}_n = \sum_{j=1}^{N-1} \frac{A_j(\lambda_n)}{\lambda_n^j} + O_N(\lambda_n^{-N}). \tag{512}$$

The coefficient functions are the same periodic differential polynomials as in the continuous saddle because the local phases agree. In particular,

$$\log F(2^{-n}) = -\frac{L}{2}\lambda_n^2 + \left(1 + \frac{L}{2}\right)\lambda_n - \frac{1}{2}\log\lambda_n + C_F + \Psi(\lambda_n) + O(n^{-1}), \tag{513}$$

which is (454).

The coefficient saddle provides a particularly direct answer to the original question. Starting from the exact q-binomial value, (460) isolates a_n; coefficient extraction of that coefficient from the exact product produces the lower-Lambert asymptotic. The endpoint–Laplace route and the coefficient route are not the same proof in disguise: one compares $(1-X)^n$ with e^{-nX} on the real line, while the other analyzes a complex coefficient integral. Their agreement fixes the normalization and phase very rigidly.

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36 Fixed and moderately growing dyadic numerators

The master coefficient extends Route II from reciprocal powers to rays $x = m2^{-n}$. Factor

$$B_m(z) = \mathrm{e}^{(m-1)z}\mathcal{S}_m(z), \qquad \mathcal{S}_m(z) = \sum_{h=0}^{m-1} \varepsilon_h \mathrm{e}^{-hz}. \tag{514}$$

For $\Re z = r > 0$, $|\mathcal{S}_m(z) - 1| \le \mathrm{e}^{-r}/(1-\mathrm{e}^{-r})$; thus a fixed prefix is dominated uniformly on vertical lines by its first exponential.

Let $m \ge 1$ be fixed, put $x = m2^{-n}$, let $\lambda = \lambda(x)$ satisfy $x = \lambda 2^{-\lambda}$, and set $r = \lambda/m$. Then

$$n = mr - \log_2 r, \qquad \log_2 r = \lambda - n, \qquad \Psi(\log_2 r) = \Psi(\lambda). \tag{515}$$

Define

$$\mathcal{M}(x) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2}\log\lambda + C_F + \Psi(\lambda). \tag{516}$$

For fixed m, the estimate above gives $\mathcal{S}_m(r) > 0$ once n is sufficiently large, so all real logarithms below are defined. Vertical extraction of $[z^n](B_m A)$ at $z = r(1+\mathrm{i}\theta)$ gives the normalized mass

$$\mathcal{C}_{m,n} = \sqrt{\frac{\lambda}{2\pi}} \int_{-\infty}^{\infty} \frac{B_m(r(1+\mathrm{i}\theta))A(r(1+\mathrm{i}\theta))}{B_m(r)A(r)} (1+\mathrm{i}\theta)^{-n-1}\,\mathrm{d}\theta. \tag{517}$$

The coefficient identity identifies $\mathcal{C}_{m,n}$ with a positive normalization of $F(m2^{-n})$, so it is positive in the same eventual range. Taking real logarithms then gives the exact separation

$$\boxed{\log F(m2^{-n}) - \mathcal{M}(m2^{-n}) = \mathcal{E}(r) + \log\mathcal{S}_m(r) + \log\mathcal{C}_{m,n}.} \tag{518}$$

On the central window, the prefix estimate reduces the exponent to (509) with an error whose fixed derivatives are $O_m(\mathrm{e}^{-\lambda/m})$. The same vertical-tail bound therefore yields

$$\log\mathcal{C}_{m,n} = \sum_{j=1}^{N-1} \frac{A_j(\lambda)}{\lambda^j} + O_{m,N}(\lambda^{-N}). \tag{519}$$

Hence the full coefficient expansion uses the same periodic A_j on every fixed-numerator ray.

The proof remains uniform for a growing numerator $m = m(n)$ whenever

$$\frac{\lambda/m}{\log\lambda} \longrightarrow \infty, \qquad \text{equivalently } m = o(\lambda/\log\lambda). \tag{520}$$

Indeed, each fixed family of prefix derivatives is $O_N(r^{K_N}\mathrm{e}^{-r})$, smaller than every required power of $1/\lambda$ in this regime. This does not reach the large numerators needed for sharp interpolation of every real argument; that two-parameter boundary is described below.

37 Exact Bromwich-to-coefficient collapse

The continuous endpoint integral and the coefficient integral are connected before any asymptotic approximation. Iterating

$$A(-2z) = \frac{1-\mathrm{e}^{-z}}{z} A(-z) \tag{521}$$

gives

$$A(-2^n z) = 2^{-\binom{n}{2}} z^{-n} A(-z) \prod_{j=0}^{n-1} (1 - \mathrm{e}^{-2^j z}). \tag{522}$$

The finite product is the complete Thue–Morse block

$$P_n(z) = \prod_{j=0}^{n-1}(1 - \mathrm{e}^{-2^j z}) = \sum_{h=0}^{2^n-1} \varepsilon_h \mathrm{e}^{-hz}. \tag{523}$$

Starting from the Laplace–Stieltjes inversion formula for F, putting $x = 2^{-n}$, substituting $s = 2^n z$, and using $A(z) = \mathrm{e}^z A(-z)$ yields

$$F(2^{-n}) = \frac{2^{-\binom{n}{2}}}{2\pi\mathrm{i}} \int \frac{A(z)P_n(z)}{z^{n+1}}\,\mathrm{d}z. \tag{524}$$

With $A(z) = \mathbb{E}\mathrm{e}^{zX}$, inverse Laplace transforms this exactly into

$$F(2^{-n}) = 2^{-\binom{n}{2}}\mathbb{E}\left[\sum_{h=0}^{2^n-1} \varepsilon_h \frac{(X-h)_+^n}{n!}\right]. \tag{525}$$

Because $0 \le X \le 1$, every $h \ge 1$ term vanishes almost surely and the remaining term is $2^{-\binom{n}{2}} a_n$. On the saddle line $\Re z = \lambda_n$ one also has $P_n(z) = 1 + O(\mathrm{e}^{-\lambda_n})$, while the continuous radius $2^n\lambda_n = 2^{\lambda_n}$ maps exactly to the coefficient radius λ_n. Thus the rescaled continuous integrand and the coefficient integrand differ only by a beyond-all-orders finite block.

38 Consequences for every exact arithmetic representation

38.1 The normalized recurrence and composition sum

Since $a_n = 2^{\binom{n}{2}} F(2^{-n})$, (455) immediately gives

$$\begin{aligned}\log a_n = {} & -n\log n + n - \frac{\log^2 n}{2L} + C_F \\ & - \frac{\log^2 n}{2L^2 n} + \Psi(\lambda_n) + \mathcal{O}(n^{-1}).\end{aligned} \tag{526}$$

Thus the exact composition sum (319) has the asymptotic

$$\begin{aligned}&\sum_{\rho\vDash n}\prod_{j=1}^{\ell(\rho)} \frac{1}{(2^{s_j}-1)(r_j+1)!} \\ &= \exp\left(-n\log n + n - \frac{\log^2 n}{2L} + C_F - \frac{\log^2 n}{2L^2 n} + \Psi(\lambda_n) + \mathcal{O}(n^{-1})\right).\end{aligned} \tag{527}$$

The recurrence therefore has a moving periodic multiplier; no ansatz with a single fixed constant can be sharp.

38.2 The half moments

Adding Stirling's expansion to (526),

$$\begin{aligned}\log d_n = {} & \frac{1}{2}\log(2\pi n) + C_F + \Psi(\lambda_n) - \frac{\log^2 n}{2L} \\ & - \frac{\log^2 n}{2L^2 n} + \mathcal{O}(n^{-1}).\end{aligned} \tag{528}$$

In particular,

$$d_n \sim \sqrt{2\pi n}\,\exp\left(C_F + \Psi(\lambda_n) - \frac{\log^2 n}{2L}\right). \tag{529}$$

The notation "$\sim$" is legitimate because the omitted logarithmic error tends to zero. The periodic factor $\exp\Psi(\lambda_n)$ remains part of the equivalent.

38.3 The literal q-binomial numerator

By (460),

$$\mathcal{N}_n = \mathfrak{m}_n a_n. \tag{530}$$

Furthermore,

$$\mathfrak{m}_n = 2^{\binom{n+1}{2}}(1/2;1/2)_n \tag{531}$$

and

$$\log(1/2;1/2)_n = \log(1/2;1/2)_\infty + \mathcal{O}(2^{-n}). \tag{532}$$

Consequently

$$\begin{aligned}\log \mathcal{N}_n = \frac{L}{2}n^2 - n\log n + \left(1+\frac{L}{2}\right)n - \frac{\log^2 n}{2L} \\ + C_F + \log(1/2;1/2)_\infty - \frac{\log^2 n}{2L^2 n} + \Psi(\lambda_n) + \mathcal{O}(n^{-1}).\end{aligned} \tag{533}$$

This formula is the sharp asymptotic of the finite signed q-binomial/Thue–Morse sum itself. It makes the scale cancellation visible: the numerator grows like $\exp(Ln^2/2)$, while the explicit denominator in (299) contains 2^{n^2}. The residual triangular factor is exactly the decay $2^{-\binom{n}{2}}$ in (457).

38.4 A first all-orders corollary

The general small-argument saddle expansion supplies a first correction

$$A_1(u) = \frac{1}{24} + \frac{1}{2L} - \frac{\Psi''(u) + \Psi'(u)^2}{2L^2}. \tag{534}$$

At reciprocal powers this gives

$$\log F(2^{-n}) = \mathcal{M}_n + \frac{A_1(\lambda_n)}{\lambda_n} + \mathcal{O}(\lambda_n^{-2}). \tag{535}$$

The same correction can be transported to a_n, d_n, the composition sum, and $\mathcal{N}_n$ by the exact multiplicative normalizations above. Higher orders have the same structure: a Poincaré series in inverse powers of λ_n with smooth periodic coefficient functions evaluated at the exact phase.

39 What each exact formula contributes asymptotically

It is useful to separate exact equivalence from asymptotic convenience. The formulae compute the same numbers, but they expose different structures.

Exact representation	What it exposes	Best asymptotic use
Endpoint identity $F(2^{-n}) = d_n/(n!2^{\binom{n}{2}})$	The triangular dyadic factor, factorial drift, and one unknown half moment	Immediately proves the coefficient $-L/2$ of n^2; becomes sharp after d_n is compared with $A(-n)$
Recurrence for a_n	A positive triangular coefficient recursion	Reassemble into $A(2z) = E(z)A(z)$; direct iteration term by term obscures the moving saddle
Composition sum	A positive sum over all ordered descents of the recurrence	Identifies a_n exactly; asymptotics are best obtained after resummation into A, not by selecting a few compositions

Exact representation	What it exposes	Best asymptotic use
Entire product $A(z) = \prod E(z/2^j)$	The dyadic scale at every analytic level	Primary source for both the negative-Laplace harmonic sum and the coefficient saddle
Thue–Morse block	A zero of exact order k and a finite product over dyadic scales	Converts the inner signed power sum into a coefficient of $A(-2^k X)/A(-X)$
Half-base q-binomial row	A divided difference on $-1, -2, \dots, -2^n$	Extracts the leading coefficient a_n after the block cancellation; termwise asymptotics are badly conditioned
q-Pochhammer normalization	The Mersenne product $\mathfrak{m}_n$ in normalized form	Contributes the elementary limit $(1/2; 1/2)_n \to (1/2; 1/2)_\infty$ and cancels $\mathfrak{m}_n$ exactly
Denominator and valuation formulae	Prime-adic arithmetic of the exact rationals	Strong arithmetic information, but by themselves insufficient to determine Archimedean magnitude or the periodic saddle phase

39.1 Why direct termwise analysis of the q-sum is misleading

The formula

$$F(2^{-n}) = \frac{\mathcal{N}_n}{2^{n^2}(1/2; 1/2)_n}$$

contains an explicit logarithmic contribution $-Ln^2$, whereas the final leading term is only $-Ln^2/2$. The missing $+Ln^2/2$ is generated inside the alternating finite numerator. There is no contradiction: the outer Gaussian-binomial row is designed to annihilate all lower scale powers exactly, and the surviving leading coefficient is exponentially larger than a naive bounded-summand estimate would suggest.

The cancellation can be quantified without inspecting any individual summand. From (533),

$$\log \mathcal{N}_n = \frac{L}{2} n^2 + \mathcal{O}(n \log n), \tag{536}$$

while

$$-\log\bigl(2^{n^2}(1/2; 1/2)_n\bigr) = -Ln^2 - \log(1/2; 1/2)_\infty + o(1). \tag{537}$$

Their sum is $-Ln^2/2 + \mathcal{O}(n \log n)$, as required.

Remark 39.1 (Numerical conditioning). A floating-point implementation of the literal signed formula is normally inferior to the positive recurrence. It must resolve the two exact annihilations before dividing by a factor of size 2^{n^2}. The decompressed identity $\mathcal{N}_n = \mathfrak{m}_n a_n$ is therefore not only a proof device; it is the stable evaluation strategy.

40 How much of the full small-argument asymptotic is encoded by inverse dyadics?

40.1 The arbitrary-argument formula

For $x \downarrow 0$, let

$$\lambda(x) = -\frac{1}{L} W_{-1}(-Lx), \qquad \lambda(x) 2^{-\lambda(x)} = x. \tag{538}$$

The full endpoint saddle has the same compact main term:

$$\log F(x) = -\frac{L}{2}\lambda(x)^2 + \left(1+\frac{L}{2}\right)\lambda(x) - \frac{1}{2}\log\lambda(x) + C_F + \Psi(\lambda(x)) + \mathcal{O}(\lambda(x)^{-1}). \tag{539}$$

At $x = 2^{-n}$ this becomes (454). Thus the exact dyadic arithmetic recovers the restriction of the entire arbitrary-argument asymptotic, including the same constant and the same periodic function at the same exact phase.

40.2 Dense sampling of the periodic correction

The sequence of inverse dyadics does not freeze the periodic phase. Put

$$y_n = \lambda_n - n = \log_2 \lambda_n. \tag{540}$$

Then y_n is increasing and unbounded, while

$$y_{n+1} - y_n \longrightarrow 0. \tag{541}$$

Indeed, $\lambda_n \sim n$ and $y_{n+1} - y_n = \log_2(\lambda_{n+1}/\lambda_n) = \mathcal{O}(n^{-1})$.

Lemma 40.1 (Density of the dyadic Lambert phases)**.** *The fractional parts* $\{\lambda_n\} = \{y_n\}$ *are dense in* $[0, 1]$.

Proof. Fix $\theta \in [0, 1]$. For every sufficiently large integer m, let $n(m)$ be the first index for which $y_{n(m)} \geq m + \theta$. Then

$$0 \leq y_{n(m)} - (m+\theta) < y_{n(m)} - y_{n(m)-1}.$$

The right side tends to zero by (541). Hence the fractional parts of $y_{n(m)}$ tend to θ. □

After removing the nonperiodic terms in (455), the exact rational sequence therefore determines Ψ uniquely. More precisely, if

$$\begin{aligned} R_n = {} & \log F(2^{-n}) + \frac{L}{2}n^2 + n\log n - \left(1+\frac{L}{2}\right)n + \frac{\log^2 n}{2L} \\ & - C_F + \frac{\log^2 n}{2L^2 n}, \end{aligned} \tag{542}$$

then

$$R_n = \Psi(\lambda_n) + \mathcal{O}(n^{-1}). \tag{543}$$

For any $u \in [0, 1]$, choose a subsequence with $\{\lambda_{n_j}\} \to u$; continuity gives $R_{n_j} \to \Psi(u)$. In this precise sense the inverse-power values encode the whole periodic correction, not just a single phase sample.

40.3 What inverse powers alone do not prove

The inverse-power sequence does recover the universal leading quadratic law. If $x = 2^{-t}$ and $n \leq t < n + 1$, monotonicity gives $F(2^{-(n+1)}) \leq F(x) \leq F(2^{-n})$; dividing the coarse endpoint estimates by t^2 yields

$$\log F(2^{-t}) \sim -\frac{L}{2}t^2. \tag{544}$$

It does not, however, extend (454) uniformly to all $x \downarrow 0$. The same bracket places a point $2^{-(n+1)} \leq x \leq 2^{-n}$ between two endpoint values whose logarithms differ by order n:

$$\log F(2^{-n}) - \log F(2^{-(n+1)}) = Ln + \mathcal{O}(\log n). \tag{545}$$

That uncertainty is enormous compared with the desired $\mathcal{O}(n^{-1})$ remainder.

The global formula (299) contains every dyadic rational and could, in principle, be used on meshes much finer than one reciprocal-power interval. To derive (539) by that route one would need a two-parameter asymptotic uniform in both numerator and denominator. The arbitrary-argument Bromwich saddle avoids that difficult finite-sum uniformity. Accordingly, the exact inverse-power formulae recover the full phase function and all constants, but a genuinely uniform analytic argument remains necessary to pass from the dense arithmetic set to a sharp theorem on a continuum.

More concretely, for $x_n = m_n 2^{-n}$ the master coefficient contains the prefix factor

$$\mathcal{S}_{m_n}(r_n) = \sum_{h<m_n} \varepsilon_h \mathrm{e}^{-hr_n}, \qquad r_n = \lambda(x_n)/m_n. \tag{546}$$

The fixed/moderate analysis requires r_n large, whereas a mesh fine enough to control an additive $o(1)$ error in $\log F$ needs $m_n \gg \lambda$. In that latter regime the prefix cannot be replaced by one and depends on a long stretch of numerator digits. A uniform theorem there is a two-parameter de-Poissonization of the exact dyadic formula, essentially the continuous Bromwich problem in another coordinate system.

41 Numerical verification from exact rational values

The recurrence (330) computes $a_n = d_n/n!$, hence $F(2^{-n})$, in exact rational arithmetic. For example,

$$F(2^{-10}) = \frac{134926369}{1246394851358539387238350848000}, \tag{547}$$

so

$$\log F(2^{-10}) = -50.5775682823421608125\ldots.$$

Let

$$\rho_n = \log F(2^{-n}) - \mathcal{M}_n. \tag{548}$$

The leading theorem predicts $\rho_n = \mathcal{O}(1/\lambda_n)$, and the first correction predicts $\lambda_n\rho_n = A_1(\lambda_n) + \mathcal{O}(1/\lambda_n)$. The table was computed by exact rational recurrence followed by one high-precision logarithm; P and its derivatives were evaluated from the absolutely convergent product and forward-tail series.

n	λ_n	ρ_n	$\lambda_n\rho_n$	$A_1(\lambda_n)$	$\lambda_n^2(\rho_n - A_1/\lambda_n)$
10	13.78503056	0.05801333439	0.7997155875	0.7629678468	0.5065687288
20	24.62186833	0.03182756966	0.7836542295	0.7629268731	0.5103462408
40	45.50804986	0.01701388954	0.7742689337	0.7629490545	0.5151456258
80	86.43351899	0.008896709914	0.7689739453	0.7629823191	0.5178773369
160	167.3870441	0.004576872612	0.7661091776	0.7630070725	0.5192522062

Table 17: Exact reciprocal-power values against the compact Lambert main and first correction.

The fourth and fifth columns approach one another, while the last column remains bounded and appears to settle, exactly as (535) predicts.

The fixed-ray coefficient theorem has an independent exact-rational check. With $\rho_{m,n} = \log F(m2^{-n}) - \mathcal{M}(m2^{-n})$, the following rows retain the nontrivial numerators from the original bridge calculation.

A second check displays the cancellation in the literal q-formula. Since

$$\log F(2^{-n}) = \log \mathcal{N}_n - n^2 L - \log(1/2;1/2)_n,$$

m	n	λ	$\lambda\rho_{m,n}$	$\lambda^2(\rho_{m,n} - A_1(\lambda)/\lambda)$
3	20	22.93448405	0.7855611616	0.5164
3	40	43.87020711	0.7748252957	0.5182
3	80	84.82139378	0.7691025485	0.5188
5	20	22.14711904	0.7864070706	0.5162
5	40	43.10795409	0.7751701258	0.5203
5	80	84.07161885	0.7692992153	0.5214

Table 18: Exact coefficient checks on the fixed rays $m = 3, 5$.

n	$\log \mathcal{N}_n$	n^2L	$\log(1/2;1/2)_n$	$\log F(2^{-n})$
10	17.4960644003	69.3147180560	-1.2410853733	-50.5775682823
20	95.4684669707	277.2588722240	-1.2420611411	-180.5483441121
40	447.4055728988	1109.0354888959	-1.2420620948	-660.3878539023
80	1957.8744511397	4436.1419555836	-1.2420620948	-2477.0254423491

Table 19: Logarithmic cancellation in the finite q-binomial formula.

one obtains:

The Pochhammer logarithm has already converged to

$$\log(1/2;1/2)_\infty = -1.2420620948124149458\ldots. \tag{549}$$

The nontrivial compensation is between $\log \mathcal{N}_n \sim Ln^2/2$ and the explicit $-Ln^2$ denominator.

42 Correspondence with the Lean development

The relevant modules in the 25 August 2026 repository snapshot [14] form a fairly literal proof graph. The table records the role of each group rather than every import edge.

Module	Role in the bridge
`FabiusRecurrenceSequence.lean`	Defines $a_n = d_n/n!$, proves its recurrence, identifies $F(2^{-n}) = 2^{-\binom{n}{2}} a_n$, and connects its ordinary series to the analytic generating function and dyadic product.
`FabiusInverseDyadicClosedForm.lean`	Unfolds the inverse-dyadic recurrence into the exact composition formula.
`FabiusRawQBinomialFormula.lean`, `FabiusQBinomialFormula.lean`, `FabiusQBinomialScalarFormula.lean`, `FabiusDyadicQBinomialScalar.lean`	Develop the Thue–Morse block factorization, half-base Gaussian interpolation, scalar translation invariance, and the global exact dyadic formula. The coefficient-extraction argument in section 21 is the ordinary-mathematics unpacking of this layer.
`FabiusDyadicLogBounds.lean`	Combines the exact endpoint identity with elementary half-moment bounds to prove the $-Ln^2/2$ law and an explicit $3n\log(n+1)$ error.
`NegativeLaplace.lean`, `PeriodicCorrection.lean`, `PeriodicFourier.lean`	Construct the negative-Laplace logarithm, its exact quadratic-plus-periodic decomposition, the forward-tail bound, regularity, mean normalization, and Fourier data.
`EndpointLaplaceComparison.lean`	Proves the second-order pointwise and integrated comparison between $(1-x)^n$ and $\mathrm{e}^{-nx}(1-nx^2/2)$ and transfers the estimate through the logarithm.
`DyadicSharpDecomposition.lean`, `FabiusDyadicSharpCumulant.lean`	Assemble the exact logarithmic decomposition of $F(2^{-n})$, including Stirling, the periodic term, the endpoint/Laplace error, and the exact second tilted cumulant.
`LaplacePeriodicSecondOrder.lean`	Differentiates the periodic Laplace decomposition and proves the explicit $\Psi'(\log_2 n)$ correction with an $\mathcal{O}(1/n)$ remainder.
`FabiusLambertPhase.lean`, `FabiusDyadicSharpAsymptotic.lean`	Construct the exact lower-Lambert phase, transfer the periodic derivative term to $\Psi(\lambda_n)$, and prove the corrected sharp dyadic theorem.
`FabiusSharpLambertMain.lean`, `FabiusSharpAsymptotic.lean`, `FabiusFullAsymptoticExpansion.lean`	Prove the arbitrary-small-argument Lambert main and continue the saddle expansion to all orders.

The coefficient-saddle derivation in Section 35 is deliberately marked as an alternative exposition. The formal dyadic proof uses the endpoint–Laplace comparison; the formal arbitrary-argument proof uses the Bromwich/saddle kernel. No claim is made that the Cauchy coefficient route appears as a separate theorem under that exact name.

43 Conclusions

The conspicuous gap between exact dyadic evaluation and endpoint asymptotics can be closed without adding a new special function or an independent heuristic ansatz.

First, the exact reciprocal-power values are controlled by a single coefficient sequence:

$$F(2^{-n}) = 2^{-\binom{n}{2}} a_n, \qquad A(z) = \sum_{n\geq 0} a_n z^n = \prod_{j\geq 1} E(z/2^j).$$

The recurrence and composition formula are simply two expansions of this identity.

Second, the q-binomial formula reduces exactly to the same sequence. The inner Thue–Morse sum is a coefficient of $A(-2^k X)/A(-X)$, and the outer half-base Gaussian row is the leading-coefficient functional on the nodes -2^k. Therefore

$$\mathcal{N}_n = \mathfrak{m}_n a_n$$

with no approximation. This is the direct algebraic connection that makes asymptotic analysis of the finite q-formula feasible.

Third, the negative-axis product of A satisfies an exact dyadic dilation equation. Removing its quadratic drift leaves a smooth periodic function. A second-order comparison between the endpoint kernel $(1-X)^n$ and the Laplace kernel e^{-nX} supplies the first cumulant correction; differentiating the exact periodic decomposition identifies that correction; and the lower-Lambert fixed point moves it to the exact phase. The result is

$$\log F(2^{-n}) = -\frac{L}{2}\lambda_n^2 + \left(1 + \frac{L}{2}\right)\lambda_n - \frac{1}{2}\log\lambda_n + C_F + \Psi(\lambda_n) + \mathcal{O}(n^{-1}).$$

Finally, the inverse dyadic sequence is richer than a sparse one-phase sample: the fractional parts of λ_n are dense, so the exact rational values determine the entire continuous periodic correction after renormalization. What they do not provide by monotonicity alone is a uniform continuum theorem at the sharp error scale; that step still requires a uniform saddle argument or a two-parameter analysis of the general dyadic formula.

The answer to the motivating question is therefore affirmative in a strong sense. The small-argument asymptotic can be re-derived from the exact calculation formulae, including the constant, the logarithmic correction, the exact lower-Lambert phase, and the nonconstant periodic fluctuation. The only essential move is to respect the exact cancellations built into those formulae rather than attempting to estimate their finite summands independently.

Supplementary material from this synthesis

44 Algebra behind the expanded dyadic formula

This appendix records the expansion needed to pass from the compact Lambert form to the explicit expression in n. Put $\ell = \log n$ and seek

$$\lambda_n = n + \frac{\ell}{L} + \frac{c_1}{n} + \frac{c_2}{n^2} + \mathcal{O}(\ell^3/n^3). \tag{550}$$

The fixed-point equation is

$$\lambda_n = n + \frac{1}{L}\log\lambda_n. \tag{551}$$

Writing

$$\lambda_n = n\left(1 + \frac{\ell}{Ln} + \frac{c_1}{n^2} + \frac{c_2}{n^3} + \cdots\right)$$

and expanding the logarithm gives

$$\log\lambda_n = \ell + \frac{\ell}{Ln} + \frac{c_1 - \ell^2/(2L^2)}{n^2} + \mathcal{O}(\ell^3/n^3).$$

Comparison in (551) yields

$$c_1 = \frac{\ell}{L^2}, \qquad c_2 = \frac{\ell - \ell^2/2}{L^3}, \tag{552}$$

which gives the first four terms of (494).

Now set

$$H(\lambda) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2}\log\lambda.$$

Substitution of (550) with (552) and collection through order $1/n$ gives

$$\begin{aligned} H(\lambda_n) = & -\frac{L}{2}n^2 - n\ell + \left(1 + \frac{L}{2}\right)n - \frac{\ell^2}{2L} \\ & - \frac{\ell^2}{2L^2 n} + \mathcal{O}(n^{-1}). \end{aligned} \tag{553}$$

The notation $\mathcal{O}(n^{-1})$ in (553) includes bounded multiples of $1/n$; the displayed ℓ^2/n term is separated because it is asymptotically larger. Adding $C_F + \Psi(\lambda_n)$ proves (501).

45 A stable numerical recipe

The following procedure reproduces Table 17 without evaluating the unstable signed q-sum.

Algorithm 45.1 (Exact arithmetic plus convergent products)**.**

1. Compute $a_0 = 1$ and, for $1 \le j \le n$,

   $$a_j = \frac{1}{2^j - 1}\sum_{k<j}\frac{a_k}{(j-k+1)!}$$

   in rational arithmetic.

2. Form the exact rational

   $$F(2^{-n}) = 2^{-\binom{n}{2}}a_n$$

   and take its logarithm only after conversion to the desired floating-point precision.

3. Solve $\lambda - \log\lambda/L = n$ on the large branch, either by $\lambda = -W_{-1}(-L2^{-n})/L$ or by Newton iteration initialized with $n + \log n/L$.

4. Reduce $u = \lambda - \lfloor\lambda\rfloor$ and put $s = 2^u \in [1, 2)$. Evaluate

$$P(u) = \sum_{j\geq 1} \log \frac{1 - \mathrm{e}^{-s/2^j}}{s/2^j} + \frac{L}{2}(u^2 - u) + \sum_{m\geq 0} \log(1 - \mathrm{e}^{-s2^m}).$$

   The first series has a geometric small-argument tail and the second a doubly exponential tail. Use stable primitives such as `expm1` for the first factors.

5. Subtract $\overline{P}$ from (476) to obtain $\Psi(u)$. Differentiate the locally uniformly convergent series termwise, or use the rapidly convergent Fourier series (478), to obtain Ψ' and Ψ''.

6. Evaluate $\mathcal{M}_n$ from (500) and, when desired, A_1 from (534).

For a direct verification of the q-formula one may compute $\mathcal{N}_n = \mathfrak{m}_n a_n$ exactly and compare it with the finite signed expression. The equality is a better test than comparing floating-point approximations of the two sides because it checks both cancellation layers over the rationals.

46 Notation cross-reference

Symbol	Meaning
L	$\log 2$
F	bounded Fabius function / distribution function on $[0, 1]$
up	Rvachev up-function, with $\mathrm{up}(t) = F(1 - \lvert t\rvert)$ on $[-1, 1]$
X	random variable with distribution function F
d_n	half moment $\mathbb{E}(1 - X)^n$
a_n	factorially normalized half moment $d_n/n!$
$A(z)$	$\sum a_n z^n = \mathbb{E}\mathrm{e}^{zX}$
$E(z)$	$(\mathrm{e}^z - 1)/z$
$\mathfrak{m}_n$	$\prod_{j=1}^{n}(2^j - 1)$
$\mathcal{N}_n$	finite q-binomial/Thue–Morse numerator for $F(2^{-n})$
$B(s)$	$A(-s) = \mathbb{E}\mathrm{e}^{-sX}$
$\Lambda(s)$	$\log B(s)$
P	exact one-periodic correction in Λ
Ψ	$P - \overline{P}$, mean-zero periodic correction
C_F	constant $\overline{P} - \frac{1}{2}\log(2\pi)$
λ_n	large solution of $\lambda - \log\lambda/L = n$
$\mathcal{M}_n$	compact Lambert main term for $\log F(2^{-n})$

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Part V
Signed Thue–Morse Prefix Sums and the Fabius Function

Synthesized provenance and status. This part merges the former `Fabius_Thue_Morse_Convergence_Rate/Fabius_Thue_Morse_Convergence_Rate.tex` and `Fabius_Thue_Morse_Convergence_Rate-2/Fabius_Thue_Morse_Convergence_Rates.tex`. The Fourier/deconvolution source supplies the canonical sharp and all-orders results; the elementary source contributes zero-run identities, an explicit finite-level curvature certificate, binary-phase and endpoint formulas. The qualitative bridge reuses formal inputs, but all quantitative rates, constants, deconvolution expansions, and acceleration results remain natural-language proofs and formal obligations.

Former source SHA-256 values: `577c5f3426def68a774fedf3fce61552d32200c8da526ace44178f2a8995a6d3` and `384d69b461cb94af33f1c080703e269ea77104405d1940413f1944172b3312c0`.

Topic abstract

Let $\varepsilon_m = (-1)^{\mathrm{wt}_2(m)}$ be the signed Thue–Morse sequence and let $\sigma_{k+1}(m) = \sum_{j=0}^{m} \sigma_k(j)$ be its inclusive iterated prefix sums. The formal development in `Analysis/FabiusFunction` proves that a shifted and normalized order-$(n+1)$ prefix row is exactly the coefficient row of a finite product of discrete uniform laws, and that its associated histogram converges pointwise to Rvachev's compactly supported function up. Sampling the same row at $\lfloor 2^n x \rfloor$ consequently converges to the bounded Fabius function $F(x)$. The existing argument is deliberately qualitative and supplies no rate.

This report makes the convergence quantitative. At the natural cell centers, the normalized prefix coefficients admit a uniform deconvolution expansion

$$H_n(\lambda) = \mathrm{up}(\lambda) - \frac{3n+4}{72\,4^n}\,\mathrm{up}''(\lambda) + \left(\frac{15n+16}{43200\,16^n} + \frac{(3n+4)^2}{10368\,16^n}\right)\mathrm{up}^{(4)}(\lambda) + \mathcal{O}\!\left(\frac{n^3}{64^n}\right).$$

It follows that the sharp uniform center-grid error is $\frac{1}{3}n4^{-n}(1+o(1))$; the same rate holds after piecewise-linear interpolation. The piecewise-constant histogram is less accurate because of cell quantization: $\|\mathcal{W}_n - \mathrm{up}\,\|_\infty = 2^{-n} + \mathcal{O}(n4^{-n})$. Most strikingly, the literal dyadic-floor approximation

$$A_n(x) = \frac{\sigma_{n+1}(\lfloor 2^n x \rfloor)}{2^{\binom{n}{2}}}$$

has a still larger, entirely geometric first-order error. Uniformly on $[0,1]$,

$$A_n(x) = F(x) + \frac{n+2-2\,\{2^n x\}}{2^{n+1}}F'(x) + \mathcal{O}\!\left(\frac{n^2}{4^n}\right),$$

and therefore

$$\left\|\frac{2^{n+1}}{n}(A_n - F) - F'\right\|_\infty \longrightarrow 0, \qquad \frac{2^n}{n}\|A_n - F\|_\infty \longrightarrow 1.$$

The factor n comes from the exact inward displacement of the finite support, not from slow weak convergence. At $x = 1/2$ the phenomenon has the closed form $A_n(1/2) - 1/2 = (n+2)2^{-n} - 2^{-\binom{n}{2}}$.

The bridge identities used as input are formalized in Lean. The Fourier asymptotic expansion, its sharp constants, and the accelerated finite-difference constructions developed here are new analytic deductions and are not claimed to have been formalized in the repository.

47 Scope, conventions, and the answer in one table

The source article [4] isolates the connection between iterated prefix sums and up, corrects two index/normalization defects in a naive formulation, and proves pointwise convergence. It also says explicitly that its monotonicity-and-continuity argument gives no uniform rate. The purpose of this supplement is to determine that rate, including its leading constant, and to explain why several superficially similar readings of the same integer row have different orders of accuracy.

We use the source article's normalization throughout:

- F is the bounded distribution function, equal to 0 on $(-\infty, 0]$ and to 1 on $[1, \infty)$;
- up is the even compactly supported density on $[-1, 1]$, with $\mathrm{up}(x-1) = F(x)$ for $0 \le x \le 1$;
- the Fourier transform is $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) \mathrm{e}^{-2\pi \mathrm{i} x \xi} \, \mathrm{d}x$.

The sharp derivative theorem from the source gives

$$\|F^{(r)}\|_\infty = \|\,\mathrm{up}^{(r)}\,\|_\infty = 2^{\binom{r+1}{2}}. \tag{554}$$

In particular $\|F'\|_\infty = \|\,\mathrm{up}'\,\|_\infty = 2$ and $\|\,\mathrm{up}''\,\|_\infty = 8$.

The three natural rates. Put $h_n = 2^{-n}$. The same normalized prefix row produces three different errors.

Reading of the row	Sharp uniform error	Dominant mechanism
Natural center values, or their linear interpolant	$\frac{1}{3} n h_n^2 (1 + o(1))$	Deconvolution of the omitted mean-zero Bernoulli tail; its variance is of order nh_n^2.
Piecewise-constant histogram $\mathcal{W}_n$	$h_n + \mathcal{O}(nh_n^2)$	Horizontal quantization by half a cell, amplified by the optimal slope 2.
Dyadic-floor Fabius sample A_n	$nh_n(1 + o(1))$	The finite support is shorter than $[-1, 1]$ by $(n+2)h_n/2$ at each end, producing an inward drift.

Thus the pointwise floor formula is not the most accurate reconstruction of the function from the prefix row. It is worse than the histogram by a factor asymptotic to n, while a centered linear reconstruction is better than the histogram by a factor asymptotic to $3 \cdot 2^n / n$.

47.1 What is formalized and what is new here

The exact prefix convolution law, Prouhet cancellation, zero runs, generating-series identity, coefficient bridge, corrected cell centers, and pointwise limit are developed in `ThueMorsePrefix.lean` and `ThueMorseApproximation.lean`; the finite atomic and step approximants are developed in `EarlyApproximants.lean`, `EarlyMeasureBridge.lean`, `StepMeasureBridge.lean`, and `StepApproximationLimit.lean` [5]. These are the algebraic and measure-theoretic inputs used below.

The new ingredient is a quantitative inversion of the finite cosine product on its natural Nyquist interval. It yields a complete asymptotic expansion in even derivatives of up. The first two nonzero terms are enough to determine all sharp rates stated in the abstract. No part of that new expansion is represented here as already machine-checked.

48 The exact discrete bridge

48.1 Signed Thue–Morse prefixes

Let

$$\varepsilon_m = (-1)^{\mathrm{wt}_2(m)}, \qquad m \geq 0, \tag{555}$$

where $\mathrm{wt}_2(m)$ is the number of ones in the binary expansion of m. Define inclusive iterated prefix sums by

$$\sigma_0(m) = \varepsilon_m, \qquad \sigma_{k+1}(m) = \sum_{j=0}^{m} \sigma_k(j). \tag{556}$$

The word *inclusive* matters: one has

$$\sigma_{k+1}(m+1) - \sigma_{k+1}(m) = \sigma_k(m+1), \tag{557}$$

with the lower-order value at the new index.

Repeated summation is convolution with the discrete truncated-power kernel. For $k \geq 1$,

$$\sigma_k(m) = \sum_{r=0}^{m} \binom{m-r+k-1}{k-1} \varepsilon_r. \tag{558}$$

The signed generating series is

$$\sum_{m\geq 0} \varepsilon_m z^m = \prod_{j\geq 0}(1 - z^{2^j}), \qquad \sum_{m\geq 0} \sigma_k(m) z^m = \frac{\prod_{j\geq 0}(1 - z^{2^j})}{(1-z)^k}. \tag{559}$$

All identities in this paragraph are formal power-series identities over $\mathbb{Z}$.

The order-r zero of a complete block at $z = 1$ gives a useful exact boundary certificate. For $1 \leq k \leq r$,

$$\sigma_k(m) = 0 \qquad (2^r - k \leq m \leq 2^r - 1), \tag{560}$$

$$\sigma_k(2^r - k - 1) = (-1)^{r-k}, \qquad \sigma_k(2^r) = -1. \tag{561}$$

These dyadic zero runs explain both the long flat stretches of the prefix rows and the exceptional right endpoint of the floor sample.

48.2 The finite polynomial and the one-order shift

For $n \geq 0$ put

$$P_n(z) = \prod_{i=1}^{n} (1 + z + \cdots + z^{2^i - 1}), \qquad g_n = \deg P_n = 2^{n+1} - n - 2. \tag{562}$$

The cancellation of the first $n+1$ factors in (559) gives the exact coefficient identity

$$[z^m]P_n(z) = \sigma_{n+1}(m), \qquad 0 \leq m < 2^{n+1}. \tag{563}$$

This is the source of the order shift: the polynomial indexed by n corresponds to the $(n+1)$st, not the nth, prefix row.

Every factor of P_n is palindromic, hence

$$[z^m]P_n = [z^{g_n - m}]P_n, \qquad P_n(1) = 2^{\binom{n+1}{2}}. \tag{564}$$

The coefficients are nonnegative integers. Thus, after normalization, they form a symmetric probability mass function.

48.3 Natural centers, atomic law, and histogram

Set

$$h_n = 2^{-n}, \qquad \lambda_{n,m} = \frac{2m - g_n}{2^{n+1}} = h_n\left(m - \frac{g_n}{2}\right). \tag{565}$$

The points $\lambda_{n,m}$ lie on a translate of the lattice $h_n\mathbb{Z}$. Define

$$p_{n,m} = 2^{-\binom{n+1}{2}}[z^m]P_n, \qquad \mu_n = \sum_{m\in\mathbb{Z}} p_{n,m}\,\delta_{\lambda_{n,m}}, \tag{566}$$

where coefficients outside $0 \le m \le g_n$ are understood to be zero. Then μ_n is a probability measure. Its mass density relative to the lattice spacing is

$$H_n(\lambda_{n,m}) = \frac{p_{n,m}}{h_n} = \frac{[z^m]P_n}{2^{\binom{n}{2}}}. \tag{567}$$

By (563), this is exactly

$$H_n(\lambda_{n,m}) = \frac{\sigma_{n+1}(m)}{2^{\binom{n}{2}}}, \qquad 0 \le m < 2^{n+1}. \tag{568}$$

Let $I_{n,m}$ be the cell of width h_n centered at $\lambda_{n,m}$. Spreading each atom of μ_n uniformly over its cell gives the step density

$$\mathcal{W}_n(x) = \sum_m H_n(\lambda_{n,m})\mathbf{1}_{I_{n,m}}(x), \tag{569}$$

with half weight from each adjacent cell at a shared endpoint. In the interior of a cell, $\mathcal{W}_n$ is simply the corresponding normalized prefix coefficient. The formal repository proves

$$\mathcal{W}_n(x) \longrightarrow \mathrm{up}(x) \tag{570}$$

pointwise. The rest of this report quantifies (570) and the moving-center sample derived from it.

49 The hidden Bernoulli tail

49.1 Rademacher factorization

Let $R_{\ell,r}$ be independent Rademacher variables, each taking ±1 with probability $1/2$. The binary factorization of every discrete uniform variable in (562) gives an equivalent representation of (566):

$$X_n = \sum_{\ell=1}^{n}\sum_{r=1}^{\ell} 2^{-\ell-1}R_{\ell,r}, \qquad \mathcal{L}(X_n) = \mu_n. \tag{571}$$

Consequently

$$\widehat{\mu_n}(\xi) = \prod_{\ell=1}^{n}\left(\cos\frac{\pi\xi}{2^\ell}\right)^{\ell}. \tag{572}$$

The complete series

$$Y = \sum_{\ell=1}^{\infty}\sum_{r=1}^{\ell} 2^{-\ell-1}R_{\ell,r} \tag{573}$$

converges absolutely, and its density is up. Equivalently,

$$\widehat{\mathrm{up}}(\xi) = \prod_{\ell=1}^{\infty}\left(\cos\frac{\pi\xi}{2^\ell}\right)^{\ell}. \tag{574}$$

This cosine representation is equivalent to the more familiar infinite sinc product.

Write

$$T_n = Y - X_n = \sum_{\ell>n} \sum_{r=1}^{\ell} 2^{-\ell-1} R_{\ell,r}. \tag{575}$$

Then X_n and T_n are independent and

$$\mathrm{up}(x)\,\mathrm{d}x = \mu_n * \mathcal{L}(T_n). \tag{576}$$

Thus μ_n is obtained by deleting a small, symmetric smoothing tail from the target law. The center-grid asymptotics are an exact lattice version of deconvolving that tail.

49.2 Exact support radius and variance

Proposition 49.1 (Geometry and second moment of the omitted tail)**.** *The tail T_n has mean zero, support contained in $[-r_n, r_n]$, and variance v_n, where*

$$r_n = \sum_{\ell>n} \ell 2^{-\ell-1} = \frac{n+2}{2^{n+1}}, \tag{577}$$

$$v_n = \sum_{\ell>n} \ell 2^{-2\ell-2} = \frac{3n+4}{36\,4^n}. \tag{578}$$

Both support endpoints occur.

Proof. The mean vanishes by symmetry. The maximal possible absolute value is obtained by taking all signs equal, so it is the first displayed sum. The geometric-tail identity

$$\sum_{\ell=n+1}^{\infty} \ell q^\ell = q^{n+1} \frac{(n+1) - nq}{(1-q)^2}$$

with $q = 1/2$ gives (577). Independence and $\mathrm{Var}(R_{\ell,r}) = 1$ give the variance sum, and the same identity with $q = 1/4$, followed by the extra factor $1/4$, gives (578). □

The support of μ_n is therefore

$$\operatorname{supp} \mu_n = [-1 + r_n, 1 - r_n] \cap \big(h_n(\mathbb{Z} - g_n/2)\big), \tag{579}$$

because $g_n/2^{n+1} = 1 - r_n$. Two scales should be distinguished:

$$r_n \sim \frac{n}{2} h_n, \qquad \sqrt{v_n} \sim \sqrt{\frac{n}{12}}\, h_n. \tag{580}$$

The support radius is a worst-case all-signs quantity, while the standard deviation is the typical scale. The floor-indexed approximation will inherit the former; the correctly centered deconvolution error will inherit the variance itself.

49.3 Why the leading center correction is second order

If a smooth density f is convolved with a small mean-zero law of variance v, then formally

$$f * \nu = f + \frac{v}{2} f'' + \cdots .$$

Solving backward gives $f - \frac{v}{2} f'' + \cdots$. Since (576) smooths the atomic row by the tail law, one should expect

$$H_n(\lambda) - \mathrm{up}(\lambda) \approx -\frac{v_n}{2}\,\mathrm{up}''(\lambda) = -\frac{3n+4}{72\,4^n}\,\mathrm{up}''(\lambda). \tag{581}$$

The sign is that of deconvolution: in a convex region the unsmoothed profile lies below the smoothed one. The next sections make (581) uniform and rigorous and compute the fourth-derivative term.

50 Fourier inversion on the moving lattice

The main technical point is that one must invert the finite atomic law on exactly one period of its lattice Fourier series. Ordinary Fourier inversion of a density cannot be applied directly to μ_n.

50.1 A shifted-lattice inversion formula

Lemma 50.1 (Nyquist inversion for a shifted lattice). *Let $\mu = \sum_{k\in\mathbb{Z}} p_k\delta_{a+hk}$ be a finite measure on a translate of $h\mathbb{Z}$, and let $\widehat{\mu}(\xi) = \sum_k p_k \mathrm{e}^{-2\pi\mathrm{i}(a+hk)\xi}$. Then, for every $m\in\mathbb{Z}$,*

$$\frac{p_m}{h} = \int_{-1/(2h)}^{1/(2h)} \widehat{\mu}(\xi)\mathrm{e}^{2\pi\mathrm{i}(a+hm)\xi}\,\mathrm{d}\xi. \tag{582}$$

Proof. Insert the finite sum defining $\widehat{\mu}$ and integrate term by term. The kth term is

$$p_k\int_{-1/(2h)}^{1/(2h)} \mathrm{e}^{2\pi\mathrm{i}h(m-k)\xi}\,\mathrm{d}\xi,$$

which is p_m/h when $k=m$ and zero otherwise. The shift a cancels from the phase. □

For μ_n, the interval in (582) is

$$B_n = [-2^{n-1}, 2^{n-1}]. \tag{583}$$

Combining (567) and (582) gives

$$H_n(\lambda_{n,m}) = \int_{B_n} \widehat{\mu_n}(\xi)\mathrm{e}^{2\pi\mathrm{i}\lambda_{n,m}\xi}\,\mathrm{d}\xi. \tag{584}$$

50.2 The exact deconvolution multiplier

On B_n, all factors in the omitted cosine tail are positive: for $\ell > n$, $|\pi\xi/2^\ell| \le \pi/4$. Hence one may divide (574) by (572) and write

$$\widehat{\mu_n}(\xi) = \widehat{\mathrm{up}}(\xi)R_n(\xi), \qquad R_n(\xi) = \prod_{\ell>n}\left(\sec\frac{\pi\xi}{2^\ell}\right)^\ell, \qquad \xi\in B_n. \tag{585}$$

The multiplier R_n is the reciprocal characteristic function of the omitted tail on the frequency range where lattice inversion needs it.

There is a useful global bound. Put $t = \pi\xi/2^n$, so $|t|\le\pi/2$. Reindexing $\ell = n+r$ and using Viète's product $\prod_{r\ge1}\cos(t/2^r) = \sin t/t$ gives

$$R_n(\xi) = \left(\frac{t}{\sin t}\right)^n \prod_{r=1}^{\infty}\sec(t/2^r)^r. \tag{586}$$

The second product is bounded uniformly for $|t|\le\pi/2$, while $t/\sin t\le\pi/2$. Thus there is an absolute C_* such that

$$1 \le R_n(\xi) \le C_*\left(\frac{\pi}{2}\right)^n, \qquad \xi\in B_n. \tag{587}$$

This exponential growth is harmless because $\widehat{\mathrm{up}}$ decays faster than every power. Indeed, compact support and smoothness give, for each $M\ge1$,

$$|\widehat{\mathrm{up}}(\xi)| \le \frac{\|\,\mathrm{up}^{(M)}\,\|_1}{(2\pi|\xi|)^M} \le \frac{2^{1+\binom{M+1}{2}}}{(2\pi|\xi|)^M} \qquad (\xi\ne0), \tag{588}$$

where (554) was used in the second inequality.

51 Sharp center-grid asymptotics

51.1 Expansion of the reciprocal tail product

For $|u| < \pi/2$,

$$\log\sec u = \frac{u^2}{2} + \frac{u^4}{12} + \mathcal{O}(u^6). \tag{589}$$

The two geometric tails needed below are

$$\frac{1}{2}\sum_{\ell>n} \ell 4^{-\ell} = \frac{3n+4}{18\,4^n} =: \alpha_n, \tag{590}$$

$$\frac{1}{12}\sum_{\ell>n} \ell 16^{-\ell} = \frac{15n+16}{2700\,16^n} =: \beta_n. \tag{591}$$

Therefore, on a low-frequency range growing sufficiently slowly with n,

$$\log R_n(\xi) = \alpha_n(\pi\xi)^2 + \beta_n(\pi\xi)^4 + \mathcal{O}\left(\frac{n}{64^n}|\xi|^6\right), \tag{592}$$

and exponentiation gives

$$R_n(\xi) = 1 + \alpha_n(\pi\xi)^2 + \left(\beta_n + \frac{\alpha_n^2}{2}\right)(\pi\xi)^4 + \mathcal{O}\left(\frac{n^3}{64^n}|\xi|^6\right). \tag{593}$$

51.2 The uniform expansion

Theorem 51.1 (Two-term lattice deconvolution expansion)**.** *Uniformly for all lattice centers $\lambda = \lambda_{n,m}$,*

$$\begin{aligned} H_n(\lambda) = {} & \mathrm{up}(\lambda) - \frac{3n+4}{72\,4^n}\,\mathrm{up}''(\lambda) \\ & + \left(\frac{15n+16}{43200\,16^n} + \frac{(3n+4)^2}{10368\,16^n}\right)\mathrm{up}^{(4)}(\lambda) + \mathcal{O}\left(\frac{n^3}{64^n}\right). \end{aligned} \tag{594}$$

The implicit constant is absolute and effective. In particular,

$$H_n(\lambda) - \mathrm{up}(\lambda) = -\frac{3n+4}{72\,4^n}\,\mathrm{up}''(\lambda) + \mathcal{O}\left(\frac{n^2}{16^n}\right) \tag{595}$$

uniformly on the moving lattice.

Proof. Use (584) and (585) and split the integral at $|\xi| = 2^{n/4}$. On the inner interval, (593) is uniform. Since $\int_{\mathbb{R}} |\xi|^6 |\widehat{\mathrm{up}}(\xi)|\,\mathrm{d}\xi < \infty$, its integrated remainder is $\mathcal{O}(n^3 64^{-n})$ uniformly in the phase λ.

On the complementary part of B_n, combine (587) with (588). Taking, for example, $M = 32$ makes the resulting integral $o(64^{-n})$. The tails of the three polynomial Fourier integrals outside B_n are even smaller. We may therefore replace the truncated integrals of the displayed polynomial terms by integrals over all of $\mathbb{R}$.

With the chosen Fourier convention,

$$\int_{\mathbb{R}} (\pi\xi)^2 \widehat{\mathrm{up}}(\xi) \mathrm{e}^{2\pi\mathrm{i}\lambda\xi}\,\mathrm{d}\xi = -\frac{1}{4}\,\mathrm{up}''(\lambda), \qquad \int_{\mathbb{R}} (\pi\xi)^4 \widehat{\mathrm{up}}(\xi) \mathrm{e}^{2\pi\mathrm{i}\lambda\xi}\,\mathrm{d}\xi = \frac{1}{16}\,\mathrm{up}^{(4)}(\lambda).$$

Substituting (590) and (591) gives (594). Omitting the fourth-derivative term and using (554) gives (595). ☐

Remark 51.2 (Variance form). Because $\alpha_n = 2v_n$, the leading term is exactly

$$H_n = \mathrm{up} - \frac{v_n}{2}\,\mathrm{up}'' + \cdots, \tag{596}$$

which confirms the deconvolution heuristic (581).

51.3 Sharp norm constant

Corollary 51.3 (Sharp center-grid rate)**.** *One has the uniform profile limit*

$$\sup_{m\in\mathbb{Z}}\left|\frac{4^n}{n}\big(H_n(\lambda_{n,m})-\mathrm{up}(\lambda_{n,m})\big)+\frac{1}{24}\,\mathrm{up}''(\lambda_{n,m})\right|\longrightarrow 0. \tag{597}$$

Consequently,

$$\lim_{n\to\infty}\frac{4^n}{n}\sup_{m\in\mathbb{Z}}|H_n(\lambda_{n,m})-\mathrm{up}(\lambda_{n,m})|=\frac{1}{3}. \tag{598}$$

Proof. The coefficient $(3n+4)/(72n)$ tends to $1/24$, while the remainder in (595), after multiplication by $4^n/n$, tends uniformly to zero. The lattices become dense, and up'' is continuous with exact supremum 8. This supremum is attained; for example, the attained-point form of the sharp derivative theorem gives $\mathrm{up}''(-3/4)=F''(1/4)=8$. Hence the norm limit is $8/24=1/3$. □

51.4 A continuous reconstruction with the same rate

Let L_n be the piecewise-linear interpolant through the points $(\lambda_{n,m},H_n(\lambda_{n,m}))$, with zero values continued outside the finite support. The ordinary interpolation error for a C^2 function is at most $h_n^2\|\,\mathrm{up}''\,\|_\infty/8=h_n^2$. This is smaller by a factor $1/n$ than the leading center bias.

Corollary 51.4 (Sharp continuous linear reconstruction)**.** *Uniformly on* $\mathbb{R}$*,*

$$\frac{4^n}{n}(L_n-\mathrm{up})\longrightarrow-\frac{1}{24}\,\mathrm{up}'', \tag{599}$$

in the supremum norm. In particular,

$$\|L_n-\mathrm{up}\,\|_\infty=\frac{n}{3\,4^n}(1+o(1)). \tag{600}$$

Thus iterated prefix sums do not merely converge to up after a qualitative passage through histograms. Read at their actual centers and interpolated linearly, they give a continuous approximation with a fully determined, curvature-shaped leading error.

51.5 The all-orders pattern

The same proof is not tied to fourth order. Write

$$\log\sec z=\sum_{r\geq 1}c_r z^{2r}\quad (c_1=1/2,\ c_2=1/12,\ c_3=1/45,\ldots) \tag{601}$$

and put

$$s_{r,n}=c_r\sum_{\ell>n}\ell 2^{-2r\ell}. \tag{602}$$

For a fixed M, expand

$$\exp\left(\sum_{r=1}^{M-1}s_{r,n}z^{2r}\right)=\sum_{j=0}^{M-1}b_{j,n}z^{2j}+\mathcal{O}(z^{2M}). \tag{603}$$

Then the lattice inversion argument gives

$$H_n(\lambda)=\sum_{j=0}^{M-1}\frac{(-1)^j b_{j,n}}{4^j}\,\mathrm{up}^{(2j)}(\lambda)+\mathcal{O}_M\left(n^M 4^{-Mn}\right), \tag{604}$$

uniformly on the center lattice. Each $b_{j,n}$ is an explicit polynomial in the geometric tails $s_{r,n}$. Thus the connection admits a full asymptotic differential expansion, not just a big-O convergence estimate.

51.6 An elementary finite-level certificate

The Fourier expansion above determines the sharp asymptotics. A complementary argument gives explicit, if nonsharp, bounds at every level $n \geq 3$. Let $\beta_h = h^{-1}\mathbf{1}_{[-h/2,h/2]}$ and define the mass-preserving dilation

$$(D_h\,\mathrm{up})(x) = h^{-1}\,\mathrm{up}(x/h).$$

Put $q_n = \beta_{h_n}^{*n} * D_{h_n}\,\mathrm{up}$, and rename the cardinal weights

$$\pi_{n,k} = h_n q_n(kh_n), \qquad k \in \mathbb{Z}. \tag{605}$$

They form a symmetric probability distribution and satisfy the exact identities

$$\mathrm{up}(\lambda_{n,m}) = \sum_{k\in\mathbb{Z}} \pi_{n,k} H_n(\lambda_{n,m-k}), \tag{606}$$

$$\sum_{k\in\mathbb{Z}} k^2\pi_{n,k} = \frac{n}{12} + \frac{1}{9}. \tag{607}$$

The second formula follows by differentiating the Poisson identity twice: the n box factors kill every nonzero integer frequency, leaving the variance $n/12 + 1/9$ at the origin.

Writing $w_{n,m} = H_n(\lambda_{n,m})$, the factorization $(1-z)^2P_n = (1-z^{2^{n-1}})(1-z^{2^n})P_{n-2}$ and the coefficient bound for P_{n-2} give

$$|w_{n,m} - 2w_{n,m-1} + w_{n,m-2}| \leq 32\,4^{-n}. \tag{608}$$

A telescoping symmetric-difference argument upgrades this to $|w_{n,m+k} + w_{n,m-k} - 2w_{n,m}| \leq 32\,4^{-n}k^2$. Combining it with (606)– (607) yields the explicit certificate

$$|H_n(\lambda_{n,m}) - \mathrm{up}(\lambda_{n,m})| \leq \left(\frac{4n}{3} + \frac{16}{9}\right)4^{-n}. \tag{609}$$

Consequently

$$\|\mathcal{W}_n - \mathrm{up}\,\|_\infty \leq 2^{-n} + \left(\frac{4n}{3} + \frac{16}{9}\right)4^{-n}, \tag{610}$$

$$\|L_n - \mathrm{up}\,\|_\infty \leq \left(\frac{4n}{3} + \frac{25}{9}\right)4^{-n}. \tag{611}$$

These are effective envelopes, not replacements for the sharper limiting constants proved above.

52 The histogram: a sharp first-order quantization error

The center values are exceptionally accurate, but $\mathcal{W}_n$ holds each value constant across a cell. That reconstruction injects a larger first-order horizontal error.

For a point x in the interior of a cell, let $c_n(x)$ denote that cell's center and define the sawtooth phase

$$\rho_n(x) = \frac{c_n(x) - x}{h_n}, \qquad -\frac{1}{2} < \rho_n(x) < \frac{1}{2}. \tag{612}$$

Then (595) and Taylor's theorem give, uniformly away from the shared cell endpoints,

$$\mathcal{W}_n(x) - \mathrm{up}(x) = h_n\rho_n(x)\,\mathrm{up}'(x) + \mathcal{O}(nh_n^2). \tag{613}$$

At a shared endpoint, the half-weight convention averages the two adjacent center values and cancels the first-order odd displacement, leaving only $\mathcal{O}(nh_n^2)$ there.

Theorem 52.1 (Sharp histogram rate). *The step approximants satisfy*

$$\|\mathcal{W}_n - \operatorname{up}\|_\infty = 2^{-n} + \mathcal{O}\Big(\frac{n}{4^n}\Big). \tag{614}$$

Equivalently,

$$\lim_{n\to\infty} 2^n\|\mathcal{W}_n - \operatorname{up}\|_\infty = 1. \tag{615}$$

Proof. For an interior point, combine (595) with the optimal Lipschitz bound $\|\operatorname{up}'\|_\infty = 2$:

$$|\mathcal{W}_n(x) - \operatorname{up}(x)| \le \mathcal{O}(nh_n^2) + 2|c_n(x) - x| \le h_n + \mathcal{O}(nh_n^2).$$

At a shared endpoint the averaged value obeys the same bound. There remains only the thin uncovered strip between the support of $\mathcal{W}_n$ and an endpoint of $[-1, 1]$. Its width is at most $r_n + h_n/2 = \mathcal{O}(nh_n)$. Since up and all of its derivatives vanish at ± 1, Taylor's theorem with the sharp bounded third derivative gives

$$|\operatorname{up}(\pm 1 \mp s)| \le \frac{\|\operatorname{up}^{(3)}\|_\infty}{6}s^3 = \mathcal{O}(n^3h_n^3) = o(nh_n^2) \qquad (0 \le s \le r_n + h_n/2).$$

Outside $[-1, 1]$ both functions vanish. This completes the upper estimate globally.

For the lower estimate, choose cells whose centers tend to $-1/2$, where $\operatorname{up}'(-1/2) = 2$, and approach one edge of each chosen cell from its interior. The center error is $\mathcal{O}(nh_n^2)$, while the change of up across half a cell is $h_n + \mathcal{O}(h_n^2)$. Taking the supremum gives $\|\mathcal{W}_n - \operatorname{up}\|_\infty \ge h_n - \mathcal{O}(nh_n^2)$. ☐

Remark 52.2. The pointwise proof in the source article could not reveal (614): weak convergence plus unimodality controls values through fixed comparison points but does not track a moving point at distance $h_n/2$ from a cell center. The rate is a local geometric phenomenon, and the sharp constant comes from the sharp derivative bound.

53 The floor-prefix approximation to the bounded Fabius function

53.1 Definition and exact moving-center displacement

The source theorem obtains F from the prefix row by evaluating its up approximation at half the argument. Define

$$A_n(x) = \mathcal{J}_n(x/2) = \frac{\sigma_{n+1}(\lfloor 2^n x\rfloor)}{2^{\binom{n}{2}}}, \qquad 0 \le x \le 1. \tag{616}$$

Let

$$\theta_n(x) = \{2^n x\} = 2^n x - \lfloor 2^n x\rfloor. \tag{617}$$

The normalized prefix value in (616) is not located at $x - 1$. Its actual center is

$$\begin{aligned}\lambda_n(x) &= \lambda_{n,\lfloor 2^n x\rfloor}\\ &= x - 1 + \delta_n(x),\end{aligned} \tag{618}$$

$$\delta_n(x) = \frac{n + 2 - 2\theta_n(x)}{2^{n+1}}. \tag{619}$$

This identity, not merely the fact that $\lambda_n(x) \to x - 1$, determines the dominant error. It also gives

$$\frac{n}{2^{n+1}} < \delta_n(x) \le \frac{n+2}{2^{n+1}}. \tag{620}$$

In terms of the omitted-tail support radius (577),

$$\delta_n(x) = r_n - \theta_n(x)h_n. \tag{621}$$

Thus the index $\lfloor 2^n x\rfloor$ compensates for the ordinary fractional cell position but not for the shortening of the finite support. The residual is roughly $r_n \sim nh_n/2$, which is much larger than either the cell width error of the histogram or the variance-scale center bias.

53.2 First- and second-order expansions

Theorem 53.1 (Uniform floor-prefix expansion). *Uniformly for $x \in [0,1]$,*

$$A_n(x) = F(x) + \delta_n(x)F'(x) + \mathcal{O}\left(\frac{n^2}{4^n}\right). \tag{622}$$

More precisely,

$$\begin{aligned} A_n(x) &= F(x) + \delta_n(x)F'(x) \\ &\quad + \left(\frac{\delta_n(x)^2}{2} - \frac{3n+4}{72\,4^n}\right)F''(x) + \mathcal{O}\left(\frac{n^3}{8^n}\right). \end{aligned} \tag{623}$$

All remainder constants are independent of x and n.

Proof. By the exact coefficient identification, $A_n(x) = H_n(\lambda_n(x))$. Apply (595) at $\lambda_n(x) = x - 1 + \delta_n(x)$, use $\mathrm{up}(x-1) = F(x)$ and the same identity for derivatives, and Taylor-expand around $x - 1$. Since $\delta_n = \mathcal{O}(n2^{-n})$, the first omitted Taylor term in (622) is $\mathcal{O}(n^2 4^{-n})$.

For (623), retain the quadratic Taylor term and the leading center-bias term. The remainders are $\mathcal{O}(\delta_n^3)$, $\mathcal{O}(n4^{-n}\delta_n)$, and $\mathcal{O}(n^2 16^{-n})$, all absorbed by $\mathcal{O}(n^3 8^{-n})$. □

The earlier elementary route supplies constants at finite level rather than only uniform asymptotic notation. They are weaker than the Fourier expansion above but useful as a direct certificate.

Proposition 53.2 (Effective floor-profile certificate). *For every $n \geq 3$ and $x \in [0,1]$,*

$$A_n(x) = F(x) + F'(x)\delta_n(x) + \frac{1}{2}F''(x)\delta_n(x)^2 + R_n^{(2)}(x), \tag{624}$$

$$|R_n^{(2)}(x)| \leq \left(\frac{4n}{3} + \frac{16}{9}\right)4^{-n} + \frac{4}{3}(n+2)^3 8^{-n}, \tag{625}$$

$$A_n(x) = F(x) + F'(x)\delta_n(x) + R_n^{(1)}(x), \tag{626}$$

$$|R_n^{(1)}(x)| \leq \left(n^2 + \frac{16}{3}n + \frac{52}{9}\right)4^{-n} < (n+3)^2 4^{-n}. \tag{627}$$

Consequently the normalized profile has the explicit uniform bound

$$\sup_{x\in[0,1]} \left|\frac{2^n}{n}\big(A_n(x) - F(x)\big) - \frac{1}{2}F'(x)\right| \leq \frac{2}{n} + \left(n + \frac{16}{3} + \frac{52}{9n}\right)2^{-n}. \tag{628}$$

Proof. Replace $A_n(x) = H_n(x - 1 + \delta_n(x))$ by up using (609), then apply Taylor's theorem with the sharp bounds $\|F''\|_\infty \leq 8$ and $\|F'''\|_\infty \leq 64$ and $\delta_n(x) \leq (n+2)2^{-n-1}$. This gives the two displayed remainders. Finally use $2^n\delta_n/n = 1/2 + (1-\theta_n)/n$, $0 < 1 - \theta_n \leq 1$, and $0 \leq F' \leq 2$ to obtain (628). □

Expanding the exact displacement $\delta_n = (n/2 + 1 - \theta_n)2^{-n}$ makes the binary phase visible.

Corollary 53.3 (Three-scale binary-phase expansion). *Uniformly for $x \in [0,1]$,*

$$\begin{aligned} A_n(x) - F(x) &= \frac{1}{2}F'(x)n2^{-n} + F'(x)\big(1 - \theta_n(x)\big)2^{-n} \\ &\quad + \frac{1}{8}F''(x)n^2 4^{-n} + \mathcal{O}(n4^{-n}). \end{aligned} \tag{629}$$

For dyadic x, the phase $\theta_n(x)$ vanishes eventually. For a general real x, it is the shifted binary tail rather than a periodic function of n.

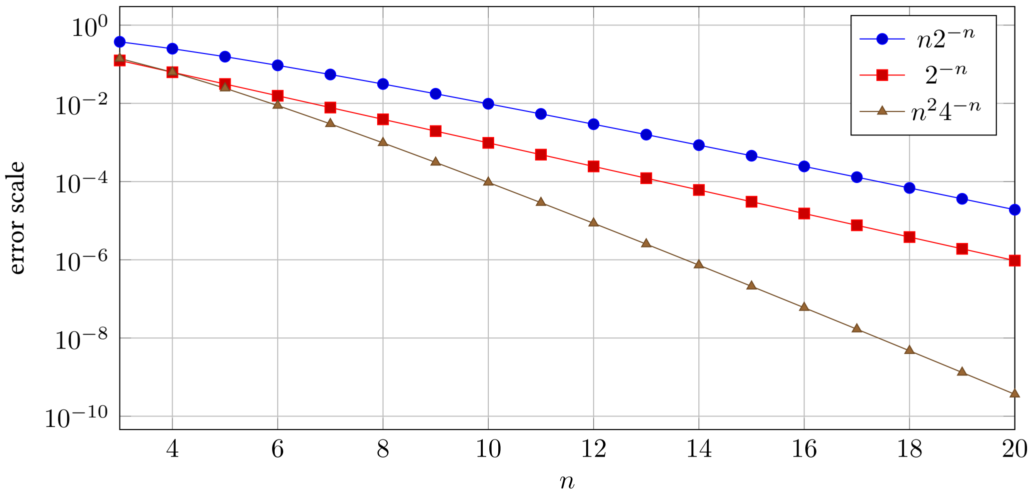

Figure 4: The three scales in (629). The deterministic center drift $n2^{-n}$ dominates the one-cell scale 2^{-n} and the quadratic Taylor scale $n^2 4^{-n}$.

Corollary 53.4 (Uniform profile and sharp norm rate)**.** *The entire rescaled error profile converges uniformly:*

$$\left\| \frac{2^{n+1}}{n}(A_n - F) - F' \right\|_\infty \longrightarrow 0. \tag{630}$$

Consequently,

$$\lim_{n\to\infty} \frac{2^n}{n} \|A_n - F\|_\infty = 1. \tag{631}$$

For each fixed x,

$$\lim_{n\to\infty} \frac{2^{n+1}}{n} \big(A_n(x) - F(x)\big) = F'(x). \tag{632}$$

Proof. By (619),

$$\frac{2^{n+1}}{n}\delta_n(x) = 1 + \frac{2 - 2\theta_n(x)}{n} \longrightarrow 1$$

uniformly in x. Multiplying the remainder in (622) by $2^{n+1}/n$ gives $\mathcal{O}(n2^{-n})$. This proves (630). Taking norms and using $\|F'\|_\infty = 2$ gives (631). □

53.3 The equivalent statement for the up sample

The source notation uses

$$\mathcal{J}_n(u) = \frac{\sigma_{n+1}(\lfloor 2^{n+1}u \rfloor)}{2^{\binom{n}{2}}}. \tag{633}$$

For $0 \le u < 1$ put $\vartheta_n(u) = \{2^{n+1}u\}$ and

$$d_n(u) = \frac{n + 2 - 2\vartheta_n(u)}{2^{n+1}}. \tag{634}$$

Then

$$\mathcal{J}_n(u) = \mathrm{up}(2u - 1) + d_n(u)\,\mathrm{up}'(2u - 1) + \mathcal{O}\left(\frac{n^2}{4^n}\right). \tag{635}$$

At $u = 1$, the prefix coefficient identity is outside its range and the exact zero-run value is $-2^{-\binom{n}{2}}$; since $\mathrm{up}(1) = \mathrm{up}'(1) = 0$, this exceptional endpoint is much smaller than the uniform principal scale and does not change the norm asymptotic. At the symmetric center one has the exact value

$$\mathcal{J}_n(1/2) = 1 - 2^{-\binom{n}{2}}. \tag{636}$$

The bounded floor sample has superexponentially accurate endpoints:

$$A_n(0) = 2^{-\binom{n}{2}}, \qquad A_n(1) = 1 - 2^{-\binom{n}{2}}, \tag{637}$$

$$A_n(0) - F(0) = 2^{-\binom{n}{2}}, \qquad A_n(1) - F(1) = -2^{-\binom{n}{2}}. \tag{638}$$

The first identity is the constant coefficient of P_n; the second is the sharp zero-run boundary value (561).

54 An exact midpoint identity

The sharp constant in (631) can be seen without asymptotic analysis at the single point where F' attains its maximum.

Theorem 54.1 (Exact floor-prefix error at $x = 1/2$)**.** *For every $n \geq 2$,*

$$A_n\left(\frac{1}{2}\right) = \frac{1}{2} + \frac{n+2}{2^n} - 2^{-\binom{n}{2}}. \tag{639}$$

Equivalently,

$$\frac{2^n}{n}\left(A_n\left(\frac{1}{2}\right) - \frac{1}{2}\right) = 1 + \frac{2}{n} - \frac{2^{n-\binom{n}{2}}}{n}. \tag{640}$$

Proof. Put $M = 2^{n-1}$ and write $Q = P_{n-1} = \sum_{r=0}^{d} q_r z^r$, where $d = g_{n-1} = 2^n - n - 1$, extending q_r by zero outside $0 \leq r \leq d$. Since $M < 2^n$, multiplication by the last factor of P_n gives

$$[z^M]P_n = \sum_{r=0}^{M} q_r =: C. \tag{641}$$

The polynomial Q is palindromic and has total mass

$$T = Q(1) = 2^{\binom{n}{2}}. \tag{642}$$

By symmetry,

$$2C - T = \sum_{r=d-M}^{M} q_r. \tag{643}$$

The band in (643) has $n+2$ coefficients. To evaluate them, write $Q = P_{n-2}(1 + \cdots + z^{2^{n-1}-1})$. Put $T_0 = P_{n-2}(1) = 2^{\binom{n-1}{2}}$. The n interior coefficients of the band are all T_0, because the convolution window contains all coefficients of P_{n-2}. Each of the two boundary coefficients is $T_0 - 1$: its window omits only the monic leading coefficient of P_{n-2}, and palindromicity gives the same value at the opposite boundary. Hence

$$2C - T = (n+2)T_0 - 2. \tag{644}$$

Since $T/T_0 = 2^{n-1}$,

$$\frac{C}{T} = \frac{1}{2} + \frac{n+2}{2^n} - \frac{1}{T}.$$

Finally, (563) and (616) identify C/T with $A_n(1/2)$, proving (639). □

Combining this exact lower witness with the elementary center certificate and Taylor's theorem gives a completely explicit finite-level bracket, complementary to the sharp asymptotic profile:

$$-2^{-\binom{n}{2}} \leq \|A_n - F\|_\infty - (n+2)2^{-n} \leq (n+3)^2 4^{-n} \qquad (n \geq 3). \tag{645}$$

The exact identity also explains an otherwise puzzling numerical observation: after a few levels, the error is almost indistinguishable from the tangent displacement $2\delta_n(1/2) = (n+2)2^{-n}$. The remaining defect is the super-exponentially small term $2^{-\binom{n}{2}}$.

Table 22: Exact midpoint error and its sharp-rate normalization.

n	$A_n(1/2) - 1/2$	$2^n\big(A_n(1/2) - 1/2\big)/n$
2	0.5	1.000000000000
3	0.5	1.333333333333
4	0.359375	1.437500000000
5	0.2177734375	1.393750000000
6	0.124969482421875	1.333007812500
8	0.0390624962747097	1.249999880791
10	0.0117187499999716	1.199999999997
12	0.0034179687500000	1.166666666667
16	0.0002746582031250	1.125000000000
20	0.00002098083496094	1.100000000000

54.1 Correct centering can be exact at the same point

The large error in (639) is not an unavoidable defect of the prefix row. It is caused by reading the wrong index as the point $-1/2$ on the up axis. When n is even, $-1/2$ itself is a lattice center, with index

$$m_n^* = 2^{n-1} - \frac{n+2}{2}. \tag{646}$$

For this index, $m_n^* = (g_{n-1} - 1)/2$. Since P_{n-1} is palindromic of odd degree, exactly half of its total coefficient mass lies at indices at most m_n^*. The last-factor prefix identity therefore gives

$$H_n(-1/2) = \frac{1}{2} = \mathrm{up}(-1/2) \qquad (n \text{ even}). \tag{647}$$

The floor rule instead uses $M = 2^{n-1}$, which is $(n+2)/2$ whole cells to the right of m_n^*. This is the discrete form of the support drift.

55 Bias removal and accelerated prefix reconstructions

The expansion (594) can be used constructively. It tells us how to remove the leading deconvolution bias using only neighboring prefix coefficients.

For a function on the center lattice, define the centered second difference

$$\Delta_h^2 H(\lambda) = \frac{H(\lambda + h_n) - 2H(\lambda) + H(\lambda - h_n)}{h_n^2}. \tag{648}$$

Put

$$c_n = \frac{3n+4}{72} h_n^2 \tag{649}$$

and define the corrected grid

$$H_n^{[1]}(\lambda) = H_n(\lambda) + c_n \Delta_h^2 H_n(\lambda). \tag{650}$$

Every value in (650) is a rational linear combination of three normalized iterated-prefix values.

Theorem 55.1 (One-step deconvolution acceleration)**.** *Uniformly on the center lattice,*

$$H_n^{[1]}(\lambda) - \mathrm{up}(\lambda) = -\frac{n^2}{1152\,16^n}\,\mathrm{up}^{(4)}(\lambda) + o\left(\frac{n^2}{16^n}\right). \tag{651}$$

Consequently,

$$\lim_{n\to\infty} \frac{16^n}{n^2} \sup_{\lambda \in h_n(\mathbb{Z} - g_n/2)} |H_n^{[1]}(\lambda) - \mathrm{up}(\lambda)| = \frac{8}{9}. \tag{652}$$

Proof. For a smooth function,

$$\Delta_h^2 f = f'' + \frac{h_n^2}{12} f^{(4)} + \mathcal{O}(h_n^4). \tag{653}$$

Write (594) as $H_n = \mathrm{up} - c_n \, \mathrm{up}'' + d_n \, \mathrm{up}^{(4)} + \mathcal{O}(n^3 h_n^6)$, where

$$d_n = \left(\frac{15n+16}{43200} + \frac{(3n+4)^2}{10368} \right) h_n^4.$$

Substitute this into (650) and use (653). If the uniform remainder is denoted by R_n, then $\|\Delta_h^2 R_n\|_\infty \leq 4h_n^{-2}\|R_n\|_\infty$; hence $c_n \Delta_h^2 R_n = \mathcal{O}(n^4 h_n^6) = o(n^2 h_n^4)$. Thus the remainder remains negligible at the claimed scale. The second-derivative terms cancel, and the coefficient of $\mathrm{up}^{(4)}$ is

$$d_n - c_n^2 + \frac{c_n h_n^2}{12} = -\frac{n^2}{1152} h_n^4 + \mathcal{O}(n h_n^4).$$

This proves (651). By (554), $\| \, \mathrm{up}^{(4)} \, \|_\infty = 2^{10} = 1024$, so the sharp norm constant is $1024/1152 = 8/9$. □

Higher corrections follow by replacing the reciprocal-tail multiplier with successively longer polynomials in the lattice Laplacian. The all-orders expansion (604) supplies the coefficients recursively. To preserve the accelerated rate in a continuous reconstruction, one should use interpolation of sufficiently high order: piecewise-linear interpolation has an $\mathcal{O}(h_n^2)$ geometric error and would mask the $\mathcal{O}(n^2 h_n^4)$ corrected lattice error, whereas local cubic interpolation has $\mathcal{O}(h_n^4)$ error and retains the latter as the dominant term.

56 Conceptual synthesis

56.1 One row, three coordinate choices

The normalized integer row

$$m \longmapsto \frac{\sigma_{n+1}(m)}{2^{\binom{n}{2}}} \tag{654}$$

contains highly accurate samples, but the meaning of m must be specified.

1. The intrinsic coordinate is the center $\lambda_{n,m} = h_n(m - g_n/2)$. At that coordinate the row differs from up by a curvature term of order nh_n^2.
2. Holding a center value constant across its width-h_n cell changes the coordinate by at most $h_n/2$. The target has maximal slope 2, so the histogram error is h_n.
3. Replacing the intrinsic coordinate by the naive dyadic coordinate implicit in $m = \lfloor 2^n x \rfloor$ overlooks the support deficit r_n. It changes the coordinate by roughly $nh_n/2$, so the Fabius error is roughly $nh_n F'(x)/2$.

The three rates are therefore compatible, not contradictory. They measure distinct reconstructions of the same discrete data.

56.2 Variance versus support radius

The center expansion sees the omitted tail only through its even cumulants. Its mean is zero, so the first nonzero effect is its variance $v_n \asymp nh_n^2$. The floor-coordinate drift, by contrast, sees the extreme support loss $r_n \asymp nh_n$. This is why a centered analytic argument is exponentially more accurate than a support-aligned but miscentered index rule.

This distinction is common in lattice approximation. Weak convergence controls averages, and moment matching controls smooth centered observables, but an endpoint-derived coordinate can retain a worst-case support defect long after the bulk distribution has become extremely close.

56.3 Why the n is structural

The degree

$$g_n = 2^{n+1} - n - 2 \tag{655}$$

is shorter than the naive value 2^{n+1} by exactly $n + 2$. After scaling by 2^{n+1} and centering, half of this deficit appears at each endpoint. Hence the factor n in the floor rate is already encoded in the elementary degree formula for P_n. No delicate property of F is needed to predict its presence; the analytic work is needed to turn the horizontal deficit into the sharp vertical profile F' and to control the smaller terms.

56.4 A practical recommendation

For numerical reconstruction from a computed prefix row:

- do not use $m/2^n$ as the physical coordinate;
- attach each value to $\lambda_{n,m}$ from (565);
- use at least linear interpolation rather than a histogram;
- if another factor of roughly $4^n/n$ is valuable, apply the three-point correction (650) and use cubic interpolation.

The coefficients remain exact rationals until the final interpolation or evaluation stage.

57 A Lean formalization roadmap

The new proof separates naturally into reusable lemmas. A formalization need not begin with the full asymptotic theorem.

1. **Shifted-lattice Fourier inversion.** Formalize theorem 50.1 for a finitely supported measure on $a + h\mathbb{Z}$. This is a finite-sum orthogonality calculation.
2. **Rademacher realization.** Refine the existing finite-measure bridge to the triangular Rademacher array (571); obtain (572) and the independent tail decomposition (575).
3. **Exact geometric tails.** Add the closed forms (577), (578), (590), and (591). These reduce to finite manipulations of geometric series.
4. **Uniform $\log\sec$ remainder.** Prove a sixth-order Taylor bound on $[-1/4, 1/4]$ and use it scale by scale. The low-frequency cutoff can be chosen as $2^{n/4}$ or replaced by any convenient dyadic integer bound.
5. **Rapid Fourier decay.** Package repeated integration by parts for compactly supported C^∞ functions and instantiate it with the already formalized sharp derivative bounds.
6. **Center expansion.** Combine the preceding lemmas with Fourier derivative identities. The first-order version (595) is substantially shorter than the fourth-order theorem and already proves all three basic rates.
7. **Exact drift.** Formalize (618) by integer-floor arithmetic. Then (622) is a direct Taylor estimate.
8. **Midpoint identity.** Prove theorem 54.1 independently of Fourier analysis from palindromicity and the central coefficient plateau. It is an especially useful regression test for index conventions.

The formal theorem statements should distinguish three domains: values at the center lattice, the step function on all reals, and the floor sample on $[0, 1]$. Combining them into one statement would obscure exactly the coordinate issue responsible for the different rates.

58 Conclusion

Iterated summation of the signed Thue–Morse sequence does more than produce a sequence that happens to resemble the Fabius function. After the correct one-order shift, its finite row is an exact lattice deconvolution of up. The omitted part is an explicit triangular Bernoulli tail with support radius $(n+2)2^{-n-1}$ and variance $(3n+4)/(36\,4^n)$. Those two exact numbers govern two different approximation regimes.

At the natural centers, the mean-zero tail leaves a variance-scale curvature bias, yielding the sharp rate $n/(3\,4^n)$. Holding the values constant across cells introduces the larger sharp rate 2^{-n}. Reading the same row at the naive dyadic floor leaves the full support deficit as a coordinate drift and yields the still larger sharp rate $n2^{-n}$. The latter has the uniform error profile F' and an exact midpoint formula.

The main conceptual correction is therefore simple but consequential: the integer index is not itself the continuum coordinate. Once the genuine center map is retained, the Thue–Morse prefix construction is far more accurate than its original pointwise formulation suggests, and its error admits a complete, explicitly computable differential expansion.

Supplementary material from this synthesis

59 Elementary geometric-tail formulas

For $|q| < 1$ and $n \geq 0$,

$$\sum_{\ell=n+1}^{\infty} \ell q^{\ell} = q \frac{\mathrm{d}}{\mathrm{d}q} \left(\frac{q^{n+1}}{1-q} \right) = q^{n+1} \frac{(n+1) - nq}{(1-q)^2}. \tag{656}$$

The specializations used in the text are

$$\sum_{\ell > n} \ell 2^{-\ell} = (n+2) 2^{-n}, \tag{657}$$

$$\sum_{\ell > n} \ell 4^{-\ell} = \frac{3n+4}{9\, 4^n}, \tag{658}$$

$$\sum_{\ell > n} \ell 16^{-\ell} = \frac{15n+16}{225\, 16^n}, \tag{659}$$

$$\sum_{\ell > n} \ell 64^{-\ell} = \frac{63n+64}{3969\, 64^n}. \tag{660}$$

These identities make every coefficient in the all-orders expansion elementary: no unevaluated infinite tail sum is necessary.

60 A short exact-arithmetic algorithm

The coefficient row $[z^m]P_n$ can be built without polynomial multiplication. Multiplication by $1 + \cdots + z^{2^i - 1}$ is a sliding-window sum. The following pseudocode maintains exact integers and costs $\mathcal{O}(2^n)$ arithmetic operations per level.

```
coeff := [1]
for i := 1,...,n:
    width := 2^i
    nextLength := length(coeff) + width - 1
    next := array(nextLength, 0)
    window := 0
    for m := 0,...,nextLength-1:
        if m < length(coeff):
            window := window + coeff[m]
        if m >= width:
            window := window - coeff[m-width]
        next[m] := window
    coeff := next

# Normalized prefix sample:
H_n(m) := coeff[m] / 2^(n(n-1)/2)
# Physical coordinate:
lambda_n(m) := (2m - (2^(n+1)-n-2)) / 2^(n+1)
```

The midpoint identity (639), symmetry $\texttt{coeff}[\mathtt{m}] = \texttt{coeff}[\mathtt{g_n} - \mathtt{m}]$, the total mass $2^{\binom{n+1}{2}}$, and the central maximum $2^{\binom{n}{2}}$ provide independent exact checks of an implementation.

Topic references

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Part VI
Inverse and Small-Argument Saddle Analysis

Source provenance and status. This topic synthesizes the former `Fabius_Inverse_and_Saddle_Research_Frontiers/Fabius_Inverse_and_Saddle_Research_Frontiers.tex` (historically recorded SHA-256 `f9d8605761aaaa1b2c2af83e3c5c55dcd6acfc847402163458b03b68c7b35ff8`; the audit did not reproduce it from a reachable whole-file blob, whose authoritative archived Git identifier is `7b5cd9f7887de95236045de21c176d77962721bb`) and `Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex` (SHA-256 `85f51a20fc7b6bdf3b1d049ec4506f508aee3c4cd70554e6a68cbcc30977cb0b`). The inverse dossier supplies the canonical mathematical backbone; the specialist phase-polynomial and right-endpoint transfers, exact coefficient means, higher coefficients, Fourier-amplitude analysis, numerical checks, traps, and open obligations are retained from the small-argument notebook. Formalized inputs are distinguished from derived and numerical consequences throughout.

Topic abstract

The Fabius function is unusually rich: a single compactly supported C^∞ density produces a self-similar distribution function, exact dyadic arithmetic, Fourier products, a probability model, and a small-argument expansion whose coefficients are smooth periodic functions. The companion article *The Fabius Function and Rvachev's Up-Function* develops most of that landscape. The present paper is deliberately different. It does not repeat the introductory theory. Instead it fills two identifiable gaps in the exposition by turning recently formalized material into ordinary mathematics.

The first gap is the inverse. We construct the totalized inverse $G = F^{-1}$ on the whole real line, prove its global monotonicity, continuity, reflection law, comparison calculus, and exact values at all outputs of the dyadic evaluator, and show how every flatness estimate for F becomes an effective steepness estimate for G. We then derive the classical interior calculus of the inverse and invert the sharp Lambert-phase asymptotic. The result is the explicit endpoint law

$$G(y) \sim \frac{\sqrt{\log(1/y)}}{\mathrm{e}\sqrt{\log 2}} \exp\Big(-\sqrt{2\log 2\, \log(1/y)}\Big) \qquad (y \downarrow 0),$$

together with its reflected analogue at 1. This explains in one formula why the inverse is steeper than every positive power of y, although it still tends to zero faster than every negative power of $\log(1/y)$.

The second gap concerns the internal machinery of the all-orders saddle expansion. We give complete derivations of the closed periodic jet formula, its shifted signed-Stirling conversion, the collapse of the centered exponent coefficients, the formal mass-to-logarithm recursions, the leading-derivative law, and the full second logarithmic correction. Finally, we give a proof-level account of the arbitrary-order analytic remainder: the order-dependent central window, vertical Taylor control, finite exponential substitution, Gaussian contraction, polynomial-tail replacement, minor-arc estimates, logarithmic transfer, and return from the dyadic logarithmic scale to $x \downarrow 0$.

Statements explicitly represented by theorem declarations in the Lean development are marked Lean. Consequences proved here from those statements by standard real or analytic arguments are marked Derived.

Formalization boundary. This document is a research-frontier notebook, not part of the project's formalization-backed primary exposition. It deliberately preserves a mixture of proved Lean inputs, ordinary mathematical deductions, symbolic coefficient calculations, and numerical evidence so that the unformalized steps are not lost. A statement here is a project theorem only when an exact current Lean declaration is named for the full statement; the labels LEAN and DERIVED record the draft's provenance but do not themselves certify that every displayed consequence has been formalized. In particular, Hölder consequences beyond the now-formalized inverse-curvature identities, inverse endpoint asymptotics, the ordered-composition coefficient formulas and the logarithmic highest-derivative law remain research obligations.

61 Purpose, scope, and status conventions

61.1 A supplement rather than a second survey

Let F denote the bounded, totalized Fabius function: $F(x) = 0$ for $x \leq 0$, $F(x) = 1$ for $x \geq 1$, and on $[0, 1]$ it is the unique strictly increasing C^∞ function associated with Rvachev's compactly supported up-function. We use only a small portion of the foundational theory:

$$F(0) = 0, \qquad F(1) = 1, \qquad F(1-x) = 1 - F(x), \tag{661}$$

$$F'(x) = 2\,\mathrm{up}(2x-1), \qquad 0 < x < 1, \tag{662}$$

where up is positive on $(-1, 1)$ and $\mathrm{up}(0) = 1$. Consequently F is a continuous strictly increasing bijection of $[0, 1]$ onto itself. The exact second derivative is now also a named formal input:

$$F''(x) = 8\big(\mathrm{up}(4x-1) - \mathrm{up}(4x-3)\big).$$

Its positive, negative, and zero loci are respectively $(0, 1/2)$, $(1/2, 1)$, and $\mathbb{R} \setminus (0, 1) \cup \{1/2\}$, by `deriv_deriv_fabiusReal_pos_iff`, `deriv_deriv_fabiusReal_neg_iff`, and `deriv_deriv_fabiusReal_eq_zero_iff`.

The main companion article already explains the random-series model, infinite convolutions, the Fourier transform, moment recurrences, Thue–Morse identities, dyadic rationality, and the broad shape of the Lambert saddle analysis. Repeating those chapters would obscure the missing material. The division of labor adopted here is summarized in table 23.

Table 23: Expository audit and the role of this companion.

Topic	Situation in the main article	Treatment here
Inverse function	The current article summarizes the totalized inverse and its endpoint estimates.	Full construction, exact identities, dyadic inversion, effective steepness, endpoint limits, regularity, and inverse asymptotics.
Closed saddle jets	Closed formulas are displayed, but the operator calculus and the shifted Stirling conversion are compressed.	Complete induction, forcing-term propagation, coefficient table, and the reason for the index shift $s(n+1, m+1)$.

Topic	Situation in the main article	Treatment here
Mass and logarithmic coefficients	The all-orders result is stated and some coefficients are listed.	Formal exponential recursion, Gaussian contraction, composition formulas, structural bounds, and the leading derivative law.
Second correction	The final polynomial appears, while much of the order-four algebra is easy to miss.	A line-by-line calculation from $E_1, \ldots, E_4$ through B_4, Gaussian moments, a_2, and A_2.
Arbitrary-order remainder	The main article gives a high-level proof architecture.	The order-dependent contour window and every analytic error mechanism are separated and justified.
Probability, Fourier theory, dyadic arithmetic	Already developed in substantial depth.	Used as input only; not duplicated.

61.2 What the labels mean

A paragraph headed Lean restates a mathematical theorem whose content is represented directly in the Lean files under

`Analysis/FabiusFunction/Lean/FabiusFunction`.

A paragraph headed Derived gives a paper-level consequence. Its proof uses formalized results plus a standard theorem such as the inverse function theorem, ordinary asymptotic inversion, or elementary formal power-series algebra. The distinction matters especially in sections 65 and 69: those sections contain new synthesis, but they do not pretend that every displayed corollary has a declaration of its own.

> The exact Lambert phase is never silently replaced inside a periodic coefficient. If $\lambda = \lambda(x)$ is the exact lower solution of $x = \lambda 2^{-\lambda}$, then a term such as $A_j(\lambda)$ remains at that exact phase. Replacing λ by an asymptotic approximation inside A_j is a separate composition problem, because a bounded phase error does not become small modulo the period.

61.3 Local saddle notation and its crosswalk

Three namings for the same principal saddle objects appear across the specialist source, the Wikipedia draft, and the Lean development:

this synthesis / the primary article	Wikipedia draft	Lean object
$\lambda(x)$	λ	`fabiusLambertPhasex`
Ψ	Ψ	`negativeLaplacePsi`
$\mathcal{M}(x)$	main term	`fabiusSharpLambertMainx`
A_j	P_j	`fabiusSaddleLogCoefficientj`
a_j	b_j	`fabiusSaddleMassCoefficientj`
A_1	P_1	`fabiusFirstSaddleCorrection`
A_2	P_2	`fabiusSecondSaddleCorrection`
C_F	C_F	`fabiusSharpAsymptoticConstant`

The Wikipedia draft calls the coefficient functions merely continuous. The direct regularity result theorem 69.7 makes them smooth, one-periodic, and bounded, while sections 67 and 69 constructs them from smooth periodic jets.

The local jets and exponent coefficients have no counterpart in the Wikipedia draft. The primary article gives them different letters because p, d, and E are already used there: p_j is the Lambert phase polynomial and p_m the binary prefix, d_n is the half-moment sequence, and E denotes both the tail error and the exponential quotient. The exact correspondence is

Local symbol	Main-exposition symbol	Lean object
p_n	Z_n	`negativeLaplacePeriodicJet`
d_n	$\widetilde{Z}_n$	`negativeLaplaceBoundedExponentJet`
E_m	Ξ_m	`negativeLaplaceExponentCoefficient`
$\mathrm{d}/\,\mathrm{d}t$	D	—

The tilde in $\widetilde{Z}_n$ is apt: the periodic and bounded jet families differ by the single constant $(-1)^{n+1}n!$, by (686). In the derivative row, t is the independent variable used by the former specialist source; the consolidated synthesis otherwise writes that variable as u and defines $D = \mathrm{d}/\,\mathrm{d}u$. In this final part, d_n is a bounded saddle jet and is not the half-moment sequence $d_n = \mathbb{E}X^n$ used in the probability and dyadic parts. When numerical work later returns to half moments, it writes them as h_n to keep the two roles visibly separate.

62 The inverse as a global mathematical object

62.1 The order isomorphism on the unit interval

Set

$$I = [0, 1], \qquad \mathrm{clamp}_{[0,1]}(y) = \min\{1, \max\{0, y\}\}.$$

Because $F|_I$ is a continuous strictly increasing bijection $I \to I$, it is an order isomorphism. Its ordinary inverse will first be denoted by $F_I^{-1} : I \to I$.

Definition 62.1 (Totalized inverse)**.** The totalized inverse of the Fabius function is

$$G(y) = F_I^{-1}\big(\mathrm{clamp}_{[0,1]}(y)\big), \qquad y \in \mathbb{R}.$$

Thus the inverse is extended by constants outside its natural range, exactly as F itself is extended by constants outside its natural domain.

Theorem 62.2 (Global inverse package; Lean)**.** *The function G has the following properties.*

1. $0 \le G(y) \le 1$ *for every* $y \in \mathbb{R}$.
2. *If* $0 \le y \le 1$, *then*
$$F(G(y)) = y.$$
If $0 \le x \le 1$, *then*
$$G(F(x)) = x.$$
3. G *is strictly increasing on* $[0, 1]$, *monotone on* $\mathbb{R}$, *and continuous on* $\mathbb{R}$.
4. *The restriction* $G|_I$ *is a bijection* $I \to I$.

Proof. The clamp maps $\mathbb{R}$ continuously and monotonically into I. The inverse of an order isomorphism between compact real intervals is itself an order isomorphism, hence is continuous and strictly increasing. Their composition is therefore continuous and monotone on the whole line and takes values in I. If $y \in I$, then $\mathrm{clamp}_{[0,1]}(y) = y$, so the usual right-inverse identity for F_I^{-1} gives $F(G(y)) = y$. Likewise, $F(x) \in I$ for $x \in I$, and the left-inverse identity gives $G(F(x)) = x$. Strict increase and bijectivity on I are inherited from F_I^{-1}. ☐

The totalization is not cosmetic. It produces a globally continuous monotone function and makes the reflection law valid without restricting the variable to $[0, 1]$.

62.2 A four-way comparison calculus

The order-isomorphism viewpoint turns inverse inequalities into direct inequalities without any numerical inversion.

Theorem 62.3 (Comparison equivalences; Lean)**.** *For $x, y \in [0, 1]$,*

$$G(y) \le x \iff y \le F(x), \tag{663}$$
$$F(x) \le y \iff x \le G(y), \tag{664}$$
$$G(y) < x \iff y < F(x), \tag{665}$$
$$F(x) < y \iff x < G(y). \tag{666}$$

Proof. Apply the strictly increasing function F to the two sides of an inequality involving $G(y)$, then use $F(G(y)) = y$. For example,

$$G(y) \le x \iff F(G(y)) \le F(x) \iff y \le F(x).$$

The other three equivalences are identical, with the order reversed in placement but not in direction because F and G are increasing. ☐

Remark 62.4. The equivalences are often more useful than an explicit inverse formula. They let one convert a known dyadic or asymptotic bound for F into a location bound for G with no loss in constants.

62.3 Tails, endpoints, and reflection

Theorem 62.5 (Constant tails and symmetry; Lean)**.** *For every $y \in \mathbb{R}$,*

$$y \le 0 \Longrightarrow G(y) = 0, \tag{667}$$
$$y \ge 1 \Longrightarrow G(y) = 1, \tag{668}$$
$$G(1 - y) = 1 - G(y). \tag{669}$$

In particular,

$$G(0) = 0, \qquad G(1) = 1, \qquad G(1/2) = 1/2.$$

Proof. The two tail identities follow immediately from the clamp. Suppose first that $0 \le y \le 1$. Then $G(y) \in [0, 1]$, and by the symmetry of F,

$$F(1 - G(y)) = 1 - F(G(y)) = 1 - y.$$

The point $1 - G(y)$ lies in $[0, 1]$, so uniqueness of the inverse gives $G(1 - y) = 1 - G(y)$. If $y \le 0$ or $y \ge 1$, the same equation follows from the two constant tails. Setting $y = 0, 1,$ and $1/2$ gives the stated special values. ☐

63 Exact inversion of the dyadic evaluator

63.1 The exact statement

Let $\mathsf{F}_n(a)$ denote the exact bounded dyadic evaluator from the formal development: for $0 \le a \le 2^n$,

$$\mathsf{F}_n(a) = F\left(\frac{a}{2^n}\right),$$

and for larger a it is clamped at the endpoint value 1.

Theorem 63.1 (Dyadic inversion; Lean). *For all* $n, a \in \mathbb{N}$,

$$G\big(\mathsf{F}_n(a)\big) = \frac{\min\{a, 2^n\}}{2^n}.$$

In particular, if $0 \le a \le 2^n$*, then*

$$G\left(F\left(\frac{a}{2^n}\right)\right) = \frac{a}{2^n}.$$

Proof. When $a \le 2^n$, the argument $a/2^n$ lies in I, so the left-inverse identity in theorem 62.2 applies. When $a \ge 2^n$, the bounded evaluator equals 1, the minimum equals 2^n, and both sides are 1. □

The theorem is stronger than a table of examples: every exact rational produced by the dyadic algorithm immediately gives an exact algebraic input for the inverse.

63.2 A small exact table

x	$F(x)$	exact inverse statement
0	0	$G(0) = 0$
$\frac{1}{16}$	$\frac{143}{2073600}$	$G\left(\frac{143}{2073600}\right) = \frac{1}{16}$
$\frac{1}{8}$	$\frac{1}{288}$	$G\left(\frac{1}{288}\right) = \frac{1}{8}$
$\frac{1}{4}$	$\frac{5}{72}$	$G\left(\frac{5}{72}\right) = \frac{1}{4}$
$\frac{3}{8}$	$\frac{73}{288}$	$G\left(\frac{73}{288}\right) = \frac{3}{8}$
$\frac{1}{2}$	$\frac{1}{2}$	$G\left(\frac{1}{2}\right) = \frac{1}{2}$
$\frac{5}{8}$	$\frac{215}{288}$	$G\left(\frac{215}{288}\right) = \frac{5}{8}$
$\frac{3}{4}$	$\frac{67}{72}$	$G\left(\frac{67}{72}\right) = \frac{3}{4}$
$\frac{7}{8}$	$\frac{287}{288}$	$G\left(\frac{287}{288}\right) = \frac{7}{8}$
$\frac{15}{16}$	$\frac{2073457}{2073600}$	$G\left(\frac{2073457}{2073600}\right) = \frac{15}{16}$

Table 24: Exact inverse values obtained from the dyadic evaluator and reflection.

Remark 63.2. The threshold $F(1/4) = 5/72$ will reappear in the effective endpoint steepness theorem. Its occurrence is structural rather than arbitrary: $1/4 = 2^{-2}$ is precisely the scale on which the quadratic flatness bound is valid.

64 Flatness of F becomes steepness of G

64.1 The transported power inequalities

Write

$$C_n = 2^{\binom{n+1}{2}}.$$

The effective flatness theorem for F says, in one of its standard forms, that

$$0 \le x \le 2^{-n} \quad \Longrightarrow \quad F(x) \le C_n x^n.$$

The sharp version improves the right-hand side by the factor $n!$.

Theorem 64.1 (Inverted effective and sharp flatness; Lean)**.** *Let $n \in \mathbb{N}$ and $0 \le y \le 1$. If*

$$2^n G(y) \le 1,$$

then

$$y \le C_n G(y)^n, \tag{670}$$

$$y \le \frac{C_n}{n!} G(y)^n. \tag{671}$$

Moreover, the scale condition follows from the argument condition

$$0 \le y \le F(2^{-n}).$$

Proof. Put $x = G(y)$. The scale hypothesis says $0 \le x \le 2^{-n}$, while $F(x) = y$. Substituting x into the effective and sharp flatness bounds for F gives (670) and (671).

Now suppose $y \le F(2^{-n})$. By (663), $G(y) \le 2^{-n}$, which is exactly the required scale condition. □

For $n \ge 1$, taking positive nth roots yields the more geometric form.

Corollary 64.2 (Root lower bounds; Lean input, Derived form)**.** *If $n \ge 1$, $0 < y \le F(2^{-n})$, then*

$$G(y) \ge C_n^{-1/n} y^{1/n}, \tag{672}$$

$$G(y) \ge \left(\frac{n!}{C_n}\right)^{1/n} y^{1/n}. \tag{673}$$

Proof. All quantities are nonnegative. Rearrange equations (670) and (671) and apply the monotonicity of the positive nth-root function. □

64.2 An effective bound against every line

The quadratic case already detects infinite endpoint slope.

Theorem 64.3 (Effective linear steepness; Lean)**.** *Let $M, y \in \mathbb{R}$. If*

$$0 < y, \qquad y \le F(1/4) = \frac{5}{72}, \qquad 8M^2 y < 1,$$

then

$$My < G(y).$$

Proof. The comparison theorem gives $G(y) \le 1/4$, so the quadratic inverse flatness estimate applies:

$$y \le 8G(y)^2.$$

If $M \le 0$, then $My \le 0 < G(y)$ because $y > 0$ and G is strictly increasing on $[0, 1]$. Suppose $M > 0$ and, toward a contradiction, $G(y) \le My$. Then

$$y \le 8G(y)^2 \le 8M^2 y^2.$$

Dividing by $y > 0$ gives $1 \le 8M^2 y$, contrary to the last hypothesis. □

64.3 Infinite one-sided slope

Theorem 64.4 (Endpoint quotient divergence; Lean). *As $y \downarrow 0$,*

$$\frac{G(y)}{y} \longrightarrow +\infty.$$

By reflection, as $y \uparrow 1$,

$$\frac{1 - G(y)}{1 - y} \longrightarrow +\infty.$$

Proof. Fix a real target A. Put $B = |A| + 1$. For all sufficiently small positive y we have simultaneously

$$y < F(1/4), \qquad 8B^2 y < 1.$$

By theorem 64.3, $By < G(y)$, and hence

$$A \le B < \frac{G(y)}{y}.$$

This is exactly the filter definition of divergence to $+\infty$.

For the right endpoint use (669):

$$\frac{1 - G(y)}{1 - y} = \frac{G(1 - y)}{1 - y},$$

and let $1 - y \downarrow 0$. □

64.4 Failure of every positive Hölder estimate

Theorem 64.5 (Stronger than non-Lipschitz; Derived). *For every $\alpha > 0$,*

$$\frac{G(y)}{y^\alpha} \longrightarrow +\infty \qquad (y \downarrow 0).$$

Consequently there are no $C, \delta > 0$ for which

$$G(y) \le C y^\alpha \qquad (0 \le y \le \delta).$$

The reflected assertion holds at 1:

$$\frac{1 - G(y)}{(1 - y)^\alpha} \longrightarrow +\infty \qquad (y \uparrow 1).$$

Proof. Choose an integer $n > 1/\alpha$. On the window $0 < y \le F(2^{-n})$, equation (672) gives

$$\frac{G(y)}{y^\alpha} \ge C_n^{-1/n} y^{1/n - \alpha}.$$

The exponent $1/n - \alpha$ is negative, so the right-hand side tends to $+\infty$. A local Hölder upper bound would keep the quotient bounded and is therefore impossible. The statement at 1 follows from reflection. □

Each fixed n supplies a root lower bound only on the shrinking window $y \le F(2^{-n})$. There is no bound uniform in n. This is not a defect: it is exactly the scale dependence needed to prove that G outruns *every fixed* positive power.

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65 Classical calculus and the exact regularity of the inverse

This section uses the formalized inverse package together with standard one-variable analysis. Smoothness, the reciprocal first-derivative formula, the exact second-derivative formula, and the strict closed-half shape are now marked LEAN; the general higher inverse-derivative formulas remain DERIVED.

65.1 Smoothness and differential identities in the interior

Theorem 65.1 (Interior inverse calculus through first order; LEAN). *The inverse G is C^∞ on $(0,1)$. For $0<y<1$,*

$$G'(y) = \frac{1}{F'(G(y))} = \frac{1}{2\,\mathrm{up}(2G(y)-1)} > 0. \tag{674}$$

Proof. For $x \in (0,1)$, (662) and positivity of up on $(-1,1)$ imply $F'(x) > 0$. The C^∞ inverse function theorem therefore applies at every interior point and gives a smooth local inverse. Uniqueness identifies all these local inverses with G. Differentiating $F(G(y)) = y$ and substituting (662) proves (674). □

The corresponding Lean declarations are `fabiusInv_contDiffOn_Ioo`, `fabiusInv_hasDerivAt`, `deriv_fabiusInv_eq_inv_two_mul_rvachevUp`, and `deriv_fabiusInv_pos`.

Remark 65.2 (Current global smoothness locus; LEAN). The totalized inverse has locally constant tails. The newer declarations `fabiusInv_differentiableAt_iff` and `fabiusInv_contDiffAt_infty_iff` sharpen the interior statement: the inverse is differentiable, and indeed C^∞, exactly at the points $y \neq 0, 1$. At each clamping endpoint it is not differentiable. More generally, `fabiusInv_contDiffAt_iff` says that order-zero smoothness is global, whereas every positive finite order and C^∞ have precisely those two exceptions. Its order side condition deliberately excludes Mathlib's stronger analytic order; the analytic locus is formalized separately by `fabiusInv_analyticAt_iff` in `InverseNotElementary.lean`.

Proposition 65.3 (Second inverse derivative; LEAN). *For $0<y<1$,*

$$G''(y) = -\frac{F''(G(y))}{F'(G(y))^3}. \tag{675}$$

Proof. Differentiating the identity behind (674) once more gives

$$F''(G(y))G'(y)^2 + F'(G(y))G''(y) = 0.$$

Substitution of (674) yields (675). This is `deriv_fabiusInv_hasDerivAt` and its pointwise corollary `deriv_deriv_fabiusInv`. □

Remark 65.4 (Higher inverse derivatives; DERIVED). Repeated differentiation expresses every higher derivative of G as a rational polynomial in the derivatives of F evaluated at $G(y)$, with a power of $F'(G(y))$ in the denominator. These general Faà di Bruno inverse-derivative polynomials are not separately packaged in the Lean API.

Corollary 65.5 (Midpoint differential data; LEAN). *At the fixed point $1/2$,*

$$G(1/2) = 1/2, \qquad G'(1/2) = \frac{1}{2}, \qquad G''(1/2) = 0.$$

Proof. The value follows from reflection. Since $\mathrm{up}(0) = 1$, (674) gives $G'(1/2) = 1/2$. The up-function is even, so $\mathrm{up}'(0) = 0$, hence $F''(1/2) = 4\,\mathrm{up}'(0) = 0$; now use (675). The three exact Lean declarations are `fabiusInv_half`, `deriv_fabiusInv_half`, and `deriv_deriv_fabiusInv_half`. □

Proposition 65.6 (Exact interior curvature loci; Lean). *For* $0 < y < 1$,

$$G''(y) < 0 \iff 0 < y < 1/2, \qquad G''(y) > 0 \iff 1/2 < y < 1,$$

and

$$G''(y) = 0 \iff y = 1/2.$$

These are `deriv_deriv_fabiusInv_neg_iff`, `deriv_deriv_fabiusInv_pos_iff`, *and* `deriv_deriv_fabiusInv_eq_zero_iff`. *Their interior hypothesis is part of the statement: G is not differentiable at the clamping points.*

65.2 Shape reversal under inversion

The formal development proves that F is strictly convex on $(0, 1/2)$ and strictly concave on $(1/2, 1)$. Inversion reverses these two curvature signs.

Proposition 65.7 (Concavity and convexity of the inverse; Lean). *The function G is strictly concave on $[0, 1/2]$ and strictly convex on $[1/2, 1]$.*

Proof. The inverse is strictly increasing and preserves the midpoint. It therefore maps the first half into itself, where F' is strictly increasing and positive. Its reciprocal in (674) is strictly decreasing, so the derivative criterion and continuity give strict concavity on the closed first half. On the second half F' is strictly decreasing and positive, hence the reciprocal derivative of the inverse is strictly increasing and the same criterion gives strict convexity.

The closed-interval conclusions are `strictConcaveOn_fabiusInv_firstHalf` and `strictConvexOn_fabiusInv_secondHalf`. Their derivative criteria inspect only the two open interval interiors, while global continuity supplies the endpoints; no finite endpoint derivative is asserted. ☐

Remark 65.8 (Interior derivative blow-up; Lean). The secant slopes diverge by theorem 64.4. More strongly, the positive interior derivative satisfies

$$G'(y) \longrightarrow +\infty \quad (y \downarrow 0), \qquad G'(y) \longrightarrow +\infty \quad (y \uparrow 1).$$

These are `tendsto_deriv_fabiusInv_atTop_at_zero_right` and `tendsto_deriv_fabiusInv_atTop_at_one_left`. They are one-sided limits through the open interval, not derivative values at the nondifferentiable clamping points.

65.3 The C^∞ and analytic loci

Theorem 65.9 (Exact regularity locus; Lean). *The totalized inverse has the following regularity.*

1. *G is C^∞ precisely on*
$$\mathbb{R} \setminus \{0, 1\}.$$
2. *It has no finite derivative at* 0 *or at* 1.
3. *It is real analytic precisely on*
$$\mathbb{R} \setminus [0, 1].$$

Proof. On $(-\infty, 0)$ and $(1, \infty)$ the function is constant, hence smooth and analytic. On $(0, 1)$ smoothness follows from theorem 65.1. At 0, the left derivative is 0 because the function is constant to the left, whereas the right difference quotient $G(y)/y$ tends to $+\infty$. Thus no finite derivative exists. Reflection gives the same conclusion at 1.

It remains to exclude analyticity in the open unit interval. Suppose G were analytic near some $y_0 \in (0, 1)$. Its derivative there is nonzero by (674); the analytic inverse function theorem would then make its local inverse F analytic near $x_0 = G(y_0)$. But the Fabius function is nowhere analytic on the interior of its transition interval. This contradiction proves the claim.

The exact smoothness and differentiability classifications are `fabiusInv_contDiffAt_infty_iff` and `fabiusInv_differentiableAt_iff`; the analytic classification is `fabiusInv_analyticAt_iff`. □

65.4 Analytic-density and non-elementarity consequences

The generic analytic-locus obstruction is proved in sections 2.2 and 2.3; applying it to the exact locus above gives a fully formal inverse conclusion. The theorem `not_eqOn_fabiusInv_of_dense_analyticLocus` excludes agreement with G on any subset of $[0, 1]$ having nonempty interior whenever the comparison function has dense analytic locus. Thus `fabiusInv_not_isElementary` proves that the totalized inverse is not elementary. The stronger `not_isElementaryOrInverse_eqOn_fabiusInv` and `fabiusInv_not_isElementaryOrInverse` reach the same conclusion for the class generated by elementary functions and admissible continuous inverse branches. The branch constructor requires analyticity of the chosen total extension on the interior of the complement of its open branch domain; its conditional Lambert-style criterion is not an instantiated Mathlib Lambert W declaration. These domain qualifications are essential at branch boundaries.

66 Inverting the sharp small-argument asymptotic

The power inequalities show qualitative infinite steepness. The sharp Lambert expansion gives the actual scale. This section extracts that scale without needing the detailed value of the periodic fluctuation.

66.1 The exact lower Lambert phase

Put

$$L = \log 2.$$

For sufficiently small $x > 0$, let $\lambda = \lambda(x)$ be the large positive solution of

$$x = \lambda 2^{-\lambda} = \lambda e^{-L\lambda}. \tag{676}$$

Equivalently,

$$\lambda(x) = -\frac{1}{L} W_{-1}(-Lx),$$

but (676) is the identity used in the proofs.

The sharp expansion established for F has the form

$$\log F(x) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2}\log\lambda + C_F + \Psi(\lambda) + \mathcal{O}(\lambda^{-1}), \tag{677}$$

This is the general-real endpoint statement (539), restated here in the inverse notation. Here Ψ is smooth, bounded, and one-periodic. The exact value of C_F and the Fourier description of Ψ are important for F itself, but surprisingly they do not affect the leading equivalent of the inverse.

66.2 Asymptotic solution for the phase

Let $y = F(x)$ and

$$T = \log\frac{1}{y}.$$

Negating (677) gives

$$T = \frac{L}{2}\lambda^2 - B\lambda + \frac{1}{2}\log\lambda - C_F - \Psi(\lambda) + \mathcal{O}(\lambda^{-1}), \qquad B = 1 + \frac{L}{2}. \tag{678}$$

Lemma 66.1 (Asymptotic inversion of the quadratic phase; Derived). *As* $T \to +\infty$,

$$\lambda = \sqrt{\frac{2T}{L}} + \frac{B}{L} + \mathcal{O}\left(\frac{\log T}{\sqrt{T}}\right).$$

Proof. Since Ψ is bounded, (678) first gives

$$T = \frac{L}{2}\lambda^2 + \mathcal{O}(\lambda + \log \lambda).$$

Hence $\lambda \to \infty$ and, with

$$q = \sqrt{\frac{2T}{L}},$$

we have $\lambda/q \to 1$. Dividing

$$T - \frac{L}{2}\lambda^2 = -\frac{L}{2}(\lambda - q)(\lambda + q)$$

by $\lambda + q \asymp q$ and using the preceding error estimate improves this to $\lambda = q + \mathcal{O}(1)$.

Write

$$\lambda = q + \frac{B}{L} + \rho.$$

Insert this into (678). The terms proportional to q cancel because $L(B/L)q = Bq$. What remains is

$$Lq\rho = -\frac{1}{2}\log\left(q + \frac{B}{L} + \rho\right) + \mathcal{O}(1) + \mathcal{O}(\rho^2).$$

The already known bound $\rho = \mathcal{O}(1)$ makes the final term bounded. Since $q \asymp \sqrt{T}$ and $\log q \asymp \log T$, division by Lq gives

$$\rho = \mathcal{O}\left(\frac{\log T}{\sqrt{T}}\right).$$

□

66.3 The endpoint equivalent for the inverse

Theorem 66.2 (Sharp inverse asymptotic; Derived). *As* $y \downarrow 0$,

$$\boxed{G(y) \sim \frac{\sqrt{\log(1/y)}}{\mathrm{e}\sqrt{\log 2}} \exp\left[-\sqrt{2\log 2\,\log(1/y)}\right].} \tag{679}$$

Equivalently,

$$\log G(y) = -\sqrt{2L\log(1/y)} + \frac{1}{2}\log\log(1/y) - 1 - \frac{1}{2}\log L + o(1). \tag{680}$$

At the right endpoint,

$$1 - G(y) \sim \frac{\sqrt{\log(1/(1-y))}}{\mathrm{e}\sqrt{\log 2}} \exp\left[-\sqrt{2\log 2\,\log(1/(1-y))}\right] \quad (y \uparrow 1). \tag{681}$$

Proof. Let $x = G(y)$. Then $y = F(x)$ and x is related to its exact phase by $x = \lambda\mathrm{e}^{-L\lambda}$. Therefore

$$\log x = \log \lambda - L\lambda.$$

With $T = \log(1/y)$, theorem 66.1 gives

$$\lambda = q + \frac{B}{L} + o(1), \qquad q = \sqrt{\frac{2T}{L}},$$

because $(\log T)/\sqrt{T} = o(1)$. Also $\log\lambda = \log q + o(1)$. Consequently

$$\begin{aligned}\log x &= \log q - Lq - B + o(1)\\ &= -\sqrt{2LT} + \frac{1}{2}\log T + \frac{1}{2}\log\frac{2}{L} - 1 - \frac{L}{2} + o(1)\\ &= -\sqrt{2LT} + \frac{1}{2}\log T - 1 - \frac{1}{2}\log L + o(1),\end{aligned}$$

because $L = \log 2$. This is (680). Exponentiating proves (679). Finally, (669) gives $1-G(y) = G(1-y)$ and hence (681). □

Remark 66.3 (Why the periodic fluctuation disappears at leading order). The bounded term $\Psi(\lambda)$ changes the solution λ of (678) only by $\mathcal{O}(1/\lambda)$. Since $\log x = \log\lambda - L\lambda$, this changes $\log x$ by $o(1)$ and therefore changes x by a factor tending to 1. Periodicity returns in the next relative correction, but not in the leading equivalent (679).

66.4 The complete hierarchy of elementary scales

Corollary 66.4 (Between powers and logarithms; Derived)**.** *For every $\alpha > 0$ and every $m > 0$,*

$$y^{\alpha} = o(G(y)), \tag{682}$$

$$G(y) = o\big((\log(1/y))^{-m}\big) \tag{683}$$

as $y \downarrow 0$.

Proof. Set $T = \log(1/y)$. Formula (680) gives

$$\log\frac{G(y)}{y^{\alpha}} = \alpha T - \sqrt{2LT} + \mathcal{O}(\log T) \longrightarrow +\infty,$$

which proves (682). Likewise,

$$\log(G(y)T^{m}) = -\sqrt{2LT} + \mathcal{O}(\log T) \longrightarrow -\infty,$$

which proves (683). □

The inverse endpoint is therefore neither power-like nor logarithmic. It lies in the stretched-logarithmic regime $\exp(-c\sqrt{\log(1/y)})$ with an essential $\sqrt{\log(1/y)}$ prefactor.

67 Closed periodic saddle jets

We now turn from the inverse to the coefficient machinery hidden inside the all-orders endpoint expansion. Throughout this part,

$$L = \log 2, \qquad D = \frac{\mathrm{d}}{\mathrm{d}u},$$

and Ψ is the smooth one-periodic function occurring in the sharp Lambert main term.

67.1 The recursive jets

Define the periodic jets p_n by

$$p_0(u) = \frac{1}{2} + \frac{\Psi'(u)}{L}, \tag{684}$$

$$p_{n+1}(u) = \left(\frac{D}{L} - (n+1)\right) p_n(u) + \frac{(-1)^{n+1} n!}{L}. \tag{685}$$

The bounded exponent jets are

$$d_n(u) = p_n(u) + (-1)^{n+1} n!. \tag{686}$$

The subtraction in (686) removes the factorial part that would otherwise obscure the centered saddle expansion.

For $n \geq 0$, write

$$H_n = \sum_{k=1}^{n} \frac{1}{k}, \qquad H_0 = 0,$$

and define integers $c(n, m)$ by

$$\prod_{k=1}^{n} (z - k) = \sum_{m=0}^{n} c(n, m) z^m. \tag{687}$$

Theorem 67.1 (Closed periodic jets; Lean). *For every $n \geq 0$,*

$$p_n(u) = (-1)^n n! \left(\frac{1}{2} + \frac{H_n}{L}\right) + \sum_{m=0}^{n} c(n, m) \frac{\Psi^{(m+1)}(u)}{L^{m+1}}, \tag{688}$$

$$d_n(u) = (-1)^n n! \left(\frac{H_n}{L} - \frac{1}{2}\right) + \sum_{m=0}^{n} c(n, m) \frac{\Psi^{(m+1)}(u)}{L^{m+1}}. \tag{689}$$

Proof. Let

$$T_k = \frac{D}{L} - k.$$

The operators T_k commute because each is a polynomial in D. Iterating (685) gives

$$p_n = T_n T_{n-1} \cdots T_1 p_0 + \sum_{j=0}^{n-1} T_n T_{n-1} \cdots T_{j+2} \left(\frac{(-1)^{j+1} j!}{L}\right), \tag{690}$$

where the product is empty when $j = n - 1$.

First consider the homogeneous term. Since

$$T_n \cdots T_1 = \prod_{k=1}^{n} \left(\frac{D}{L} - k\right),$$

its action on the constant $1/2$ is

$$\frac{1}{2} \prod_{k=1}^{n} (-k) = \frac{(-1)^n n!}{2}.$$

Its action on Ψ'/L is found by expanding the operator polynomial with (687):

$$\prod_{k=1}^{n} \left(\frac{D}{L} - k\right) \frac{\Psi'}{L} = \sum_{m=0}^{n} c(n, m) \frac{\Psi^{(m+1)}}{L^{m+1}}.$$

Now propagate the forcing term inserted at stage j. It is constant, so each subsequent T_k acts on it as multiplication by $-k$. Hence its contribution at stage n is

$$\frac{(-1)^{j+1}j!}{L}\prod_{k=j+2}^{n}(-k) = \frac{(-1)^{j+1}j!}{L}\frac{(-1)^{n-j-1}n!}{(j+1)!}$$
$$= \frac{(-1)^n n!}{(j+1)L}.$$

Summing over $j = 0, \dots, n-1$ gives

$$\frac{(-1)^n n!}{L}\sum_{j=0}^{n-1}\frac{1}{j+1} = \frac{(-1)^n n! H_n}{L}.$$

Combining the constant, derivative, and forcing contributions proves (688). Finally,

$$p_n + (-1)^{n+1}n! = (-1)^n n!\left(\frac{1}{2} + \frac{H_n}{L} - 1\right) + \text{derivative part},$$

which is exactly (689). □

67.2 Why the Stirling indices are shifted

Let $s(n,k)$ be the signed Stirling number of the first kind, defined by

$$z(z-1)\cdots(z-n+1) = \sum_{k=0}^{n} s(n,k)z^k.$$

Proposition 67.2 (Shifted Stirling conversion; Lean)**.** *For* $0 \le m \le n$*,*

$$c(n,m) = s(n+1,m+1).$$

Proof. Apply the defining identity with $n+1$:

$$z(z-1)\cdots(z-n) = \sum_{k=0}^{n+1} s(n+1,k)z^k.$$

Both sides have zero constant term, so divide by z:

$$(z-1)\cdots(z-n) = \sum_{m=0}^{n} s(n+1,m+1)z^m.$$

Comparison with (687) proves the formula. □

Warning 67.3. The coefficients are not $s(n,m)$. The product starts with $(z-1)$ rather than z, and this forces the simultaneous shift $(n,m) \mapsto (n+1,m+1)$. In particular, the nonzero constant column $c(n,0) = (-1)^n n!$ is essential.

67.3 Coefficient rows and the first jets

The first rows of $c(n, m)$, listed from $m = 0$ to $m = n$, are

n	$c(n,0)$	$c(n,1)$	$c(n,2)$	$c(n,3)$	$c(n,4)$	$c(n,5)$
0	1					
1	-1	1				
2	2	-3	1			
3	-6	11	-6	1		
4	24	-50	35	-10	1	
5	-120	274	-225	85	-15	1

and (689) gives

$$d_0 = -\frac{1}{2} + \frac{\Psi'}{L}, \tag{691}$$

$$d_1 = \frac{1}{2} - \frac{1}{L} - \frac{\Psi'}{L} + \frac{\Psi''}{L^2}, \tag{692}$$

$$d_2 = -1 + \frac{3}{L} + \frac{2\Psi'}{L} - \frac{3\Psi''}{L^2} + \frac{\Psi'''}{L^3}, \tag{693}$$

$$d_3 = 3 - \frac{11}{L} - \frac{6\Psi'}{L} + \frac{11\Psi''}{L^2} - \frac{6\Psi'''}{L^3} + \frac{\Psi^{(4)}}{L^4}. \tag{694}$$

Corollary 67.4 (Structural properties; Lean). *Every p_n and d_n is smooth, one-periodic, and bounded on $\mathbb{R}$. Moreover, d_n involves derivatives of Ψ only through order $n + 1$, and the coefficient of the highest derivative is $L^{-(n+1)}$.*

Proof. The closed formulas are finite linear combinations of derivatives of the smooth periodic function Ψ and constants. Smoothness, periodicity, and boundedness follow. Since $c(n, n) = 1$, the coefficient of $\Psi^{(n+1)}$ is $L^{-(n+1)}$. □

68 The centered exponent and its algebraic collapse

68.1 From a long Taylor expansion to one short formula

At the saddle, introduce the half-power parameter

$$\varepsilon = \lambda^{-1/2}$$

and the Gaussian variable v. After removing the exact quadratic term, the centered exponent has a formal expansion

$$\mathcal{E}(u, v; \varepsilon) = \sum_{m\geq 1} E_m(u, v)\varepsilon^m.$$

The original construction of E_m contains separate contributions from the vertical negative-Laplace jet and from the Taylor expansion of $\log(1 + \mathrm{i}\varepsilon v)$. The bounded exponent jets make these two pieces collapse.

Theorem 68.1 (Closed exponent coefficient; Lean). *For every $m \geq 1$,*

$$E_m(u, v) = \mathrm{i}^m \left(\frac{d_{m-1}(u)}{m!} v^m + \frac{(-1)^{m+1}}{m+2} v^{m+2} \right). \tag{695}$$

In particular,

$$E_1 = \mathrm{i}v\,\frac{3d_0 + v^2}{3}, \tag{696}$$

$$E_2 = v^2\,\frac{-2d_1 + v^2}{4}, \tag{697}$$

$$E_3 = \mathrm{i}v^3\left(-\frac{d_2}{6} - \frac{v^2}{5}\right), \tag{698}$$

$$E_4 = \frac{v^4}{24}(d_3 - 4v^2). \tag{699}$$

Proof. Before centering, the coefficient at order m has two contributions. The vertical jet contributes $\mathrm{i}^m p_{m-1}(u)v^m/m!$, while the logarithmic saddle coordinate contributes a factorial term to the same monomial and a universal v^{m+2} term. Using

$$d_{m-1} = p_{m-1} + (-1)^m(m-1)!$$

combines the two v^m coefficients into $\mathrm{i}^m d_{m-1}/m!$. The remaining logarithmic term is

$$\frac{\mathrm{i}^{m+2}(-1)^m}{m+2}v^{m+2} = \mathrm{i}^m\frac{(-1)^{m+1}}{m+2}v^{m+2},$$

because $\mathrm{i}^2 = -1$. This is (695). Substitution of $m = 1, 2, 3, 4$ gives (696)–(699). □

Corollary 68.2 (Parity; Lean). *For every $m \geq 1$,*

$$E_m(u, -v) = (-1)^m E_m(u, v).$$

Thus the odd-indexed E_m are odd polynomials in v and the even-indexed E_m are even.

Proof. Both monomials in (695) have parity m, because $m + 2$ has the same parity as m. □

69 Formal exponential, Gaussian contraction, and logarithm

The generic exponential, logarithmic, and Gaussian engines—including their uniqueness, naturality, parity, and finite-remainder theorems—are developed in sections 5.2, 5.3 and 5.5. This section records only their Fabius-specific instantiation and the ordered-composition formulas needed for the later explicit coefficients.

69.1 The exponential recursion

Define polynomials $B_n(u, v)$ by the formal identity

$$\exp\left(\sum_{m\geq 1} E_m(u, v)\varepsilon^m\right) = \sum_{n\geq 0} B_n(u, v)\varepsilon^n. \tag{700}$$

Thus $B_0 = 1$.

Proposition 69.1 (Finite exponential recursion; Lean). *For $n \geq 1$,*

$$nB_n = \sum_{m=1}^{n} mE_m B_{n-m}. \tag{701}$$

Proof. Apply the generic coefficient identity from section 5.2 to the Fabius exponent sequence $E_m(u,v)$. □

Corollary 69.2 (Ordered-composition exponential formula; Derived). *For $n \geq 1$,*

$$B_n = \sum_{\substack{r\geq 1\\ m_1+\cdots+m_r=n}} \frac{1}{r!} E_{m_1}\cdots E_{m_r},$$

where the inner sum is over ordered compositions of n into r positive parts.

Proof of theorem 69.2. Expand the finite coefficient of the formal exponential by the number of factors. This is an algebraic consequence of the recursion in each degree; no analytic convergence is used. The repository does not currently package this ordered-composition statement as one general Lean declaration. □

Corollary 69.3 (Parity of the exponential coefficients; Lean). *For all $n \geq 0$,*

$$B_n(u,-v) = (-1)^n B_n(u,v).$$

Proof. Every product contributing to B_n has indices $m_1+\cdots+m_r = n$. By theorem 68.2, its parity is $(-1)^{m_1+\cdots+m_r} = (-1)^n$. □

69.2 Gaussian contraction and mass coefficients

For a polynomial P define normalized Gaussian contraction by

$$\mathfrak{G}[P] = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \mathrm{e}^{-v^2/2} P(v)\, \mathrm{d}v.$$

It is determined by

$$\mathfrak{G}[v^{2k}] = (2k-1)!!, \qquad \mathfrak{G}[v^{2k+1}] = 0. \tag{702}$$

The mass coefficients are

$$a_j(u) = \mathfrak{G}[B_{2j}(u,\cdot)], \qquad j \geq 0. \tag{703}$$

In particular $a_0 = 1$. The odd half-powers vanish automatically:

$$\mathfrak{G}[B_{2j+1}] = 0$$

by theorem 69.3.

Theorem 69.4 (Closed composition formula for a_j; Derived). *For $j \geq 1$,*

$$\begin{aligned} a_j(u) = (-1)^j \sum_{r=1}^{2j} \frac{1}{r!} \sum_{\substack{m_1+\cdots+m_r=2j\\ m_i\geq 1}} \sum_{S\subseteq\{1,\dots,r\}} (2j+2|S|-1)!! \\ \times \prod_{i\in S} \frac{(-1)^{m_i+1}}{m_i+2} \prod_{i\notin S} \frac{d_{m_i-1}(u)}{m_i!}. \end{aligned} \tag{704}$$

Consequently a_j is a polynomial with rational coefficients in $d_0,\dots,d_{2j-1}$.

Proof. Use the ordered-composition formula in theorem 69.2 for B_{2j}. Each factor

$$E_{m_i} = \mathrm{i}^{m_i}\left(\frac{d_{m_i-1}}{m_i!} v^{m_i} + \frac{(-1)^{m_i+1}}{m_i+2} v^{m_i+2}\right)$$

offers two choices. Let S be the set of factors from which the second, universal monomial is chosen. Since $\sum_i m_i = 2j$, the product of the powers of i is

$$\mathrm{i}^{2j} = (-1)^j,$$

and the total power of v is $2j + 2|S|$. Its Gaussian contraction is $(2j + 2|S| - 1)!!$. Multiplying the remaining scalar factors gives (704). The largest possible jet index is $m_i - 1 \le 2j - 1$. ☐

Formula (704) is a human-readable algebraic consequence of the formal exponent and exponential-recursion theorems. The repository machine-checks the underlying recursion, the structural top-jet theorem, and explicit low orders. It does not package this exact threefold summation as a single general Lean declaration.

69.3 Logarithmic coefficients

Define $A_j(u)$ by

$$1 + \sum_{j\ge1} a_j(u) z^j = \exp\left(\sum_{j\ge1} A_j(u) z^j\right). \tag{705}$$

These are the coefficients that occur in $\log F$.

Proposition 69.5 (Mass-to-logarithm conversion; Lean)**.** *For $n \ge 1$,*

$$A_n = a_n - \frac{1}{n}\sum_{k=1}^{n-1} k A_k\, a_{n-k}. \tag{706}$$

Proof. Specialize the mutually inverse formal logarithm/exponential transforms of section 5.3. The recursive form is the coefficient identity for $A' = (1 + \sum a_j z^j)'/(1 + \sum a_j z^j)$. ☐

Corollary 69.6 (Ordered-composition logarithm formula; Derived)**.** *For $n \ge 1$,*

$$A_n = \sum_{r=1}^{n} \frac{(-1)^{r-1}}{r} \sum_{\substack{q_1+\cdots+q_r=n\\ q_i\ge1}} a_{q_1}\cdots a_{q_r}. \tag{707}$$

Proof of theorem 69.6. Take the finite coefficient of $\log(1 + w)$. This is an algebraic identity in each fixed degree, but the displayed general composition sum is not currently exported as a single Lean theorem. ☐

Corollary 69.7 (Regularity and derivative order; Lean input, Derived conclusion)**.** *For every j, the functions a_j and A_j are smooth, one-periodic, and bounded. Neither involves a derivative of Ψ of order greater than $2j$.*

Proof. The function a_j is a polynomial in $d_0, \ldots, d_{2j-1}$, and d_n is smooth, periodic, and depends on Ψ only through order $n + 1$. Thus a_j has the stated properties and uses derivatives only through order $2j$. The recursion (706) transfers the same assertions to A_j. ☐

69.4 The top jet and the highest derivative

The most complicated-looking coefficient has a very simple dependence on its newest jet.

Theorem 69.8 (Leading mass-jet law; Lean input, Derived logarithmic consequence)**.** *For $j \geq 1$, the mass coefficient a_j is affine in d_{2j-1}, and the coefficient of that jet is*

$$\frac{(-1)^j}{2^j j!}. \tag{708}$$

The logarithmic coefficient A_j is derived here to have the same top-jet coefficient; this transfer is not asserted by the cited mass-coefficient theorem. Consequently the coefficient of $\Psi^{(2j)}(u)$ in $A_j(u)$ is

$$\boxed{\frac{(-1)^j}{2^j j! L^{2j}}.} \tag{709}$$

Proof. The jet d_{2j-1} occurs only in E_{2j}. Therefore the only term of B_{2j} that contains it is the linear term E_{2j} itself. From (695), its jet part is

$$\mathrm{i}^{2j}\frac{d_{2j-1}}{(2j)!}v^{2j} = (-1)^j\frac{d_{2j-1}}{(2j)!}v^{2j}.$$

Gaussian contraction multiplies this by $(2j-1)!!$. Since

$$(2j)! = 2^j j!(2j-1)!!,$$

the coefficient is (708).

In (706), every correction term $A_k a_{j-k}$ has $k < j$ and $j-k < j$; it involves jets only through d_{2j-3} and therefore cannot alter the coefficient of d_{2j-1}. Thus A_j has the same top-jet coefficient. Finally, theorem 67.4 says that the coefficient of $\Psi^{(2j)}$ in d_{2j-1} is L^{-2j}, giving (709). □

Remark 69.9. The theorem provides a quick consistency check for every explicit coefficient. For example, A_1 must contain $-\Psi''/(2L^2)$, while A_2 must contain $+\Psi^{(4)}/(8L^4)$.

70 The first and second logarithmic corrections

70.1 Order two: the first correction

From equations (696) and (697),

$$B_2 = E_2 + \frac{1}{2}E_1^2.$$

A direct expansion gives

$$B_2 = -\frac{d_0^2+d_1}{2}v^2 + \left(\frac{1}{4}-\frac{d_0}{3}\right)v^4 - \frac{1}{18}v^6.$$

Using $\mathfrak{G}[v^2]=1$, $\mathfrak{G}[v^4]=3$, and $\mathfrak{G}[v^6]=15$ yields

$$a_1 = A_1 = -\frac{d_0^2}{2} - d_0 - \frac{d_1}{2} - \frac{1}{12}. \tag{710}$$

Equivalently,

$$A_1 = -\frac{6d_0^2+12d_0+6d_1+1}{12}.$$

Substituting equations (691) and (692) recovers the first periodic correction in terms of Ψ' and Ψ''.

70.2 Order four of the formal exponential

The fourth exponential coefficient is

$$B_4 = E_4 + E_3E_1 + \frac{1}{2}E_2^2 + \frac{1}{2}E_2E_1^2 + \frac{1}{24}E_1^4. \tag{711}$$

This follows either from the composition formula or from the recurrence (701). Substitution of equations (696) to (699) and collection of even powers of v gives the following polynomial.

Proposition 70.1 (The order-four reference polynomial; Lean). *Writing* $d_k = d_k(u)$*,*

$$\begin{aligned} B_4(u,v) = {} & \left(\frac{d_0^4}{24} + \frac{d_0^2d_1}{4} + \frac{d_0d_2}{6} + \frac{d_1^2}{8} + \frac{d_3}{24}\right)v^4 \\ & + \left(\frac{d_0^3}{18} - \frac{d_0^2}{8} + \frac{d_0d_1}{6} + \frac{d_0}{5} - \frac{d_1}{8} + \frac{d_2}{18} - \frac{1}{6}\right)v^6 \\ & + \left(\frac{d_0^2}{36} - \frac{d_0}{12} + \frac{d_1}{36} + \frac{47}{480}\right)v^8 \\ & + \left(\frac{d_0}{162} - \frac{1}{72}\right)v^{10} + \frac{1}{1944}v^{12}. \end{aligned} \tag{712}$$

Proof. Let

$$P_1 = d_0v + \frac{v^3}{3}, \qquad P_3 = -\frac{d_2v^3}{6} - \frac{v^5}{5},$$

so that $E_1 = \mathrm{i}P_1$ and $E_3 = \mathrm{i}P_3$. Then $E_3E_1 = -P_3P_1$ and $E_1^4 = P_1^4$. The even terms E_2, E_4 are already real. Insert these expressions in (711). Each summand has degree at most 12 and is even. Multiplication gives

	v^4	v^6	v^8	v^{10}	v^{12}
E_4	$d_3/24$	$-1/6$	0	0	0
E_3E_1	$d_0d_2/6$	$d_2/18 + d_0/5$	$1/15$	0	0
$E_2^2/2$	$d_1^2/8$	$-d_1/8$	$1/32$	0	0
$E_2E_1^2/2$	$d_0^2d_1/4$	$d_0d_1/6 - d_0^2/8$	$d_1/36 - d_0/12$	$-1/72$	0
$E_1^4/24$	$d_0^4/24$	$d_0^3/18$	$d_0^2/36$	$d_0/162$	$1/1944$

and the two universal v^8 contributions satisfy $1/15 + 1/32 = 47/480$. Summing the columns proves (712). □

70.3 Gaussian contraction and the second mass coefficient

The relevant Gaussian moments are

$$\mathfrak{G}[v^4] = 3, \quad \mathfrak{G}[v^6] = 15, \quad \mathfrak{G}[v^8] = 105, \quad \mathfrak{G}[v^{10}] = 945, \quad \mathfrak{G}[v^{12}] = 10395.$$

Theorem 70.2 (Second mass coefficient; Lean). *Gaussian contraction of* (712) *gives*

$$\begin{aligned} a_2 = {} & \frac{d_0^4}{8} + \frac{5d_0^3}{6} + \frac{3d_0^2d_1}{4} + \frac{25d_0^2}{24} + \frac{5d_0d_1}{2} + \frac{d_0d_2}{2} + \frac{d_0}{12} \\ & + \frac{3d_1^2}{8} + \frac{25d_1}{24} + \frac{5d_2}{6} + \frac{d_3}{8} + \frac{1}{288}. \end{aligned} \tag{713}$$

Proof. Multiply the coefficients of $v^4, v^6, v^8, v^{10}, v^{12}$ in (712) by $3, 15, 105, 945, 10395$, respectively, and collect like monomials. For illustration, the coefficient of d_0 is

$$15 \cdot \frac{1}{5} + 105\left(-\frac{1}{12}\right) + 945 \cdot \frac{1}{162} = 3 - \frac{35}{4} + \frac{35}{6} = \frac{1}{12},$$

and the universal constant is

$$15\left(-\frac{1}{6}\right) + 105\frac{47}{480} + 945\left(-\frac{1}{72}\right) + 10395\frac{1}{1944} = \frac{1}{288}.$$

The remaining coefficients follow by the same arithmetic. □

70.4 The second logarithmic coefficient

Since

$$\log(1 + a_1 z + a_2 z^2 + \cdots) = a_1 z + \left(a_2 - \frac{1}{2}a_1^2\right) z^2 + \cdots,$$

we have $A_2 = a_2 - a_1^2/2$.

Theorem 70.3 (Second periodic saddle correction; Lean). *The second logarithmic coefficient is*

$$\boxed{A_2 = \frac{8d_0^3 + 12d_0^2 d_1 + 12d_0^2 + 48d_0 d_1 + 12d_0 d_2 + 6d_1^2 + 24d_1 + 20d_2 + 3d_3}{24}.} \tag{714}$$

It is smooth, one-periodic, and bounded.

Proof. Insert (710) and (713) into $A_2 = a_2 - a_1^2/2$. The quartic term $d_0^4/8$ cancels, as do the terms $d_0/12$ and $1/288$ after the square is expanded. The remaining terms reduce to (714). Smoothness, periodicity, and boundedness follow because A_2 is a polynomial in d_0, d_1, d_2, d_3. □

Corollary 70.4 (Two explicit correction orders; Lean). *Let $\lambda = \lambda(x)$ be the exact lower Lambert phase. Then, as $x \downarrow 0$,*

$$\log F(x) = \mathcal{M}(x) + \frac{A_1(\lambda)}{\lambda} + \frac{A_2(\lambda)}{\lambda^2} + \mathcal{O}(\lambda^{-3}), \tag{715}$$

where $\mathcal{M}(x)$ is the exact sharp Lambert main term. Since $\lambda \asymp -\log x$, the remainder is also $\mathcal{O}((-\log x)^{-3})$.

Warning 70.5. The coefficients in (715) are evaluated at the exact phase $\lambda(x)$. Substituting a truncated asymptotic formula for λ into A_1 or A_2 is not licensed merely by $\lambda - \lambda_{\text{approx}} = \mathcal{O}(1)$: periodic functions are sensitive to bounded phase shifts.

71 Why the all-orders remainder is actually rigorous

The formal coefficient algebra by itself does not prove an asymptotic expansion. One must control the exact contour integral uniformly on a window that grows with the requested order, prove that exponentiation does not magnify the Taylor error, replace the finite Gaussian window by the whole line, and show that the discarded contour is smaller than every retained inverse power. This section spells out that analytic chain.

71.1 The exact normalized saddle mass

For $x > 0$ small, let $\lambda = \lambda(x)$ be the exact phase (676). The exact saddle reduction can be organized as

$$\log F(x) = \mathcal{M}(x) + \log \mathcal{G}(x) + \mathcal{T}(x), \tag{716}$$

where:

- $\mathcal{M}(x)$ is the exact sharp Lambert main term, containing the quadratic phase, the Gaussian normalization, the periodic finite part, and the fixed constant;

- $\mathcal{G}(x) > 0$ is the normalized saddle-kernel mass and $\mathcal{G}(x) \to 1$;
- $\mathcal{T}(x)$ is the forward negative-Laplace tail, which is flat against every inverse power of λ.

Thus all algebraic corrections beyond $\mathcal{M}$ come from $\log \mathcal{G}$.

It is technically convenient to begin on the real dyadic logarithmic scale

$$x = 2^{-t}, \qquad t \to +\infty,$$

and to write

$$b = \lambda(2^{-t}), \qquad \varepsilon = b^{-1/2}.$$

This is not a restriction to integer t. Every sufficiently small positive x has the unique representation $x = 2^{-t}$ with $t = -\log_2 x$.

After the saddle contour is centered and rescaled by $v = \sqrt{b}\,\theta$, its central kernel has the normalized form

$$\frac{1}{\sqrt{2\pi}} \mathrm{e}^{-v^2/2} \exp R_b(v), \tag{717}$$

where R_b is the exact centered exponent. The formal expansion of R_b is

$$R_b(v) \sim \sum_{m \geq 1} E_m(b, v) b^{-m/2}. \tag{718}$$

The first variable of E_m is evaluated at b because all coefficient functions are periodic in the exact phase.

71.2 The order-dependent central window

For a requested order $N \geq 0$, define

$$A_N(b) = \sqrt{32(N+1)\log b}. \tag{719}$$

Lemma 71.1 (Scale separation; Lean). *For every fixed N,*

$$A_N(b) \longrightarrow +\infty, \qquad \varepsilon A_N(b) \longrightarrow 0 \qquad (b \to +\infty).$$

Moreover, for every fixed $r > 0$ and $\delta > 0$,

$$A_N(b)^r = o(b^\delta).$$

Proof. The first limit is immediate from $\log b \to \infty$. The second follows from

$$\varepsilon A_N(b) = \sqrt{\frac{32(N+1)\log b}{b}} \longrightarrow 0.$$

Finally, every fixed power of $\log b$ grows more slowly than every positive power of b. □

The window has two simultaneous purposes. It grows, so replacing a Gaussian integral over it by a whole-line Gaussian integral creates a tiny tail. Yet its angular width $A_N(b)/\sqrt{b}$ shrinks to zero, so all local Taylor estimates remain uniform.

71.3 All-order control of the exact centered exponent

The exact negative-Laplace factor contains a finite periodic part plus endpoint and forward-product corrections. On the vertical line through the saddle, the proof decomposes it into:

1. a Taylor polynomial in $\theta = v/\sqrt{b}$ whose coefficients are the periodic jets p_n;
2. the Taylor polynomial of $\log(1 + \mathrm{i}\theta)$;
3. two boundary-jet remainders from the finite part;
4. forward-product jets whose size is exponentially small in b;
5. explicit integral remainders for the vertical derivatives.

The factorial pieces in the first two items combine into the bounded jets d_n, producing (695).

Proposition 71.2 (Integrated exponent truncation; Lean)**.** *Fix N. The Taylor depth can be chosen, depending only on N, so that on $|v| \le A_N(b)$ the exact exponent may be replaced by*

$$R_{b,N}(v) = \sum_{m=1}^{2N+1} E_m(b, v) b^{-m/2}$$

with an error which, after multiplication by the Gaussian weight and integration over the central window, is

$$\mathcal{O}(b^{-N-1}).$$

The implied constant may depend on N but not on b.

Proof. All vertical derivatives of the periodic finite part are uniformly bounded because the periodic jets are smooth and bounded. Taylor's theorem therefore bounds a vertical remainder of order M by

$$C_M |\theta|^{M+1} = C_M b^{-(M+1)/2} |v|^{M+1}.$$

The logarithmic coordinate has an analogous remainder on the shrinking angular window, because $\varepsilon A_N(b) \to 0$ keeps $1 + \mathrm{i}\theta$ uniformly away from zero. Boundary jets satisfy the same kind of estimate, while the forward-product terms are exponentially small and hence smaller than any prescribed inverse power.

Choose M with enough surplus beyond $2N + 1$. On the central window, powers of $|v|$ are bounded by powers of $A_N(b)$, hence by powers of $\log b$. By theorem 71.1, the spare powers of $b^{-1/2}$ absorb every such logarithmic loss. The pointwise exponent error is therefore integrably $\mathcal{O}(b^{-N-1})$ against $\mathrm{e}^{-v^2/2}\,\mathrm{d}v$. Removing terms above order $2N + 1$ from the finite Taylor polynomial costs the same order because each carries at least the corresponding spare half-power. This leaves $R_{b,N}$. □

71.4 Finite exponentiation without an infinite-series argument

A common informal step is to exponentiate (718) term by term. The formal development instead uses a finite Taylor polynomial for the exponential and an explicit remainder.

Let

$$\mathcal{P}_N(b, v) = \sum_{\ell=0}^{2N+1} B_\ell(b, v) b^{-\ell/2}.$$

It is the reference polynomial obtained from the formal recursion in theorem 69.1.

Proposition 71.3 (Finite exponential substitution; Lean)**.** *For every fixed N,*

$$\frac{1}{\sqrt{2\pi}} \int_{|v| \le A_N(b)} \mathrm{e}^{-v^2/2} \left|\exp R_b(v) - \mathcal{P}_N(b, v)\right| \mathrm{d}v = \mathcal{O}(b^{-N-1}). \tag{720}$$

Proof. First replace R_b by $R_{b,N}$ using theorem 71.2. On the central window the truncated exponent tends uniformly to zero after the exact quadratic term has been removed, and it is bounded by a fixed polynomial in v times $b^{-1/2}$. Apply Taylor's formula

$$\mathrm{e}^z = \sum_{r=0}^{L-1} \frac{z^r}{r!} + \frac{z^L}{(L-1)!} \int_0^1 (1-s)^{L-1} \mathrm{e}^{sz} \, \mathrm{d}s$$

with $L = 2(N+1)$. Uniform control of the real part of $R_{b,N}$ bounds the integral remainder by a Gaussian-integrable polynomial times b^{-N-1}.

Now expand the finite Taylor polynomial in the finitely many powers $b^{-m/2}$. The coefficient of $b^{-\ell/2}$ for $\ell < L$ is exactly B_ℓ, by the same recursion that defines the formal exponential coefficients. Algebraically, the difference between the finite exponential Taylor polynomial and $\mathcal{P}_N$ factors as

$$b^{-L/2} \times \{\text{a finite quotient polynomial in } b^{-1/2}, v, E_1, \dots, E_{2N+1}\}.$$

Because $L/2 = N+1$ and the coefficient functions are bounded, Gaussian integration gives (720). □

71.5 Gaussian contraction on the whole line

By theorem 69.3, odd ℓ contribute odd polynomials. Therefore

$$\mathfrak{G}[\mathcal{P}_N(b,\cdot)] = \sum_{j=0}^{N} a_j(b) b^{-j}. \tag{721}$$

The exact central integral, however, only runs over $|v| \le A_N(b)$. We must show that the polynomial reference can be extended to the whole real line at the required order.

Lemma 71.4 (Gaussian polynomial tail; Lean)**.** *For every fixed polynomial P and every fixed N,*

$$\int_{|v|>A_N(b)} \mathrm{e}^{-v^2/2} |P(v)| \, \mathrm{d}v = \mathcal{O}(b^{-N-1}).$$

The same assertion holds for a finite family of polynomials whose coefficients are uniformly bounded in b.

Proof. For every integer $q \ge 0$ there is a constant C_q such that

$$\int_A^\infty v^q \mathrm{e}^{-v^2/2} \, \mathrm{d}v \le C_q (1 + A^q) \mathrm{e}^{-A^2/4} \qquad (A \ge 1).$$

At $A = A_N(b)$,

$$\mathrm{e}^{-A_N(b)^2/4} = \mathrm{e}^{-8(N+1)\log b} = b^{-8(N+1)}.$$

The polynomial factor in $A_N(b)$ is a power of $\log b$ and is absorbed by the enormous surplus of inverse powers. Sum over the finitely many monomials of P. □

Combining theorems 71.3 and 71.4 with (721), the central contour contributes

$$\sum_{j=0}^{N} a_j(b) b^{-j} + \mathcal{O}(b^{-N-1}). \tag{722}$$

71.6 The complementary contour

The exact contour outside the central window is split into an intermediate region and a far region. The two estimates use different mechanisms.

Proposition 71.5 (All-order minor-arc bounds; LEAN). *For every fixed N, the normalized contour outside $|v| \le A_N(b)$ contributes $\mathcal{O}(b^{-N-1})$. More precisely:*

1. *the intermediate region has a bound with fourfold power slack, of order at most a constant times $b^{-4(N+1)}$ once the relevant dyadic index is at least $b/4$;*
2. *the far region is bounded by a constant times*

$$b \exp\left(-\frac{\log 2}{4} b\right),$$

which is flat against every inverse power of b.

Proof. On the intermediate region the modulus estimate for the saddle kernel contains a decay factor whose exponent is quadratic in the enlarged radius. Substitution of (719) yields

$$\exp\{-4(N+1)\log b\} = b^{-4(N+1)}$$

after the standard comparison between the dyadic index and b. The remaining geometric and polynomial factors consume far less than the three spare blocks of inverse powers.

In the far region the dyadic product supplies geometric decay in the index. Summing the geometric tail gives a bound of the displayed exponential form. For every K,

$$b^K\, b e^{-(\log 2)b/4} \longrightarrow 0,$$

so this contribution is $\mathcal{O}(b^{-N-1})$ for the requested fixed N. □

71.7 The all-orders mass expansion

Theorem 71.6 (Normalized mass expansion; LEAN). *For every fixed $N \ge 0$, as $t \to +\infty$ with $b = \lambda(2^{-t})$,*

$$\mathcal{G}(2^{-t}) = \sum_{j=0}^{N} \frac{a_j(b)}{b^j} + \mathcal{O}(b^{-N-1}). \tag{723}$$

The coefficient functions are evaluated at the exact phase b.

Proof. The central part is (722). Add the intermediate and far parts controlled by theorem 71.5. All three errors are $\mathcal{O}(b^{-N-1})$. □

71.8 Transfer through the logarithm

Theorem 71.7 (Logarithmic expansion on the dyadic real scale; LEAN). *For every fixed $N \ge 0$,*

$$\log \mathcal{G}(2^{-t}) = \sum_{j=1}^{N} \frac{A_j(b)}{b^j} + \mathcal{O}(b^{-N-1}), \qquad b = \lambda(2^{-t}). \tag{724}$$

Proof. The boundedness of the a_j and (723) with $N = 0$ give $\mathcal{G}(2^{-t}) = 1 + \mathcal{O}(b^{-1})$, hence $\mathcal{G} > 0$ and $|\mathcal{G} - 1| < 1/2$ eventually. Taylor's theorem for $\log(1+z)$ is therefore uniform. Substitute (723) into the finite Taylor polynomial for the logarithm. The coefficient of b^{-j} is exactly A_j, either by (707) or by the recursion (706). The logarithmic Taylor remainder and the original mass remainder are both $\mathcal{O}(b^{-N-1})$. □

71.9 Adding the flat tail and returning to $x \downarrow 0$

The tail $\mathcal{T}$ in (716) satisfies, for every K,

$$\mathcal{T}(x) = \mathcal{O}(\lambda(x)^{-K}).$$

It therefore does not change any coefficient in a Poincaré expansion.

Theorem 71.8 (Full exact-phase expansion; Lean)**.** *For every $N \geq 0$,*

$$\log F(x) = \mathcal{M}(x) + \sum_{j=1}^{N} \frac{A_j(\lambda(x))}{\lambda(x)^j} + \mathcal{O}(\lambda(x)^{-N-1}) \qquad (x \downarrow 0). \tag{725}$$

Equivalently, using a truncation with indices $j < N$,

$$\log F(x) - \mathcal{M}(x) - \sum_{j=1}^{N-1} \frac{A_j(\lambda(x))}{\lambda(x)^j} = \mathcal{O}(\lambda(x)^{-N}).$$

Proof. First combine (716), (724), and the flatness of $\mathcal{T}$ on the scale $x = 2^{-t}$. Next use the exact change of variables $t = -\log_2 x$. As $x \downarrow 0$, $t \to +\infty$ and $\lambda(2^{-t}) = \lambda(x)$, so the dyadic-real estimate transports without approximation. This yields (725). □

Warning 71.9 (Poincaré does not mean convergent)*.* The theorem is fixed-order. For each N there is an error constant depending on N, and the central radius itself depends on N. No uniform control as $N \to \infty$ is asserted, and no convergence of the infinite coefficient series is implied.

71.10 Exact transfer to the right endpoint

Corollary 71.10 (Right-endpoint deficiency; Derived from formalized identities)**.** *For every $N \geq 0$, as $x \downarrow 0$,*

$$\log\bigl(1 - F(1-x)\bigr) = \mathcal{M}(x) + \sum_{j=1}^{N} \frac{A_j(\lambda(x))}{\lambda(x)^j} + \mathcal{O}(\lambda(x)^{-N-1}). \tag{726}$$

Proof. The reflection identity gives $1 - F(1-x) = F(x)$ exactly. Substitute (725); no asymptotic error is introduced by the transfer. □

71.11 The separate elementary expansion of the Lambert phase

The exact-phase theorem above does not require replacing $\lambda(x)$. For comparison, put $S = -\log x$ and $z = \log S - \log L$. There are unique polynomials ℓ_j, with $\deg \ell_j = j$ and $\ell_j(0) = 0$, such that for each fixed $N \geq 1$

$$L\lambda(x) = S + z + \sum_{j=1}^{N-1} \frac{\ell_j(z)}{S^j} + \mathcal{O}\left(\frac{(1+|z|)^N}{S^N}\right). \tag{727}$$

The letter ℓ avoids collision with the periodic saddle jets p_n of section 67. The first terms are

$$\ell_1(z) = z, \qquad \ell_2(z) = z - \frac{z^2}{2}, \qquad \ell_3(z) = z - \frac{3z^2}{2} + \frac{z^3}{3}.$$

If $\mathfrak{s}(j,k)$ denotes an *unsigned* Stirling number of the first kind, then

$$\ell_j(z) = \sum_{m=1}^{j} \frac{(-1)^{m+1}}{m!} \mathfrak{s}(j, j-m+1) z^m. \tag{728}$$

Equivalently, one may use signed first-kind Stirling numbers and absorb the sign into that convention; mixing the signed convention with the displayed exterior sign would be wrong. Substituting a finite truncation of (727) into the periodic coefficients is a separate composition problem: the quadratic main exponent amplifies phase error, and even a bounded error moves the argument modulo one. No such all-orders substitution is claimed here.

72 Higher explicit corrections and their Fourier signatures

Merged specialist source. This section preserves the independent symbolic and numerical work formerly stored at `Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex`, SHA-256 `85f51a20fc7b6bdf3b1d049ec4506f508aee3c4cd70554e6a68cbcc30977cb0b`. Its repeated derivations of the lower-Lambert phase, periodic main term, closed jets, general coefficient recursions, and first two corrections have been removed in favor of part IV and sections 67, 69 and 70. What remains is the material genuinely added by that specialist calculation: exact mean identities, expanded third and fourth corrections, derivative/Fourier forms, amplitude estimates, high-precision checks, and numerical traps. The displayed A_3 and A_4 formulas and the numerical evidence are not promoted to Lean theorems by this merge.

72.1 The third and fourth coefficients in bounded-jet coordinates

The first two jet formulas are equations (710) and (714); continuing the same symbolic exponential–Gaussian–logarithm pipeline gives the following two research-frontier coefficients.

$$\begin{aligned}A_3 = -\frac{1}{720}\big(&180d_0^4 + 720d_0^3d_1 + 120d_0^3d_2 + 240d_0^3 + 360d_0^2d_1^2 + 2520d_0^2d_1\\&+ 1440d_0^2d_2 + 180d_0^2d_3 + 2160d_0d_1^2 + 720d_0d_1d_2 + 1440d_0d_1\\&+ 2520d_0d_2 + 960d_0d_3 + 90d_0d_4 + 120d_1^3 + 1980d_1^2 + 1800d_1d_2\\&+ 180d_1d_3 + 150d_2^2 + 720d_2 + 780d_3 + 210d_4 + 15d_5 - 2\big).\end{aligned}\tag{729}$$

One order further, in the same coordinates,

$$\begin{aligned}A_4 = \frac{1}{5760}\big(&1152d_0^5 + 8640d_0^4d_1 + 2880d_0^4d_2 + 240d_0^4d_3 + 1440d_0^4\\&+ 11520d_0^3d_1^2 + 2880d_0^3d_1d_2 + 28800d_0^3d_1 + 25920d_0^3d_2\\&+ 6720d_0^3d_3 + 480d_0^3d_4 + 2880d_0^2d_1^3 + 63360d_0^2d_1^2\\&+ 46080d_0^2d_1d_2 + 4320d_0^2d_1d_3 + 17280d_0^2d_1 + 2880d_0^2d_2^2\\&+ 46080d_0^2d_2 + 29280d_0^2d_3 + 6000d_0^2d_4 + 360d_0^2d_5\\&+ 23040d_0d_1^3 + 8640d_0d_1^2d_2 + 74880d_0d_1^2 + 123840d_0d_1d_2\\&+ 30720d_0d_1d_3 + 2160d_0d_1d_4 + 21600d_0d_2^2 + 3840d_0d_2d_3\\&+ 17280d_0d_2 + 30720d_0d_3 + 14640d_0d_4 + 2400d_0d_5 + 120d_0d_6\\&+ 720d_1^4 + 30720d_1^3 + 28800d_1^2d_2 + 2160d_1^2d_3 + 14400d_1^2\\&+ 3600d_1d_2^2 + 66240d_1d_2 + 36960d_1d_3 + 6720d_1d_4 + 360d_1d_5\\&+ 25200d_2^2 + 12000d_2d_3 + 840d_2d_4 + 480d_3^2 + 5760d_3\\&+ 7392d_4 + 2720d_5 + 360d_6 + 15d_7\big).\end{aligned}\tag{730}$$

Its top term is $15d_7/5760 = d_7/384$, and $(-1)^4/(2^4 \cdot 4!) = 1/384$: a fourth independent check of theorem 69.8.

The occurrence of $d_7/384$ in A_4 is consistent with the formalized mass top-jet coefficient in theorem 69.8. Its transfer through the logarithm, and hence the general highest-derivative conclusion, remains a derived calculation as stated there.

72.2 The same coefficients in the derivatives of Ψ

Write $P_k = \Psi^{(k)}(u)$. Then

$$\begin{aligned} A_1(u) &= \frac{1}{24} + \frac{1}{2L} - \frac{P_2 + P_1^2}{2L^2}, \\ A_2(u) &= \frac{P_1}{24L} + \frac{2P_1+1}{4L^2} - \frac{P_1^3 + 3P_1^2 + 3P_1P_2 + 3P_2 + P_3}{6L^3} \\ &\quad + \frac{4P_1^2P_2 + 4P_1P_3 + 2P_2^2 + P_4}{8L^4}. \end{aligned} \tag{731}$$

The corresponding form of A_3 is long. Its *linear part* is the following; the omitted terms are quadratic and higher in the derivatives of Ψ and reach 8.9×10^{-7} over a period, which is four decades below the 6.7×10^{-2} swing of A_3 but is not negligible if six digits are wanted.

$$\begin{aligned} A_3(u) = &-\frac{7}{2880} - \frac{1}{24L} - \frac{1}{4L^2} + \frac{1}{6L^3} + \frac{P_1+P_2}{24L^2} + \frac{48P_1 + 24P_2 - P_3}{48L^3} \\ &- \frac{6P_2 + 9P_3 + P_4}{12L^4} + \frac{3P_4 + P_5}{12L^5} - \frac{P_6}{48L^6} + (\text{quadratic and higher in the } P_k). \end{aligned} \tag{732}$$

72.3 The Ψ-free parts

j	Ψ-free part of A_j	value
1	$\frac{1}{24} + \frac{1}{2L}$	0.7630141871
2	$\frac{1}{4L^2}$	0.5203422453
3	$-\frac{7}{2880} - \frac{1}{24L} - \frac{1}{4L^2} + \frac{1}{6L^3}$	−0.0824216430
4	$\frac{-L^3 - 72L^2 - 816L + 144}{1152L^4}$	−1.7167954639

The order-two value is remarkably clean: the Ψ-free part of A_2 is exactly $1/(4\log^2 2)$, with no rational term and no $1/L$ term at all.

72.4 Exact period means

The Ψ-free part is not literally the period mean, because nonlinear products of derivatives can have nonzero mean. Write $\langle H\rangle = \int_0^1 H(u)\,\mathrm{d}u$. Fourier multiplication gives the exact identity

$$\left\langle \Psi^{(a)}\Psi^{(b)} \right\rangle = \sum_{k\neq 0} (2\pi\mathrm{i}k)^a(-2\pi\mathrm{i}k)^b|\widehat{\Psi}(k)|^2. \tag{733}$$

Integration by parts over one period gives, in particular,

$$\langle \Psi'\Psi''\rangle = 0, \qquad \langle (\Psi')^2\Psi''\rangle = 0, \qquad \langle \Psi'\Psi'''\rangle = -\langle (\Psi'')^2\rangle.$$

Substitution in the exact formulas for A_1 and A_2 leaves

$$\langle A_1\rangle = \frac{1}{24} + \frac{1}{2L} - \frac{\langle (\Psi')^2\rangle}{2L^2}, \tag{734}$$

$$\langle A_2\rangle = \frac{1}{4L^2} - \frac{\langle (\Psi')^2\rangle}{2L^3} - \frac{\langle (\Psi'')^2\rangle}{4L^4} - \frac{\langle (\Psi')^3\rangle}{6L^3}. \tag{735}$$

These are identities, not truncated estimates. Numerically,

$$\langle(\Psi')^2\rangle = 8.917038\times 10^{-11},\quad \langle(\Psi'')^2\rangle = 3.520305\times 10^{-9},\quad \langle(\Psi')^3\rangle = 1.115006\times 10^{-21},$$

and hence

$$\langle A_1\rangle = 0.7630141870184,\qquad \langle A_2\rangle = 0.5203422413049.$$

Their gaps from the Ψ-free constants are respectively about -9.3×10^{-11} and -3.9×10^{-9}. The cubic term in (735) is tiny but retained because deleting it would turn an exact identity into an approximation.

The Gamma–zeta formula (478) gives, numerically,

$$\begin{aligned}
\widehat{\Psi}(1) &= (7.55864754098 - 7.47010355898\,\mathrm{i})\,10^{-7}, & |\widehat{\Psi}(1)| &= 1.06271162519\,10^{-6},\\
\widehat{\Psi}(2) &= (3.24812347723 + 5.85175898712\,\mathrm{i})\,10^{-13}, & |\widehat{\Psi}(2)| &= 6.69278636792\,10^{-13},\\
\widehat{\Psi}(3) &= (-2.81236523199 - 2.58674397916\,\mathrm{i})\,10^{-19}, & |\widehat{\Psi}(3)| &= 3.82107872359\,10^{-19}.
\end{aligned}$$

Thus Ψ itself oscillates only by a few parts in 10^6, even though repeated differentiation greatly amplifies that oscillation in the higher saddle coefficients.

73 Fourier form and the amplitude law

Because $\Psi^{(m)}(u) = \sum_{k\neq 0}(2\pi\mathrm{i}k)^m\widehat{\Psi}(k)\mathrm{e}^{2\pi\mathrm{i}ku}$, section 69 together with equation (478) turns every coefficient into an explicit Gamma–zeta Fourier object. Discarding the terms of degree two and higher in the derivatives of Ψ leaves the linearized Fourier form

$$A_j(u) = A_j^{(0)} + 2\,\mathrm{Re}\Big(\sum_{k\geq 1} c_j(k)\widehat{\Psi}(k)\mathrm{e}^{2\pi\mathrm{i}ku}\Big) + R_j(u),$$

where $A_j^{(0)}$ is the Ψ-free constant and R_j is the raw sum of terms of degree at least two in derivatives of Ψ. Thus $\langle A_j\rangle = A_j^{(0)} + \langle R_j\rangle$, as quantified above. The discarded remainder is measured at $\max_u|R_j(u)| = 1.6\times 10^{-8}$ for $j=2$ and 8.9×10^{-7} for $j=3$, growing with j like everything else here. It is negligible against the oscillation it sits inside, and it is not negligible in absolute terms. Here $c_j(k)$ is the polynomial in $w = 2\pi\mathrm{i}k$ read off from equations (731) and (732). For the first two,

$$c_1(k) = -\frac{w^2}{2L^2},\qquad c_2(k) = \frac{w}{24L} + \frac{w}{2L^2} - \frac{3w^2+w^3}{6L^3} + \frac{w^4}{8L^4}.$$

The resulting data for A_2 are all confirmed numerically in section 74. For the first Fourier mode, the linearized amplitude is $2|c_2(1)\widehat{\Psi}(1)|$. The left column is what the linearized form gives, the right what the exact A_2 gives; they are quoted separately because the difference is real, if small, and quoting ten digits of a linearization would be an overclaim.

Ψ-free constant / mean	$\frac{1}{4L^2} = 0.5203422452514$	$\langle A_2\rangle = 0.5203422413049$
$k=1$ amplitude	1.9398782×10^{-3}	1.9398782×10^{-3}
maximum at	$\{\lambda\} = 0.1011297$	$\{\lambda\} = 0.1011281$
minimum at	$\{\lambda\} = 0.6011297$	$\{\lambda\} = 0.6011313$

The exact peak-to-peak is 3.8797564×10^{-3}, twice the linearized $k=1$ amplitude to the displayed precision, and the extrema agree to six. The maximum of the exact A_2 is 0.522282117 and its minimum is 0.518402361.

73.1 The amplitude law

The top term of theorem 69.8 dominates the oscillation, and it multiplies the k-th Fourier mode by $(2\pi k)^{2j}$. Taking $k = 1$,

$$\text{amplitude of } A_j \;\approx\; 2|\widehat{\Psi}(1)|\,\frac{(2\pi^2/L^2)^j}{j!}, \qquad \frac{2\pi^2}{L^2} = 41.0845769104. \tag{736}$$

j	predicted $k = 1$ amplitude
1	8.73×10^{-5}
2	1.79×10^{-3}
3	2.46×10^{-2}
4	2.52×10^{-1}
5	2.07
6	1.42×10^{1}
7	8.33×10^{1}

These are estimates from the top term alone. Comparing them with the truth requires care about a convention: an *amplitude* is half a peak-to-peak swing, and mixing the two silently doubles every second entry. Everything below is an amplitude.

j	equation (736)	actual	ratio
1	8.73×10^{-5}	8.73×10^{-5}	1.00
2	1.79×10^{-3}	1.94×10^{-3}	1.08
3	2.46×10^{-2}	$\approx 3 \times 10^{-2}$	≈ 1.2
4	2.52×10^{-1}	≈ 0.41	≈ 1.6

At $j = 1$ the top derivative term dominates to the displayed precision, but it is not literally the only periodic term: A_1 also contains $-(\Psi')^2/(2L^2)$. The equality in the table is therefore a very accurate linearization, not an exact amplitude identity. At $j = 2$ the w^3 term of c_2 adds in phase and lifts the amplitude above the estimate; the entries at $j = 3$ and $j = 4$ are halves of the swings measured in section 74. The ratio drifts upward because the subleading terms of c_j contribute more as j grows. So equation (736) gives the growth rate and not the constant, and should never be used to size a particular coefficient. So the periodic parts of the coefficients grow rapidly with j, even though Ψ itself has amplitude only about 2×10^{-6} (roughly 4×10^{-6} peak to peak): differentiation progressively cancels the smallness of $\widehat{\Psi}(1)$. Any numerical extraction of a coefficient beyond the first must therefore control the phase; see section 75.

73.2 Why the expansion is only asymptotic

$|\widehat{\Psi}(k)|$ decays like $\exp(-\pi|\chi_k|/2) = \exp(-\pi^2 k/L)$, that is, about $\mathrm{e}^{-14.24k}$, while mode k is amplified by $(2\pi k)^{2j}/(2^j j!\, L^{2j})$. The k-th contribution therefore behaves like $\exp(2j \log k - 14.24k)$, maximized near $k^* = 2j/14.24 = 0.14j$. For j beyond about 8 the higher modes dominate, and $\sup|A_j|$ grows faster than geometrically.

What is worth noticing is *which* divergence this is. A saddle-point expansion normally diverges factorially, because the Gaussian moments grow like $(2j-1)!!$ against a fixed exponent. Here that growth is exactly cancelled: the $(2j)!$ in E_{2j} absorbs it, which is the content of theorem 69.8 and the reason the coefficient of the top jet is the mild $1/(2^j j!)$. The divergence that remains is of a different origin altogether — it is driven by the high-frequency content of Ψ, through the $(2\pi k)^{2j}$ that differentiation applies to the k-th mode. A reader who expects the usual factorial mechanism will look for it in the Gaussian integration, where it is not. (This framing is due to the parallel write-up of the main article, where it also appears.)

Remark 73.1. The expansion is a Poincaré asymptotic expansion: each truncation is valid with an $\mathcal{O}(\lambda^{-N})$ remainder for fixed N, and there is no claim, here or in the formalization, that the series converges for any fixed x. The estimate in this subsection is a heuristic size estimate, not a theorem.

74 Numerical verification

$F(2^{-n})$ is computed *exactly* as a rational number from the inverse-power recursion used by the Lean evaluator (`Arithmetic.lean`, `fabiusInversePowTwoTable`):

$$v_0 = 1, \qquad v_{n+1} = \frac{1}{2^{n+1}-1}\sum_{k=0}^{n} \frac{v_k}{2^{\binom{n+1}{2}-\binom{k}{2}}\,(n-k+2)!}, \qquad F(2^{-n}) = v_n.$$

Sanity checks: $v_1 = 1/2 = F(1/2)$ and $v_2 = 5/72 = F(1/4)$. All computations below use 260-digit working precision, with Ψ and its derivatives summed over six Gamma–zeta modes and ζ evaluated by Euler–Maclaurin (section 75).

Define the successive residuals

$$r_1 = \lambda\big(\log F - \mathcal{M}\big), \quad r_2 = \lambda^2\Big(\log F - \mathcal{M} - \frac{A_1}{\lambda}\Big), \quad r_3 = \lambda^3\Big(\log F - \mathcal{M} - \frac{A_1}{\lambda} - \frac{A_2}{\lambda^2}\Big).$$

n	λ	r_1	$A_1(\lambda)$	r_2	$A_2(\lambda)$	r_3	$A_3(\lambda)$
10	13.78503056	0.7997155875	0.7629678468	0.5065687288	0.5195595404	−0.1790787344	−0.0868938679
20	24.62186833	0.7836542295	0.7629268731	0.5103462408	0.5184188039	−0.1987615843	−0.1111875606
40	45.50804986	0.7742689337	0.7629490545	0.5151456258	0.5187247759	−0.1628801414	−0.1121601297
60	66.04538587	0.7709808331	0.7630910552	0.5210834261	0.5221643489	−0.0713899656	−0.0512594569
80	86.43351899	0.7689739453	0.7629823191	0.5178773369	0.5193822939	−0.1300787245	−0.1046130809
120	126.9885547	0.7671769004	0.7630717251	0.5213102874	0.5218167515	−0.0643151389	−0.0540815552
160	167.3870441	0.7661091776	0.7630070725	0.5192522062	0.5199082072	−0.1098060709	−0.0973365576

The decisive test is that each residual, minus its predicted coefficient, is $\mathcal{O}(1/\lambda)$. Writing $e_j = r_j - A_j$:

n	$\{\lambda\}$	λe_1	λe_2	λe_3
100	0.737929	0.51802529	−0.11213991	−1.7523898
112	0.893526	0.52019699	−0.07798492	−1.3773605
124	0.033795	0.52164121	−0.06156984	−1.3168268
136	0.161500	0.52166737	−0.06826635	−1.5329675
148	0.278716	0.52062406	−0.08861040	−1.8452719
160	0.387044	0.51925221	−0.10980607	−2.0872350

λe_1 reproduces A_2 with mean 0.5203; λe_2 reproduces A_3 with mean -0.0824; and λe_3 is bounded, locating A_4 between about -1.3 and -2.1. Each column stays bounded and oscillates with the phase rather than drifting, which is precisely the assertion that the corresponding coefficient is correct and the next one is $\mathcal{O}(1)$.

Pointwise, A_2 is confirmed at the 10^{-5} level once the next order is included. For $140 \le n \le 160$:

n	$\{\lambda\}$	$A_2(\lambda)$	r_2	$r_2 - A_3(\lambda)/\lambda$
140	0.201650	0.5219078895	0.5214039042	0.5218323910
145	0.250301	0.5214906179	0.5209456288	0.5214143255
150	0.297351	0.5209853273	0.5203979284	0.5209087825
155	0.342900	0.5204425007	0.5198167900	0.5203665207
160	0.387044	0.5199082072	0.5192522062	0.5198337121

The last two columns agree to about 7.5×10^{-5}, which is A_4/λ^2 with $A_4 \approx -2$. Note that the *variation* of A_2 over this window is 2.0×10^{-3}, twenty-six times the residual, so this is a genuine pointwise verification of the periodic shape and not merely of a mean.

Remark 74.1. An independent high-precision computation performed in a separate worktree, using half-moments to $n = 400$ at 400-digit precision and a different route to λ, located the minimum of the order-λ^{-2} residual at $\{\lambda\} = 0.602$ and measured a peak-to-peak spread of about 3×10^{-3}; equation (731) predicts a minimum at 0.6011313 and a peak-to-peak of 3.8798×10^{-3} over the full period. That is a phase prediction, not a magnitude fit, and it was made before the comparison.

74.1 The fourth coefficient

The same apparatus confirms equation (730). Writing $r_4 = \lambda^4(\log F - \mathcal{M} - A_1/\lambda - A_2/\lambda^2 - A_3/\lambda^3)$, at 300-digit precision with eight Gamma–zeta modes:

n	$\{\lambda\}$	r_4	$A_4(\lambda)$
120	0.988555	−1.299548008	−1.305123624
136	0.161500	−1.532967481	−1.506670619
152	0.315745	−1.939809221	−1.890185255
160	0.387044	−2.087234974	−2.035031382
176	0.519792	−2.166743696	−2.124617975
192	0.641266	−1.992935176	−1.970225155
200	0.698346	−1.853347748	−1.840001848

$\lambda(r_4 - A_4)$ stays inside $[-8.8, 0.8]$ over $120 \le n \le 200$ with no drift, which places A_5 and confirms A_4. The Ψ-free part -1.7167954639 sits inside the measured range, and the swing of the A_4 column, 0.82 peak to peak, is 1.6 times the leading-term estimate 0.50 of equation (736) — the same upward drift of that ratio seen at $j = 2$ (1.08) and $j = 3$ (1.24), because the subleading terms of $c_j(1)$ contribute more as j grows. Equation (736) is a leading-term estimate and should be read as one.

74.2 A worked example

It is worth seeing the expansion used once, on a value that can be written down exactly. At $x = 2^{-10}$ the evaluator gives

$$F(2^{-10}) = \frac{134\,926\,369}{1\,246\,394\,851\,358\,539\,387\,238\,350\,848\,000},$$
$$\log F(2^{-10}) = -50.57756828234216081254\ldots.$$

and $\lambda = 13.7850305606$, so $\{\lambda\} = 0.785031$. The successive truncations miss $\log F$ by

truncation	error at $n = 10$	error at $n = 160$
$\mathcal{M}(x)$	5.80×10^{-2}	4.58×10^{-3}
$+ A_1/\lambda$	2.67×10^{-3}	1.85×10^{-5}
$+ A_2/\lambda^2$	-6.84×10^{-5}	-2.34×10^{-8}
$+ A_3/\lambda^3$	-3.52×10^{-5}	-2.66×10^{-9}

where $n = 160$ has $\lambda = 167.387044$ and $\log F(2^{-160}) = -9489.629874649676879065\ldots$

Two things are visible. At $n = 160$ each order gains roughly a factor λ, as it should: 250, then 790. The last gain is only 8.8, and that is not a defect — the third-order error is already dominated by A_4/λ^4, which equation (730) puts at $-1.72/167.387^4 = -2.19 \times 10^{-9}$ against a measured -2.66×10^{-9}. So the table is measuring A_4, not a failure.

At $n = 10$, by contrast, the fourth term barely helps: the error falls only from 6.8×10^{-5} to 3.5×10^{-5}. With $\lambda = 13.8$ and $A_4 \approx -1.7$, the term A_4/λ^4 is itself 4.7×10^{-5}, comparable to the error it is meant to reduce. This is the ordinary behaviour of a Poincaré expansion near the end of its useful range, and it is the practical reason section 73 declines to claim convergence: the coefficients grow, so for a fixed x there is a best truncation, and going past it makes the approximation worse. At $x = 2^{-10}$ that optimum is already at hand around the third term.

74.3 An independent confirmation to $n = 400$

The following was produced in a separate worktree, by a different agent, from a different representation of F, and is reproduced with its permission. To avoid collision with the bounded saddle jets d_n, write h_n for the half moments in this paragraph. They were computed to $n = 400$ at 400-digit working precision from

$$h_n = \frac{1}{(n+1)(2^n - 1)} \sum_{k<n} \binom{n+1}{k} h_k,$$

whose terms are all positive, so there is no cancellation; the values agree with the exact rationals to 400 digits at $n = 5, 20, 60$. Then $\log F(2^{-n}) = \log h_n - \log n! - \binom{n}{2} \log 2$, with λ from the lower Lambert branch (satisfying $\lambda 2^{-\lambda} = x$ to 10^{-21}) and Ψ from ten Gamma–zeta modes. Nothing in this table shares an implementation with the preceding numerical tables.

n	λ	$\{\lambda\}$	r_2 measured	$A_2(\lambda)$	r_3
100	106.73793	0.7379	0.51802529	0.51907590	−0.11214
125	132.04488	0.0449	0.52169792	0.52216224	−0.06131
150	157.29735	0.2974	0.52039793	0.52098533	−0.09240
175	182.51185	0.5119	0.51801879	0.51869969	−0.12427
200	207.69835	0.6984	0.51821775	0.51875313	−0.11120
225	232.86334	0.8633	0.52015631	0.52049092	−0.07792
250	258.01129	0.0113	0.52175813	0.52198119	−0.05755
275	283.14540	0.1454	0.52199208	0.52220755	−0.06101
300	308.26804	0.2680	0.52104845	0.52130958	−0.08050
325	333.38103	0.3810	0.51967269	0.51997997	−0.10244
350	358.48577	0.4858	0.51856496	0.51889001	−0.11653
375	383.58340	0.5834	0.51810490	0.51841439	−0.11872
400	408.67481	0.6748	0.51833670	0.51860654	−0.11028

The decisive column is the last. Over this range λ quadruples, so an error of even 10^{-4} in A_2 would appear as a drift in r_3 of order 3×10^{-2} with a definite sign. Instead r_3 is -0.112 at $n = 100$ and -0.110 at $n = 400$, never leaving $[-0.125, -0.058]$ in between, and oscillating rather than drifting — which is what a periodic A_3 requires. Its sample mean over the thirteen rows is about -0.088, against the Ψ-free part -0.0824216430 of A_3; the sample is not a uniform phase average and carries an A_4/λ term, so the gap is expected.

The $A_2(\lambda)$ column also straddles the predicted mean $1/(4L^2) = 0.5203422452514019$, and the row-by-row differences $r_2 - A_2$ match the prediction $A_3(\lambda)/\lambda$ to about 10^{-6}: at $\{\lambda\} = 0.2680$ the predicted difference is -2.62×10^{-4} against a measured -2.611×10^{-4}, and at $\{\lambda\} = 0.6748$, -2.71×10^{-4} against -2.698×10^{-4}.

Neither this table nor the preceding numerical calculations is machine-checked. What is machine-checked is listed in section 77.

75 Two traps

Warning 75.1 (The phase trap). Sampling $n = 10, 20, 40, 80, 160, 320$ to estimate a coefficient is invalid. Doubling n adds about 1 to $\log_2 \lambda$, so $\{\lambda\}$ returns to nearly the same place and a geometric sample holds the phase almost fixed. Since every coefficient is one-periodic, such a sample cannot see the oscillation at all; it shows only the $\mathcal{O}(1/\lambda)$ drift in magnitude, and looks like convergence to a constant. This mistake was made independently on this project before it was caught. Always sweep n densely and report against $\{\lambda\}$, as in section 74.

Warning 75.2 (The zeta trap). Many ζ implementations route through the alternating Dirichlet eta series, dividing by $1 - 2^{1-s}$. At $s = 1 - \chi_k$ one has $2^{1-s} = \mathrm{e}^{2\pi \mathrm{i} k} = 1$ exactly, so that prefactor has a pole at precisely the arguments equation (478) needs. The singularity is removable in ζ but not in that formula, and the result is either an overflow or a plausible wrong number. Concretely, `mpmath.zeta` at these arguments returns values whose seventh significant digit drifts with the working precision:

$$\begin{aligned} \texttt{mp.dps} = 30: &\quad \widehat{\Psi}(1) = 7.55865005 \times 10^{-7} - 7.47011001 \times 10^{-7}\,\mathrm{i}, \\ \texttt{mp.dps} = 50: &\quad \widehat{\Psi}(1) = 7.55864674 \times 10^{-7} - 7.47009636 \times 10^{-7}\,\mathrm{i}, \\ \texttt{mp.dps} = 80: &\quad \widehat{\Psi}(1) = 7.55865060 \times 10^{-7} - 7.47008702 \times 10^{-7}\,\mathrm{i}, \end{aligned}$$

whereas a hand-rolled Euler–Maclaurin ζ ($N = 300$, 25 Bernoulli terms) is precision-stable and returns $7.55864754097769 \times 10^{-7} - 7.47010355898008 \times 10^{-7}\,\mathrm{i}$ at every working precision. Use Euler–Maclaurin or Hurwitz, and always check at two precisions.

76 Specialist open items

The synthesis leaves the following claim-level obligations explicit:

1. Package theorems 69.4 and 69.6 for the generic `expCoeff` and `logCoeff` recursions as exact general Lean declarations. This is the remaining formal composition layer, so it is worth recording what the implementation requires. Mathlib has `Composition n` with a `Fintype` instance. Writing $\mathcal{C}(N)$ for the finite type of ordered compositions of N into positive parts, the missing ingredient is a first-block decomposition equivalence

   $$\mathcal{C}(n+1) \simeq \coprod_{b=1}^{n+1} \mathcal{C}(n+1-b),$$

   which is the form needed for induction against the recursion.

   The route that avoids `Composition` is to work directly with the k-fold finite-convolution powers E^{*k}, as plain finite sums, and prove

   $$\mathrm{expCoeff}(E)_n = \sum_{k \le n} \frac{1}{k!} (E^{*k})_n \qquad (E_0 = 0).$$

   At coefficient level the product rule is immediate: writing $(\mathrm{D}A)_n = nA_n$,

   $$\mathrm{D}(A * B) = (\mathrm{D}A) * B + A * (\mathrm{D}B)$$

   is the identity $n = j + (n - j)$ inside the convolution sum. Promoting it to

   $$\mathrm{D}(E^{*(k+1)}) = (k+1)\, E^{*k} * \mathrm{D}E$$

   also needs associativity of convolution, a double-sum reindexing not presently packaged for bare sequences. Either route is a self-contained finite-sum combinatorics project; neither bridge is supplied merely by the displayed natural-language formulas.

2. formalize the transfer of the mass top-jet coefficient in theorem 69.8 through the logarithm and the resulting highest-Ψ-derivative coefficient at every order;

3. formalize or independently certify the displayed A_3 and A_4 formulas and their numerical checks;
4. surface effective constants in the fixed-order $O(\lambda^{-N})$ remainders; and
5. prove or refute convergence, Borel summability, or a sharper optimal-truncation law for the formal correction series discussed in section 73.

No substitution of a truncated elementary approximation for the exact phase inside the periodic coefficients is asserted. No claim is made that an individual A_j is nowhere zero, and all numerical tables remain evidence rather than proof.

77 A proof-oriented concordance with the Lean modules

The following table records where the two missing clusters are proved. Paths are relative to

`Analysis/FabiusFunction/Lean/FabiusFunction/`.

Table 25: Module concordance for this companion.

Module	Principal role	Status
`FabiusInverse.lean`	Order isomorphism of $[0,1]$, totalized inverse, global continuity and monotonicity, comparison equivalences, tails, reflection, exact dyadic inversion, reciprocal first and second derivatives, interior smoothness, strict closed-half curvature, transported flatness, effective steepness, exact global differentiability and C^∞ loci, and endpoint quotient divergence.	Lean
`InverseNotElementary.lean`	Exact analytic locus of the totalized inverse and the analytic-density obstructions used in its non-elementarity theorem.	Lean
`FabiusSaddleJet ClosedForm.lean`	Operator solution of the recursive periodic jets and the harmonic-number constant term.	Lean
`FabiusSaddleJet Stirling.lean`	Identification of the operator coefficients with $s(n+1,m+1)$ and the corresponding closed derivative formula.	Lean
`FabiusSaddleExponent ClosedForm.lean`	Collapse of each centered exponent coefficient to one bounded jet monomial plus one universal monomial; parity.	Lean
`FabiusSaddleExpansion Coefficients.lean`	Formal exponential coefficients, Gaussian contraction, mass coefficients, logarithmic transfer, periodicity, smoothness, and boundedness.	Lean
`FabiusSaddleLeading Coefficient.lean`	Affine dependence of a_j on the newest jet and coefficient $(-1)^j/(2^j j!)$.	Lean
`FabiusSecondSaddle Correction.lean`	Order-four exponential identity, explicit B_4, Gaussian contraction, a_2, A_2, and regularity of the second correction.	Lean
`FabiusSecondSaddle Expansion.lean`	Insertion of A_2 into the exact-phase asymptotic with a third-order remainder.	Lean

Module	Principal role	Status
`FabiusSaddleCentralAllOrders.lean`	Exact central decomposition, finite reference polynomial, odd-term cancellation, and integrated central remainders.	Lean
`FabiusSaddleTailAllOrders.lean`	Order-dependent radius and complementary intermediate/far contour bounds of arbitrary fixed inverse-power order.	Lean
`FabiusSaddleMassAllOrders.lean`	Assembly of central Taylor control, finite exponential substitution, whole-line Gaussian contraction, and minor arcs into the complete mass expansion.	Lean
`FabiusFullAsymptoticExpansion.lean`	Transfer from mass to logarithm, addition of the flat forward tail, and transport from $x = 2^{-t}$ to the full limit $x \downarrow 0$.	Lean

77.1 Named declarations worth knowing

The inverse interface is centered around

```
fabiusOrderIso
fabiusReal_fabiusInv
fabiusInv_fabiusReal
strictMonoOn_fabiusInv
monotone_fabiusInv
continuous_fabiusInv
fabiusInv_le_iff_le_fabiusReal
fabiusReal_le_iff_le_fabiusInv
fabiusInv_lt_iff_lt_fabiusReal
fabiusReal_lt_iff_lt_fabiusInv
fabiusInv_one_sub
fabiusInv_fabiusDyadicUnit
fabiusInv_differentiableAt_iff
fabiusInv_contDiffAt_infty_iff
fabiusInv_analyticAt_iff
fabiusInv_not_isElementaryOrInverse
le_two_pow_mul_fabiusInv_pow
le_two_pow_div_factorial_mul_fabiusInv_pow
mul_lt_fabiusInv
tendsto_fabiusInv_div_atTop
tendsto_one_sub_fabiusInv_div_one_sub_atTop
```

The all-orders endpoint interface culminates in

```
fabiusSaddleKernelMass_dyadicLambert_hasAsymptoticExpansion
fabiusSaddleRatio_dyadicLambert_hasAsymptoticExpansion
log_fabius_dyadicReal_sub_sharpLambertMain_hasAsymptoticExpansion
log_fabius_sub_sharpLambertMain_hasAsymptoticExpansion
log_fabius_sub_sharpLambertExpansion_isBigO
```

These names are not needed to read the mathematics, but they make the article useful as a map back into the source tree.

78 What is formalized and what is newly synthesized

For clarity, the principal statements can be divided into three groups.

78.1 Directly formalized

The following content is represented directly by declarations in the repository:

- the global totalized inverse package and all four comparison equivalences;
- the constant tails, reflection, midpoint, and exact dyadic inversion;
- interior C^∞ regularity, the exact formulas for G' and G'', and strict concavity on $[0, 1/2]$ and strict convexity on $[1/2, 1]$;
- the effective and sharp inverse power inequalities, effective linear steepness, and the two endpoint quotient limits;
- the exact global differentiability, C^∞, and analytic loci of the totalized inverse;
- the closed jet formulas, shifted Stirling conversion, and closed exponent coefficients;
- the formal exponential and logarithmic recurrence machinery, the mass top-jet coefficient, and the first and second corrections;
- the arbitrary-order central, tail, mass, and logarithmic asymptotic theorems.

78.2 Classical consequences derived in this article

The following statements are proved here from the formalized input:

- divergence of $G(y)/y^\alpha$ for every $\alpha > 0$;
- the sharp equivalent (679) and the scale hierarchy (682)–(683);
- the exact right-endpoint transfer (726), the separate Lambert-phase expansion (727), and the period-mean identities (734)–(735);
- the explicit all-j composition sum (704) and the highest-Ψ-derivative corollary (709).

78.3 Symbolic and numerical frontier

The expanded A_3 and A_4 formulas, their derivative forms and Ψ-free parts, the Fourier-amplitude estimates, and the residual tables in sections 72 to 74 are retained as symbolic or numerical evidence. They are calibrated against the formalized recursions and the first two corrections, but no exact current Lean declaration certifies the displayed higher formulas or their numerical verification.

78.4 Claims deliberately not made

This companion does *not* claim:

- convergence or Borel summability of the all-orders series;
- an all-orders substitution of an approximate Lambert phase inside the periodic coefficients;
- a closed form for the inverse at arbitrary rational arguments;
- analyticity of either F or G inside the transition interval;
- uniformity of the remainder constants as the truncation order tends to infinity.

79 Concluding synthesis

The inverse function and the saddle coefficient machinery illuminate opposite sides of the same endpoint phenomenon.

On the geometric side, F approaches zero more rapidly than every power. Its inverse must therefore leave zero more rapidly than every power, and the formalized comparison calculus turns this slogan into effective inequalities with exact constants. The sharp Lambert expansion goes further: the inverse lives on the stretched-logarithmic scale

$$G(y) \asymp \sqrt{\log(1/y)}\,\exp\Big(-\sqrt{2\log 2\ \log(1/y)}\Big)\,,$$

with the exact leading constant $1/(\mathrm{e}\sqrt{\log 2})$. Reflection transfers the entire picture to the endpoint 1.

On the asymptotic side, the apparent complexity of the periodic corrections is organized by three simple structures. First, the recursive jets are solved by a product of commuting first-order operators; the forcing terms sum to a harmonic number and the derivative coefficients are shifted signed Stirling numbers. Second, centering collapses every exponent coefficient to two monomials, forcing parity and eliminating all odd half-powers under Gaussian contraction. Third, the newest jet can enter only linearly, so the coefficient of the highest derivative $\Psi^{(2j)}$ is known at every order before the lower-order algebra is performed.

The analytic remainder proof then explains why these formal coefficients are not merely symbolic. The growing window $A_N(b) = \sqrt{32(N+1)\log b}$ separates the local and global tasks: locally it is still narrow enough for uniform Taylor expansion, while globally it is wide enough to make both Gaussian and contour tails smaller than the requested inverse power. Finite exponential substitution and logarithmic transfer complete a genuine Poincaré expansion at the exact Lambert phase.

Together, these missing parts close two expository loops. Flatness of F is now matched by a precise theory of the steep inverse, and the headline all-orders expansion is now matched by a transparent account of the coefficient algebra and the analytic estimates that make it true.

Supplementary material from this synthesis

80 Repository paths used in the audit

```
Analysis/FabiusFunction/
  Lean/FabiusFunction/FabiusInverse.lean
  Lean/FabiusFunction/FabiusSaddleJetClosedForm.lean
  Lean/FabiusFunction/FabiusSaddleJetStirling.lean
  Lean/FabiusFunction/FabiusSaddleExponentClosedForm.lean
  Lean/FabiusFunction/FabiusSaddleLeadingCoefficient.lean
  Lean/FabiusFunction/FabiusSecondSaddleCorrection.lean
  Lean/FabiusFunction/FabiusSecondSaddleExpansion.lean
  Lean/FabiusFunction/FabiusSaddleCentralAllOrders.lean
  Lean/FabiusFunction/FabiusSaddleTailAllOrders.lean
  Lean/FabiusFunction/FabiusSaddleMassAllOrders.lean
  Lean/FabiusFunction/FabiusFullAsymptoticExpansion.lean
  docs/Fabius_Function_and_Rvachev_Up/
    Fabius_Function_and_Rvachev_Up.tex
```

Topic references

[1] J. Fabius, A probabilistic example of a nowhere analytic C^∞-function, *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* **5** (1966), 173–174. DOI: 10.1007/BF00536652.

[2] V. A. Rvachev, Compactly supported solutions of functional-differential equations and their applications, *Russian Mathematical Surveys* **45** (1990), no. 1, 87–120.

[3] J. Arias de Reyna, Definición y estudio de una función indefinidamente diferenciable de soporte compacto, *Revista de la Real Academia de Ciencias Exactas, Físicas y Naturales de Madrid* **76** (1982), no. 1, 21–38. English translation: *An infinitely differentiable function with compact support: Definition and properties*, arXiv:1702.05442 (2017).

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[5] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey, and D. E. Knuth, On the Lambert W function, *Advances in Computational Mathematics* **5** (1996), 329–359.

[6] V. Reshetnikov, *ProveIt: Analysis/FabiusFunction*, Lean 4 formal development, repository snapshot 25 August 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction.

[7] V. Reshetnikov, *The Fabius Function and Rvachev's Up-Function*, repository document in Analysis/FabiusFunction/docs/Fabius_Function_and_Rvachev_Up, snapshot 25 August 2026.

Part VII
Primary Exposition Gap Register

Source provenance and status. Former path: `Primary_Exposition_Gap_Register/Primary_Exposition_Gap_Register.tex`. This post-consolidation register preserves smaller claims removed during the primary exposition's exact-counterpart audit. Candidate statements below are of two kinds. A candidate marked *discharged* carries a named Lean citation in the surrounding text: a public declaration now matches its hypotheses, normalization, domain and conclusion, and its former formalization obligation has been deleted. A candidate carrying no such citation is still open; it is not claimed as a Lean theorem, and it still needs a public theorem with the stated hypotheses and conclusion. Declarations identified as formal background support an open candidate without stating it.
This register is audited against a library that keeps moving. Six of its original obligations were discharged by formalization work that landed within days of the audit date: one on the audit date itself, four the day after, one the day after that. The entries below record that outcome. The remaining open entries carry the same exposure: they are accurate as of the audit date recorded under *Provenance* at the end of this part and can be overtaken in the same way, so re-check them against the current library before relying on them.

81 Fourier-decay normalization

The source proves polynomially weighted decay. The conventional shifted-weight form was removed from the primary exposition for want of a named counterpart; the library has since supplied one, so the statement is recorded here as proved rather than as a gap:

Candidate statement 81.1 (Discharged). For every $N \in \mathbb{N}$, there is $C_N > 0$ such that

$$|\widehat{\mathrm{up}}(\xi)| \le C_N(1+|\xi|)^{-N} \qquad (\xi \in \mathbb{R}).$$

It is standardly equivalent to the proved polynomially weighted estimate after a bounded-region argument and a change of constant, and it is now also a literal named Lean theorem: `rvachevFourier_real_shiftedDecay` in `PoissonSummation.lean` states exactly the displayed bound for every exponent, universally quantified over all real arguments, so the range $|\xi| \le 1$ is covered by the statement itself rather than by a separate argument. The formal library proves strictly more than the candidate: `rvachevFourier_real_iteratedDeriv_shiftedDecay` in the same module gives the same shifted-weight majorant for every iterated derivative of the transform, and the displayed estimate is its order-zero case.

82 Dyadic specializations and evaluator cost

82.1 Concrete rational values

Let F denote the bounded Fabius function and up Rvachev's compactly supported up-function. Write $c_n = \int_{-1}^{1} x^{2n}\,\mathrm{up}(x)\,\mathrm{d}x$. The general dyadic evaluator is proved correct, and these specializations now have named theorems:

Candidate statement 82.1 (Discharged).

$$c_1 = \frac{1}{9}, \qquad F(1/4) = \frac{5}{72}, \qquad F(1/8) = \frac{1}{288}, \qquad F(3/8) = \frac{73}{288}.$$

Formalization obligation 82.2 (Discharged). Prove named specializations by reducing the exact evaluator and bridge each result to the analytic function.

All four values now have named theorems in `DyadicSpecializations.lean`, each in the two requested forms. The executable reductions are `moment_one` (‘$\mu_1 = 1/9$’ from the defining recurrence), `fabiusDyadic_two_one`, `fabiusDyadic_three_one`, and `fabiusDyadic_three_three`, closed by rational normalization of the dyadic formula. The analytic bridges are `integral_sq_mul_rvachev_eq_one_ninth` together with its interval form `intervalIntegral_sq_mul_rvachev_eq_one_ninth` (the display's c_1), and `fabiusReal_one_quarter`, `fabiusReal_one_eighth`, `fabiusReal_three_eighths`, valid for every bounded Fabius function via `fabiusDyadic_cast` and `integral_even_pow_mul_rvachev_eq_moment`.

82.2 Popcount complexity

Termination is formalized, but the removed prose did not distinguish the following two possible complexity claims. For $e, a \in \mathbb{N}$ with $0 < a < 2^e$, let $D(e, a)$ be the number of recursive calls that enter the evaluator's interior branch.

Candidate statement 82.3 (Discharged). The exact-count candidate is $D(e, a) = \mathrm{wt}_2(a)$.

Candidate statement 82.4 (Discharged). The weaker upper-bound candidate is $D(e, a) \le \mathrm{wt}_2(a)$.

Formalization obligation 82.5 (Discharged). Define D in Lean, settle which statement matches the executable recursion, and prove the resulting relation to binary weight, including endpoint and saturation branches.

Both candidates are settled in `EvaluatorPopcount.lean`. The count is `dyadicInteriorCalls`, defined by literally the branch structure of `fabiusDyadicUnitAux` (the same guards and the same leading-bit-stripping recursion argument, so each interior entry is counted once). The verdict: *the exact count matches the executable recursion* — `dyadicInteriorCalls_eq_binaryWeight` proves $D(e, a) = \mathrm{wt}_2(a)$ for every $a < 2^e$, the endpoint $a = 0$ included, since each interior step strips exactly the leading binary digit (`binaryWeight_add_pow_two`). On the saturation branch $2^e \le a$ the evaluator answers without recursing, $D(e, a) = 0$ (`dyadicInteriorCalls_of_saturated`), and only there does the exact statement degrade to the inequality; the weaker candidate is the unconditional corollary `dyadicInteriorCalls_le_binaryWeight`. The all-ones worst case $D(e, 2^e - 1) = e$ is `dyadicInteriorCalls_two_pow_sub_one`, via `binaryWeight_two_pow_sub_one`.

82.3 Independent analytic derivations

Earlier prose derived the dyadic formula through Taylor integrals, block substitutions, and the exact remainder $-F(y)$. The final rational identities are formal, but those intermediate analytic equations are not public theorem-for-theorem counterparts.

Formalization obligation 82.6. Formalize named Taylor integral identities for the bounded and signed global functions, including the blockwise changes of variables and the exact remainder.

82.4 Global block fold for the signed evaluator

Let $n \in \mathbb{N}$, put $s = 2^n$, and for $a \in \mathbb{N}$ write

$$a = 2sB + R, \qquad 0 \le R < 2s, \qquad C = \begin{cases} R, & R \le s, \\ 2s - R, & R > s. \end{cases}$$

The executable definition performs exactly this quotient–remainder fold, but the following analytic formula is not exposed as a public theorem with these variables and hypotheses.

Candidate statement 82.7 (Triangular block reduction).

$$0 \le C \le s, \qquad \mathcal{F}\Big(\frac{a}{2^n}\Big) = (-1)^{\mathrm{wt}_2(B)} F\Big(\frac{C}{2^n}\Big).$$

Formalization obligation 82.8 (Discharged). Package the quotient–remainder decomposition, folded numerator, and Thue–Morse block sign as a public theorem about `extendedFabiusDyadicValue`; then compose it with `extendedFabiusDyadicValue_cast`. The evaluator definition, its cast theorem, and the private first-block fold used inside the proof do not themselves state the displayed result.

Both levels are now public in `GlobalBlockFold.lean`, with the register's explicit variables and hypotheses $a = 2sB + R$, $0 < a$, $R < 2s$: the executable fold is `extendedFabiusDyadicValue_block_fold`, the folded-numerator bound $0 \le C \le s$ is `blockFoldCore_le`, and the displayed analytic formula $\mathcal{F}(a/2^n) = (-1)^{\mathrm{wt}_2(B)} F(C/2^n)$, for every bounded Fabius function, is `extendedFabius_block_fold`, composed through `extendedFabiusDyadicValue_cast`, `fabiusDyadicUnit_eq_fabiusDyadic`, and `fabiusDyadic_cast`.

82.5 Least-common-denominator valuation

Let Δ_n be the least common multiple of the reduced denominators of $\mathrm{up}(a/2^n)$ over odd support numerators a. Lean proves that the reduced denominator of $F(2^{-n})$ divides Δ_n, and separately computes the two-adic valuation of $F(2^{-n})$. The deduction combining those facts is now a named theorem.

Candidate statement 82.9 (Discharged). For every $n \ge 1$,

$$\nu_2(\Delta_n) \ge \binom{n}{2} + 1 + \nu_2(n!).$$

This is `dyadicDenominator_padicVal_two_lower_bound` in `PaperStatements.lean`, which states the displayed inequality with exactly the hypothesis $1 \le n$. Its proof is the requested bridge: it passes from the divisibility supplied by `proposition_fifteen` to the natural two-adic valuation of Δ_n.

82.6 Rvachev rational-input wrapper

The formal API has exact reduced-rational `Option` wrappers for the bounded function and the signed global extension. For up it once had only an already decoded exponent–integer-numerator evaluator; the matching wrapper now exists.

Candidate statement 82.10 (Discharged). Define an exact rational-input evaluator for up that returns "not dyadic" exactly when the reduced denominator is not a power of two, and otherwise returns the rational value computed by `rvachevDyadic`.

The wrapper is `evalRvachevDyadic` in `Arithmetic.lean`, built on `dyadicExponent?` in the same shape as `evalFabiusDyadic` and `evalExtendedFabiusDyadic`. Both requested theorems are public: the recognition theorem `evalRvachevDyadic_eq_none_iff` in `DyadicCorrectness.lean`, and the correctness theorems `evalRvachevDyadic_eq_some_correct` and `evalRvachevDyadic_complete_correct` in `GlobalDyadic.lean`, the latter producing the rational value and its identification with analytic up for every dyadic rational input. The formalization is more general than the candidate asked for: the recognition theorem is an instance of a single evaluator-generic lemma quantified over an arbitrary $\mathbb{N} \to \mathbb{Z} \to \mathbb{Q}$ evaluator, so the same lemma also yields `evalFabiusDyadic_eq_none_iff` and `evalExtendedFabiusDyadic_eq_none_iff`.

83 Centered finite splines and local finiteness

83.1 Global positive-part form and a rejected raw-spline recurrence

Let S_p denote the finite-cutoff centered spline from (331). The following globally stated positive-part formula now has an exact public counterpart.

Candidate statement 83.1 (Discharged). For $p \geq 1$ and $x \in \mathbb{R}$,

$$S_p(x) = \frac{1}{2^{\binom{p}{2}} p!} \sum_{r \geq 0} (-1)^{\mathrm{wt}_2(r)} \left(2^p x - \frac{1}{2} - r\right)_+^p,$$

where the sum is pointwise finite.

Warning 83.2 (The formerly proposed raw recurrence is false globally). The identity

$$S_{p+1}(x) \stackrel{?}{=} \int_0^1 S_p(2x - u)\, \mathrm{d}u$$

is not a formalization target as stated. Substitution in the displayed positive-part formula at $p = 1$ and $x = 1$ gives $S_2(1) = 1$, whereas the proposed integral is $1/2$. The public Lean recurrence concerns the centered finite CDF, not this all-real raw spline.

Formalization obligation 83.3 (Discharged). Expose a public all-real theorem for the positive-part formula with its exact normalization, domain, and pointwise-finiteness interpretation. If a raw-spline smoothing law is desired, first formulate the correct identity and domain rather than reusing the refuted display above. The public declarations `uniformCenteredPartialCDF_eq_integral` and `fabiusUniformSpline_eq_centeredPartialCDF_all` are formal background; private fixed-range positive-spline lemmas and a bridge valid only on $[0, 1]$ do not state these global claims.

The all-real theorem is now public in `FabiusUniformSpline.lean`: `fabiusUniformSplin e_eq_tsum_positivePart` states, for $p \geq 1$ and every real x,

$$S_p(x) = \left(2^{\binom{p}{2}}\, p!\right)^{-1} \sum_{r \geq 0} (-1)^{\mathrm{wt}_2(r)} \left(2^p x - \tfrac{1}{2} - r\right)_+^p,$$

with $\sum_{r \geq 0}$ read as a `tsum` and the positive part as $\max(\,\cdot\,, 0)^p$; the pointwise-finiteness interpretation is its companion `positivePart_summand_support_finite`, which exhibits the summand's support inside $\{r < \lceil 2^p x \rceil\}$. The proof needs no fixed-range detour: left of the support both sides vanish, and for $x \geq 0$ the positive parts cut the `tsum` down to exactly the definitional prefix of the centered spline, the reversed power contributing the sign $(-1)^p$ absorbed by the normalizing constant. No raw-spline smoothing law is asserted, in accordance with the warning above.

83.2 What "a factor of two" is intended to compare

Let $U_p(x) = \Pr(T_p \leq x)$ be the uncentered finite CDF and $C_p(x) = U_p(x - 2^{-p-1})$ its midpoint correction.

Candidate statement 83.4 (Discharged). For every $p \in \mathbb{N}$ and $x \in \mathbb{R}$, the respective sandwich-and-Lipschitz arguments certify

$$|U_p(x) - F(x)| \leq 2^{1-p}, \qquad |C_p(x) - F(x)| \leq 2^{-p}.$$

Thus the centered construction halves this particular certified upper bound.

The two inequalities are `abs_uniformPartialCDF_sub_fabiusReal_le` and `abs_unif ormCenteredPartialCDF_sub_fabiusReal_le` in `FabiusComputability.lean`, each stated for every $p \in \mathbb{N}$ and every real x, and each deduced there from the sandwich theorems `uniformPartialCDF_sandwich` and `uniformCenteredPartialCDF_sandwich` together with the global Lipschitz bound. The factor-two comparison is `uniformCenteredPartialCDF_error_ bound_eq_half_uniformPartialCDF_error_bound` in the same module, whose documentation states the intended reading verbatim: it compares the bounds, not the actual pointwise errors, and makes no optimality claim.

83.3 Uniform CDF control and weak convergence

The centered-spline error estimate and pointwise CDF convergence are public Lean results. The removed prose also compared the uniform estimate with weak convergence for the same finite laws:

Candidate statement 83.5 (Discharged). For $p \geq 1$, let μ_p be the law of the finite weighted uniform-coordinate sum, translated to the right by 2^{-p-1}, and let μ be the weighted-infinite-sum law. Their CDFs are respectively C_p and F, and

$$\sup_{x \in \mathbb{R}} |C_p(x) - F(x)| \leq 2^{-p}.$$

On $[0, 1]$, the identification $C_p = S_p$ recovers the centered-spline estimate. Consequently μ_p converges weakly to μ, with the displayed uniform-CDF rate providing quantitative information beyond the resulting unquantified convergence statement.

Every part of this candidate is now public Lean. The translated finite law is `uniformCenteredPartialLaw` in `UniformSplineWeakConvergence.lean`, a probability measure by instance, with CDF identified as the midpoint-corrected finite CDF by `cdf_uniformCenteredPartialLaw`; the displayed all-real bound is `abs_uniformCenteredPartialCDF_sub_fabiusReal_le` in `FabiusComputability.lean`; and the weak convergence $\mu_p \to \mu$ in the probability-measure topology is `uniformCenteredPartialLaw_tendsto`, proved by Lévy's continuity theorem from the pointwise characteristic-function convergence `tendsto_charFun_uniformCenteredPartialLaw`, itself a dominated-convergence consequence of the pointwise partial-sum convergence `tendsto_uniformPartialSum`. As instructed below, the proof does not route through `finiteConvolutionProbability_tendsto`; the phrase *strictly stronger* is rendered as the neutral comparison above, so no nonconverse is required.

Formalization obligation 83.6 (Discharged). Bundle the translated finite laws and the limiting law as probability measures; identify their CDFs; combine the existing interval estimate with support facts to obtain the displayed all-real bound; and prove that the bound implies convergence in the probability-measure weak topology. If the phrase *strictly stronger* is retained, define the intended comparison and prove the required nonconverse as well. Do not use `finiteConvolutionProbability_tendsto` as this bridge: that theorem concerns a different sequence of symmetric two-atom convolution laws and the limiting measure with density up, not the continuous uniform-coordinate partial laws whose centered CDF is C_p.

83.4 Local finiteness of the signed translate family

For fixed $x \in \mathbb{R}$, consider

$$n \longmapsto (-1)^{\mathrm{wt}_2(n)}\, \mathrm{up}(x - 2n - 1).$$

Candidate statement 83.7 (Discharged). Its support in $\mathbb{N}$ is finite; equivalently, only finitely many translates in the defining series for $\mathcal{F}(x)$ are nonzero.

The finite-support statement is public as `extendedFabius_summand_hasFiniteSupport` in `GlobalExtension.lean`: for each fixed real x it asserts `Function.HasFiniteSupport` of the signed translate sequence itself, with local finiteness as its conclusion rather than as a step inside another proof. The public summability theorem `extendedFabius_summable` is now a corollary of it.

84 Step cells and corrected-prefix conventions

84.1 Exact histogram-cell tiling

Put

$$g_n = 2^{n+1} - n - 2, \qquad \lambda_{n,m} = \frac{2m - g_n}{2^{n+1}}, \qquad I_{n,m} = \left[\lambda_{n,m} - 2^{-n-1}, \lambda_{n,m} + 2^{-n-1}\right].$$

Candidate statement 84.1 (Discharged). Consecutive cells meet only at their common endpoint, and

$$\bigcup_{m=0}^{g_n} I_{n,m} = \left[-1 + \frac{n+1}{2^{n+1}}, 1 - \frac{n+1}{2^{n+1}}\right].$$

Formalization obligation 84.2 (Discharged). Package adjacency, finite interval coverage, and both endpoint calculations as a public Lean theorem. If "support" means $\mathrm{Function.support}$ of the step approximant rather than the union of its defining cells, also prove positivity of every in-range coefficient: the current support theorem is only an enclosure.

The package is public in `EarlyApproximants.lean`. Adjacency is `stepIntervalRight_eq_left_succ` together with `stepInterval_inter_succ`, which computes the intersection of consecutive cells as exactly the shared-endpoint singleton; coverage is `stepInterval_biUnion` (abutting cells tile one closed interval, for any top index); the endpoint calculations are `stepIntervalLeft_zero_eq` and `stepIntervalRight_degree_eq`; and the displayed tiling identity, packaged, is `stepInterval_tiling`. The conditional coefficient half is also done: `approximationPolynomial_coeff_pos` proves every in-range coefficient positive (each coefficient of the product of all-ones geometric factors receives at least one contributing monomial), upgrading the support enclosure to an exact description.

84.2 Why half-endpoint cells are needed

The public step approximant uses half weight at each cell endpoint. The elementary counterexample that motivated that convention was removed because it has not itself been packaged as a theorem:

Candidate statement 84.3 (Discharged). Modify the level-one step approximant by replacing its half-endpoint representatives with ordinary closed-interval indicators, leaving its coefficients and normalization unchanged. At $x = 0$, the cells $[-1/2, 0]$ and $[0, 1/2]$ then each contribute one, so the modified value is 2, rather than the intended value 1.

Formalization obligation 84.4 (Discharged). Define the ordinary-closed variant and prove its level-one value at zero. The public Lean API proves the endpoint values and adjacency law in `halfEndpointIntervalIndicator_self_left`, `halfEndpointIntervalIndicator_self_right`, and `halfEndpointIntervalIndicator_add_adjacent`, as well as the corrected value in `stepApproximant_one_zero`; it neither defines the ordinary-closed variant nor states the displayed double-counting result. A source comment explaining why the convention is useful is not a theorem.

The counterexample is now a theorem in `EarlyApproximants.lean`: `closedIntervalIndicator` is the naive representative, `naiveStepApproximant` the modified approximant with unchanged coefficients and normalization, and `naiveStepApproximant_one_zero` proves the displayed double-counting value 2 at the origin for $n = 1$, directly against `stepApproximant_one_zero`'s corrected value 1.

84.3 Independent roles of the order and coordinate corrections

The quantitative Thue–Morse synthesis already preserves both mechanisms in detail: the one-order coefficient shift and the centered affine coordinate. The stronger statement that the two repairs are independently necessary still lacks one exact theorem.

Candidate statement 84.5. For $k \geq 1$ and $0 \leq m < 2^k$, the order correction matches polynomial index $k-1$ with prefix row k:

$$[z^m]P_{k-1}(z) = \sigma_k(m),$$

whereas the centered coordinate $\lambda_{n,m} = (2m - g_n)/2^{n+1}$ independently repairs the geometric interpretation of that row. Retaining either naive convention leaves a defect that the other correction does not repair.

Formalization obligation 84.6. Package two separate failure witnesses—an explicit counterexample to matching P_k with prefix row k, and the endpoint obstruction for the naive grid coordinate—and combine them into a theorem saying that neither correction alone gives the corrected coefficient-and-coordinate interpretation.

85 Already-preserved audit dispositions

Two removed primary claims need no duplicate candidate here. Outcome-wise absolute convergence of the random coordinate series is preserved, with its geometric majorant and stronger uniform-convergence argument, in (15)–(16). Finite-Q dependence of the discrete rows and the warning that those rows are not partial sums of the binary or global q-series are preserved in section 23 and table 10. Their common-limit theorem is formalized; the finite-stage dependence and non-equality statements remain quarantined here.

86 Complex-shift spline estimate

The estimate cited by the primary exposition has the factor $\exp(|Q-\frac{1}{2}|)$. Put

$$N_p(x) = \max\left(0, \left\lfloor 2^p x + \frac{1}{2}\right\rfloor\right), \qquad S_p^{(Q)}(x) = \frac{(-1)^p}{2^{\binom{p}{2}} p!} \sum_{r=0}^{N_p(x)-1} (-1)^{\mathrm{wt}_2(r)} (r - 2^p x + Q)^p,$$

with an empty sum when $N_p(x) = 0$. The proposed sharpening was removed from the primary exposition, and has since been formalized:

Candidate statement 86.1 (Discharged). For every $p \in \mathbb{N}$ with $p \geq 1$, every $Q \in \mathbb{C}$, and every $x \in \mathbb{R}$,

$$\left| S_p^{(Q)}(x) - S_p^{(1/2)}(x) \right| \leq 2^{-(p-1)} \left(\exp\left(\left| Q - \frac{1}{2} \right| \right) - 1 \right).$$

Warning 86.2. The accompanying assertion that the relevant triangular exponent attains equality *exactly in the top term* is false when $p = 2$: both endpoint indices attain the same exponent. The formalization avoids the refuted claim by proving the exponent comparison as an inequality, from monotonicity of $d \mapsto \binom{d}{2}$, and never asserts a unique maximizing index.

The public $\exp -1$ estimate is `norm_fabiusComplexShiftSpline_sub_center_le_half_pow_mul_exp_sub_one_all` in `FabiusComplexShiftSpline.lean`, and it is more general than the candidate: it carries no $p \geq 1$ hypothesis, holding for every $p \in \mathbb{N}$ and every real x, with the naturally truncated exponent making the degenerate degree an instance rather than an exception. Its finite Taylor majorant is formalized separately, as the exact branch expansion `fabiusComplexShiftSpline_sub_center_eq_taylorBranches` in the same module together with the reflected factorial sum bounded in `FabiusDiscreteLimitComplexShift.lean`. The weaker $\exp$ form `norm_fabiusComplexShiftSpline_sub_center_le_half_pow_mul_exp_all` is derived from it.

87 Prefix-kernel interpretations

The binomial kernel and the resulting prefix-sum formula are formalized. These two interpretations are not:

Candidate statement 87.1.

- for $i, k \in \mathbb{N}$ with $k \geq 1$, $\binom{i+k-1}{k-1}$ counts ordered k-tuples of nonnegative integers summing to i;
- for $k \in \mathbb{N}$ with $k \geq 1$, the kernel is the k-fold discrete convolution power of the constant kernel.

Formalization obligation 87.2 (Discharged). Choose a finite-support or locally finite convolution model, prove the counting bijection, and connect it to the existing `iteratedPrefixKernel`.

Both interpretations are now theorems in `PrefixKernelCounting.lean`, in the finite-support model: the tuples form `Finset.Nat.antidiagonalTuple` and the convolution is the finite antidiagonal sum `natConv`. The counting bijection is `card_antidiagonalTuple` (stars and bars for weak compositions: $\binom{n+k}{k}$ ordered $(k+1)$-tuples of naturals summing to n), proved from the first-coordinate decomposition `antidiagonalTuple_succ_eq` and the hockey-stick identity; `constKernelConvPow_eq_card` identifies the k-fold convolution power of the constant kernel with the tuple count; and the connections to the kernel are `iteratedPrefixKernel_eq_card_antidiagonalTuple` and `iteratedPrefixKernel_eq_constKernelConvPow`, for every $k \geq 1$.

88 An actual infinite formal product

Lean proves coefficient stabilization of the finite products $\prod_{j<r}(1 - z^{2^j})$. The stronger construction formerly asserted was:

Candidate statement 88.1. Construct $\prod_{j\geq 0}(1 - z^{2^j})$ as a formal infinite product and prove that it equals the Thue–Morse power series.

Formalization obligation 88.2 (Discharged). Select an appropriate formal-power-series topology or summability API, define the product, and prove equality with the coefficient-stabilized series. Until then, the notation is only shorthand for finite stabilization.

The product is now genuine, in `ThueMorseFormalProduct.lean`. The chosen API is Mathlib's coefficientwise topology on $\mathbb{Z}[[X]]$ (`PowerSeries.WithPiTopology`), in which convergence is exactly coefficient stabilization. `hasProd_one_sub_X_two_pow` proves that the partial products over *arbitrary* finite sets of factors converge to the Thue–Morse series, `multipliable_one_sub_X_two_pow` packages multipliability, and `tprod_one_sub_X_two_pow` is the display: $\prod'_{j\geq 0}(1-X^{2^j})$ *equals* the Thue–Morse series as a `tprod`. The engine `coeff_prod_one_sub_X_pow` is proved over an arbitrary commutative ring and arbitrary exponent families: a finite product of factors $1 - X^{m_j}$ with all exponents above d is indistinguishable from 1 in every coefficient up to d.

89 Formal-background crosswalk

The following declarations support the proved background cited above. None of them is itself a counterpart for the adjacent candidate statement; where a candidate has since acquired an exact counterpart, that declaration is named inline with the candidate and is not repeated in this table.

Background	Lean declaration and module
Polynomially weighted Fourier decay	`rvachevFourier_real_rapidDecay` in `PoissonSummation.lean`
Even moments and exact dyadic evaluation	`moment`, `fabiusDyadicValue`, `fabiusDyadicValue_cast`, `fabiusDyadic_cast` in `Arithmetic.lean`, `PaperStatements.lean`, and `DyadicAnalytic.lean`
Complex-shift exponential bound	`norm_fabiusComplexShiftSpline_sub_center_le_half_pow_mul_exp` in `FabiusComplexShiftSpline.lean`
Prefix kernel and convolution formula	`iteratedPrefixKernel` and `iteratedPrefix_eq_sum_kernel` in `ThueMorsePrefix.lean`
Even/odd Thue–Morse recurrences	`binaryWeight_two_mul` and `binaryWeight_two_mul_add_one` in `DyadicClosedForm.lean`
Finite block product and coefficient stabilization	`thueMorseBlockPolynomial_eq_product` and `coeff_finite_thueMorse_product` in `ThueMorseGenerating.lean`
Signed dyadic evaluator and analytic cast	`extendedFabiusDyadicValue` and `extendedFabiusDyadicValue_cast` in `Arithmetic.lean` and `GlobalDyadic.lean`
Dyadic denominator and two-adic inputs	`proposition_fifteen`, `dyadicDenominator_eq_fabiusDyadicDenominator`, and `fabiusAtInverseTwoPow_padicVal_two_closed` in `PaperStatements.lean` and `TwoAdic.lean`
Dyadic recognition and decoded Rvachev evaluation	`dyadicExponent?_exists_iff` and `rvachevDyadic_cast_global` in `DyadicCorrectness.lean` and `OriginalPaperSupplement.lean`
Centered finite-CDF integral and spline bridge	`uniformCenteredPartialCDF_eq_integral` and `fabiusUniformSpline_eq_centeredPartialCDF_all` in `FabiusUniformSpline.lean`
Uncentered and centered probability sandwiches	`uniformPartialCDF_sandwich` and `uniformCenteredPartialCDF_sandwich` in `FabiusUniformSpline.lean`
Centered uniform-CDF control and pointwise convergence	`abs_fabiusUniformSpline_sub_fabiusReal_le_all`, `uniformCenteredPartialCDF_tendsto_weightedSumCDF`, and `weightedSumCDF_eq_fabiusReal` in `FabiusComputability.lean`, `FabiusUniformSpline.lean`, and `ProbabilityRepresentation.lean`
Separate symmetric-atomic-law weak limit	`finiteConvolutionProbability_tendsto` and `finiteConvolutionProbability_tendsto_fabius` in `WeakConvergence.lean`
Signed translate summability and one-block evaluation	`extendedFabius_summable` and `extendedFabius_eq_single_translate` in `GlobalExtension.lean`
Step-approximant support enclosure and cell coordinates	`support_stepApproximant_subset`, `polynomialAtomLocation`, and `stepApproximant_at_polynomialAtomLocation` in `StepApproximationLimit.lean`, `EarlyApproximants.lean`, and `ThueMorseApproximation.lean`
Half-endpoint cell gluing and the level-one value	`halfEndpointIntervalIndicator_self_left`, `halfEndpointIntervalIndicator_self_right`, `halfEndpointIntervalIndicator_add_adjacent`, and `stepIntervalRight_eq_left_succ`, together with `stepApproximant_one_zero`, in `EarlyApproximants.lean` and `StepApproximationLimit.lean`
Corrected prefix order and centered sampling	`iteratedPrefix_eq_approximationPolynomial_coeff`, `correctedPrefixCoefficient_eq_stepApproximant`, and `correctedPrefixGridSample_eq_stepApproximant` in `ThueMorseApproximation.lean`

90 Endpoint material in the consolidated volume

The combined inverse and small-argument saddle synthesis preserves the removed raw Mellin-inversion derivation, exponential Gamma–zeta decay estimates, numerical periodic profiles, proposed ordered-composition formulas for saddle coefficients, explicit higher Lambert polynomials, a proposed leading-derivative law, uniqueness and phase-density arguments, half-moment asymptotics, and strengthened endpoint comparisons.

The primary article retains only the exact negative-Laplace decomposition, the proved Gamma–

zeta Fourier series, the evaluated constant, lower-Lambert reduction, Bromwich inversion, the first two periodic saddle corrections, and the recursive all-orders theorem.

91 Provenance

This register was created from the claim-by-claim audit of the former drafts and primary exposition on 25 August 2026. Related numerical and exploratory material is preserved in the dyadic, q-binomial, integration, repeated-integration, and combined inverse/small-argument syntheses of this same research-frontier volume.

Consolidated promotion rule

When a frontier claim is formalized, promotion is claim-by-claim rather than source-by-source. The required evidence is an exact current Lean declaration whose hypotheses, normalization, domain, and conclusion match the mathematical sentence. The primary exposition should cite that name and module, while this volume should remove or relabel the discharged obligation without erasing any still-unformalized surrounding argument. Until that audit is complete, the claim remains here.